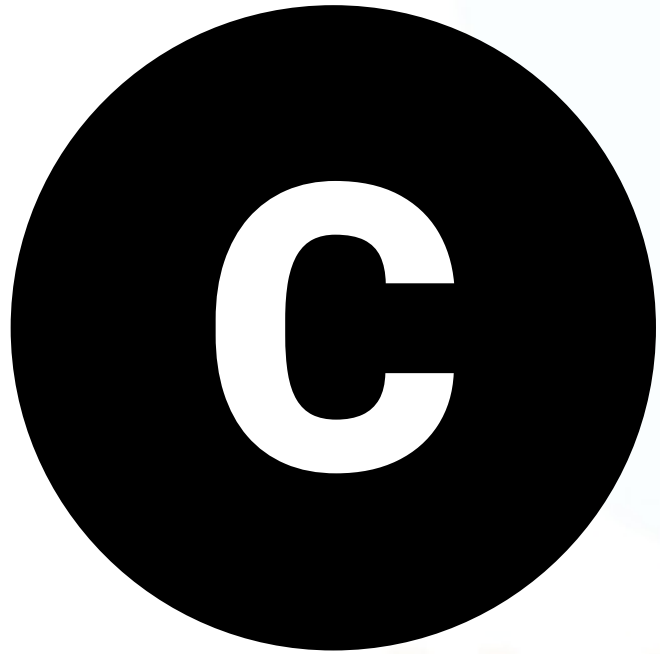


**Organic = C-H / C-C bonds**

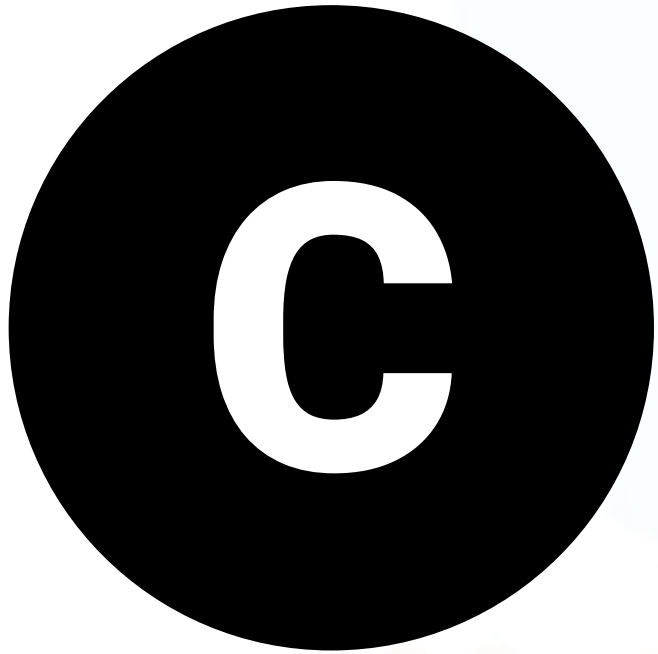
Unit Education

**Organic = C-H / C-C bonds**

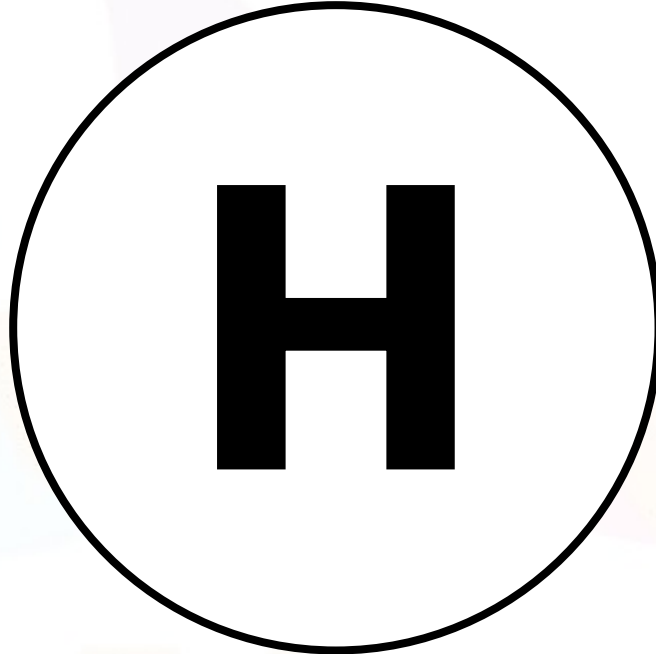


**Carbon**

**Organic = C-H / C-C bonds**

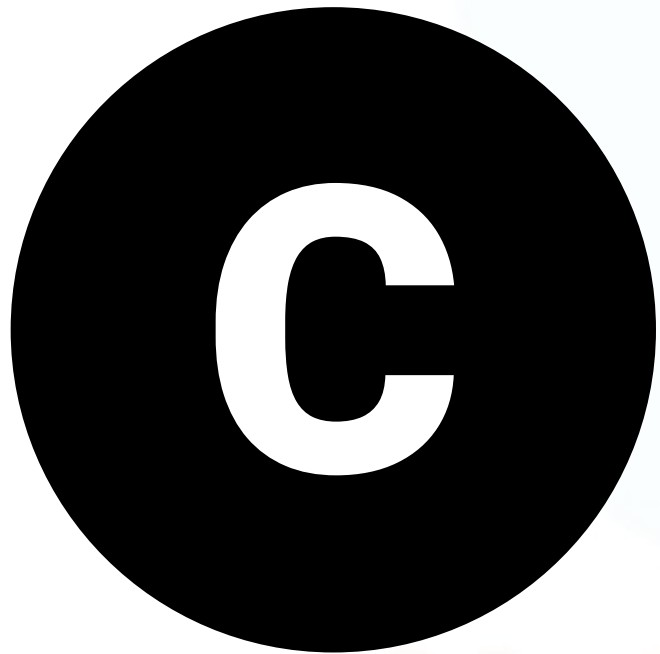


**Carbon**

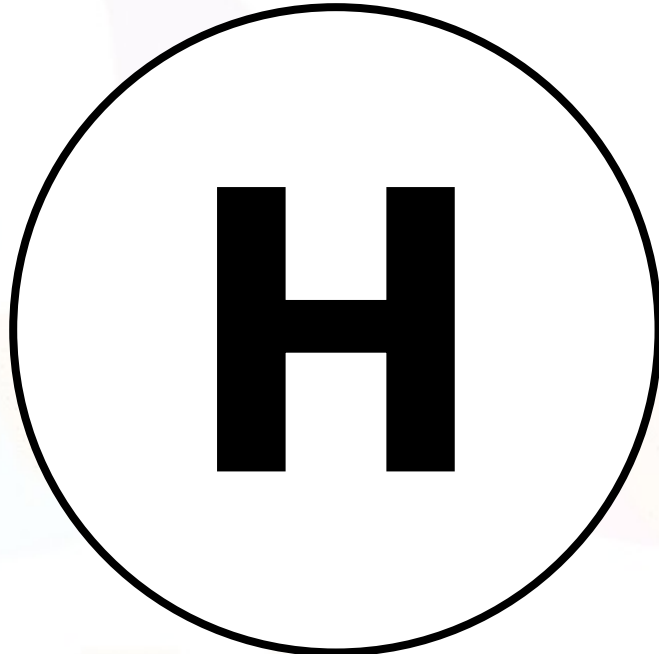


**Hydrogen**

**Organic = C-H / C-C bonds**



**Carbon**



**Hydrogen**



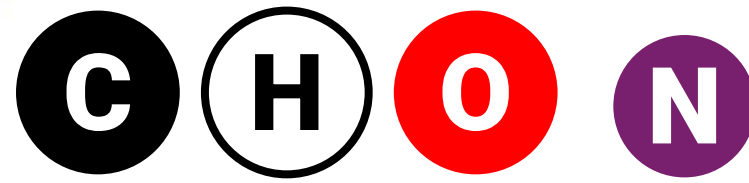
**Oxygen**

# Organic biomolecules



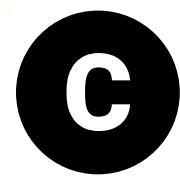
Unit Education

# Organic biomolecules



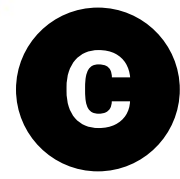
Unit Education

# Organic biomolecules



Unit Education

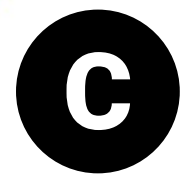
# Organic biomolecules



Unit Education



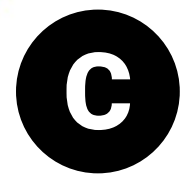
# Organic biomolecules



Nitrogen

Unit Education

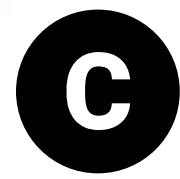
# Organic biomolecules



Nitrogen Phosphorus

Unit Education

# Organic biomolecules



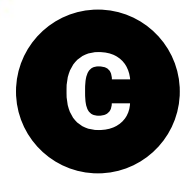
Nitrogen

Phosphorus

Sulfur

Unit Education

# Organic biomolecules

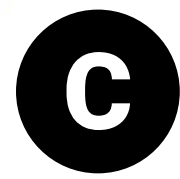


Phosphorus

Sulfur

Unit Education

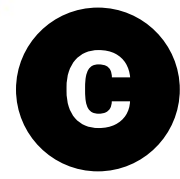
# Organic biomolecules



Sulfur

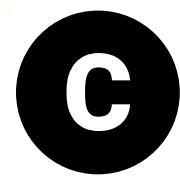
Unit Education

# Organic biomolecules



Unit Education

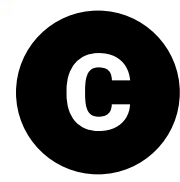
# Organic biomolecules



## 1. Carbohydrates 碳水化合物

Unit Education

# Organic biomolecules



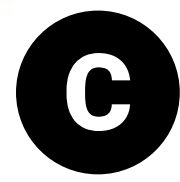
**1. Carbohydrates** 碳水化合物

**2. Lipids** 脂質

Unit Education



# Organic biomolecules



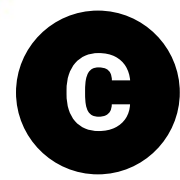
**1. Carbohydrates** 碳水化合物

**2. Lipids** 脂質

**3. Proteins** 蛋白質

Unit Education

# Organic biomolecules



**1. Carbohydrates** 碳水化合物

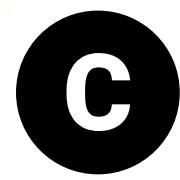
**2. Lipids** 脂質

**3. Proteins** 蛋白質

**4. Nucleic acids** 核酸

Unit Education

# Organic biomolecules



**1. Carbohydrates** 碳水化合物

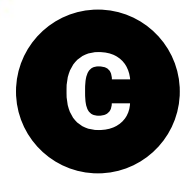
**2. Lipids** 脂質

**3. Proteins** 蛋白質

**4. Nucleic acids** 核酸

Unit Education

# Organic biomolecules



**1. Carbohydrates** 碳水化合物

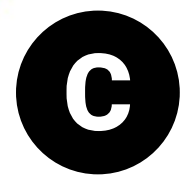
**2. Lipids** 脂質

**3. Proteins** 蛋白質

**4. Nucleic acids** 核酸

Unit Education

# Organic biomolecules



**1. Carbohydrates** 碳水化合物

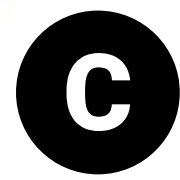
**2. Lipids** 脂質

**3. Proteins** 蛋白質

**4. Nucleic acids** 核酸

Unit Education

# Organic biomolecules



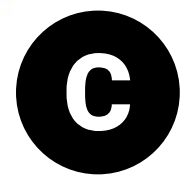
**1. Carbohydrates** 碳水化合物

**2. Lipids** 脂質

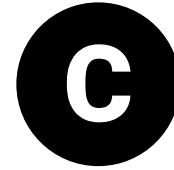
**3. Proteins** 蛋白質

**4. Nucleic acids** 核酸

# Organic biomolecules



**1. Carbohydrates** 碳水化合物

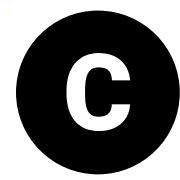


**2. Lipids** 脂質

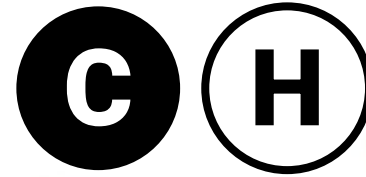
**3. Proteins** 蛋白質

**4. Nucleic acids** 核酸

# Organic biomolecules



**1. Carbohydrates** 碳水化合物



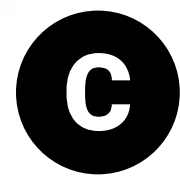
**2. Lipids** 脂質

**3. Proteins** 蛋白質

**4. Nucleic acids** 核酸



# Organic biomolecules



1. **Carbohydrates** 碳水化合物

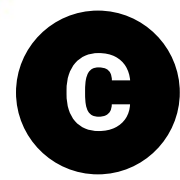


2. **Lipids** 脂質

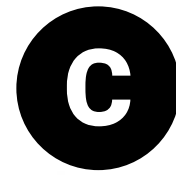
3. **Proteins** 蛋白質

4. **Nucleic acids** 核酸

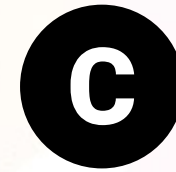
# Organic biomolecules



**1. Carbohydrates** 碳水化合物



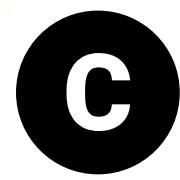
**2. Lipids** 脂質



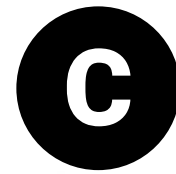
**3. Proteins** 蛋白質

**4. Nucleic acids** 核酸

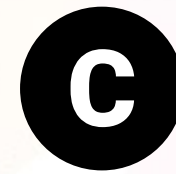
# Organic biomolecules



1. Carbohydrates 碳水化合物



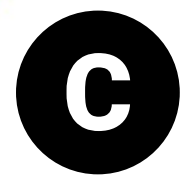
2. Lipids 脂質



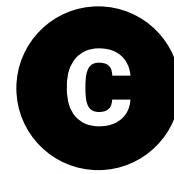
3. Proteins 蛋白質

4. Nucleic acids 核酸

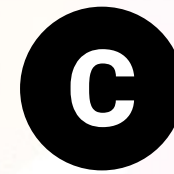
# Organic biomolecules



1. Carbohydrates 碳水化合物



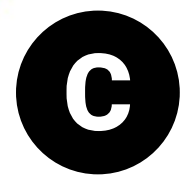
2. Lipids 脂質



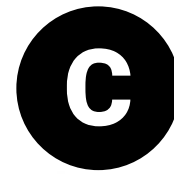
3. Proteins 蛋白質

4. Nucleic acids 核酸

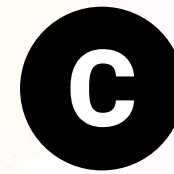
# Organic biomolecules



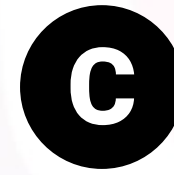
1. Carbohydrates 碳水化合物



2. Lipids 脂質

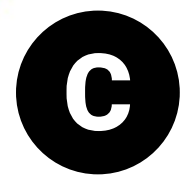


3. Proteins 蛋白質

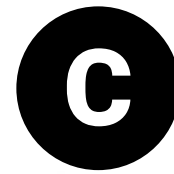


4. Nucleic acids 核酸

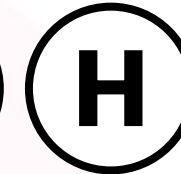
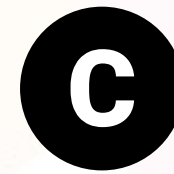
# Organic biomolecules



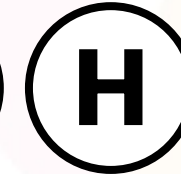
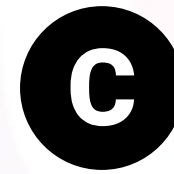
1. Carbohydrates 碳水化合物



2. Lipids 脂質

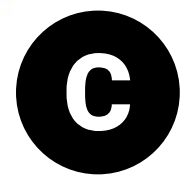


3. Proteins 蛋白質



4. Nucleic acids 核酸

# Organic biomolecules



1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質



4. Nucleic acids 核酸

# Organic biomolecules



1. Carbohydrates 碳水化合物



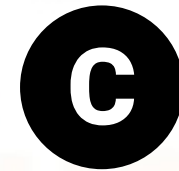
2. Lipids 脂質



3. Proteins 蛋白質



4. Nucleic acids 核酸





# Organic biomolecules



1. Carbohydrates 碳水化合物



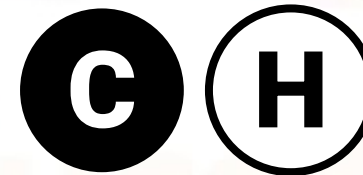
2. Lipids 脂質



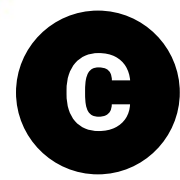
3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



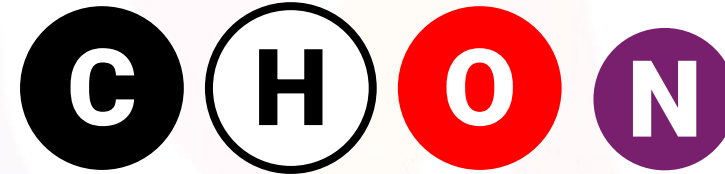
1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質



4. Nucleic acids 核酸



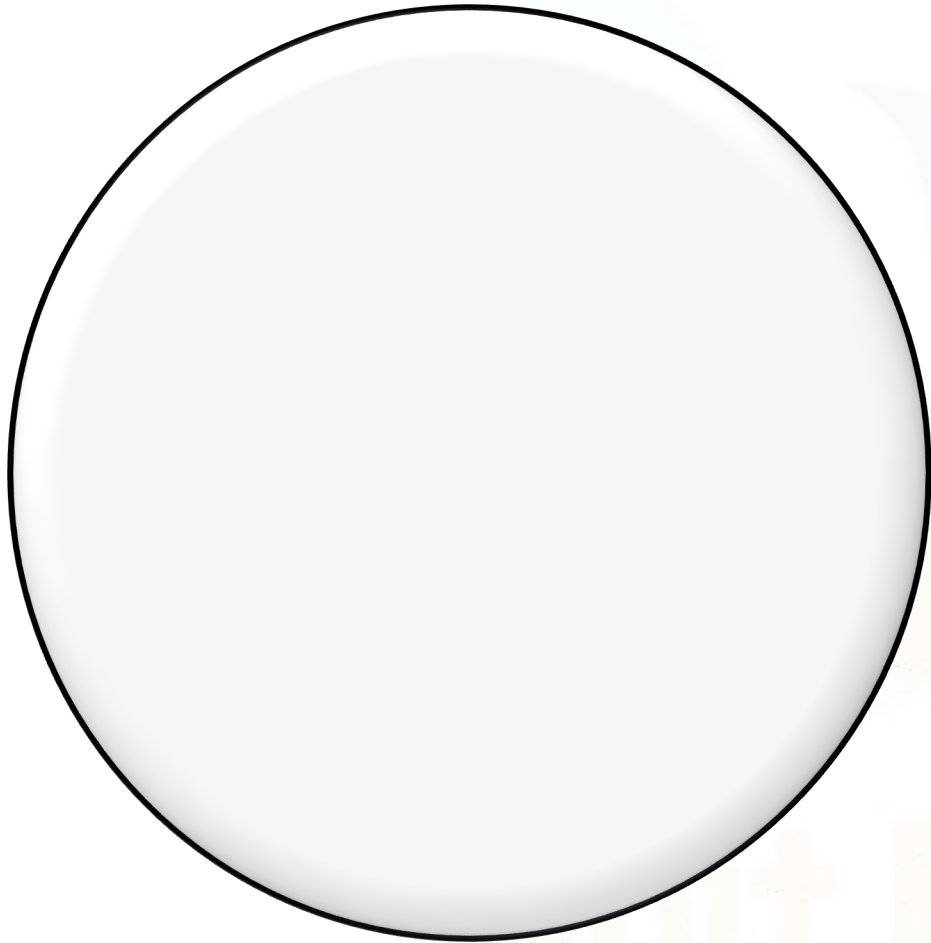
# Condensation 縮合 and Hydrolysis 水解



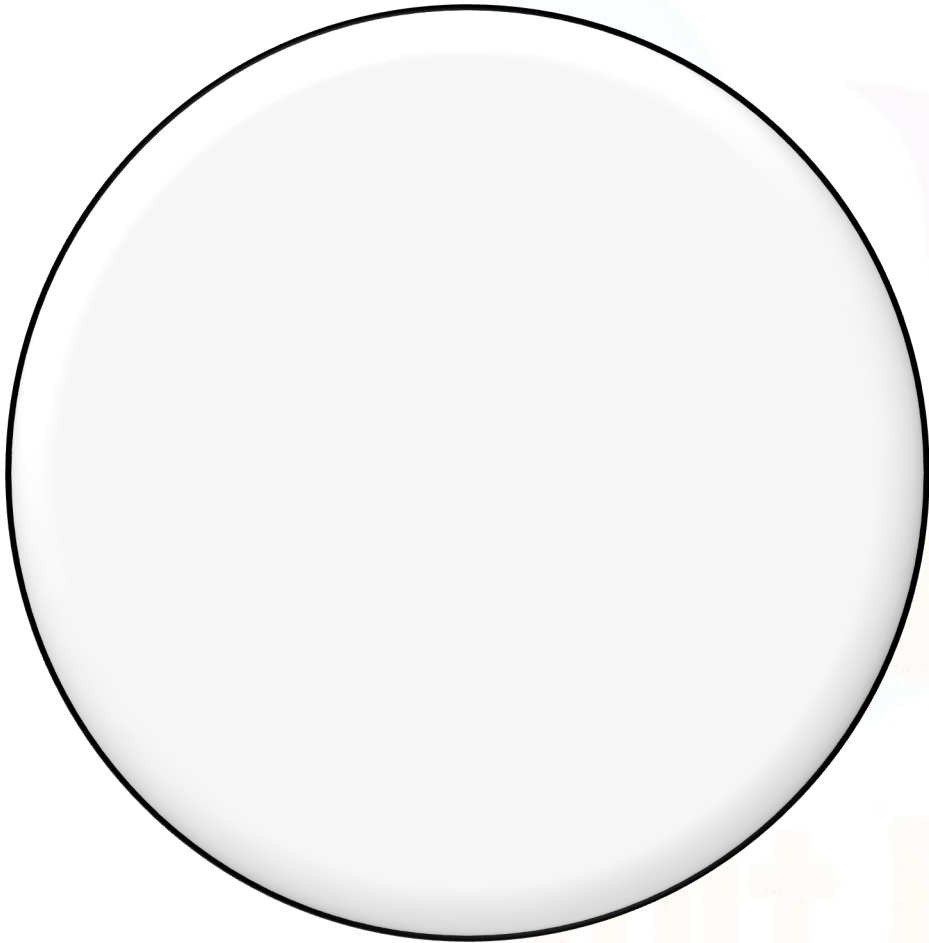
Unit Education



# Condensation 縮合 and Hydrolysis 水解

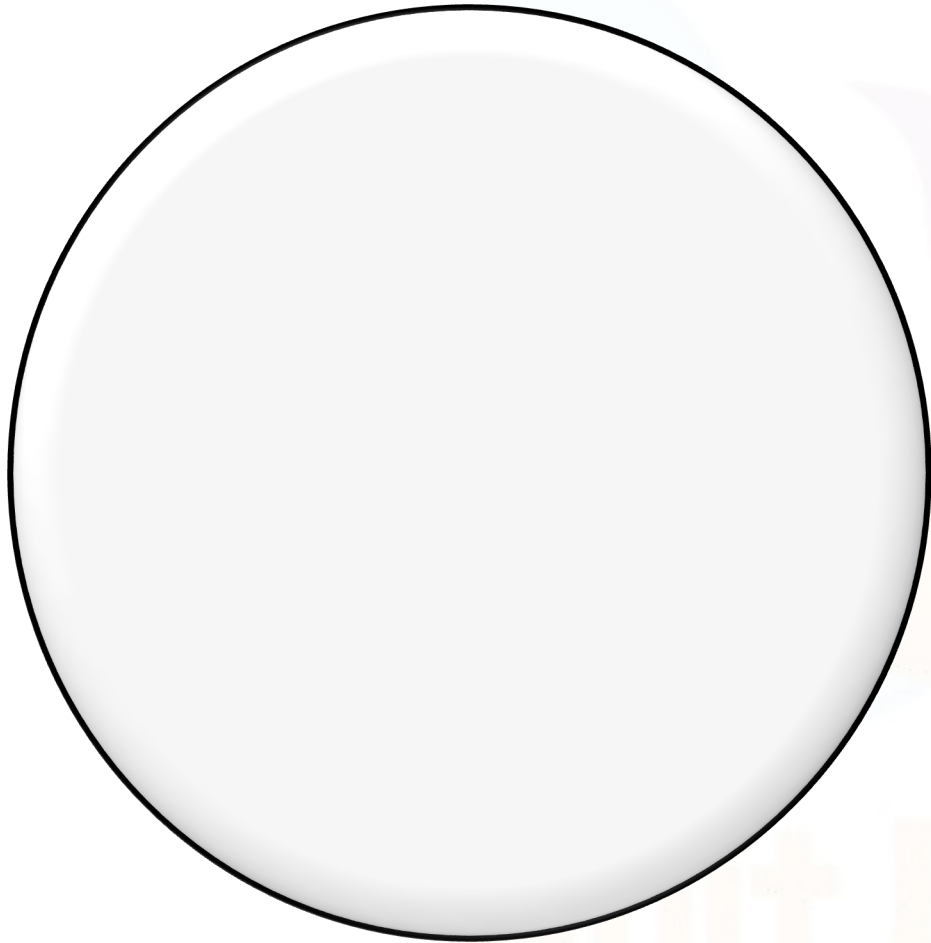


# Condensation 縮合 and Hydrolysis 水解

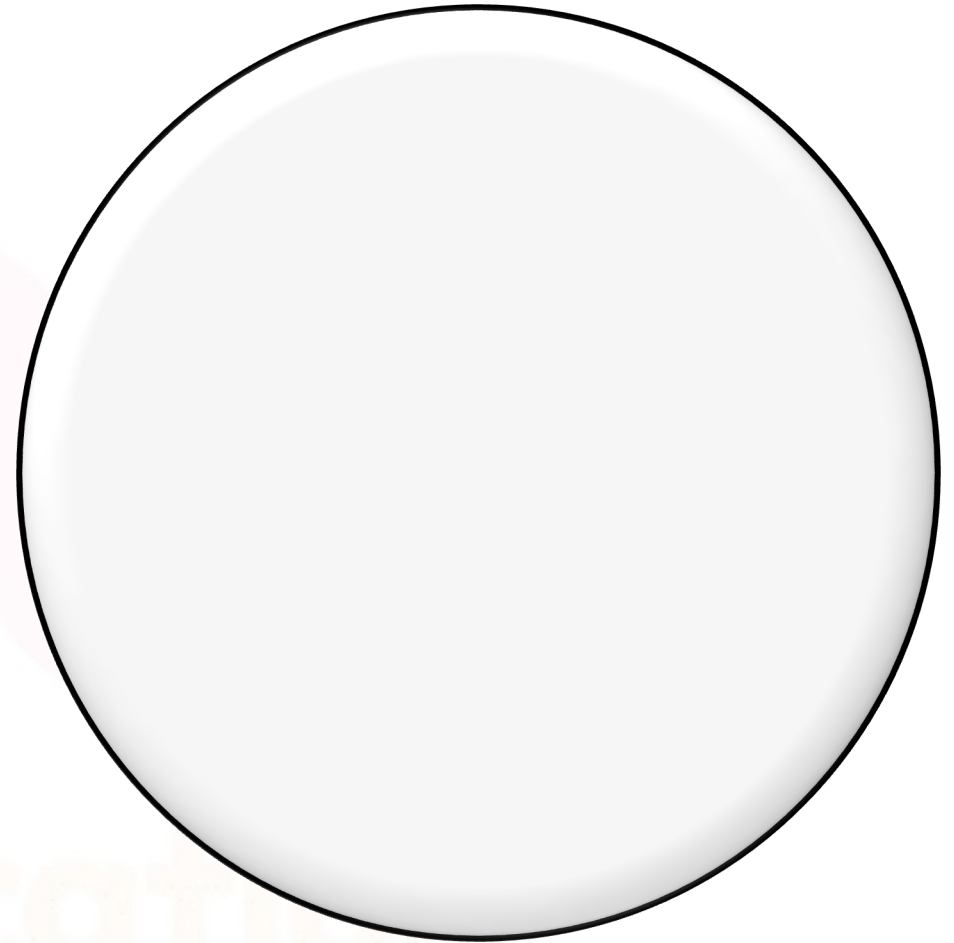


**Mono單-mer**

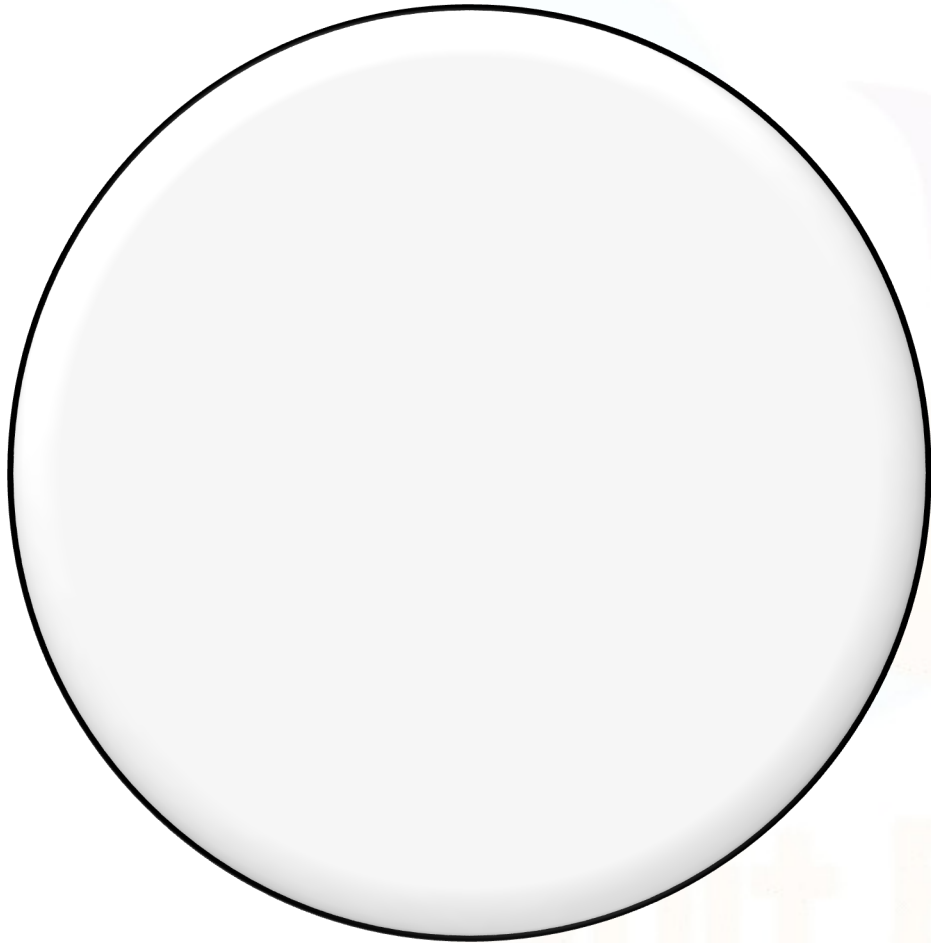
# Condensation 縮合 and Hydrolysis 水解



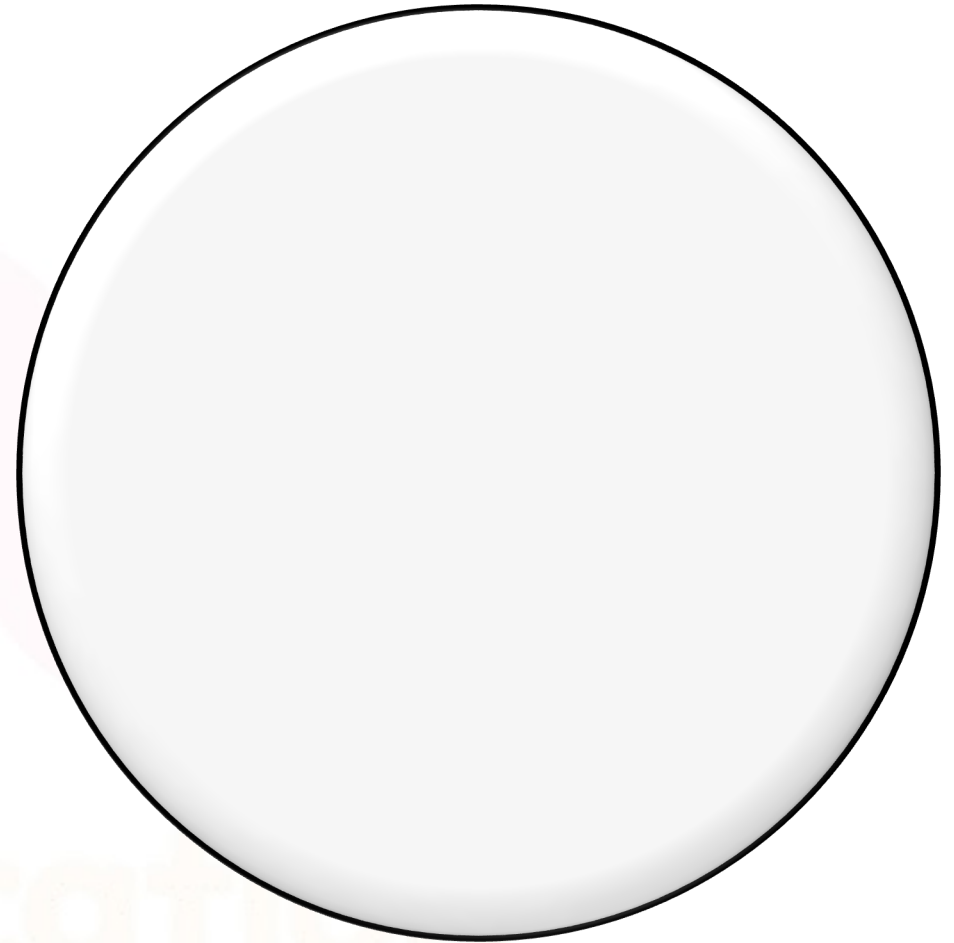
**Mono單-mer**



# Condensation 縮合 and Hydrolysis 水解

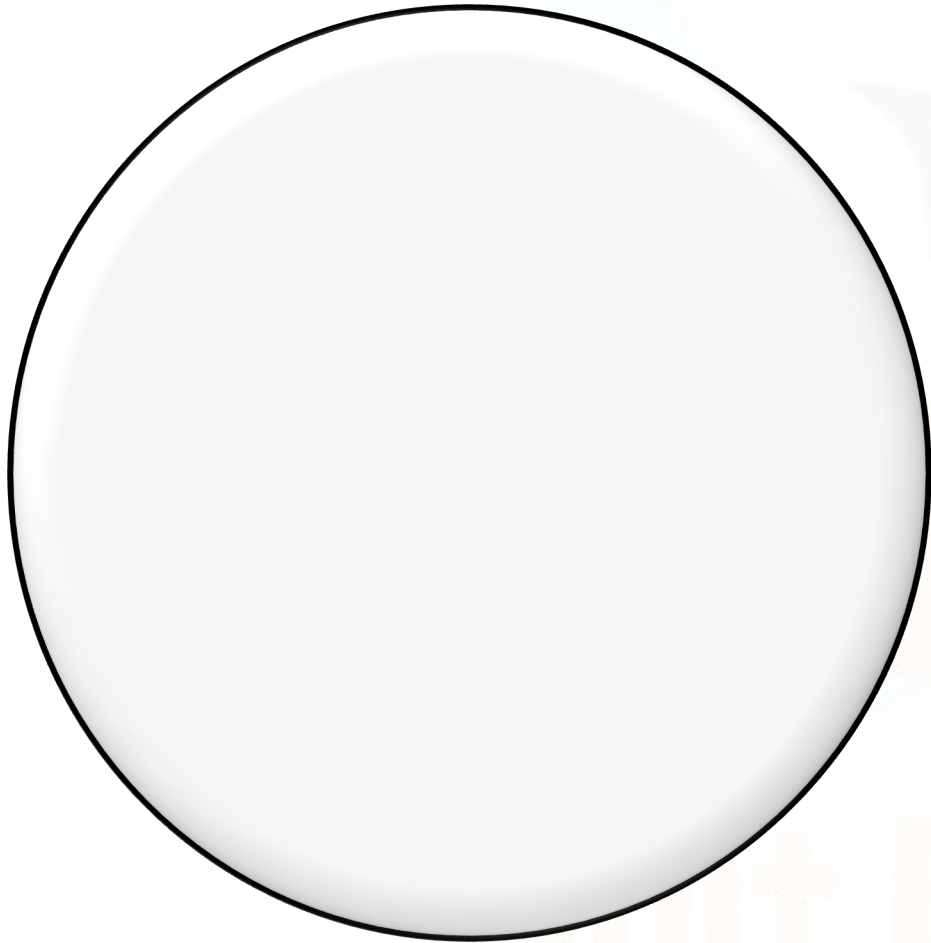


**Mono單-mer**

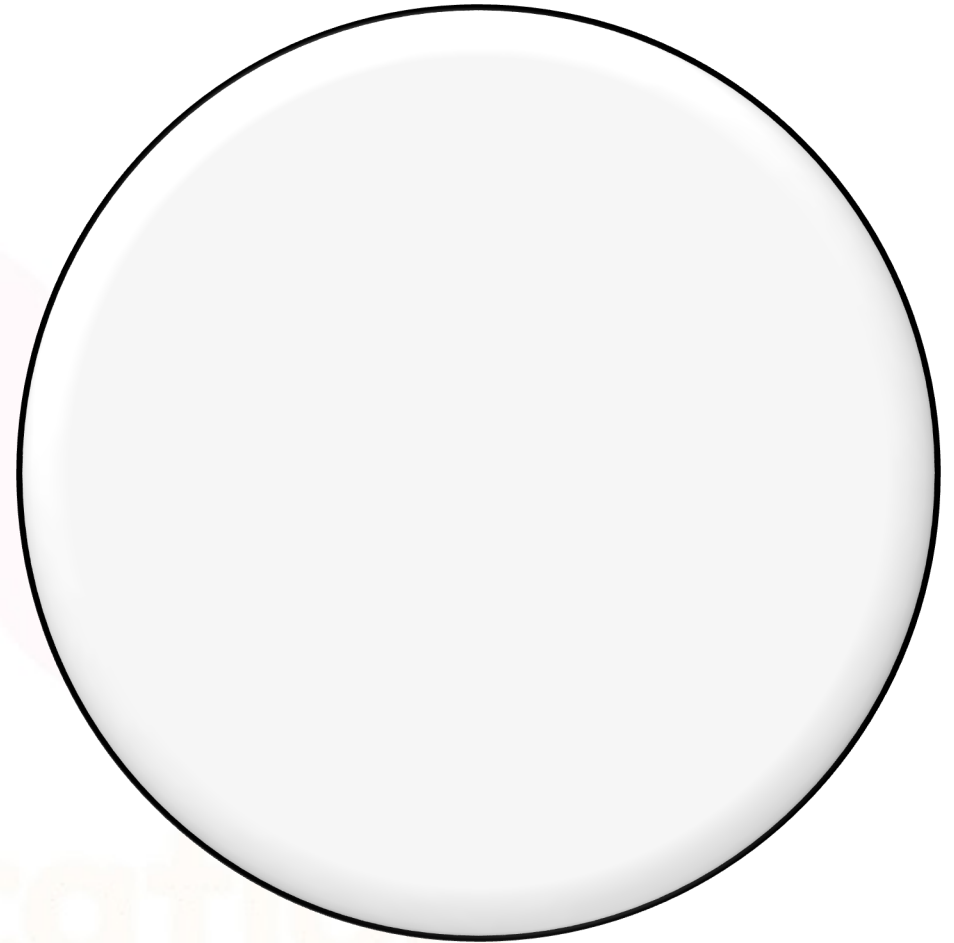


**Mono單-mer**

# Condensation 縮合

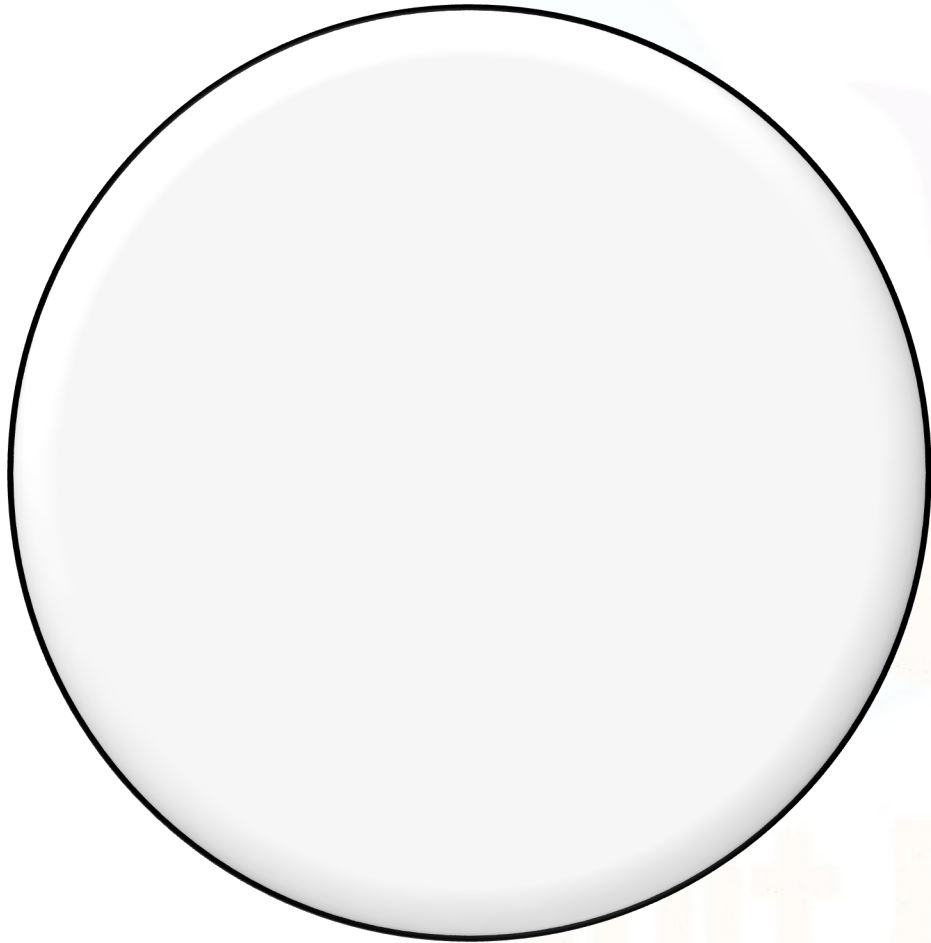


**Mono(單)-mer**

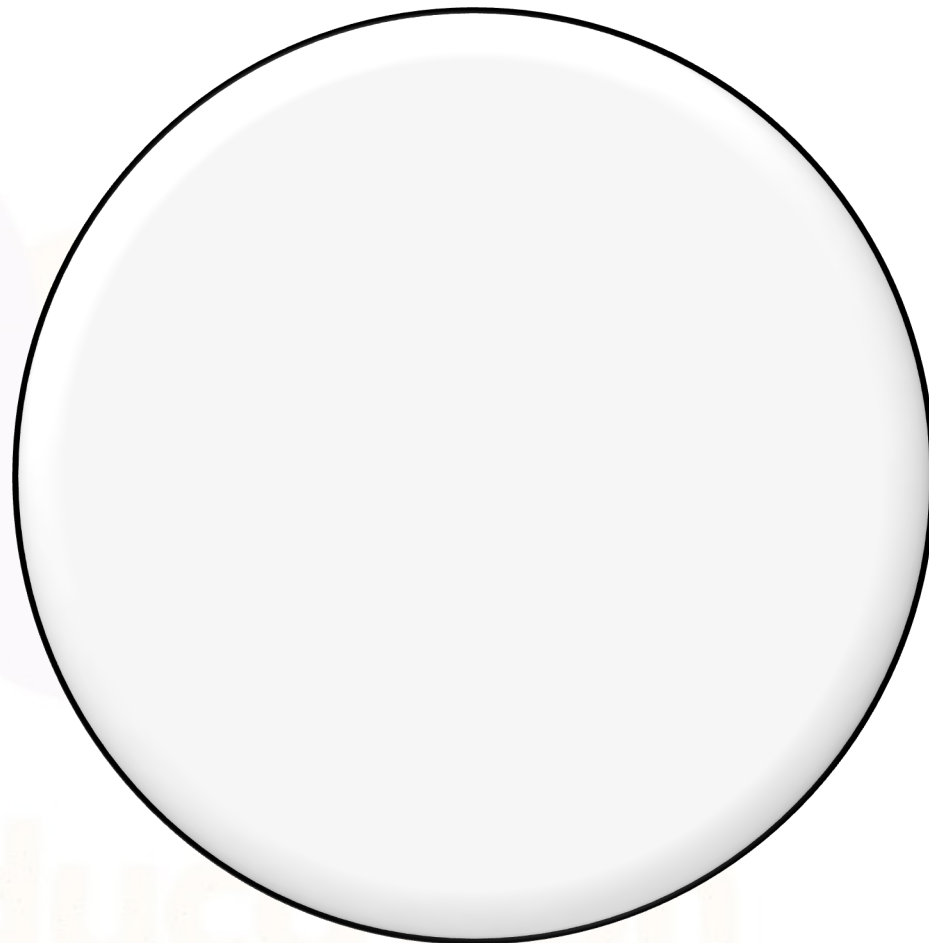


**Mono(單)-mer**

# Condensation 縮合

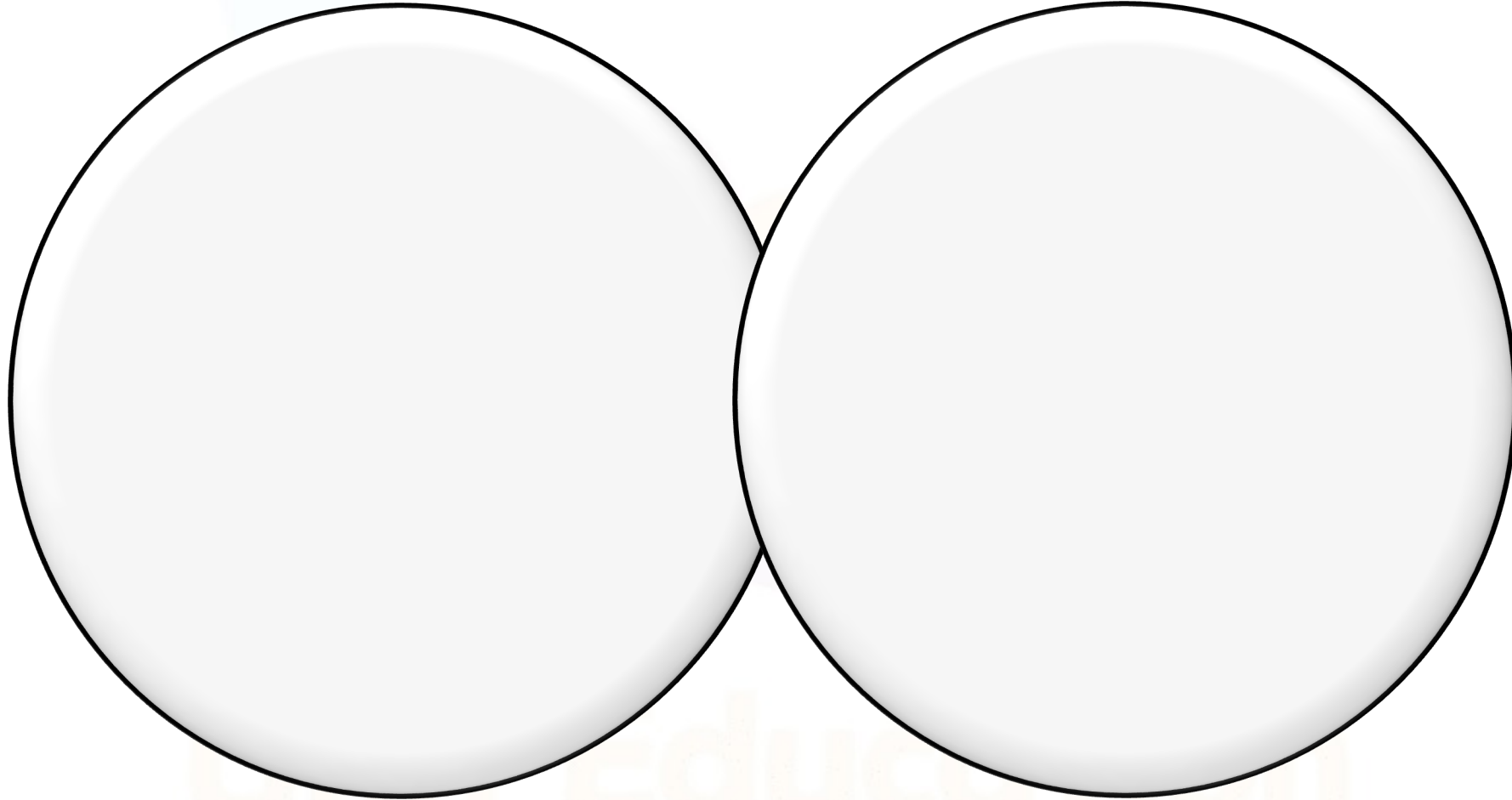


**Mono(單)-mer**



**Mono(單)-mer**

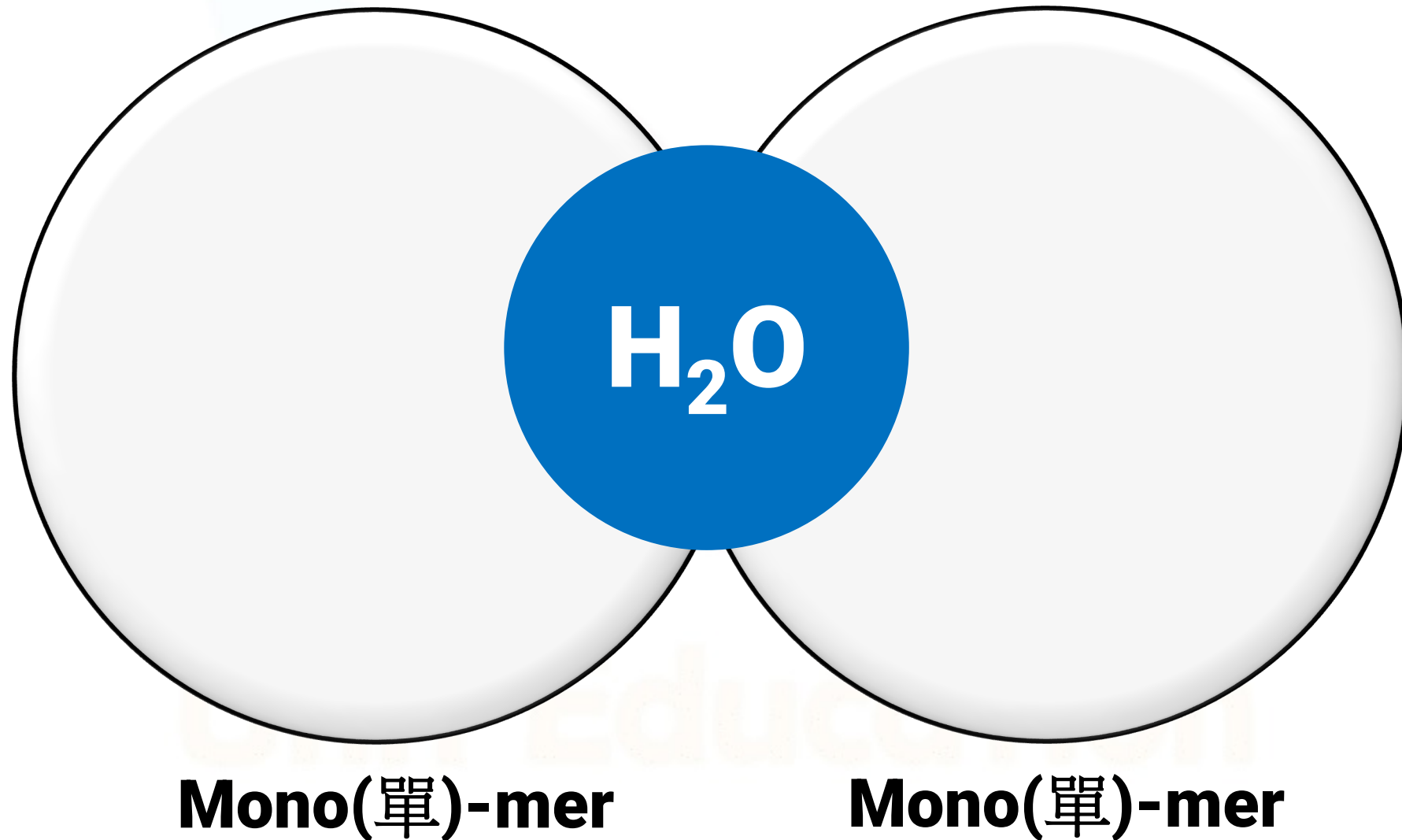
# Condensation 縮合



**Mono(單)-mer**

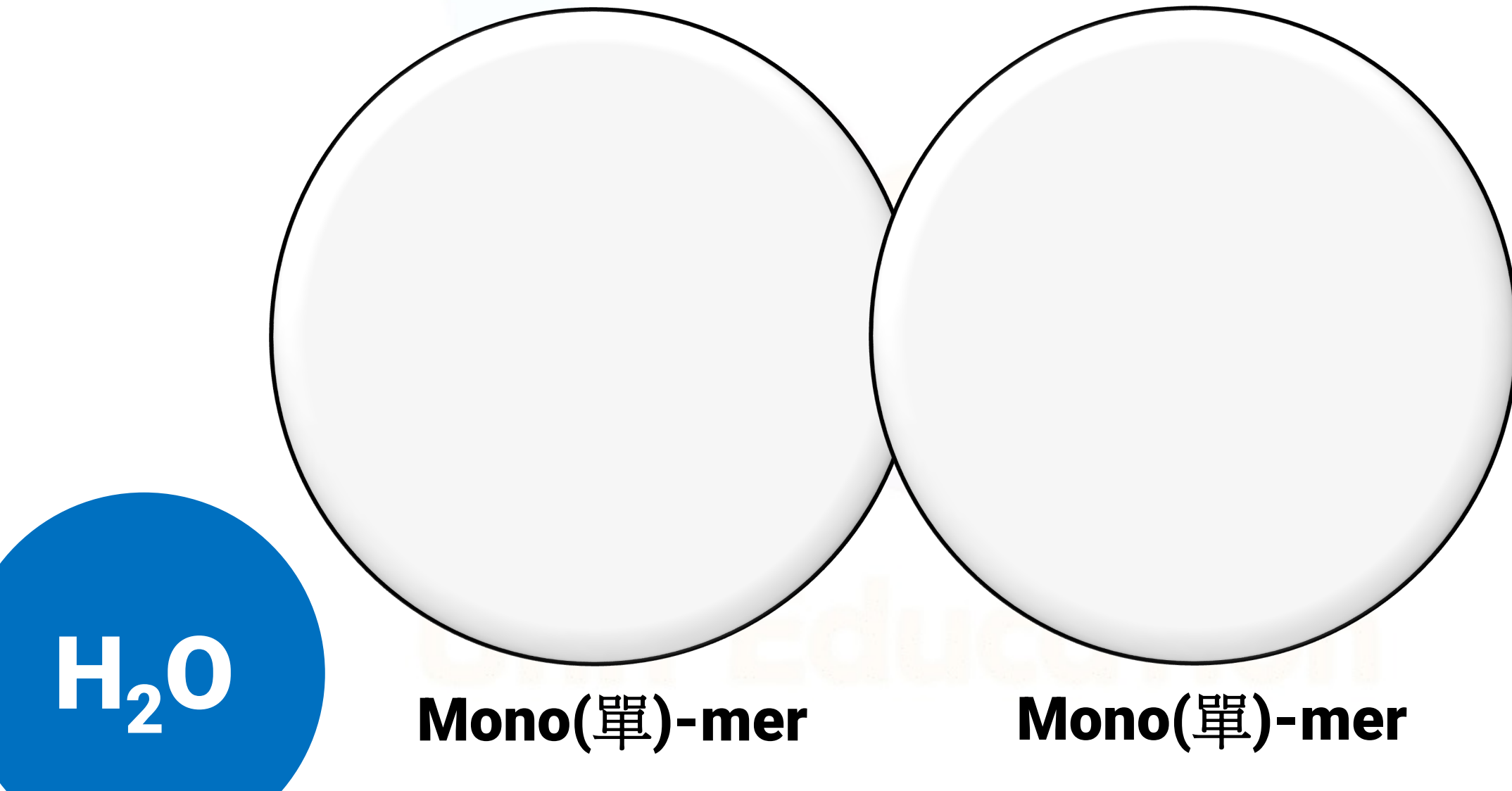
**Mono(單)-mer**

# Condensation 縮合

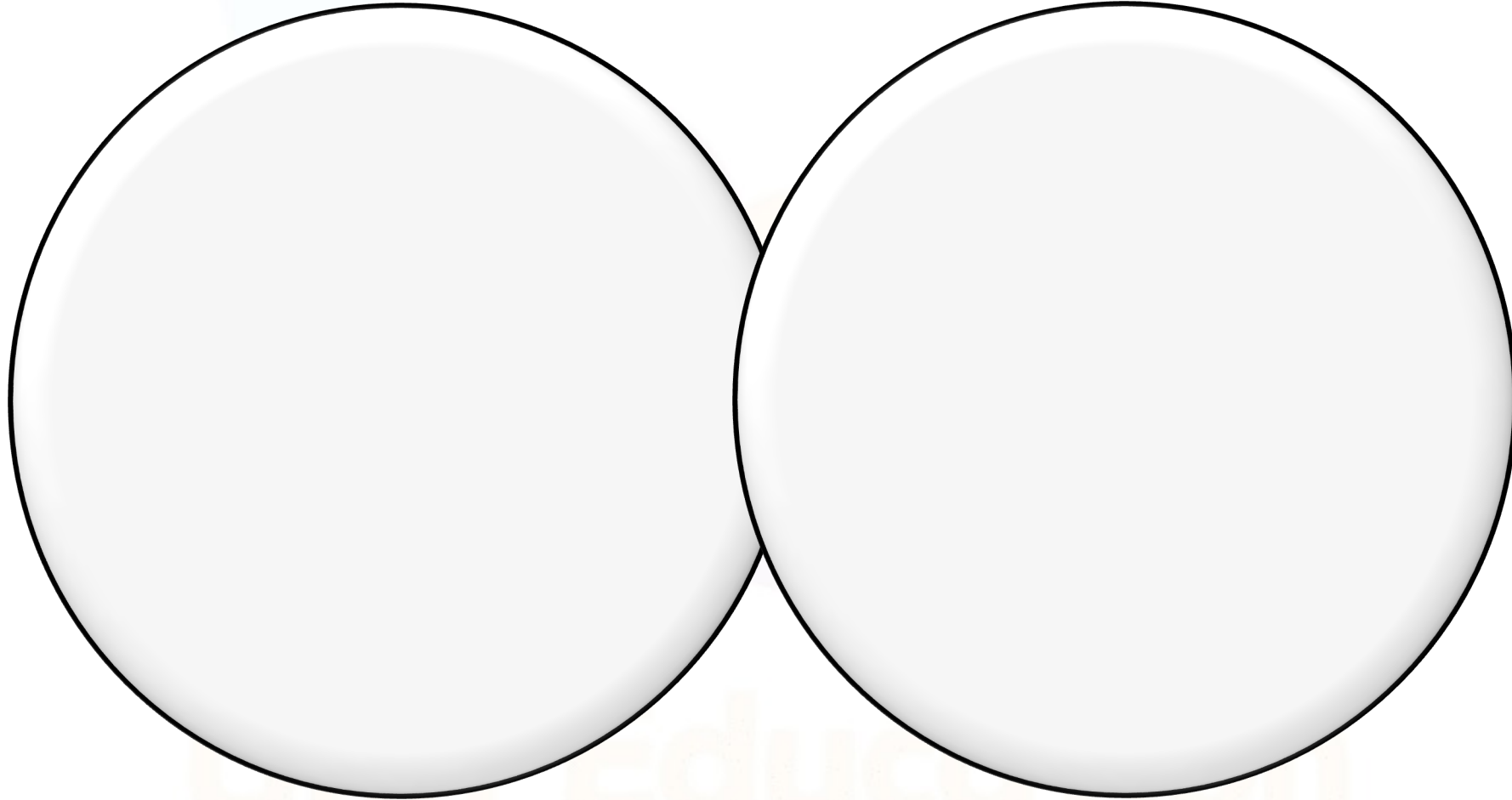
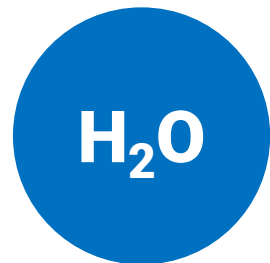




# Condensation 縮合



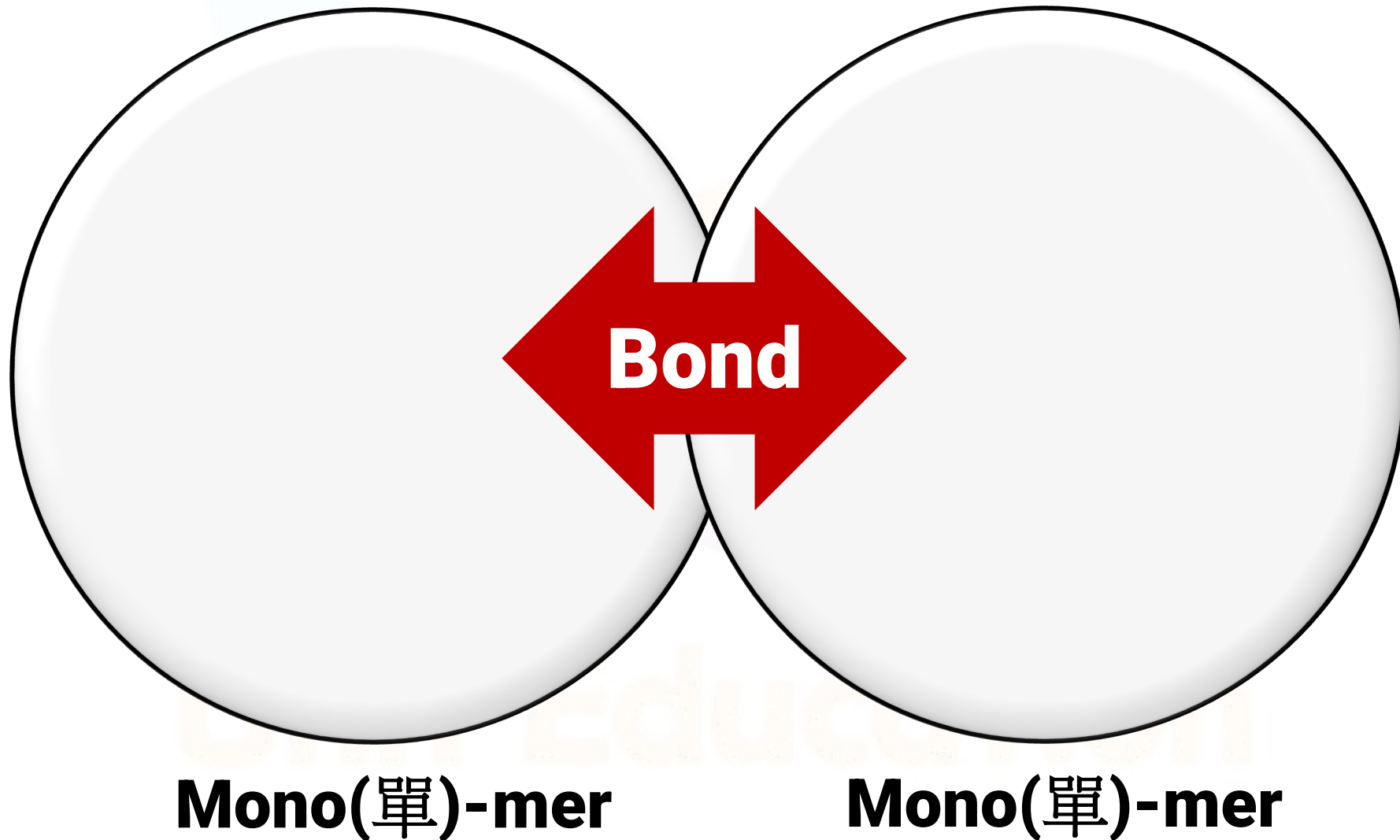
# Condensation 縮合



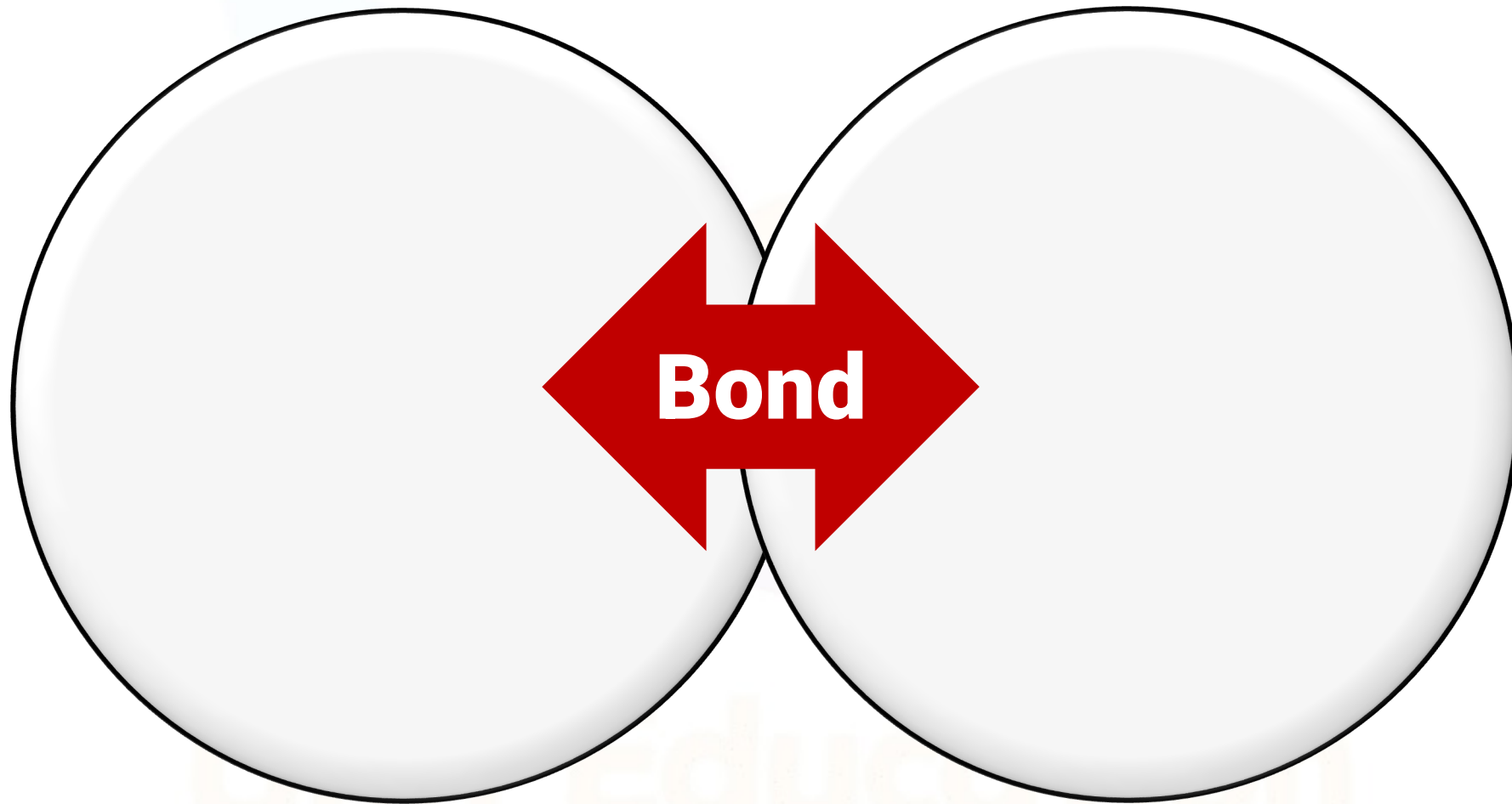
**Mono(單)-mer**

**Mono(單)-mer**

# Condensation 縮合



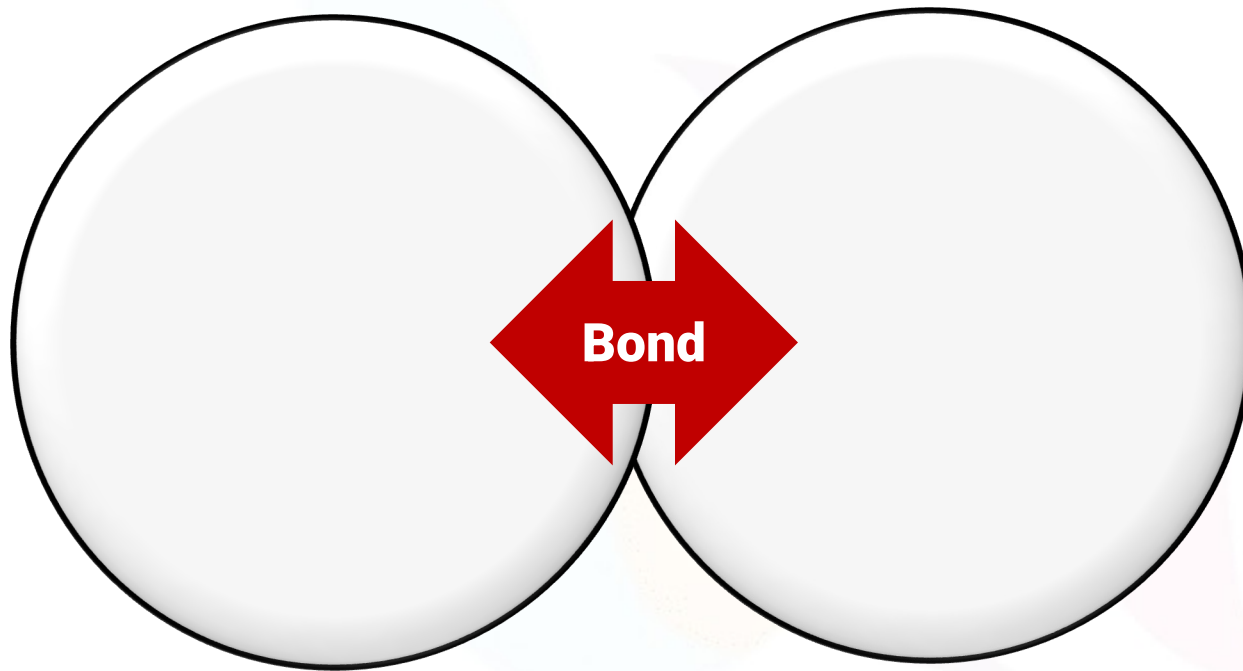
# Condensation 縮合



$\text{H}_2\text{O}$

**Di (二)-mer**

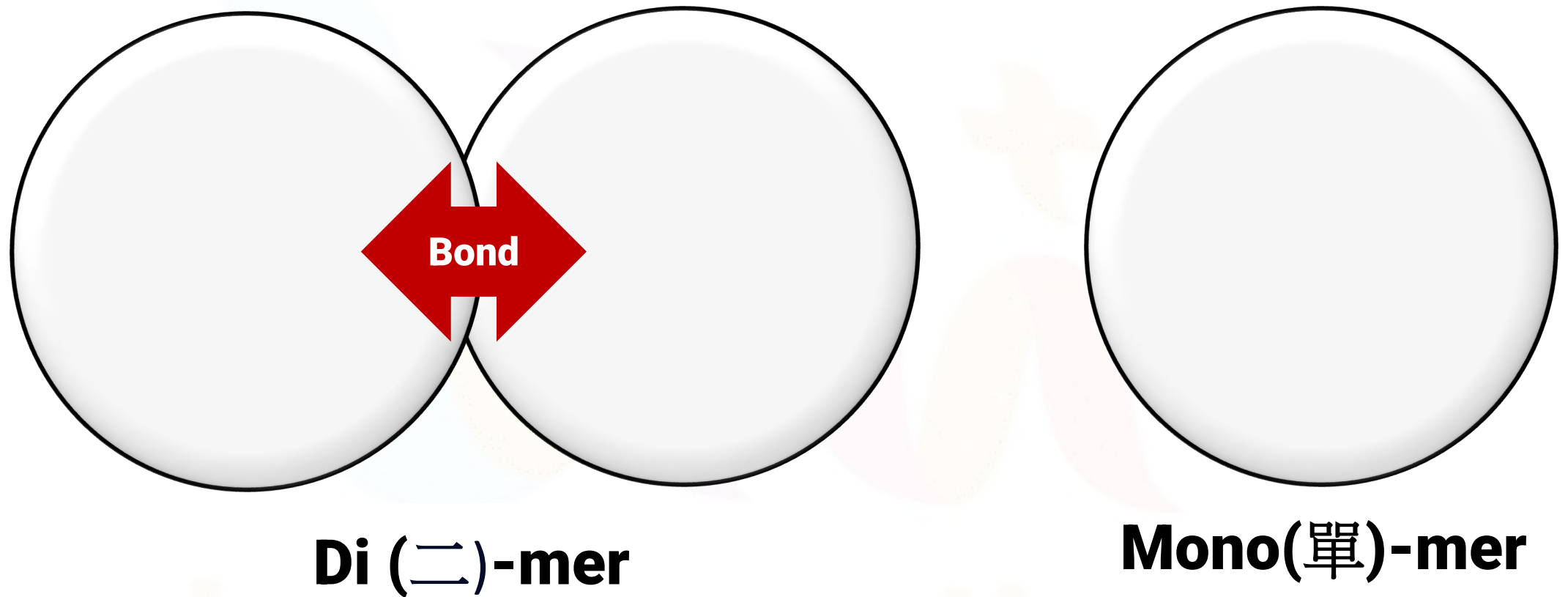
# Condensation 縮合



**Di (二)-mer**

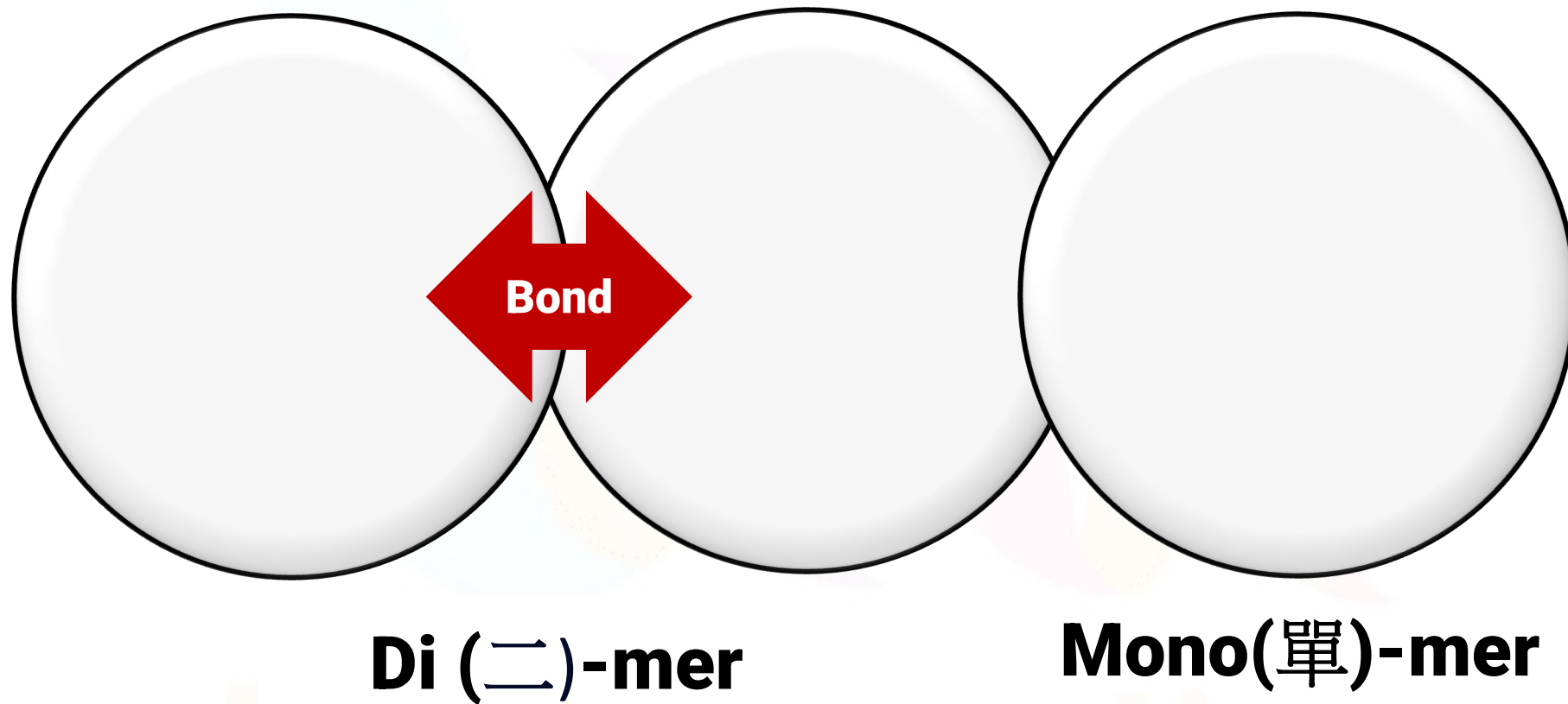
Unit Education

# Condensation 縮合



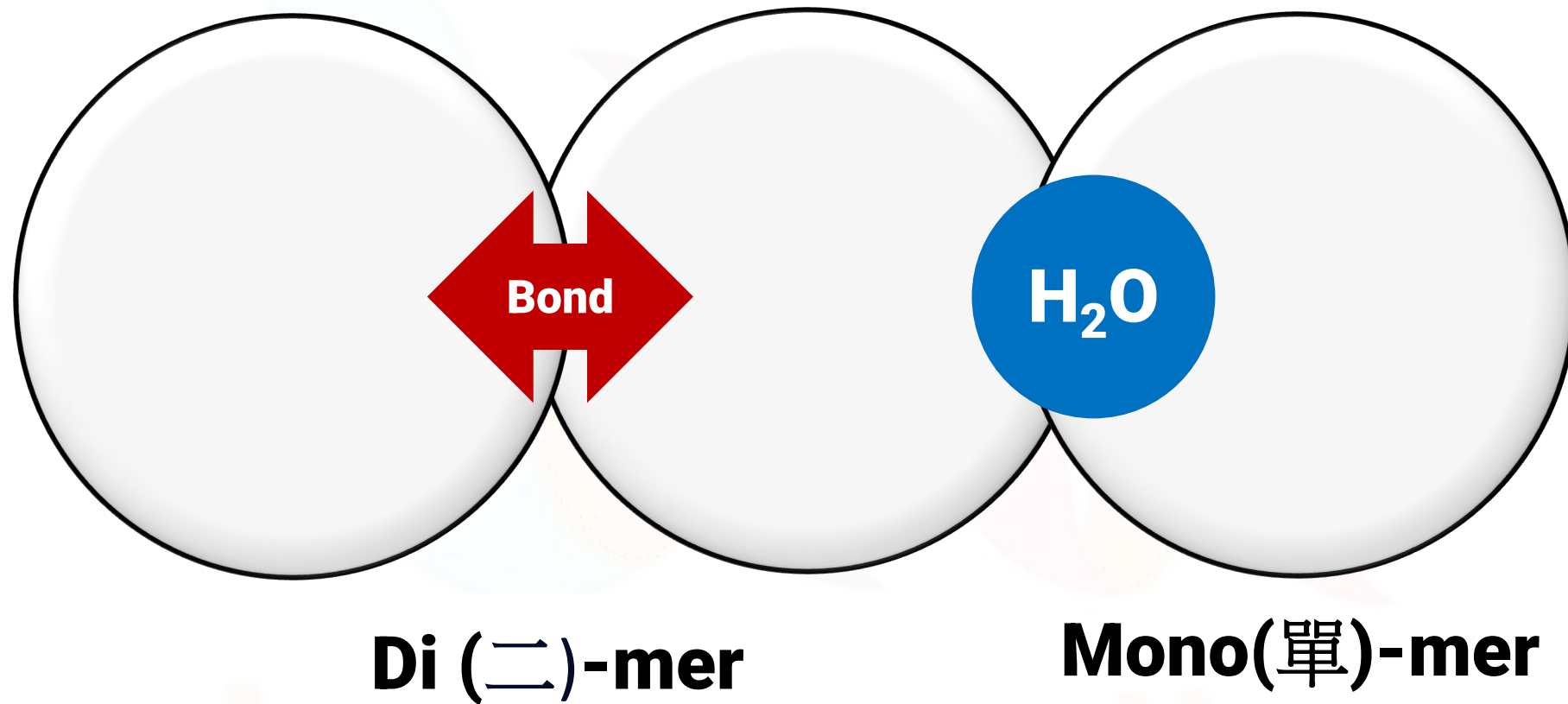
Unit Education

# Condensation 縮合



Unit Education

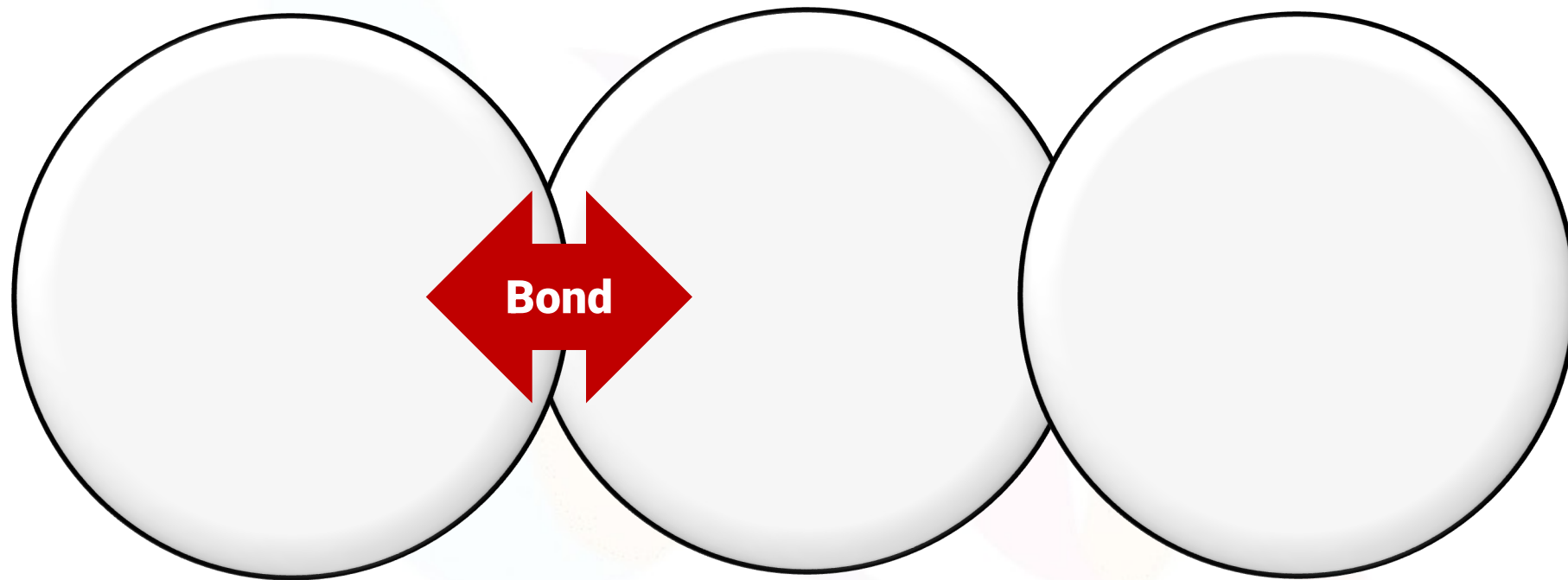
# Condensation 縮合



Unit Education



# Condensation 縮合

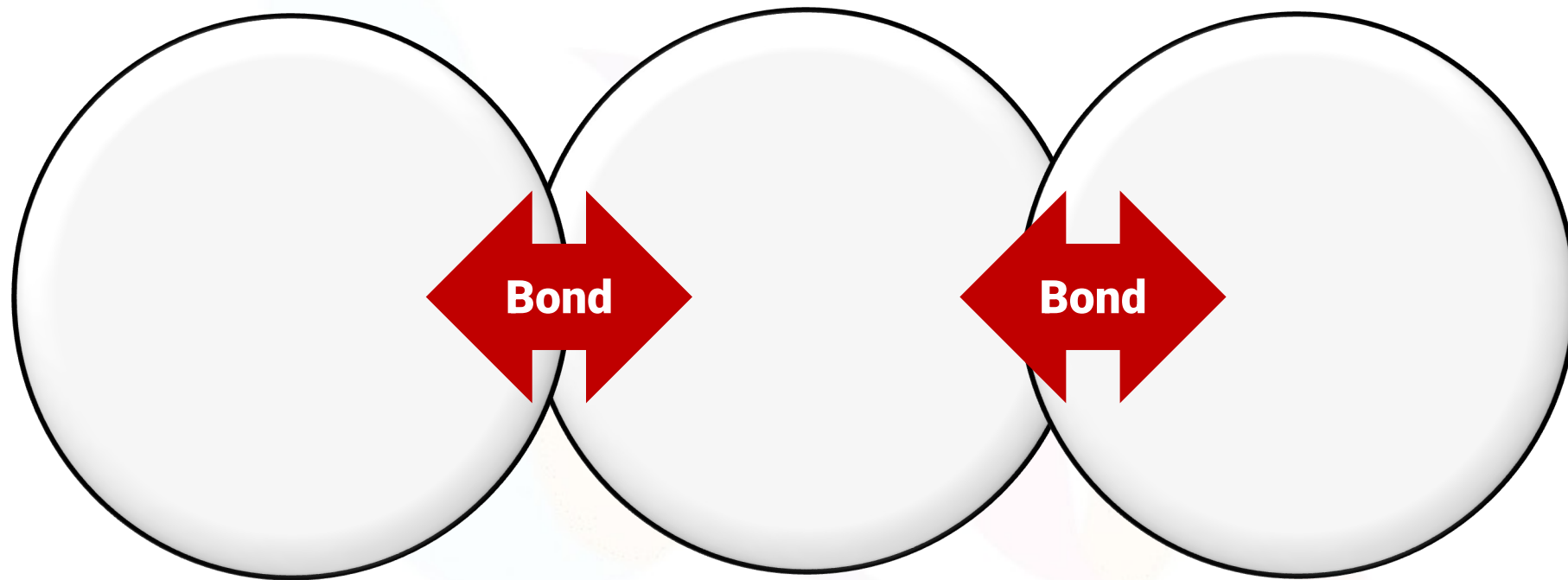


**Di (二)-mer**

**Mono(單)-mer**

**H<sub>2</sub>O**

# Condensation 縮合

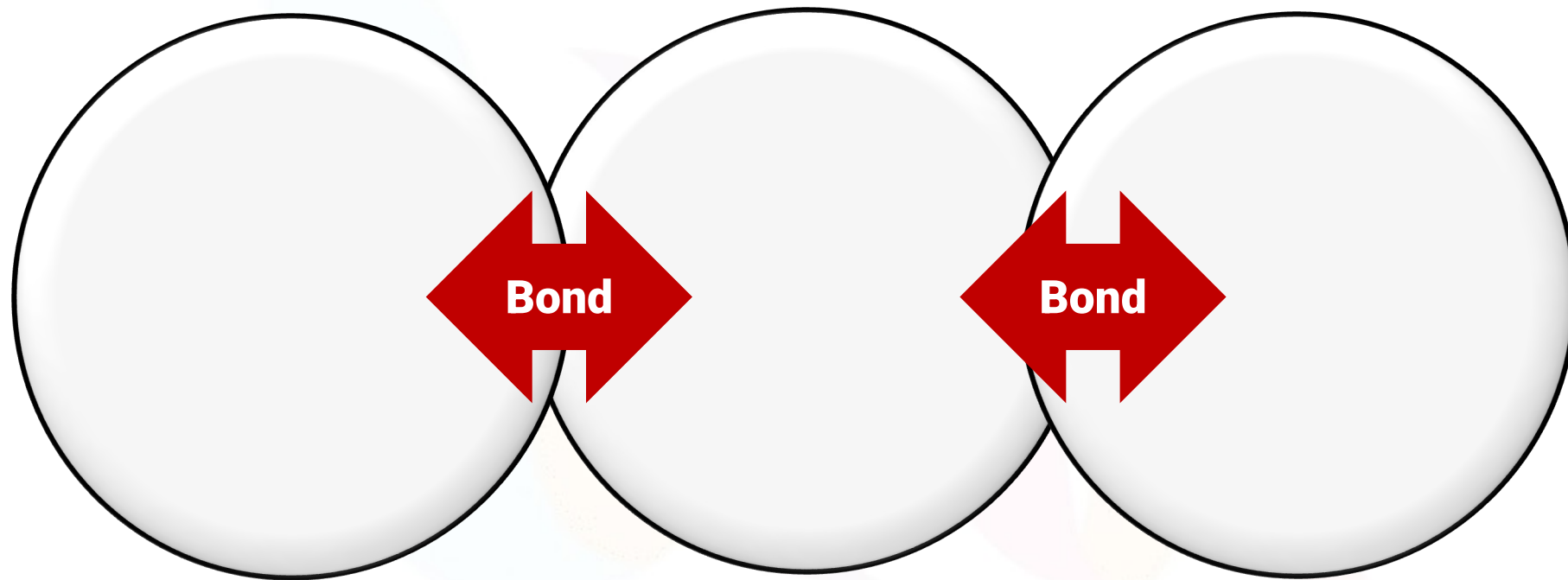


**Di (二)-mer**

**Mono(單)-mer**

**H<sub>2</sub>O**

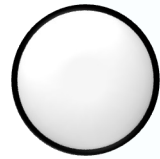
# Condensation 縮合



**Poly (多)-mer**

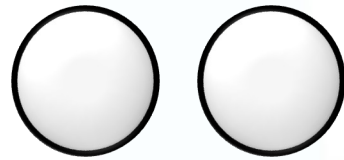
$H_2O$

# Condensation 縮合



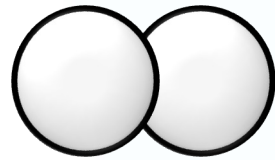
**Poly (多)-mer**

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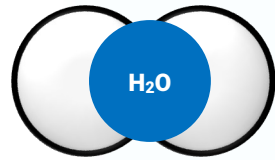
**Poly (多)-mer**

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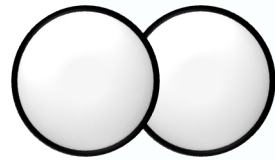
**Poly (多)-mer**

# Condensation 縮合



**Poly (多)-mer**

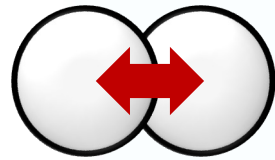
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**Poly (多)-mer**

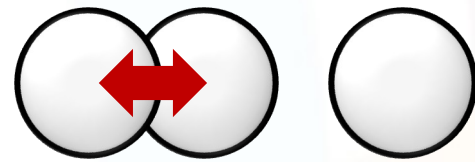


# Condensation 縮合



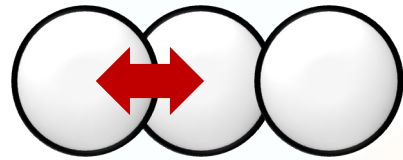
**Poly (多)-mer**

# Condensation 縮合



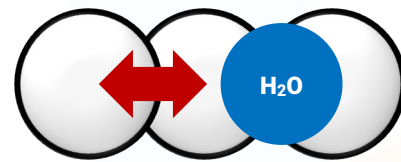
**Poly (多)-mer**

# Condensation 縮合



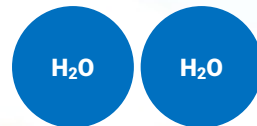
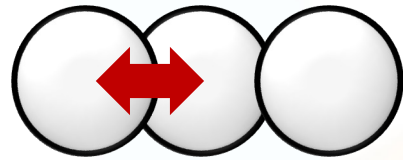
**Poly (多)-mer**

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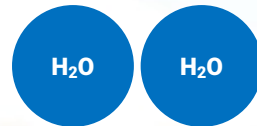
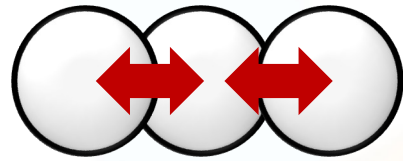
**Poly (多)-mer**

# Condensation 縮合



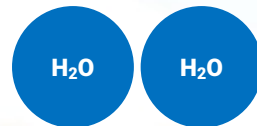
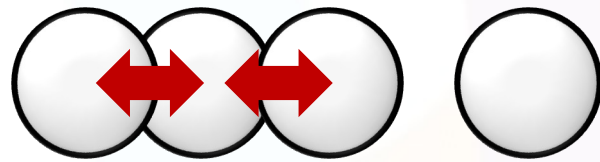
**Poly (多)-mer**

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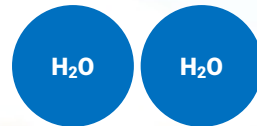
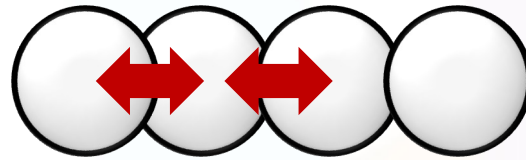
**Poly (多)-mer**

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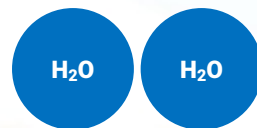
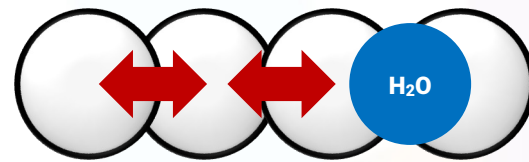
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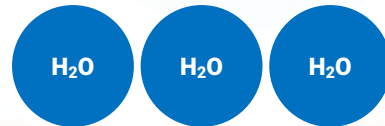
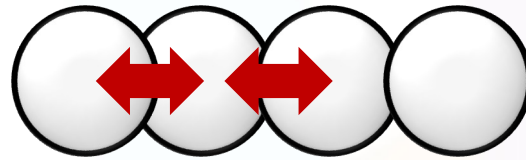


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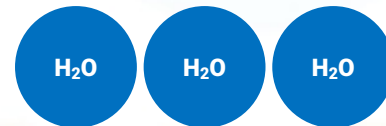
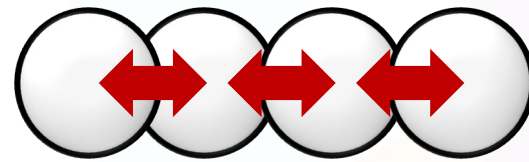
**Poly (多)-mer**

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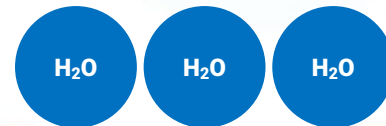
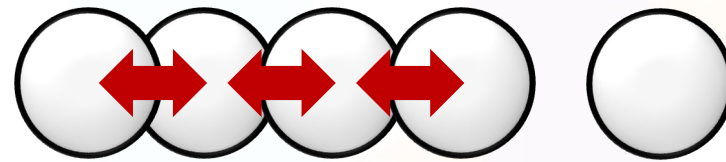
**Poly (多)-mer**

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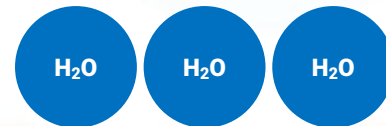
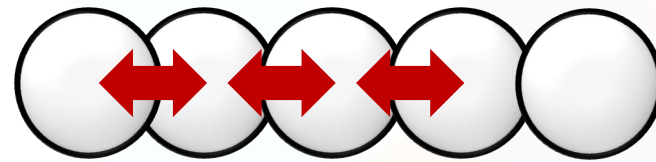
**Poly (多)-mer**

# Condensation 縮合



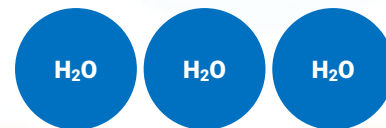
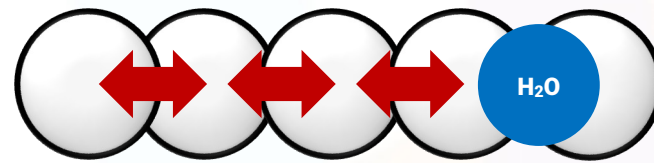
**Poly (多)-mer**

# Condensation 縮合



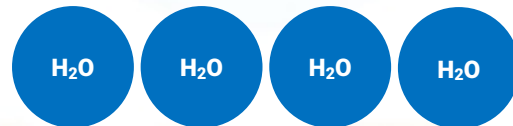
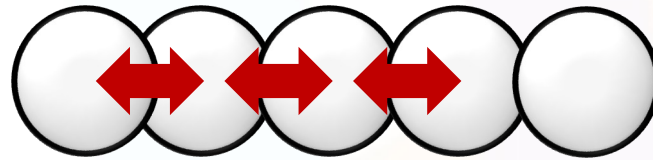
**Poly (多)-mer**

# Condensation 縮合



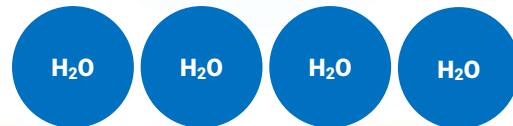
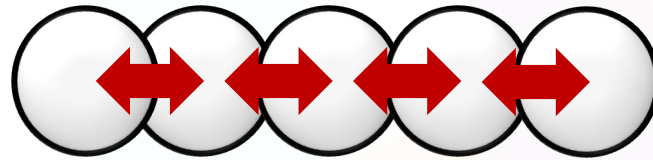
**Poly (多)-mer**

# Condensation 縮合



**Poly (多)-mer**

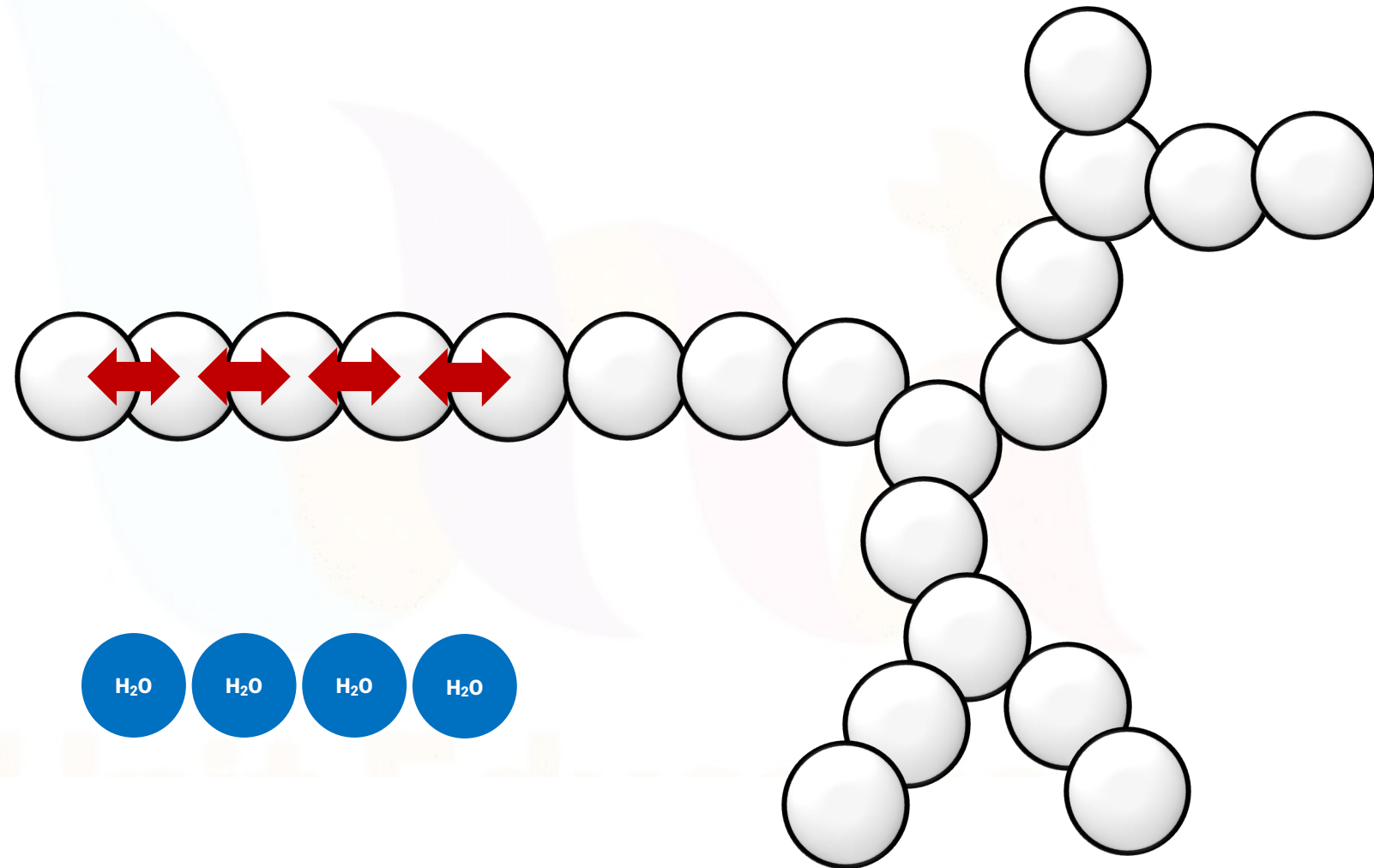
# Condensation 縮合



**Poly (多)-mer**

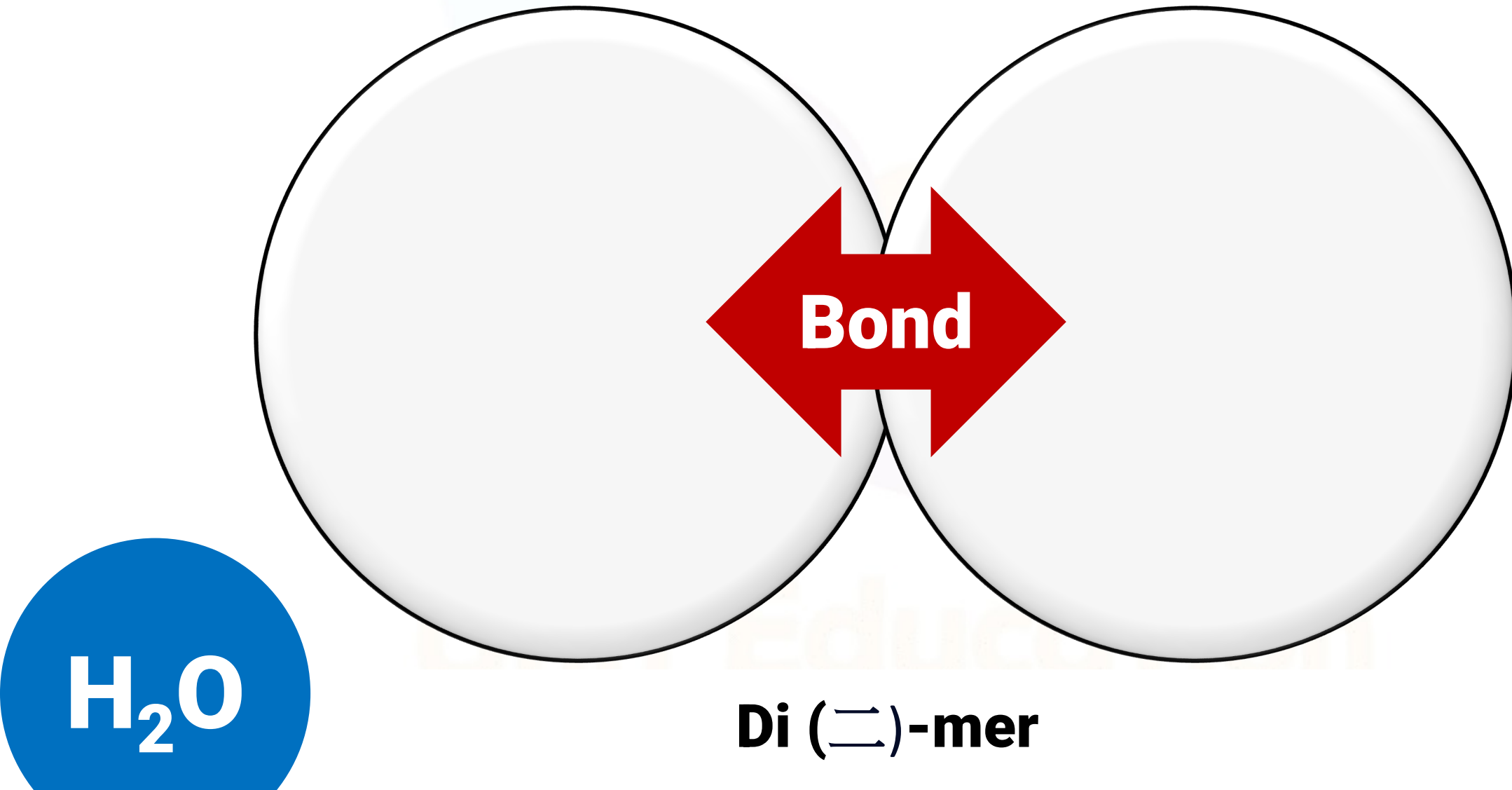


# Condensation 縮合

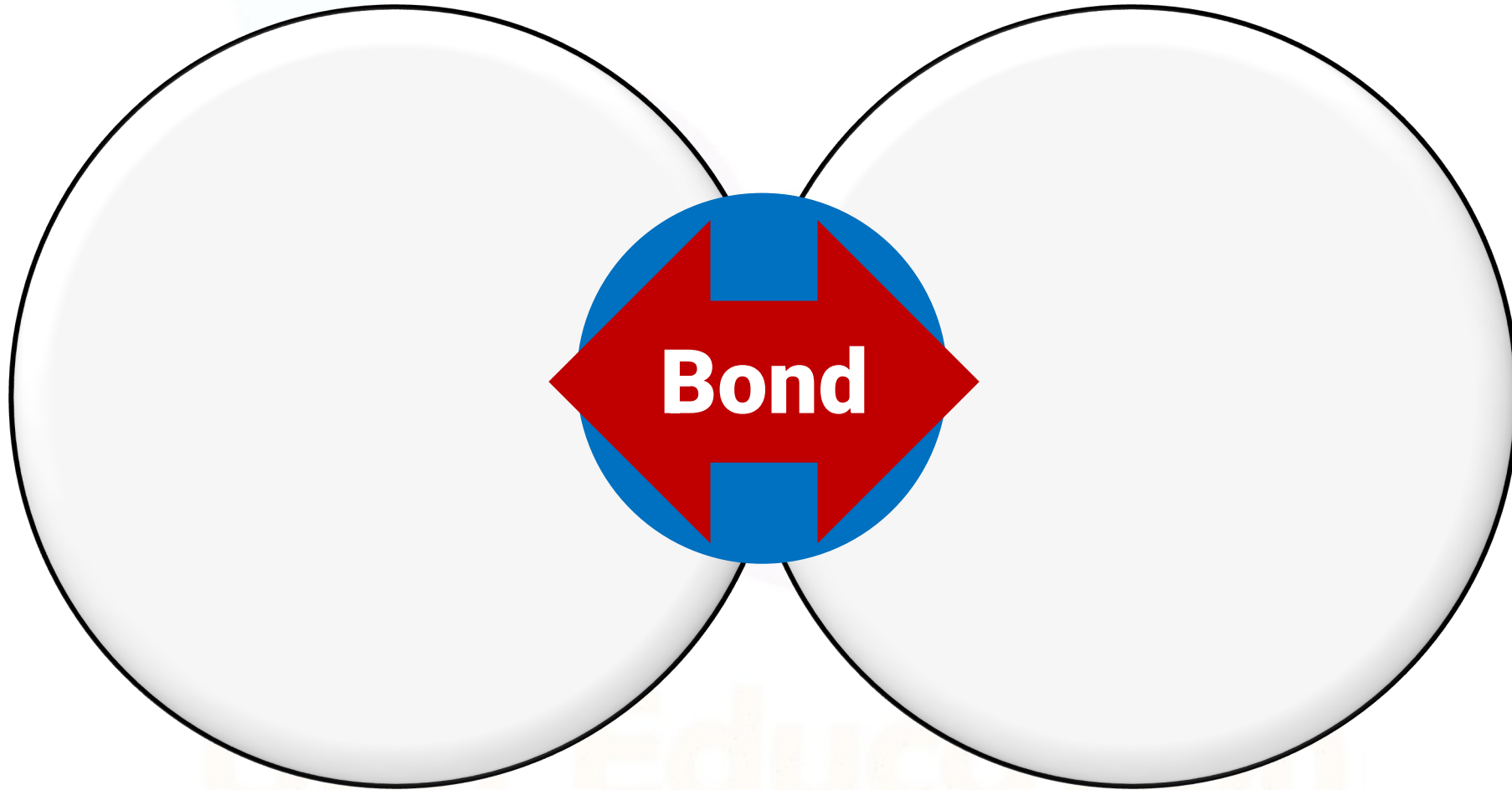


**Poly (多)-mer**

# Hydrolysis 水解

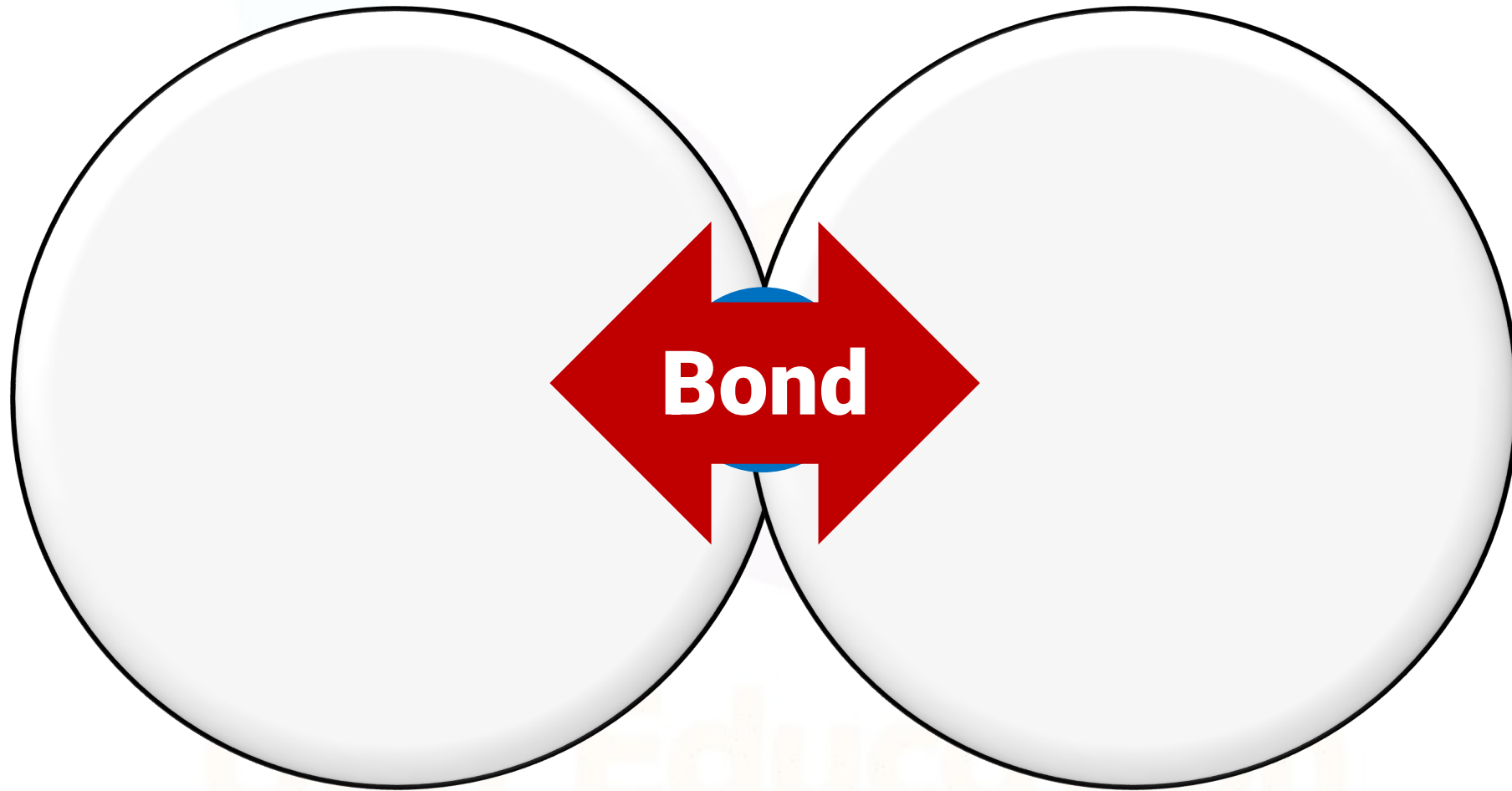


# Hydrolysis 水解



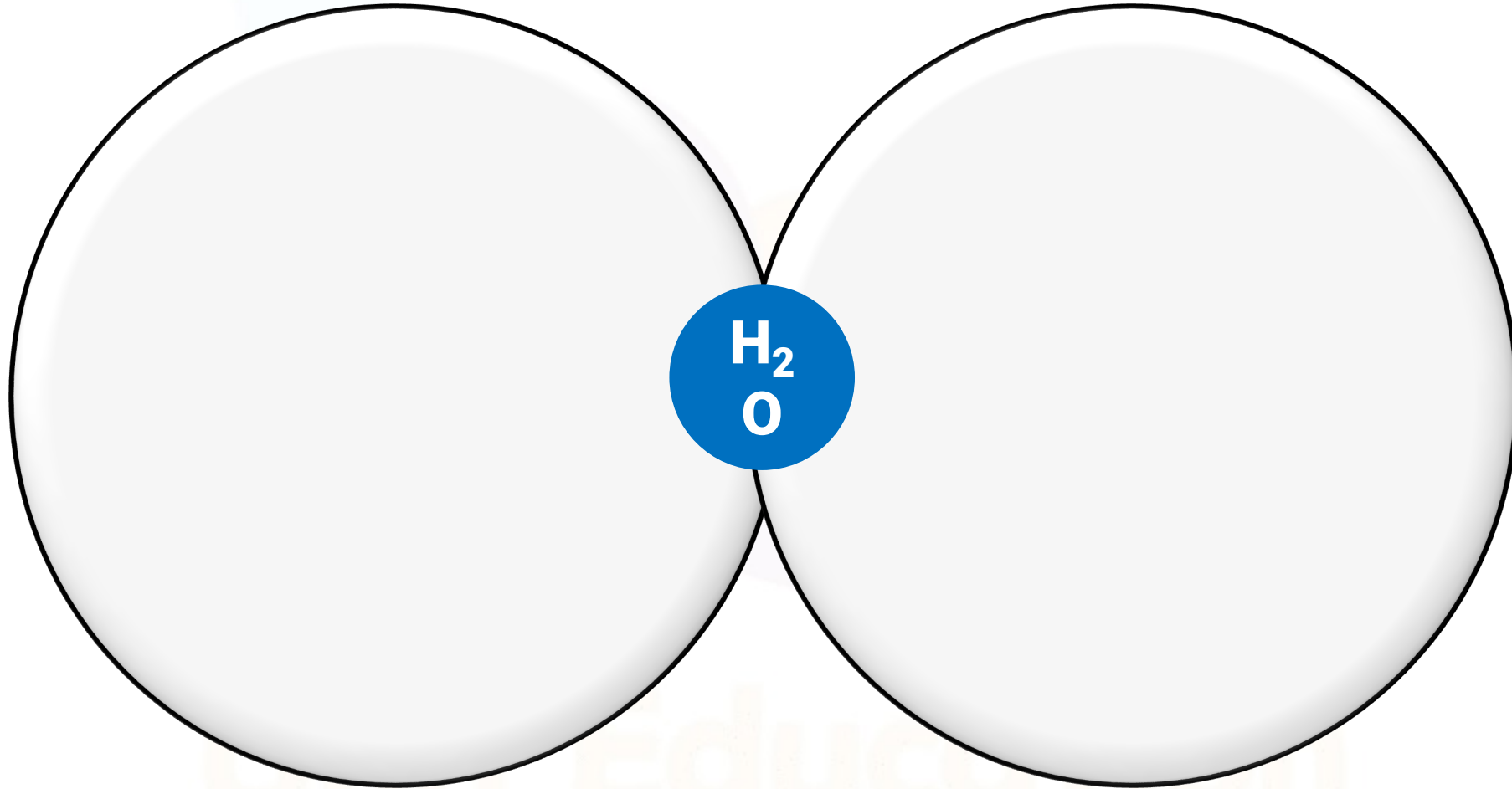
**Di (二)-mer**

# Hydrolysis 水解



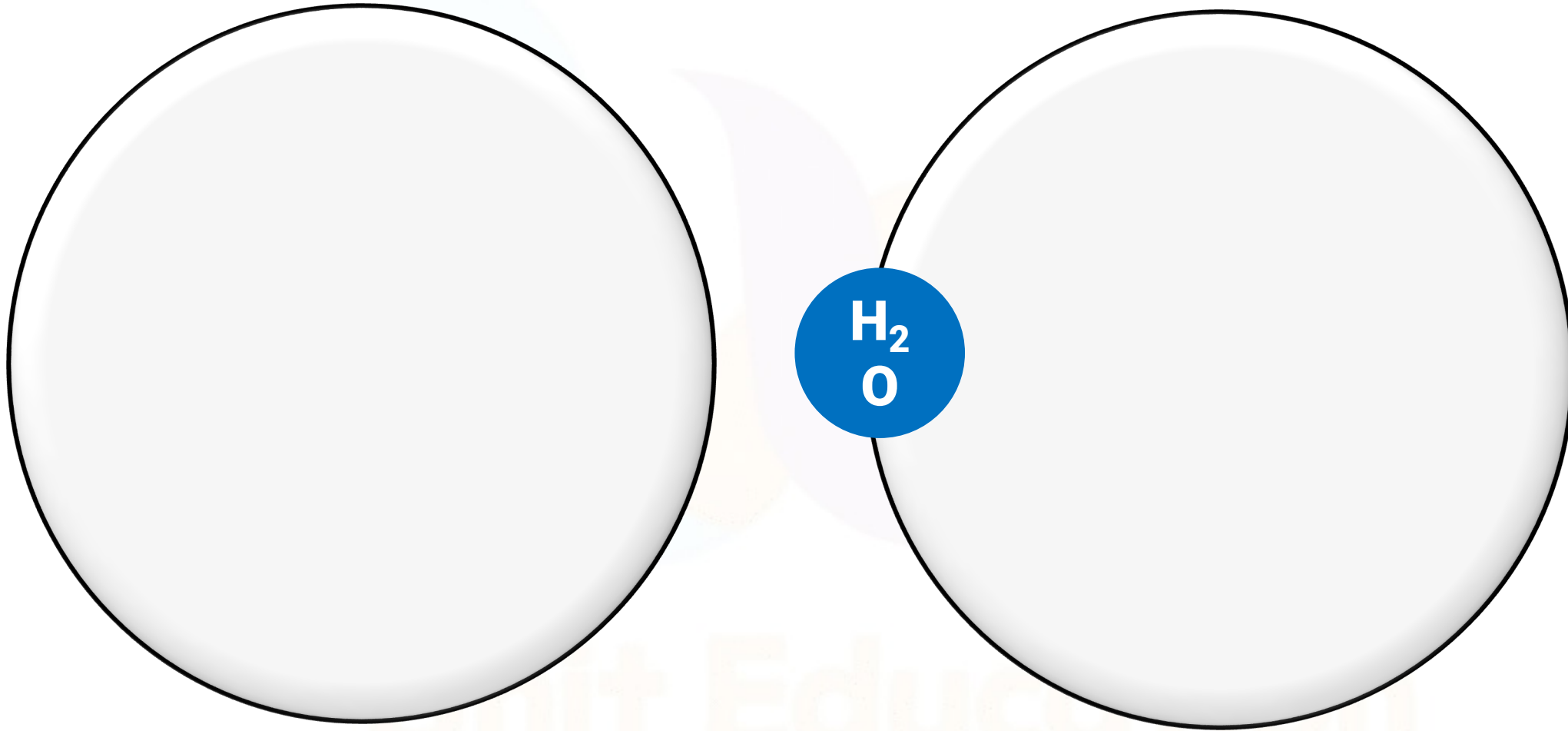
**Di (二)-mer**

# Hydrolysis 水解



**Di (二)-mer**

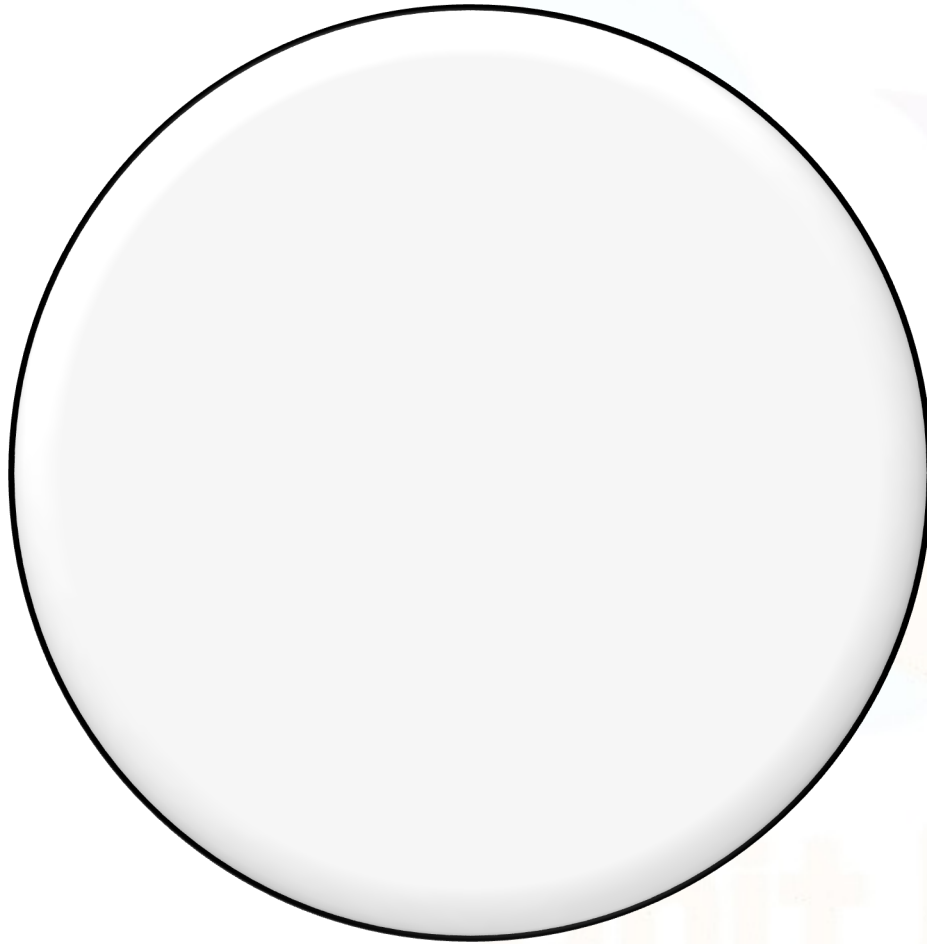
# Hydrolysis 水解



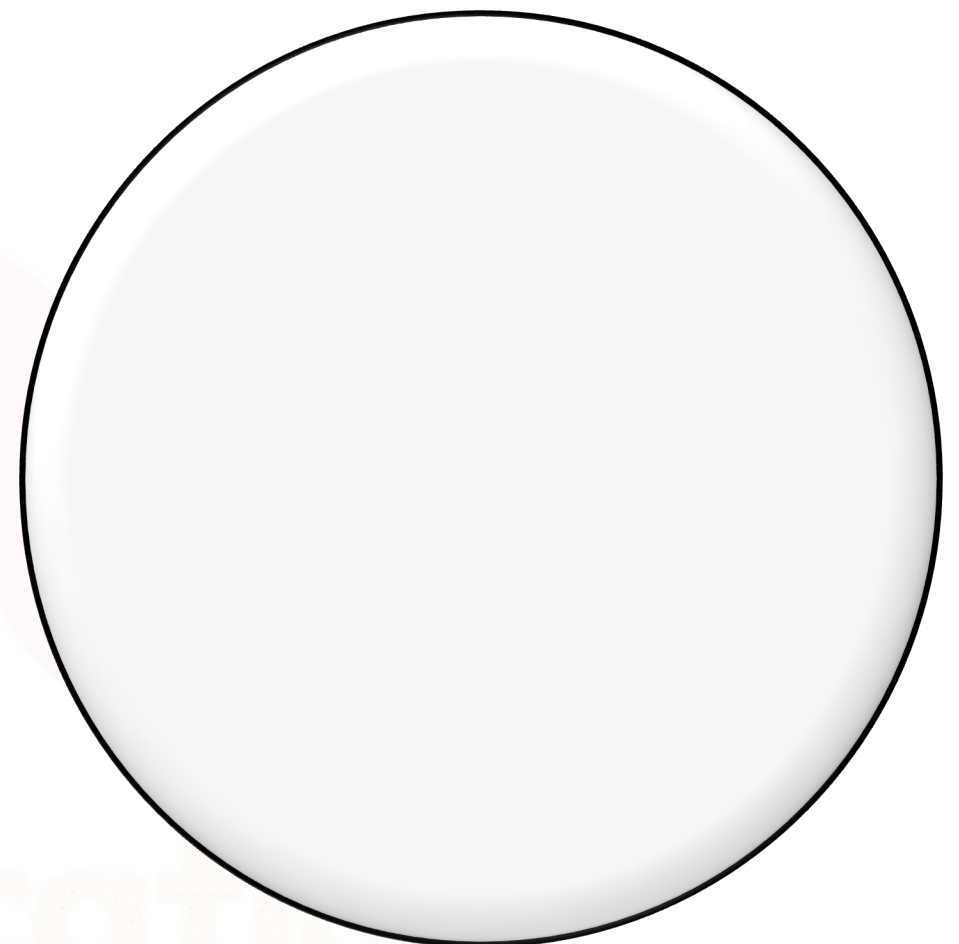
**Mono**

**Di (二)-mer**

# Hydrolysis 水解



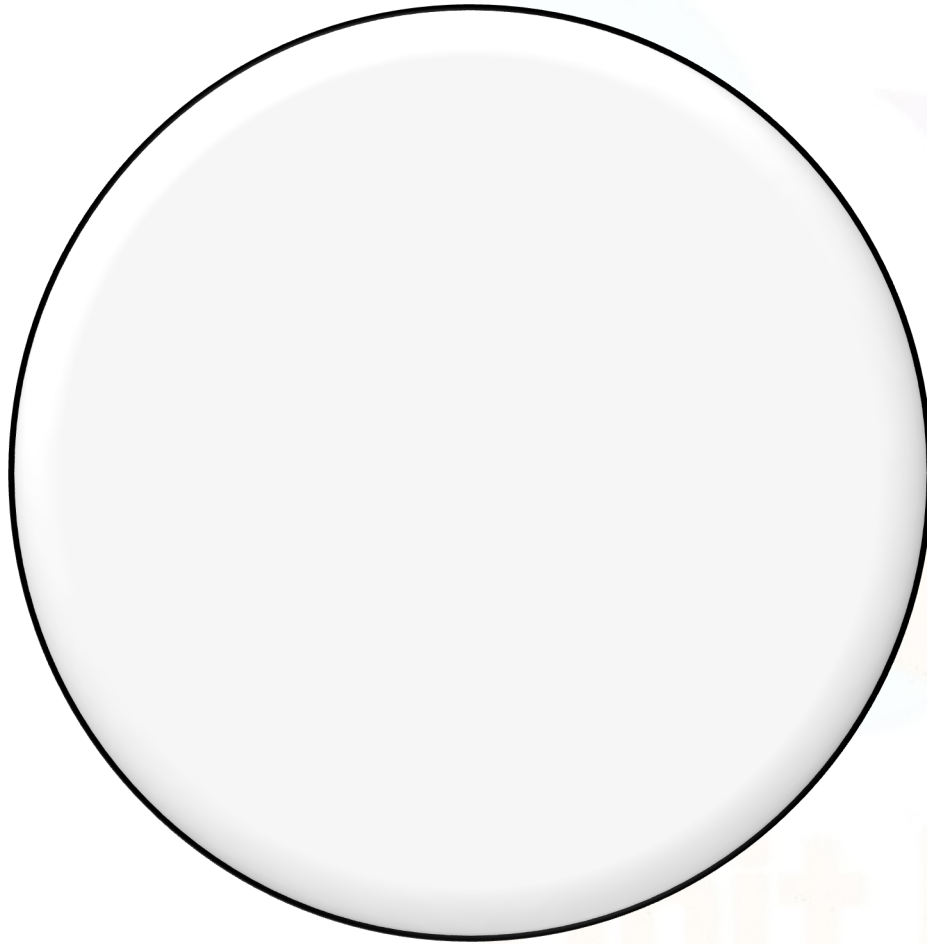
**Mono**



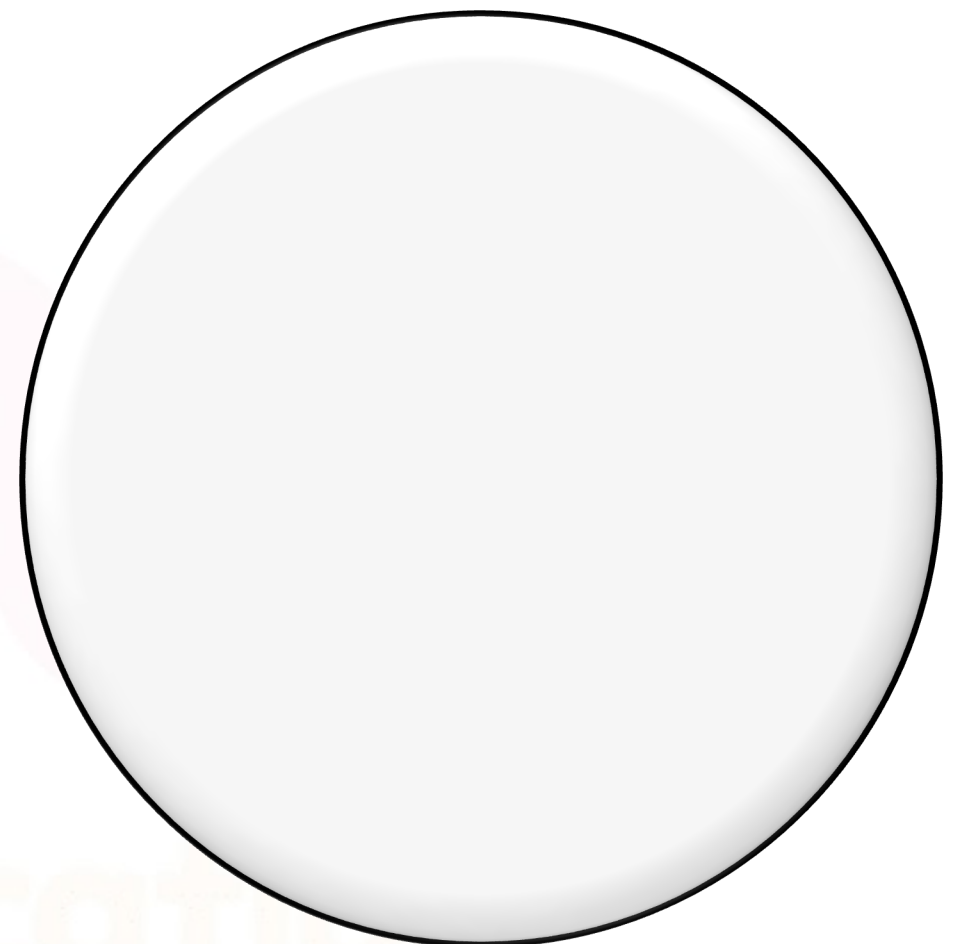
**Di (二)-mer**

**)-mer**

# Hydrolysis 水解



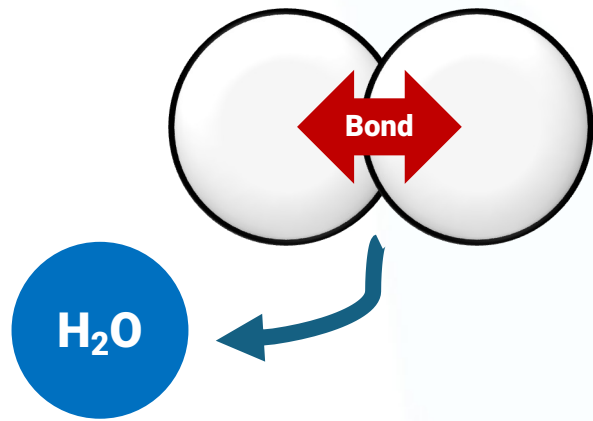
**Mono(單)-mer**



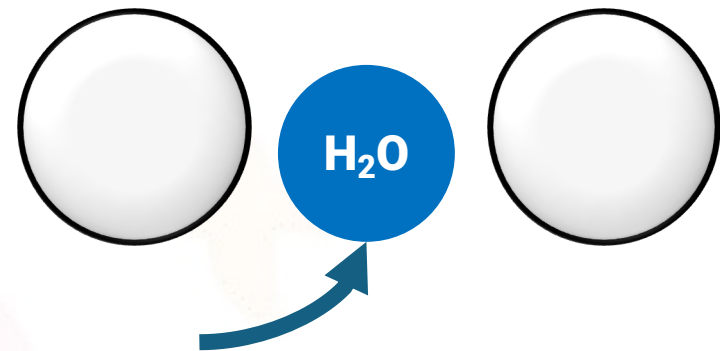
**Mono(單)-mer**



# Condensation 縮合 and Hydrolysis 水解



“Condensation 係會出水”



“水解，用水去解 (bond)”

宜家加大炸雞餐有特價，但阿媽話唔健康唔比買；點樣用生物科概念說服佢炸雞餐有營養需要？



快樂水

總糖份: ~56g

碳水化合物(Carbohydrates)  
係人不可或缺嘅**快速能量來源**  
(Fast respiration fuel)  
)，葡萄糖更係大腦必要嘅養份。



快樂條

總脂肪: ~20g

脂質(Lipids)係**高效嘅能量處**  
**存** (Efficient storage  
)，亦係組成細胞膜同部份荷爾  
蒙嘅重要成份。

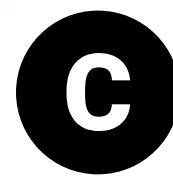


快樂雞

總蛋白質: ~32g

蛋白質(Proteins) 3D結構多變，  
**構成大部份身體組織**  
(Structural build blocks  
)以及酶、抗體、血紅素等重要  
分子。

# Organic biomolecules



Nitrogen

Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



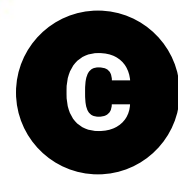
3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



Nitrogen

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Sulfur

## 1. Carbohydrates 碳水化合物



## 2. Lipids 脂質



## 3. Proteins 蛋白質

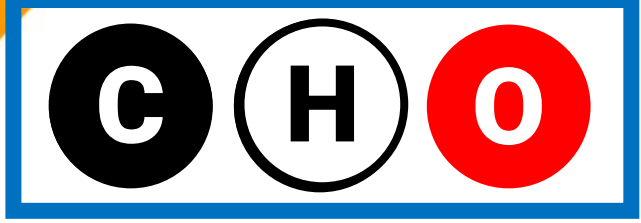


## 4. Nucleic acids 核酸



# 1. Carbohydrates

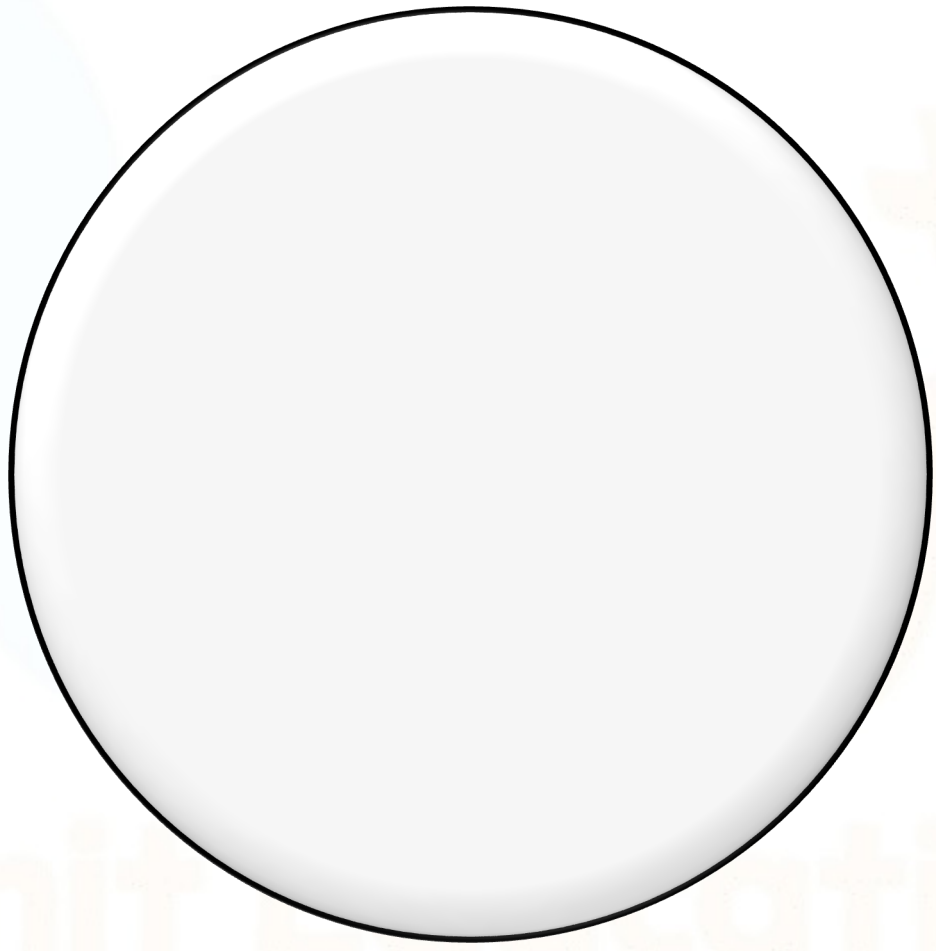
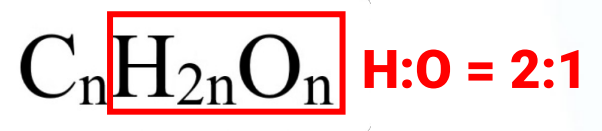
碳水化合物



Nitrogen

Phosphorus

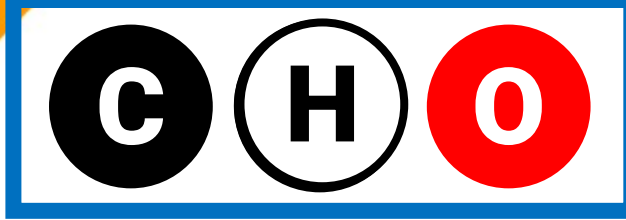
Sulfur



**Mono單-mer**

# 1. Carbohydrates

碳水化合物



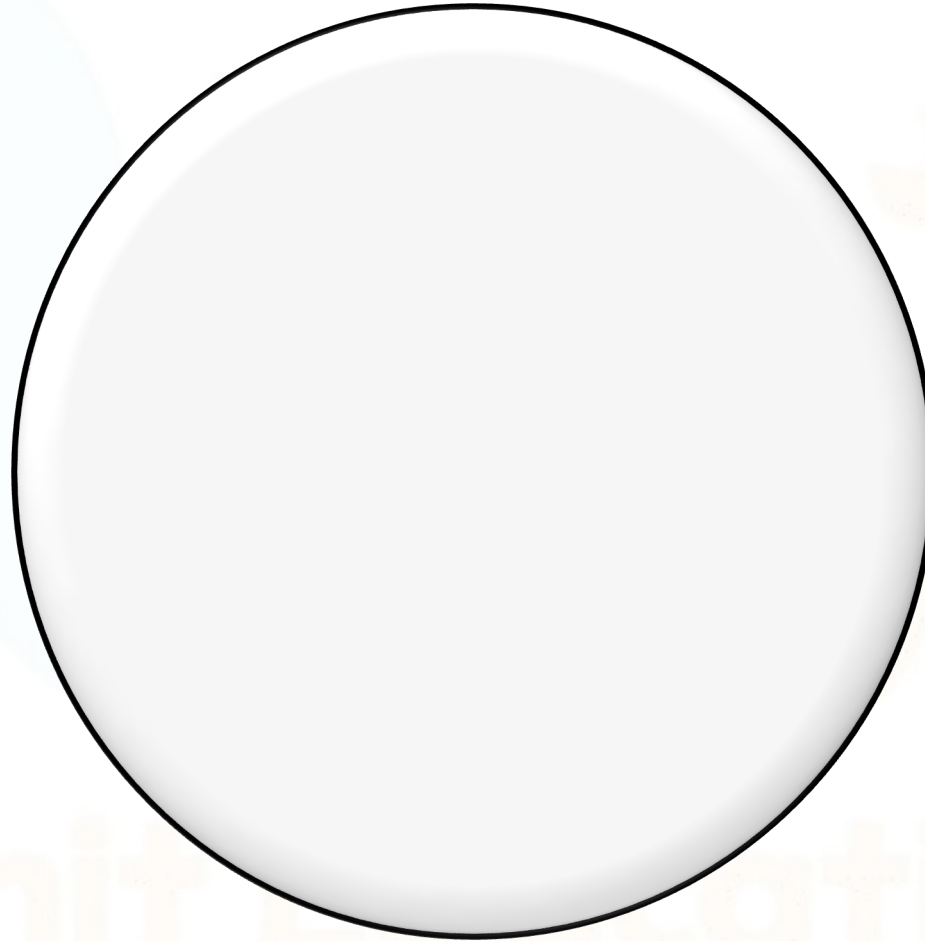
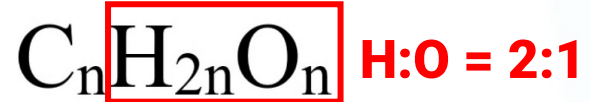
Nitrogen



Phosphorus



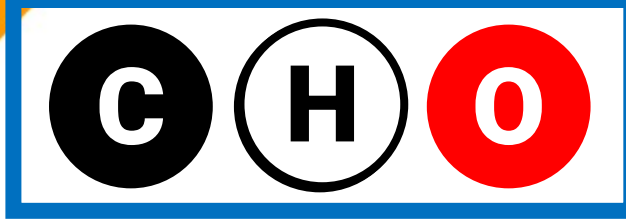
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**Mono單-mer**

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碳水化合物



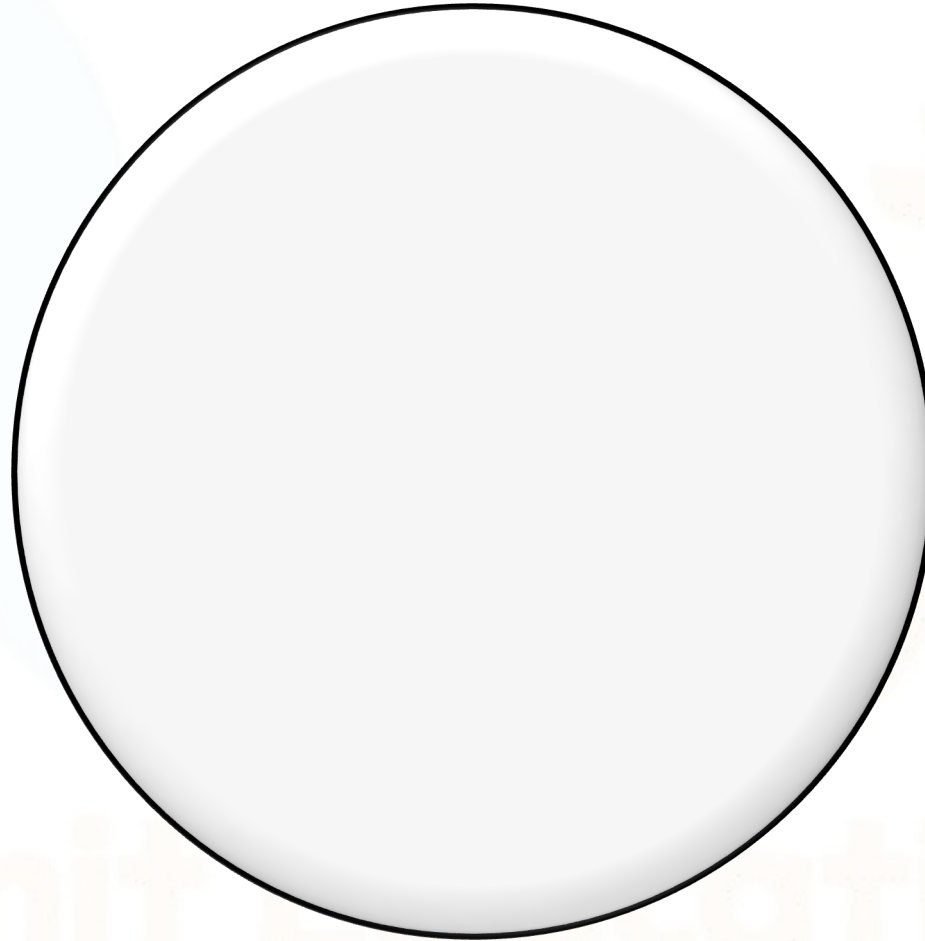
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Phosphorus



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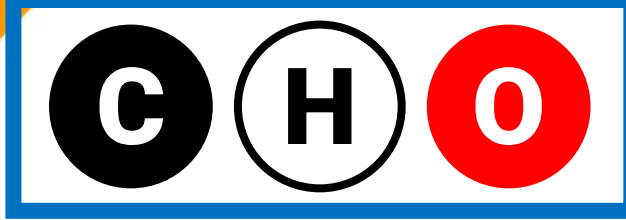


**Mono單-mer**



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碳水化合物



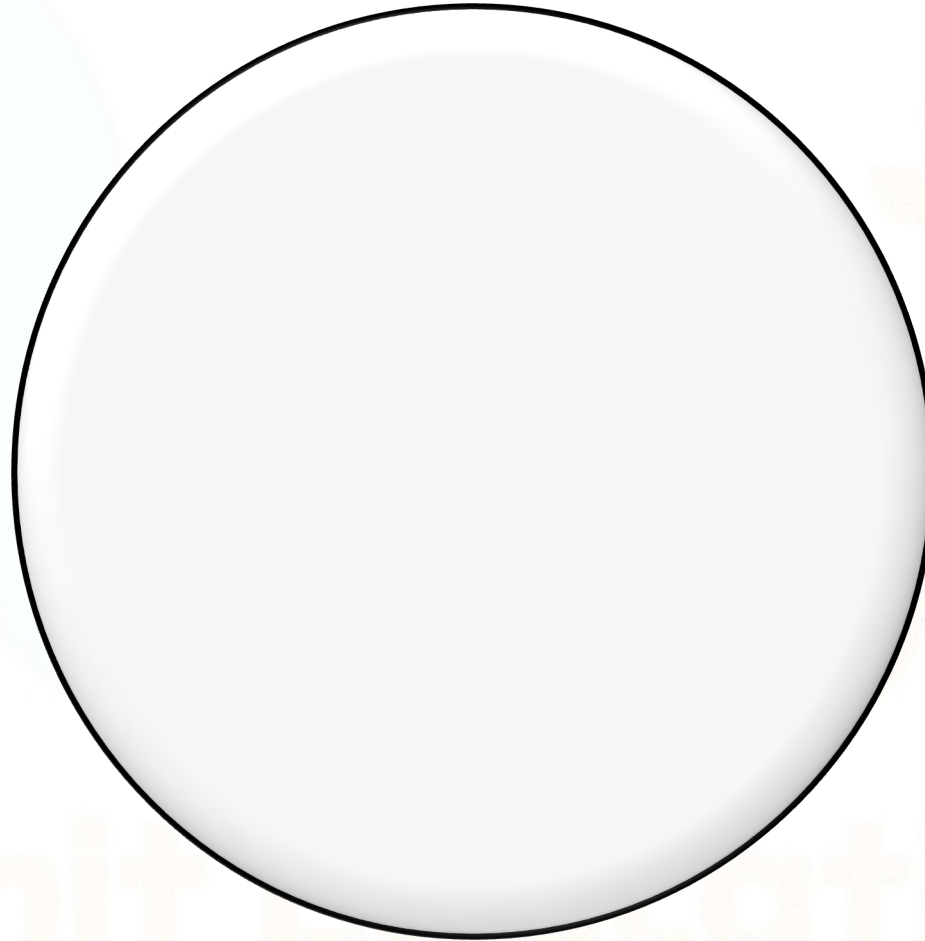
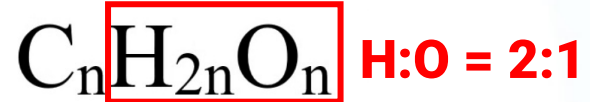
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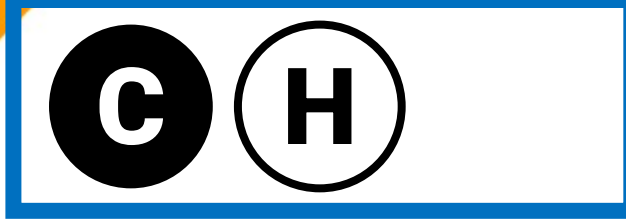


**Mono單-mer**



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碳水化合物



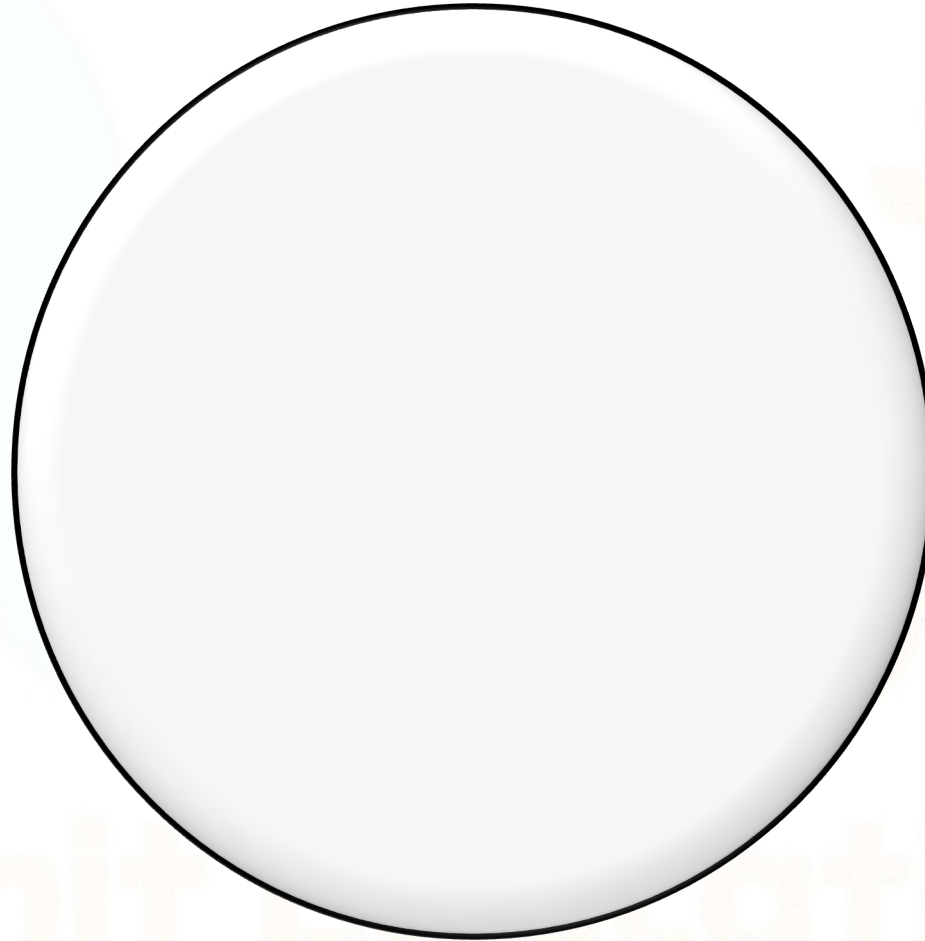
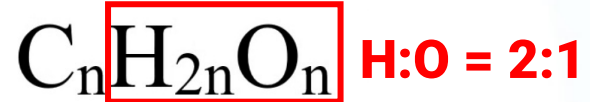
Nitrogen



Phosphorus



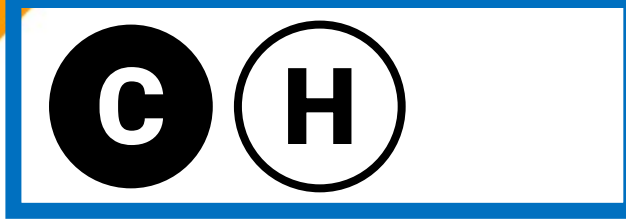
Sulfur



**Mono單-mer**

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碳水化合物



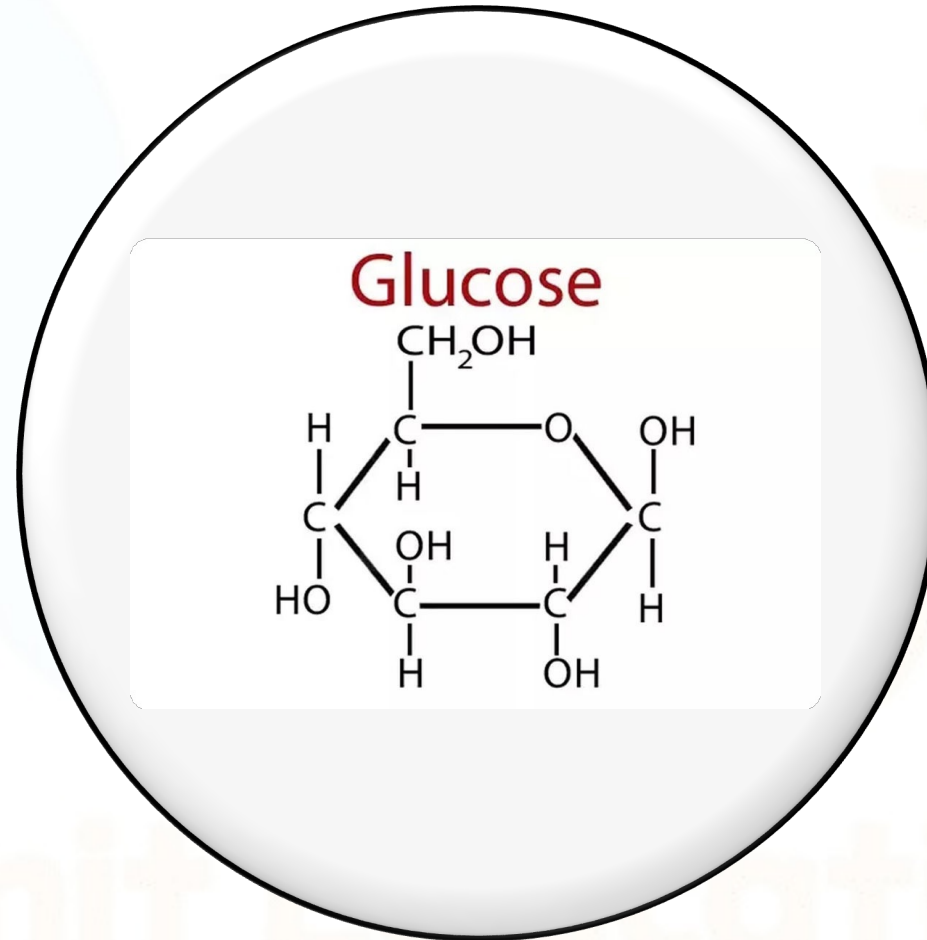
Nitrogen



Phosphorus



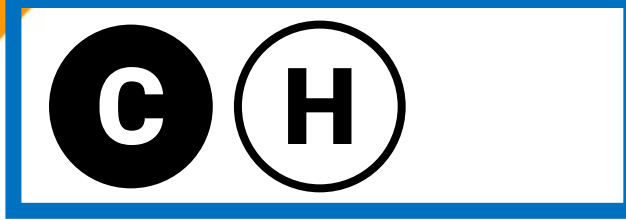
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碳水化合物



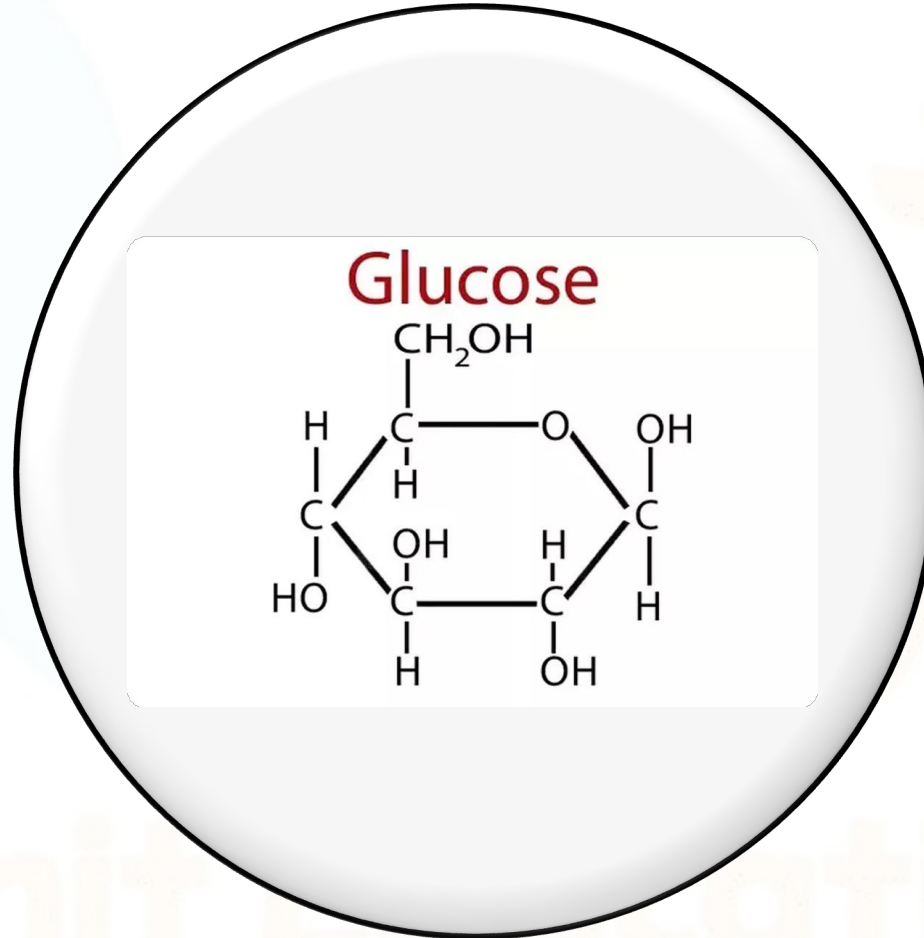
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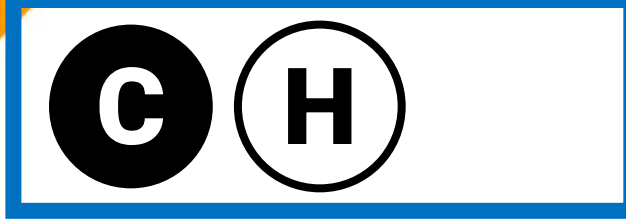
Sulfur



**Mono-saccharides (單醣)**

# 1. Carbohydrates

碳水化合物



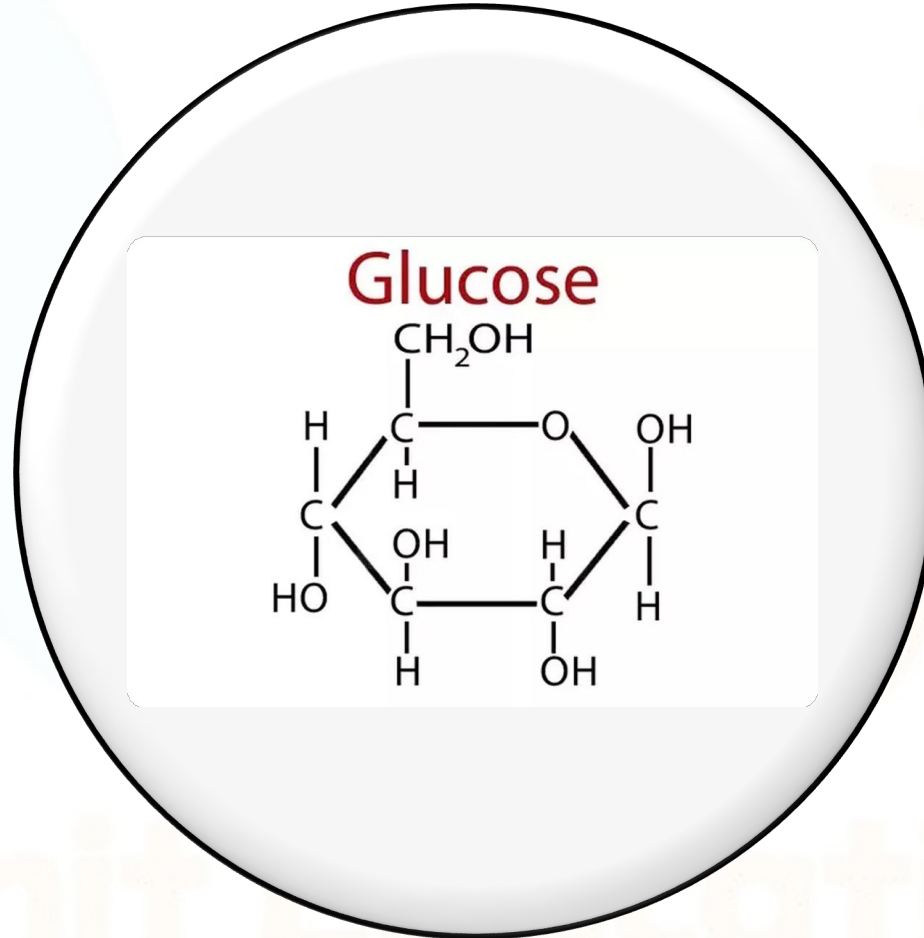
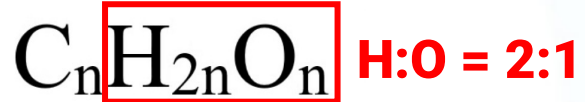
Nitrogen



Phosphorus



Sulfur



## Monosaccharides 單糖

- Soluble in water
- Taste sweet
- Reducing sugars

Examples:

Glucose 葡萄糖

Fructose 果糖

Galactose 半乳糖

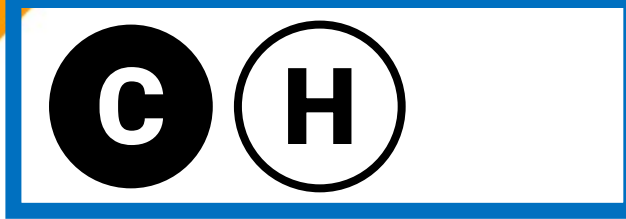
Ribose 核糖

Deoxyribose 脱氧核糖

**Mono-saccharides (單糖)**

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碳水化合物



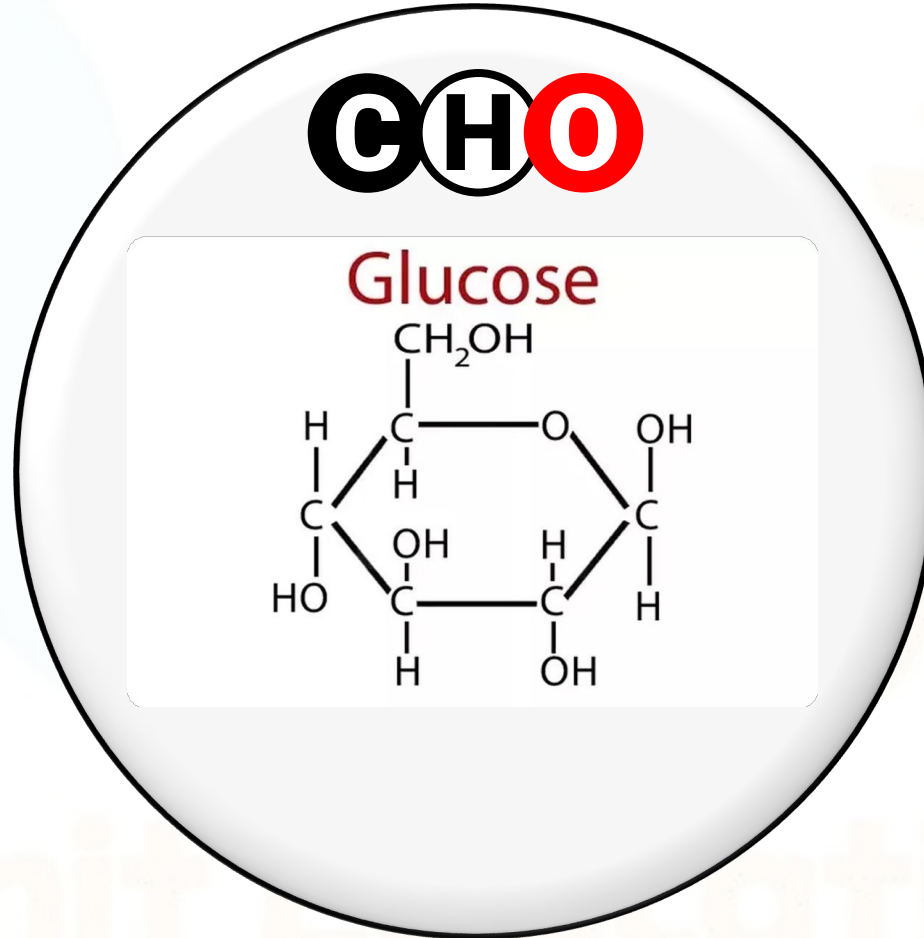
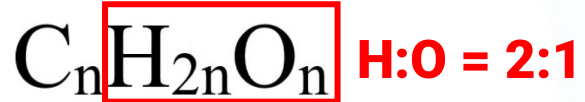
Nitrogen



Phosphorus



Sulfur



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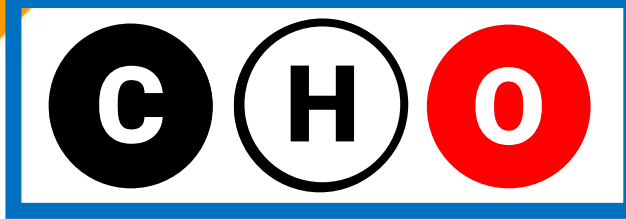
Ribose 核糖

Deoxyribose 脱氧核糖

**Mono-saccharides (單糖)**

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碳水化合物



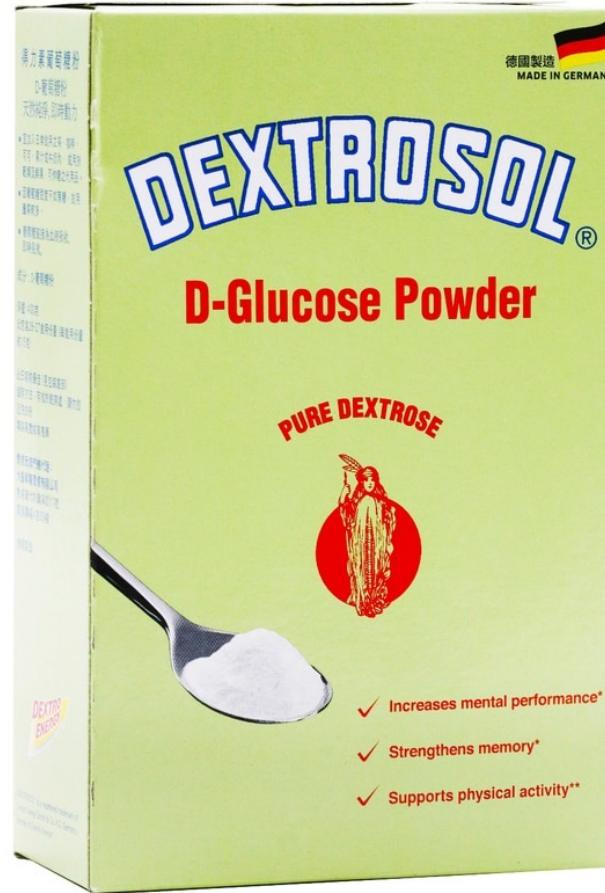
Nitrogen



Phosphorus



Sulfur



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Galactose 半乳糖

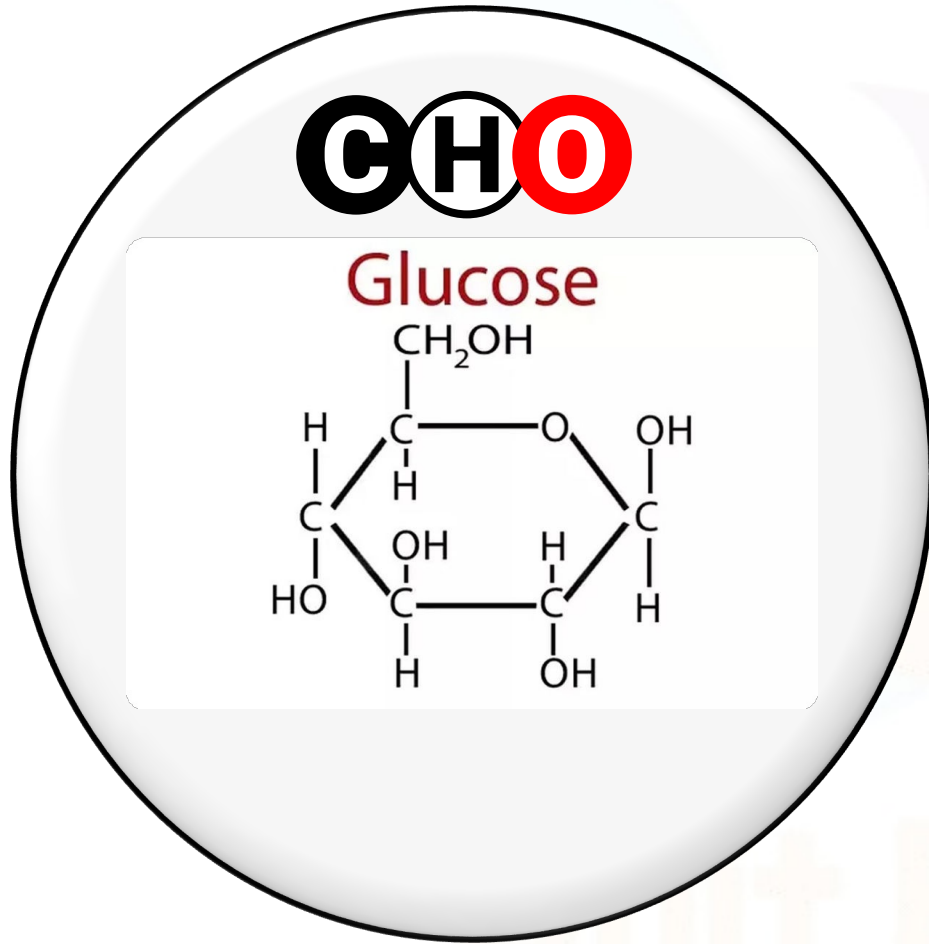
Ribose 核糖

Deoxyribose 脱氧核糖

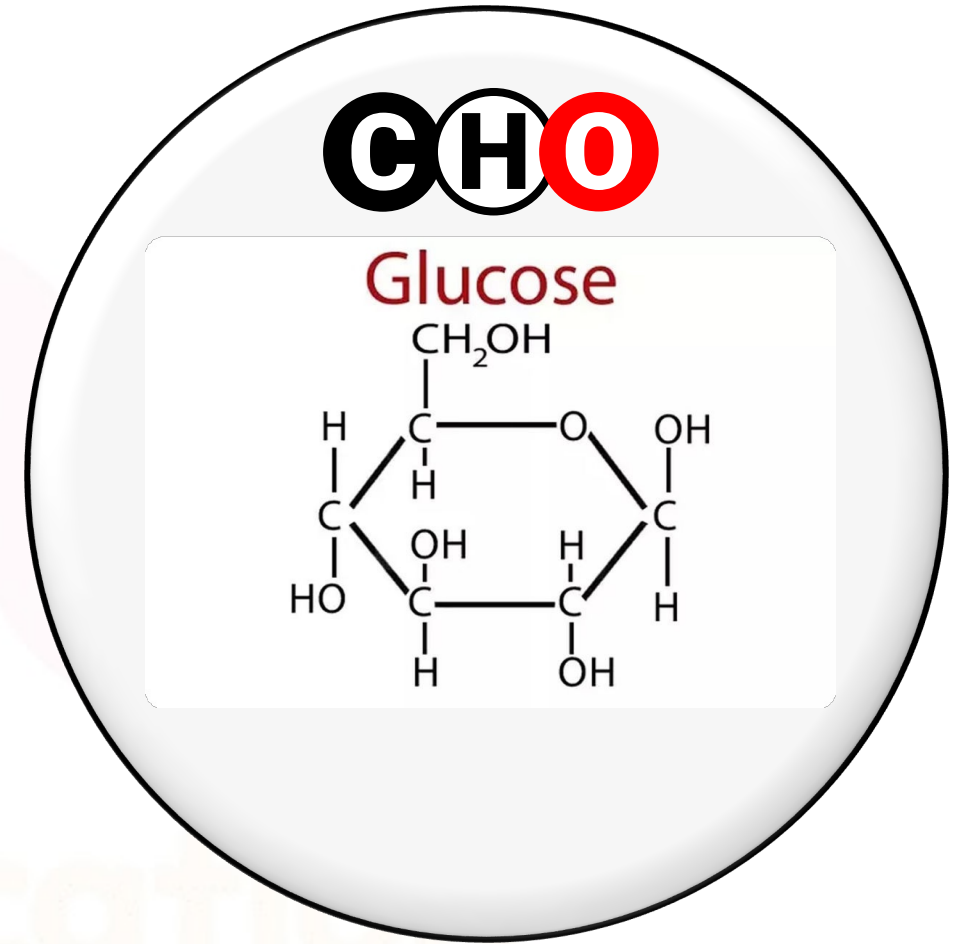
## Mono-saccharides (單糖)

# 1. Carbohydrates 碳水化合物

## Condensation 縮合



**Monosaccharides**

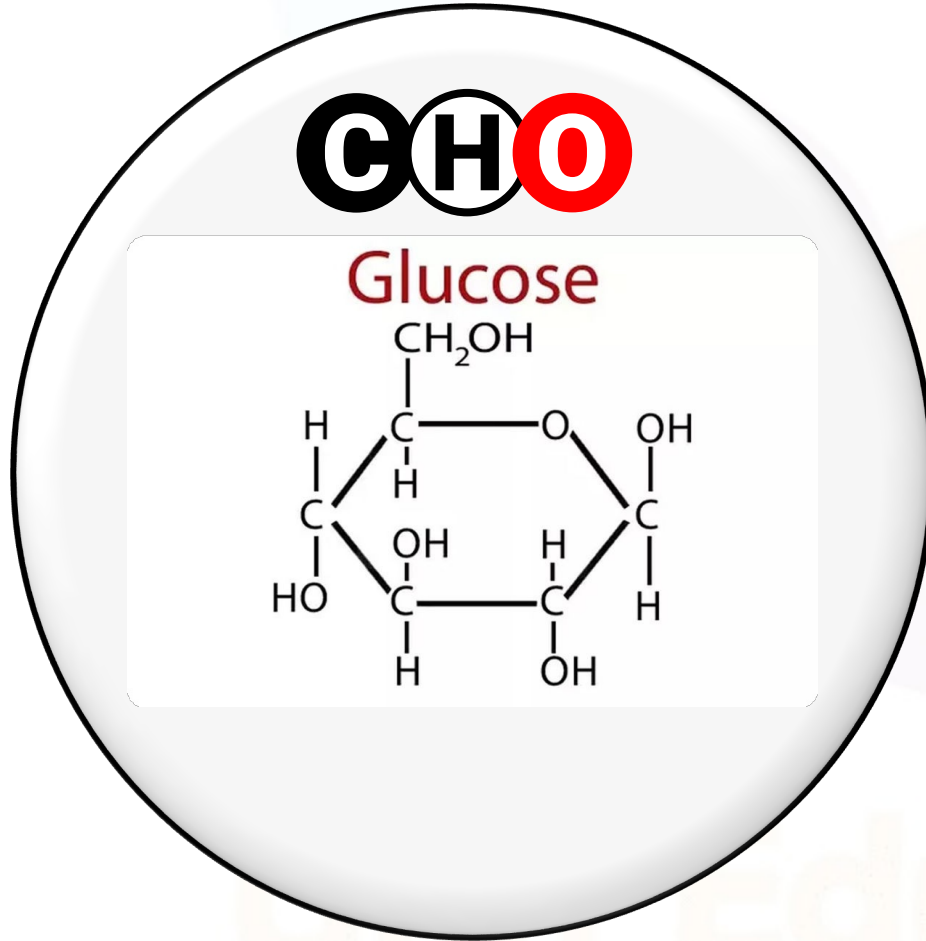


**Monosaccharides**

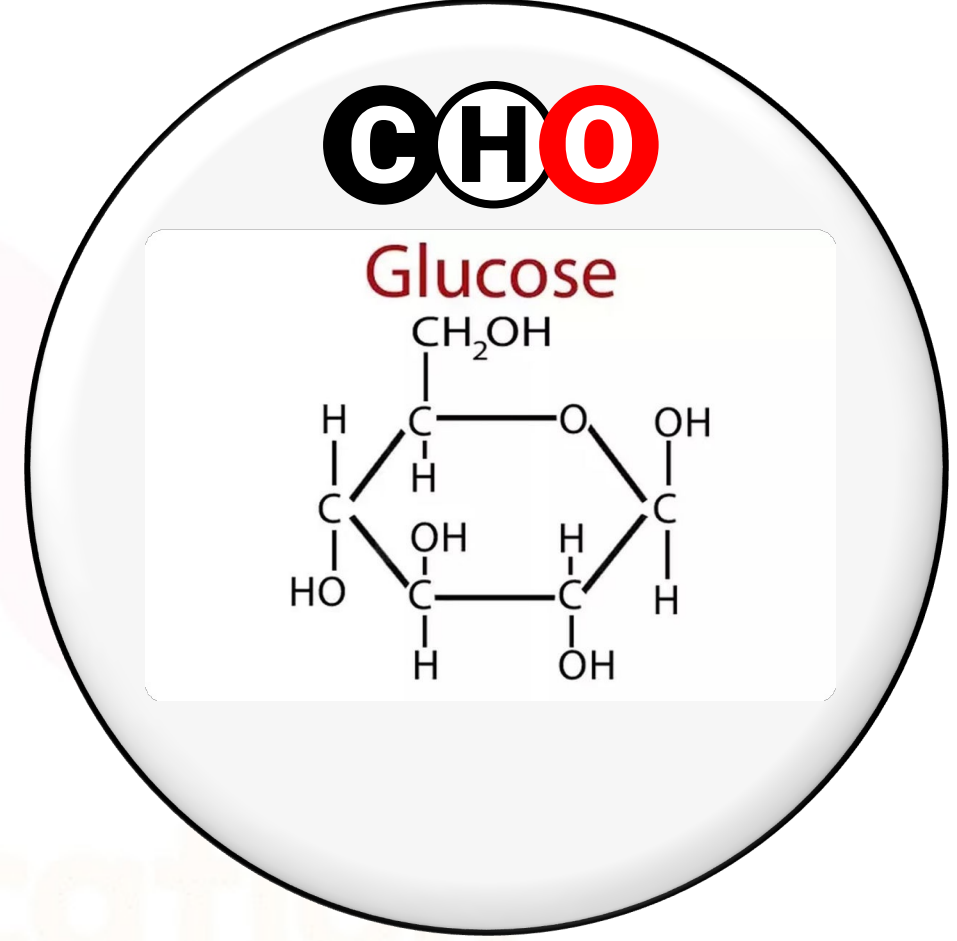
# 1. Carbohydrates

碳水化合物

## Condensation 縮合



**Monosaccharides**

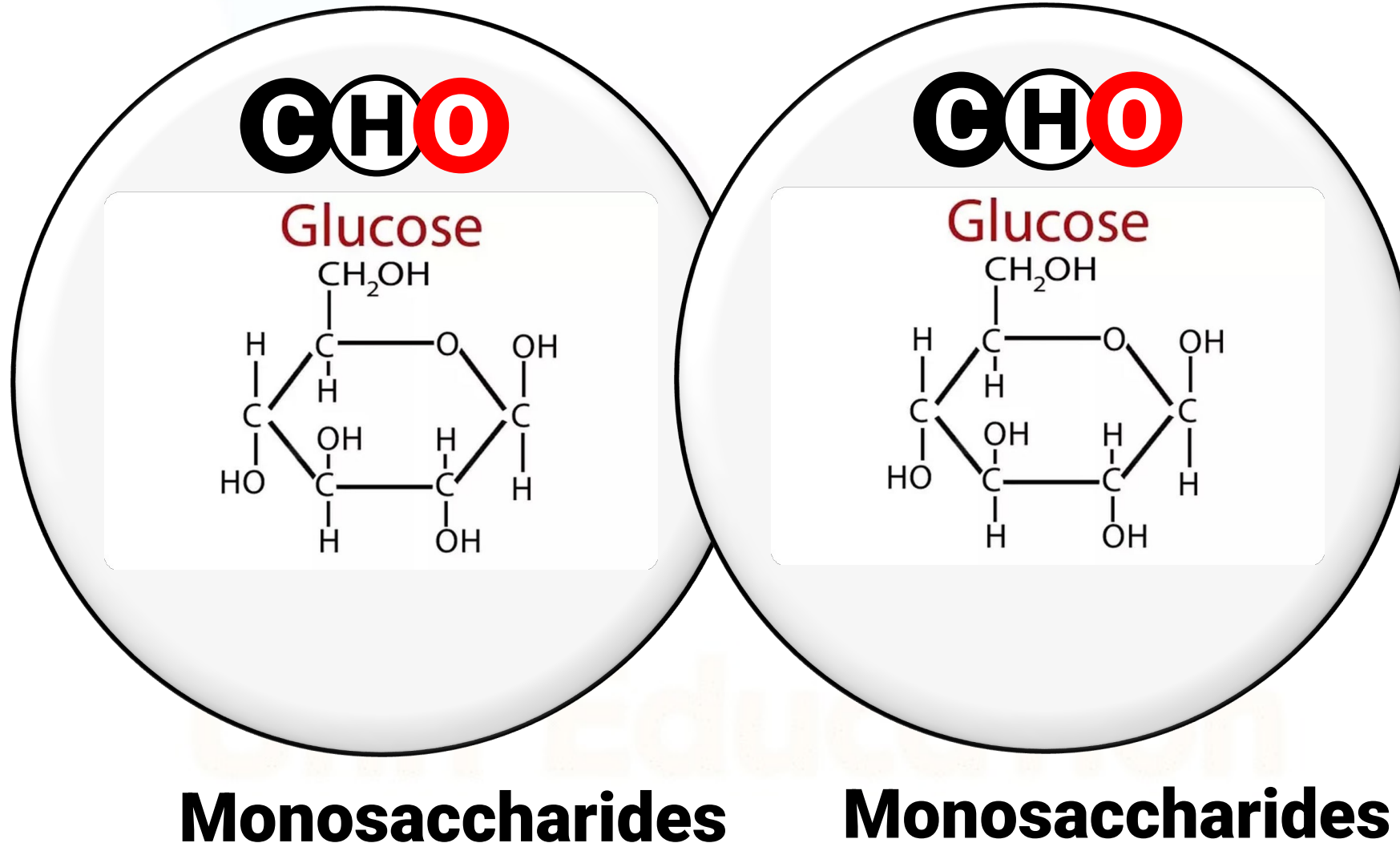


**Monosaccharides**



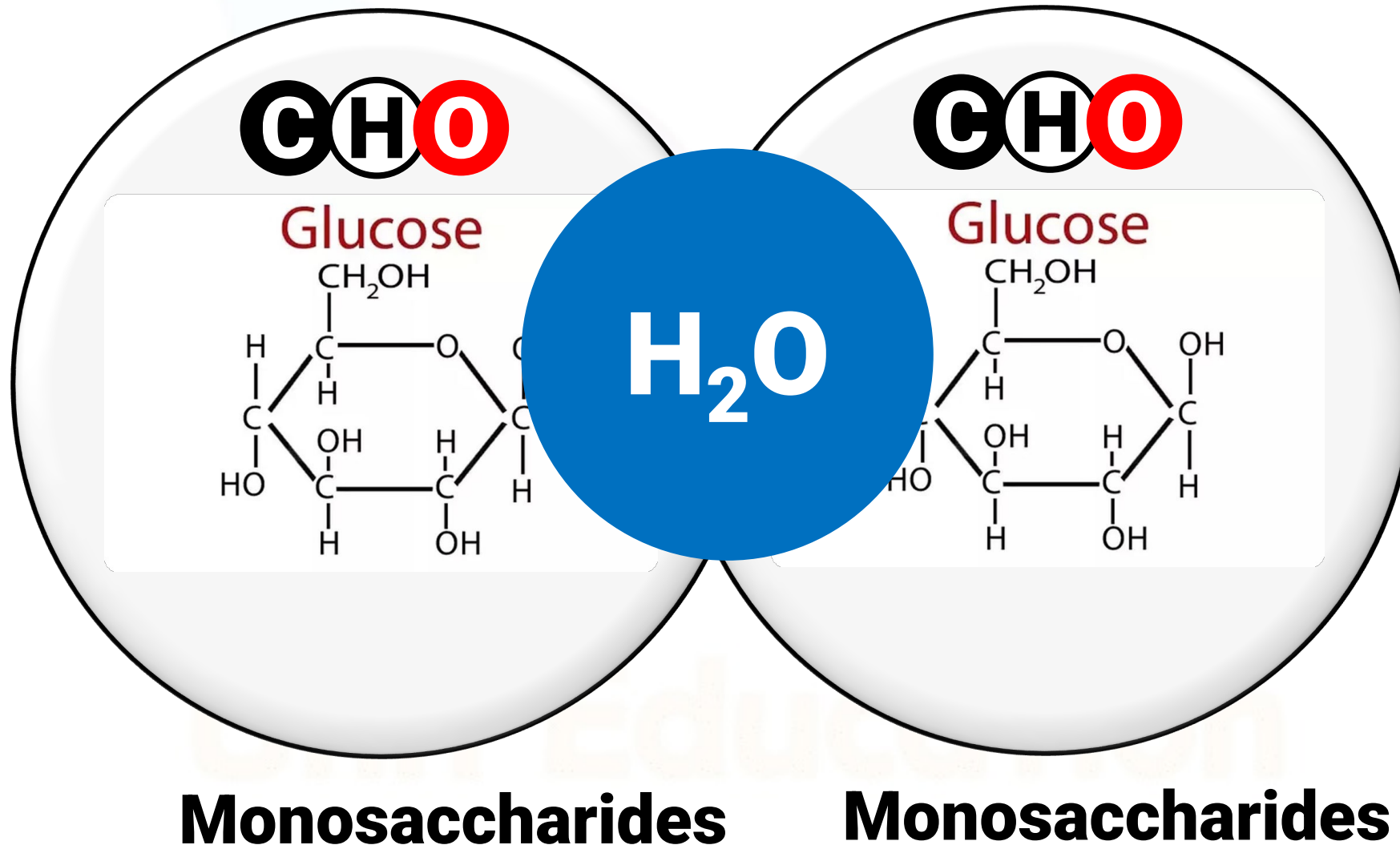
# 1. Carbohydrates 碳水化合物

## Condensation 縮合



# 1. Carbohydrates 碳水化合物

## Condensation 縮合

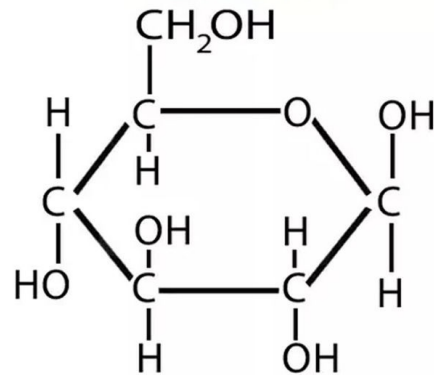


# 1. Carbohydrates 碳水化合物

## Condensation 縮合



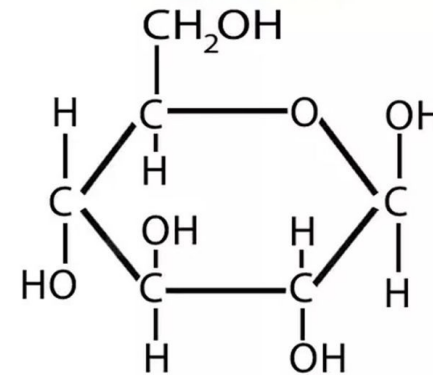
Glucose



Monosaccharides



Glucose



Monosaccharides

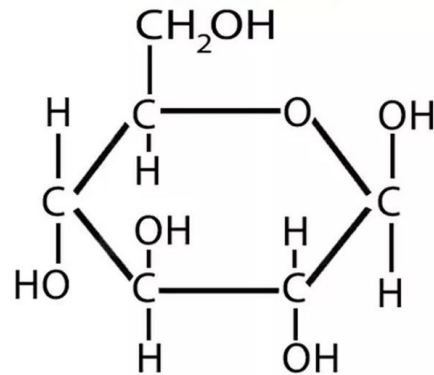


# 1. Carbohydrates 碳水化合物

## Condensation 縮合



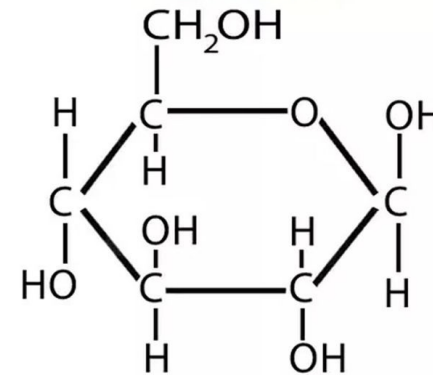
Glucose



Monosaccharides



Glucose

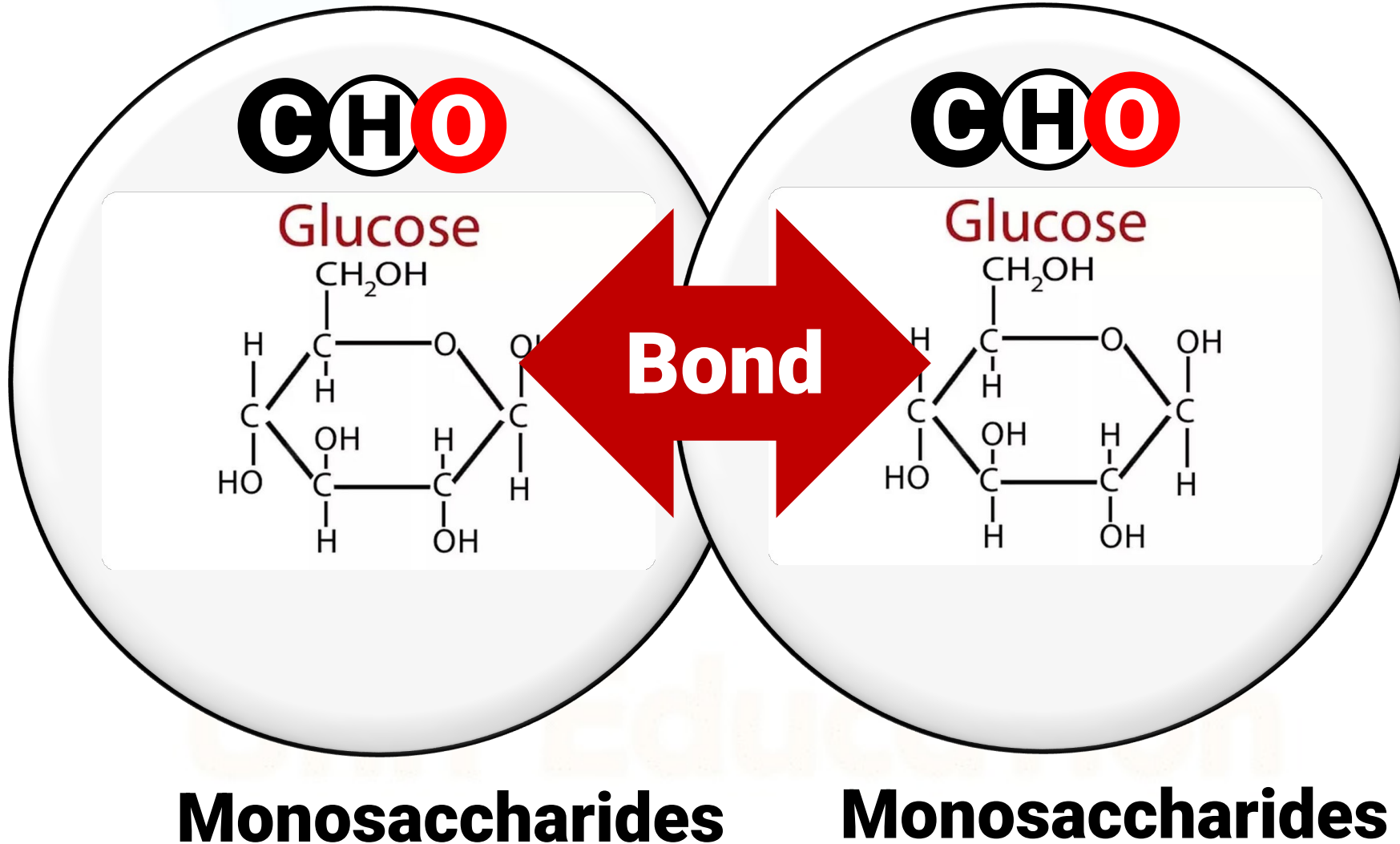
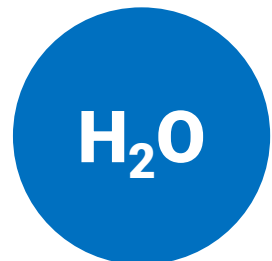


Monosaccharides



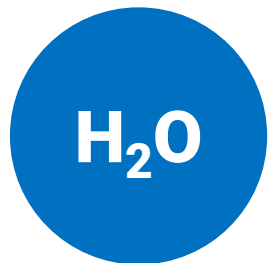
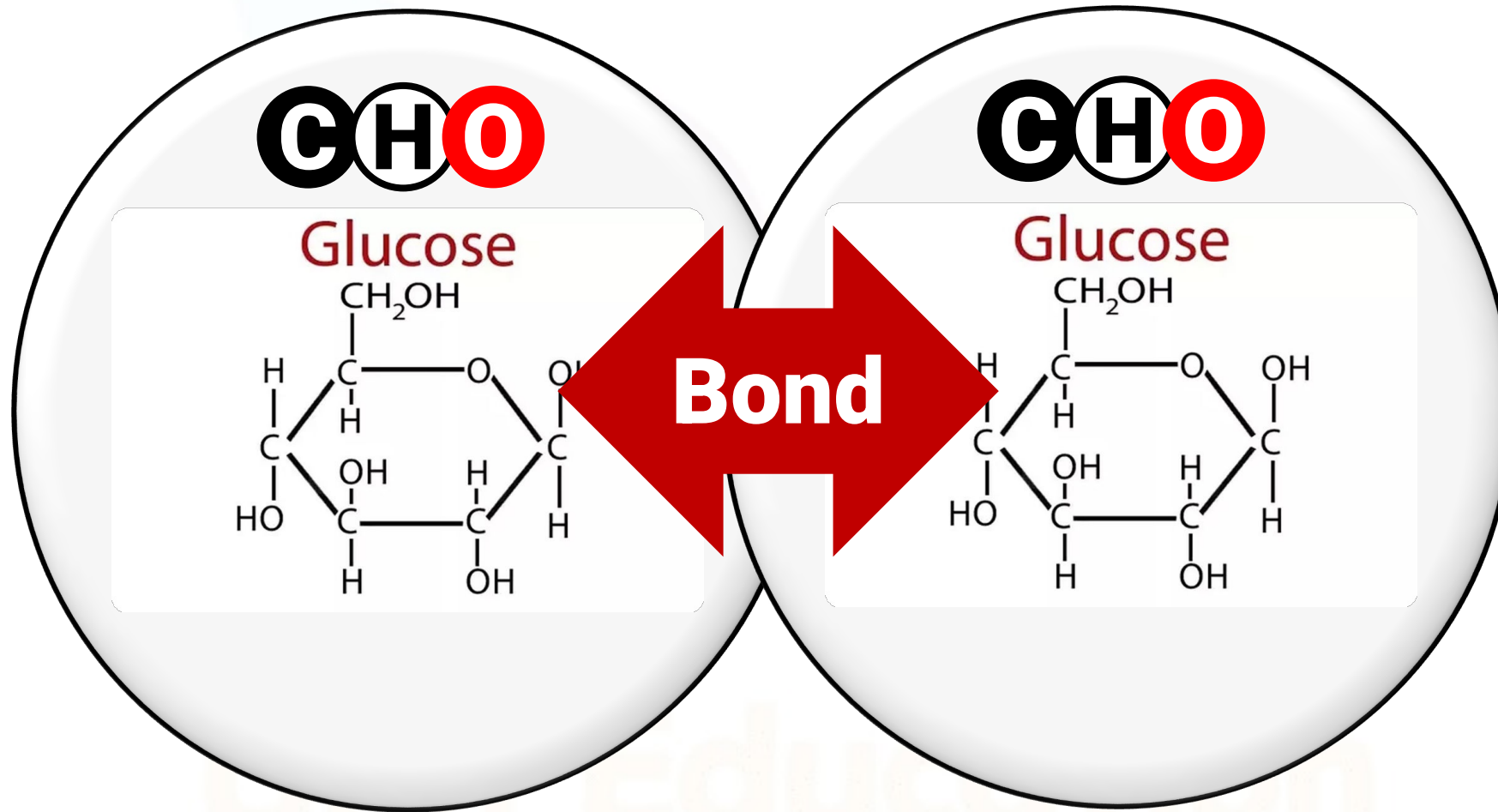
# 1. Carbohydrates 碳水化合物

## Condensation 縮合



# 1. Carbohydrates 碳水化合物

## Condensation 縮合

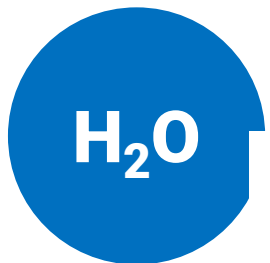
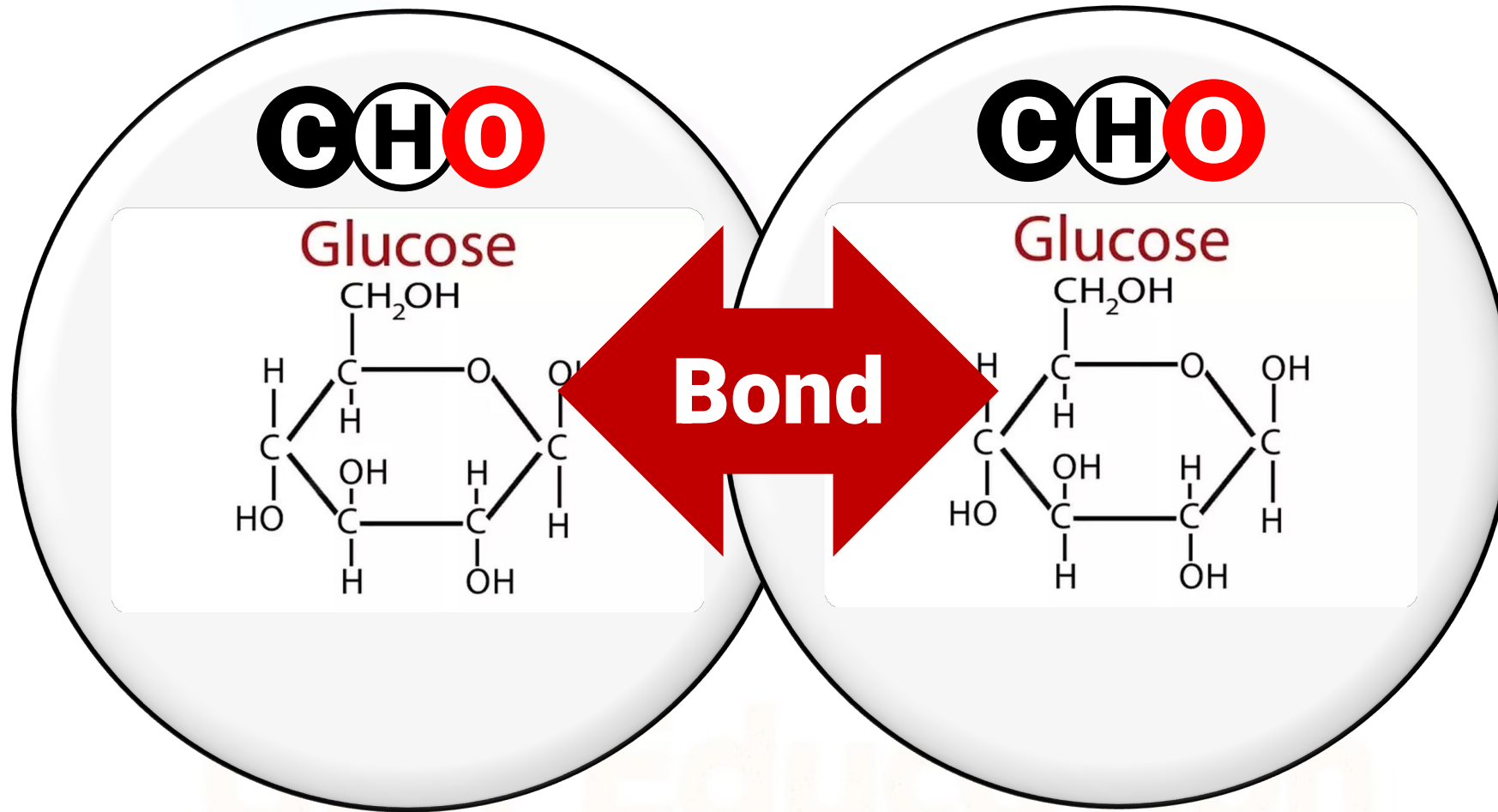


Di (二)-mer

es

# 1. Carbohydrates 碳水化合物

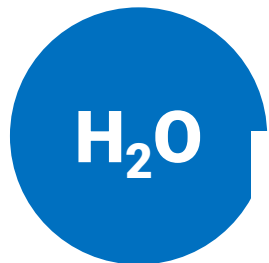
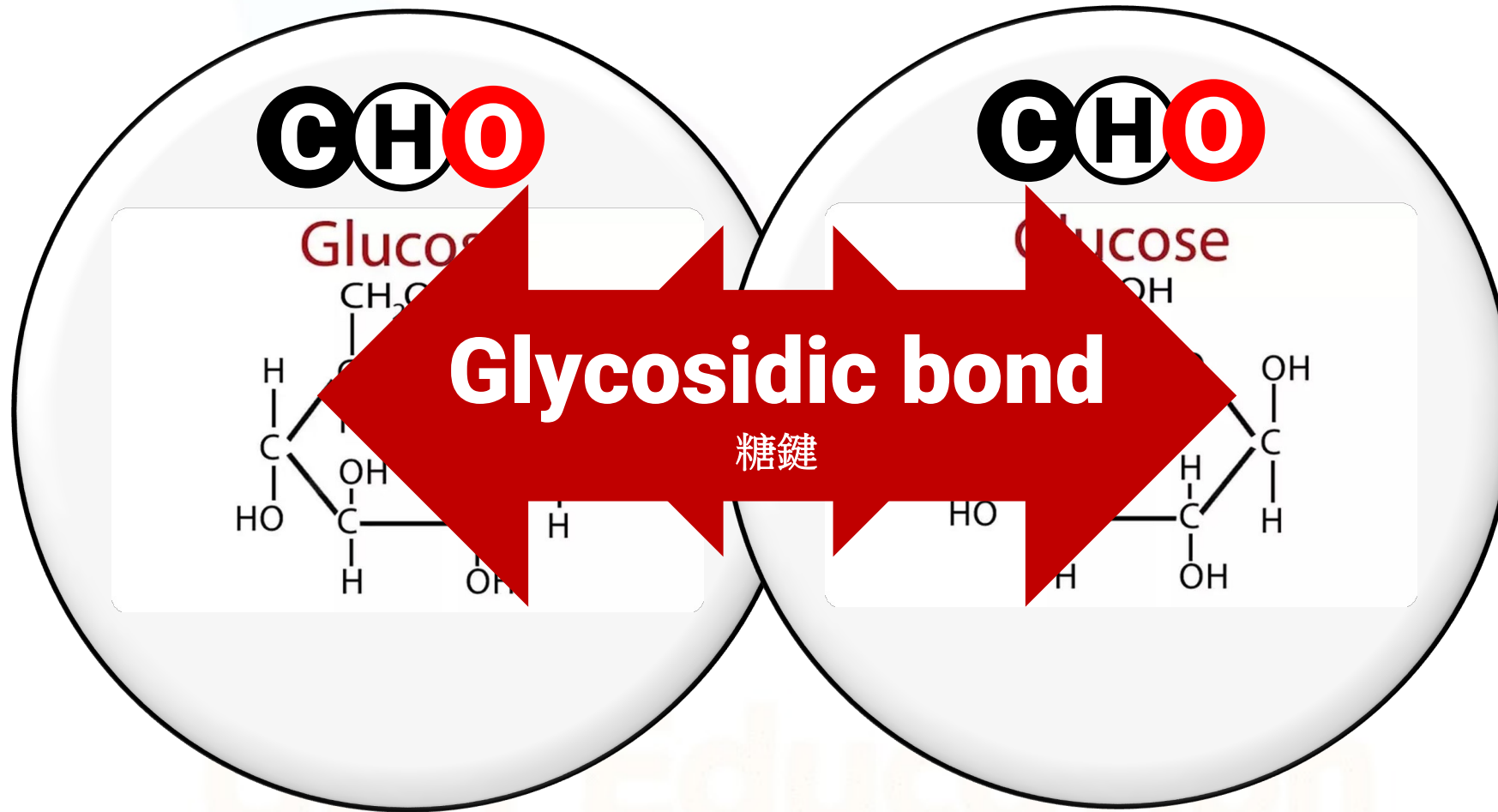
## Condensation 縮合



Disaccharides (雙糖)

# 1. Carbohydrates 碳水化合物

## Condensation 縮合



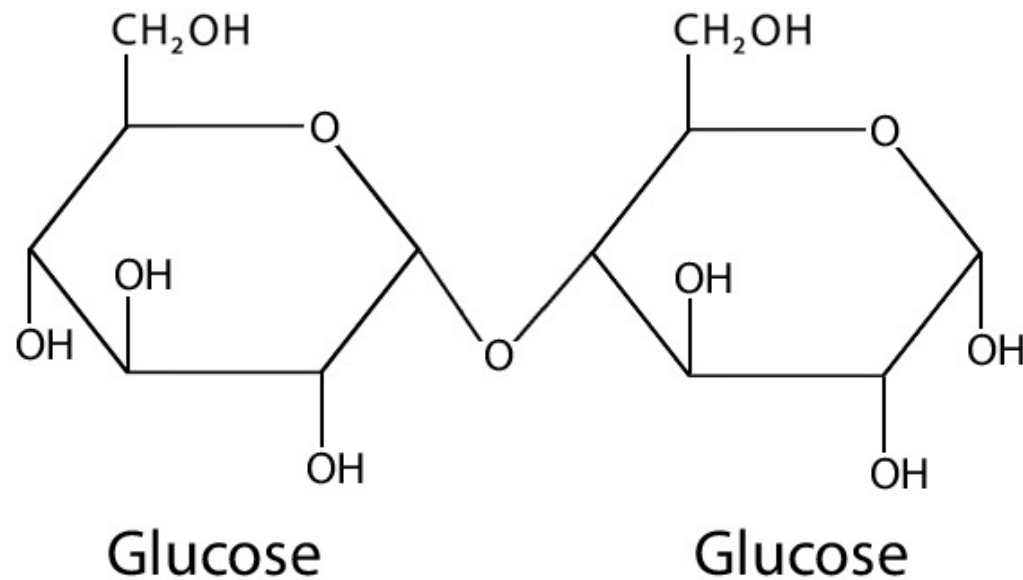
Disaccharides (雙糖)



# 1. Carbohydrates 碳水化合物

## Condensation 縮合

### Maltose



$\text{H}_2\text{O}$

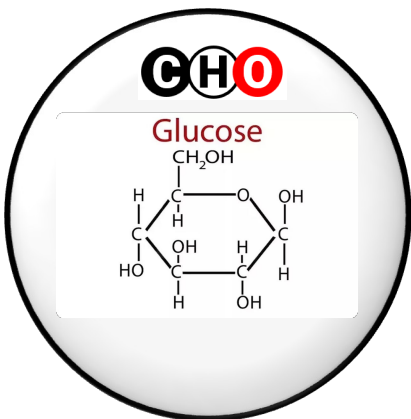
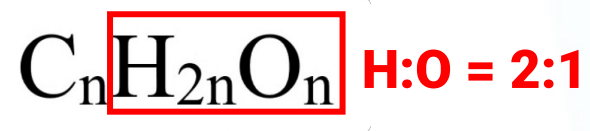
**Disaccharides (雙糖)**

# 1. Carbohydrates

## 碳水化合物



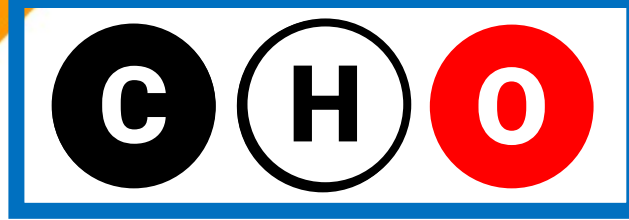
Nitrogen    Phosphorus    Sulfur



Monosaccharides  
(單糖)

# 1. Carbohydrates

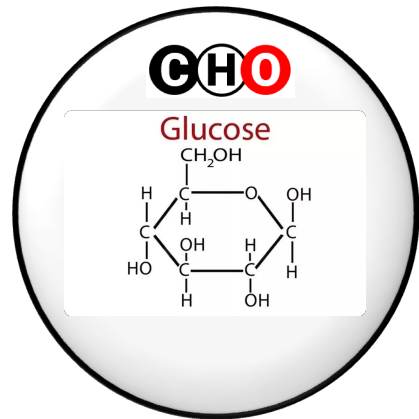
碳水化合物



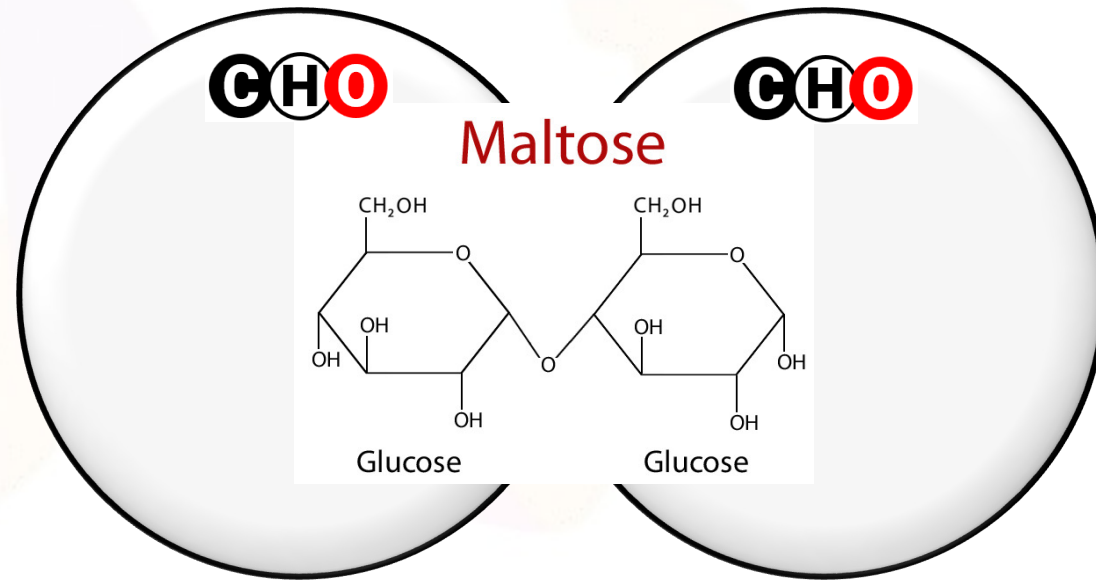
Nitrogen

Phosphorus

Sulfur



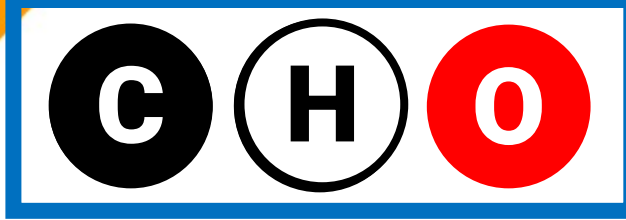
**Monosaccharides**  
(單糖)



Unit Education

# 1. Carbohydrates

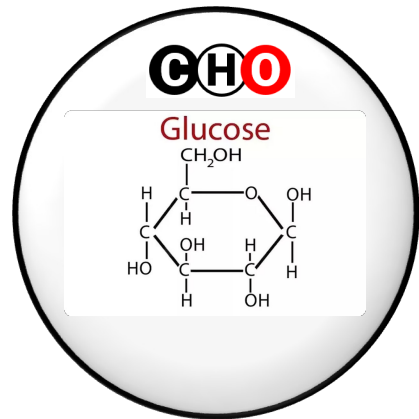
碳水化合物



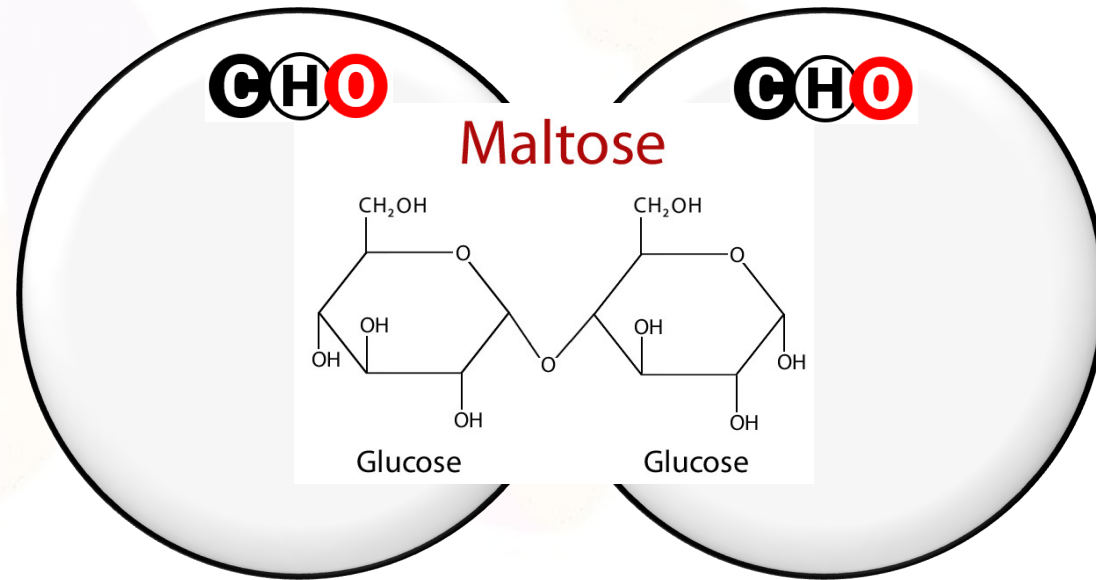
Nitrogen

Phosphorus

Sulfur



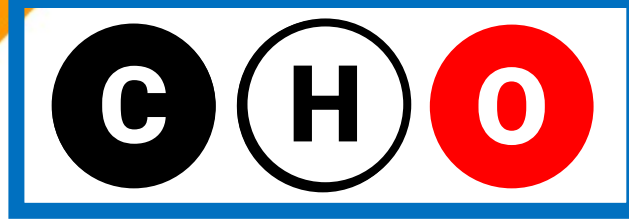
**Monosaccharides**  
(單糖)



**Disaccharides (雙糖)**

# 1. Carbohydrates

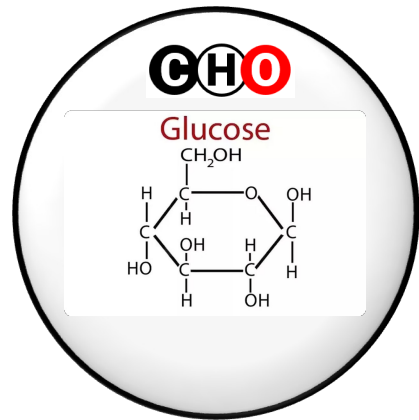
碳水化合物



Nitrogen

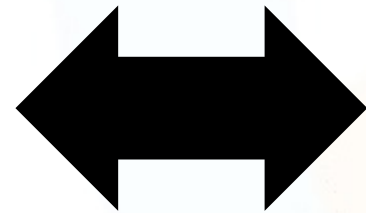
Phosphorus

Sulfur

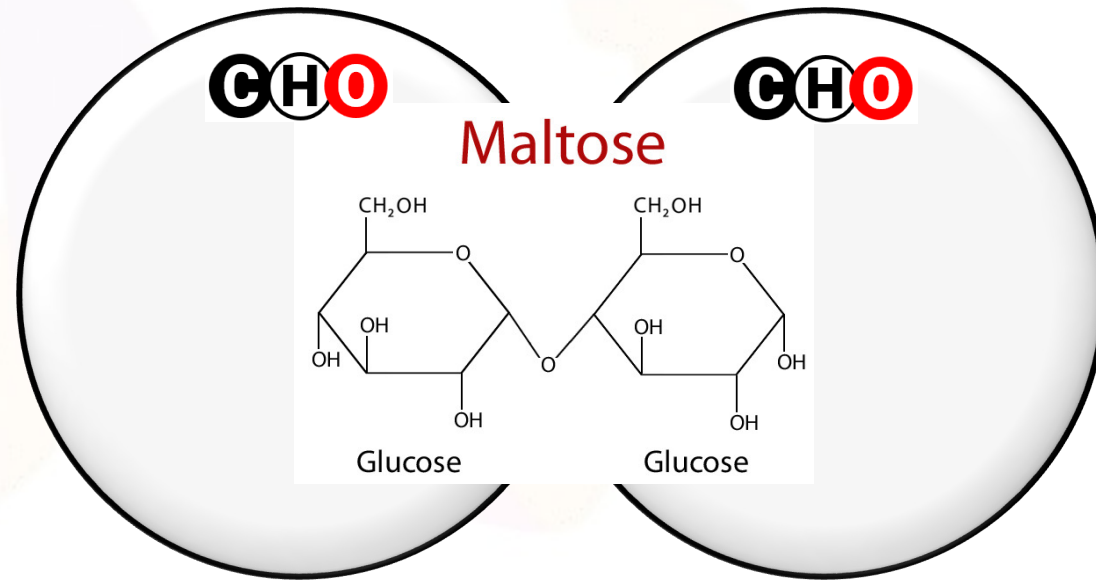


**Monosaccharides**  
(單糖)

**Hydrolysis**



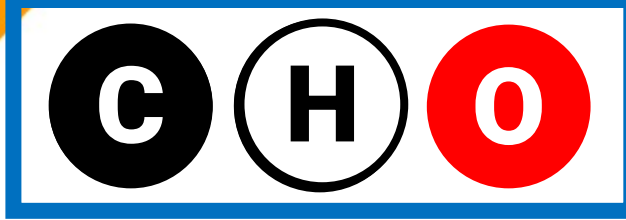
**Condensation**



**Disaccharides (雙糖)**

# 1. Carbohydrates

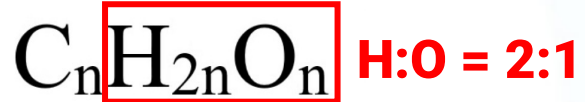
碳水化合物



Nitrogen

Phosphorus

Sulfur



**Disaccharides (雙糖)**

**Disaccharides 雙糖**

- Soluble in water
- Taste sweet
- Reducing sugars  
(except sucrose)

**Maltose 麥芽糖**



**Sucrose 蔗糖**

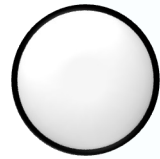


**Lactose 乳糖**



# 1. Carbohydrates 碳水化合物

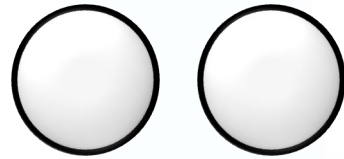
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

## Condensation 縮合

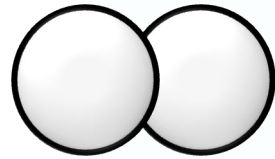


**Poly (多)-mer**



# 1. Carbohydrates 碳水化合物

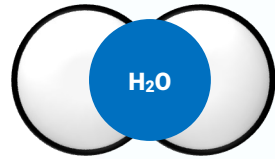
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

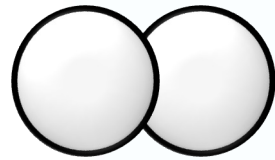
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

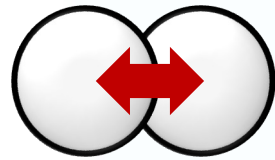
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

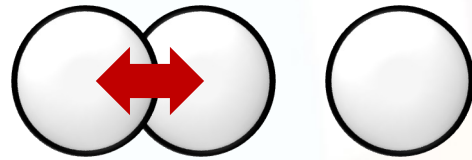
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

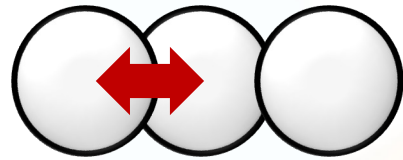
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

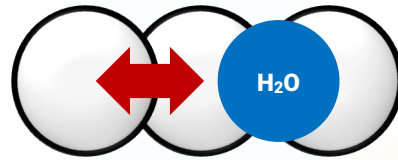
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

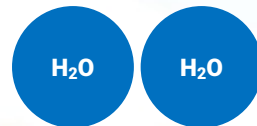
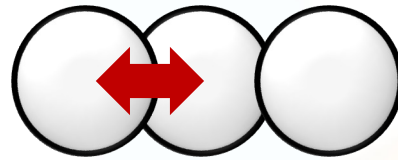
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

## Condensation 縮合

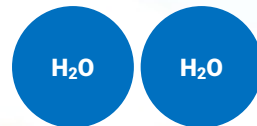
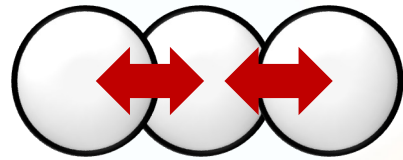


**Poly (多)-mer**



# 1. Carbohydrates 碳水化合物

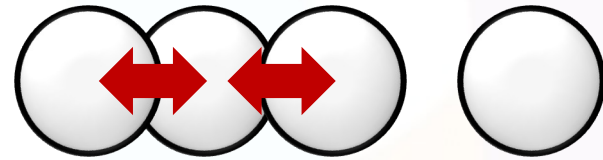
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

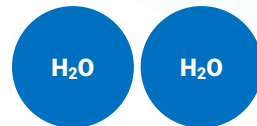
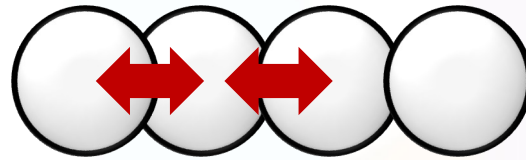
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

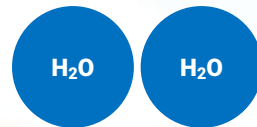
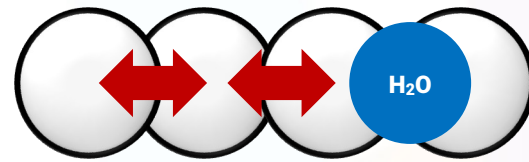
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

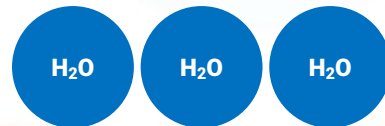
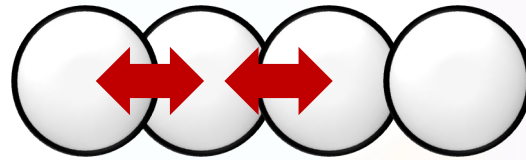
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

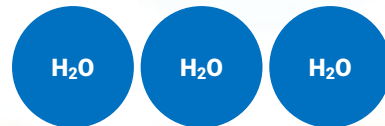
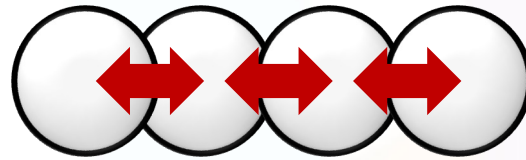
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

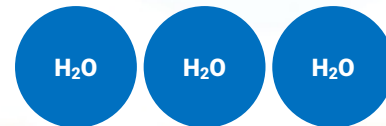
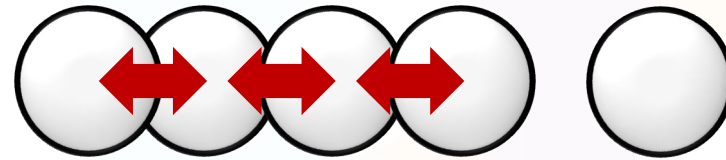
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

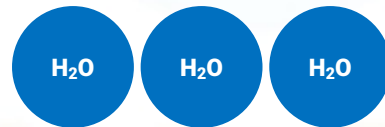
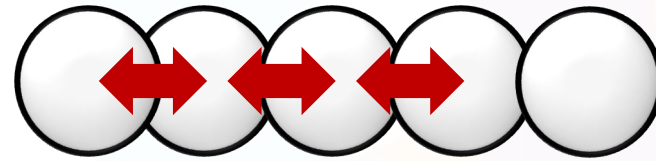
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

## Condensation 縮合

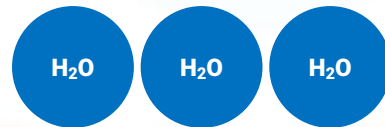
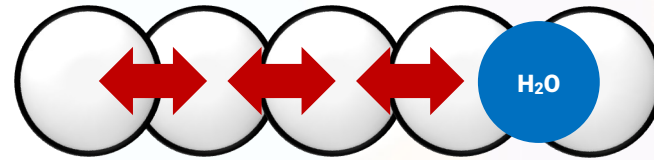


**Poly (多)-mer**



# 1. Carbohydrates 碳水化合物

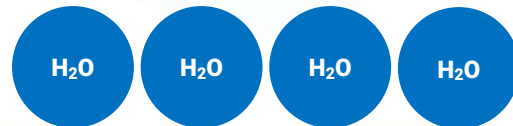
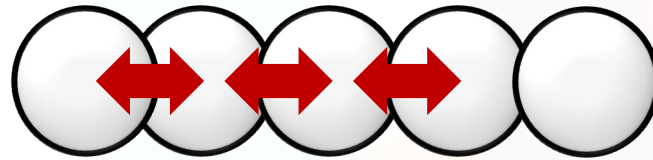
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

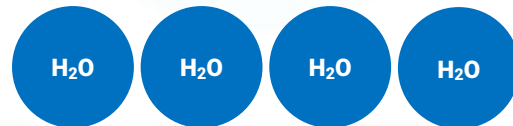
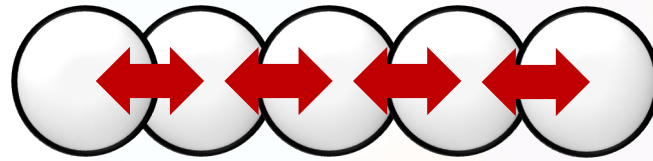
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

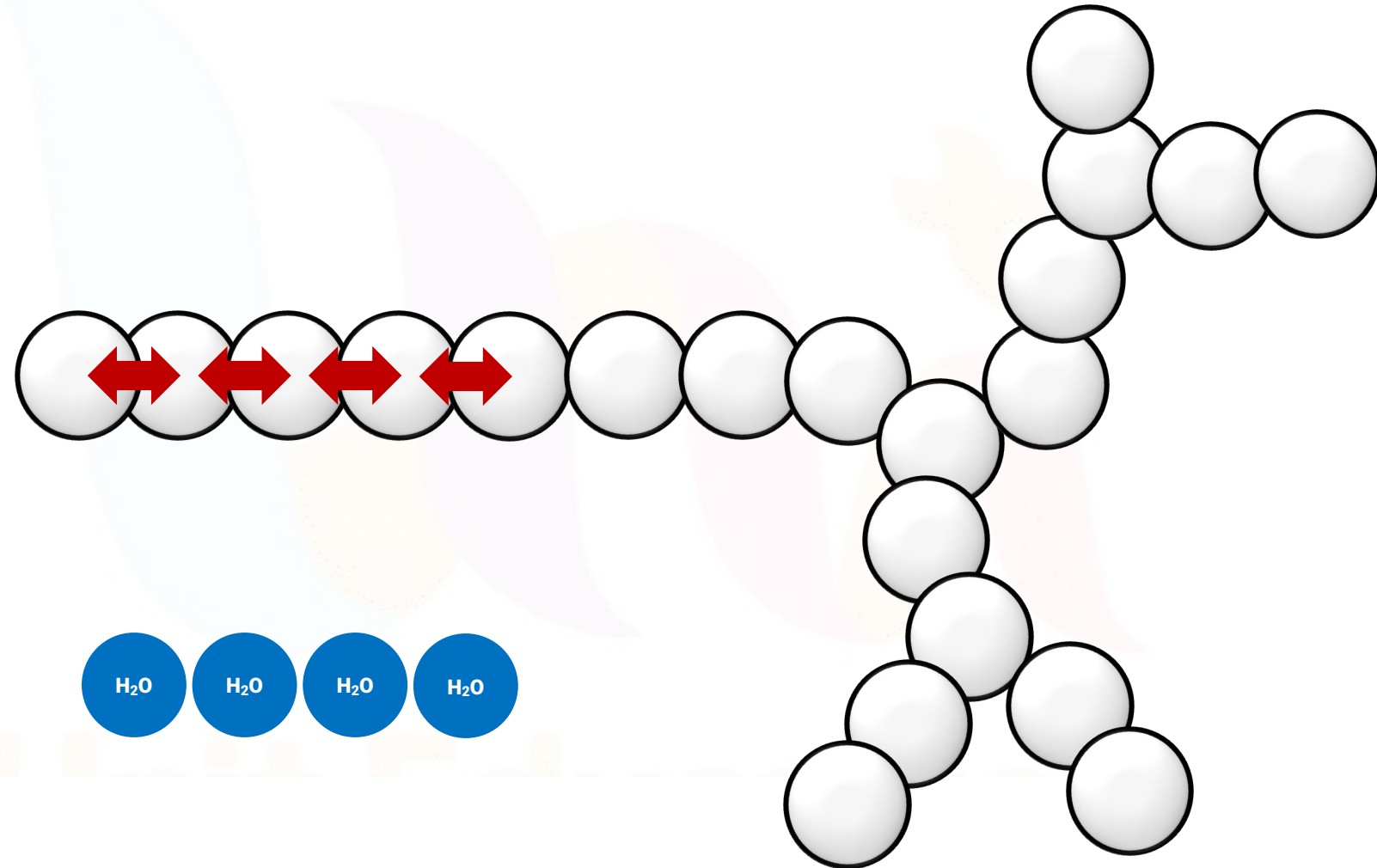
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

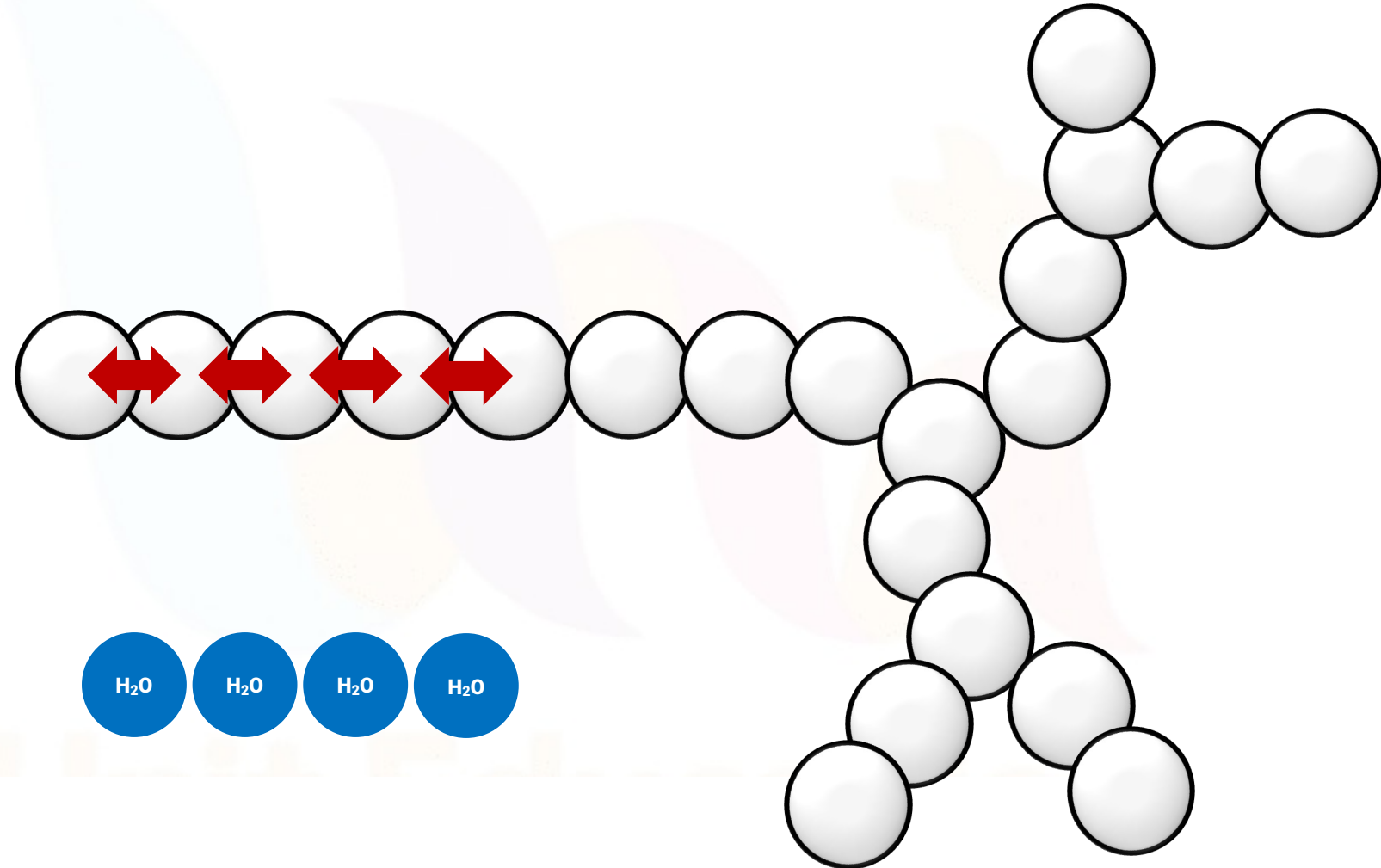
## Condensation 縮合



**Poly (多)-mer**

# 1. Carbohydrates 碳水化合物

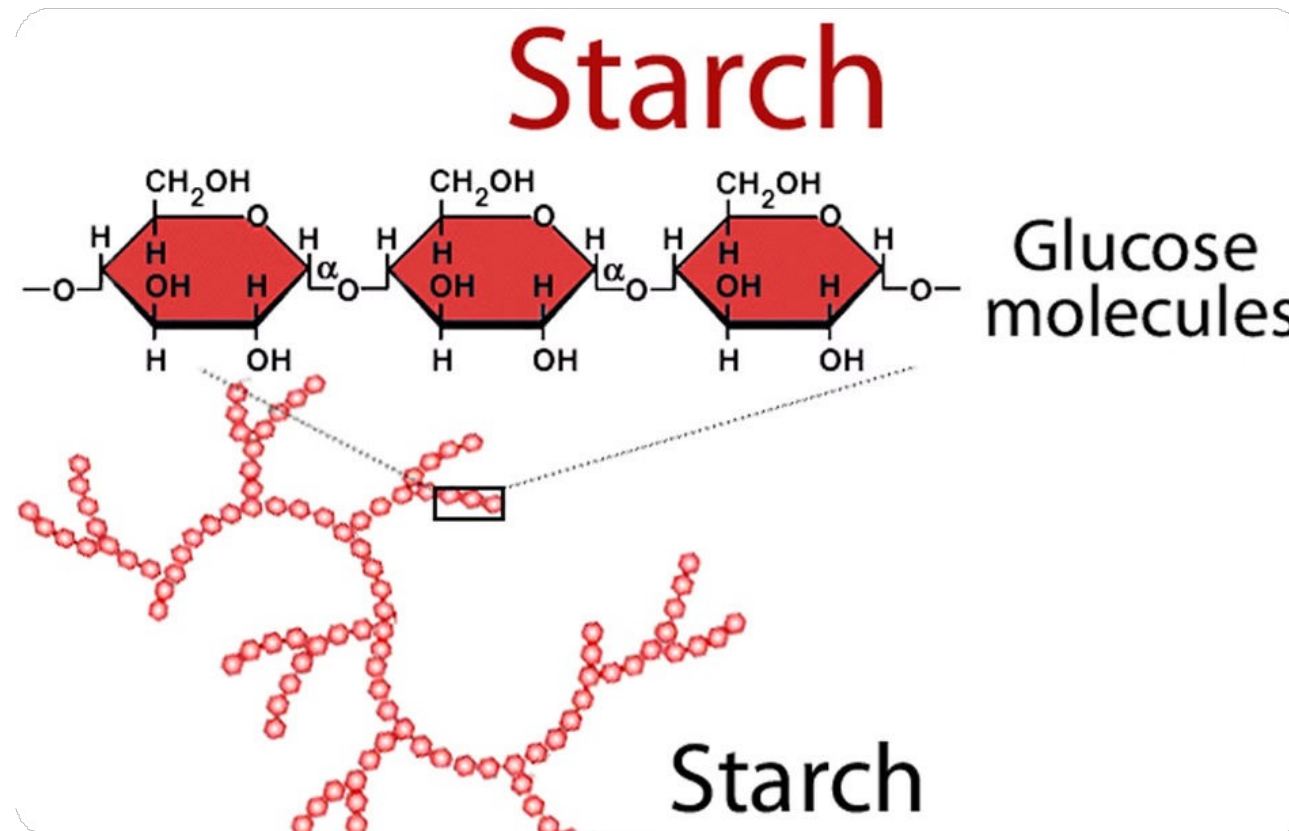
## Condensation 縮合



**Polysaccharides (多醣)**

# 1. Carbohydrates 碳水化合物

## Condensation 縮合



**Polysaccharides (多醣)**

# 1. Carbohydrates

## 碳水化合物

C

H

O

N

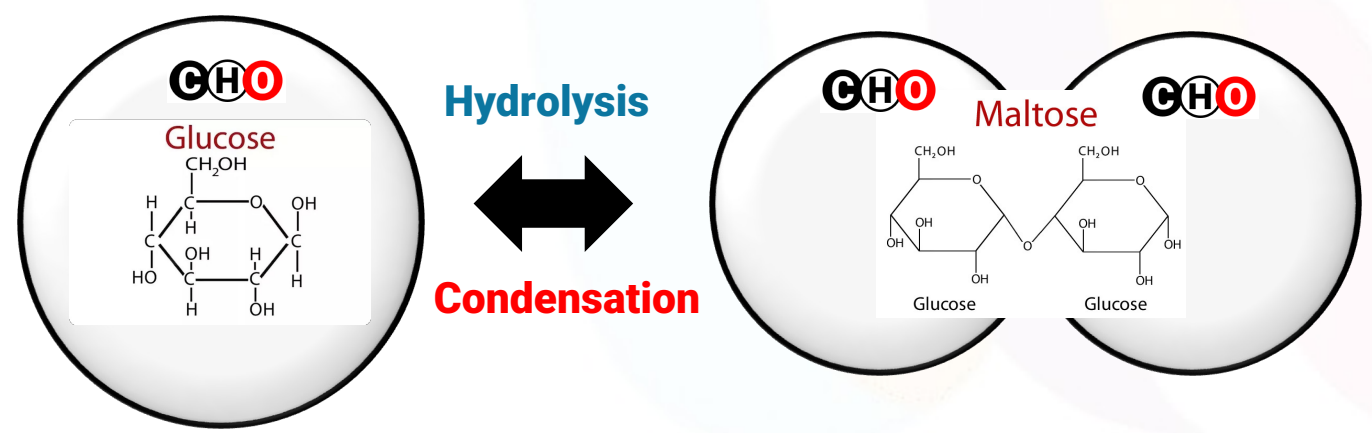
P

S

Nitrogen

Phosphorus

Sulfur



Monosaccharides  
(單糖)

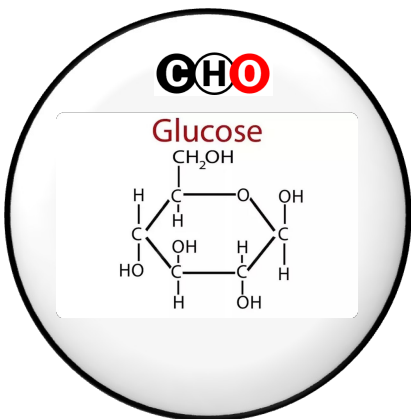
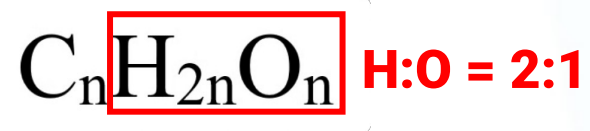
Disaccharides  
(雙糖)



Nitrogen    Phosphorus    Sulfur

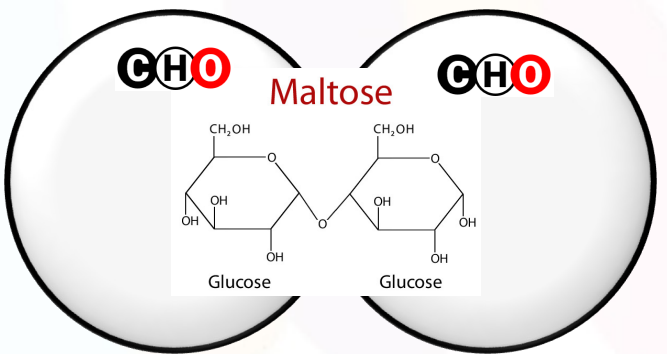
# 1. Carbohydrates

碳水化合物



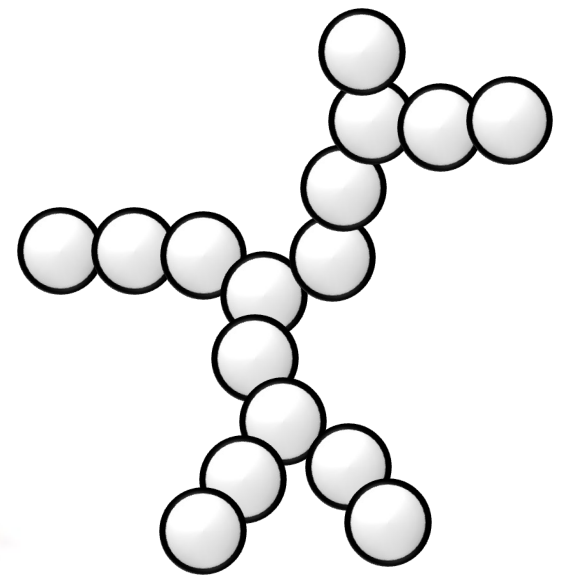
Monosaccharides  
(單糖)

Hydrolysis  
↔  
Condensation

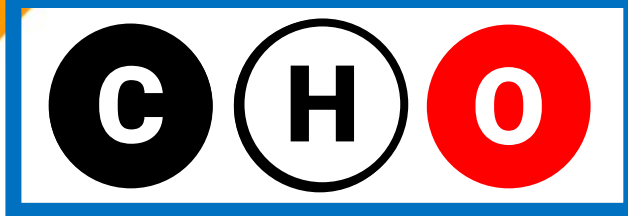


Disaccharides  
(雙糖)

Hydrolysis  
↔  
Condensation



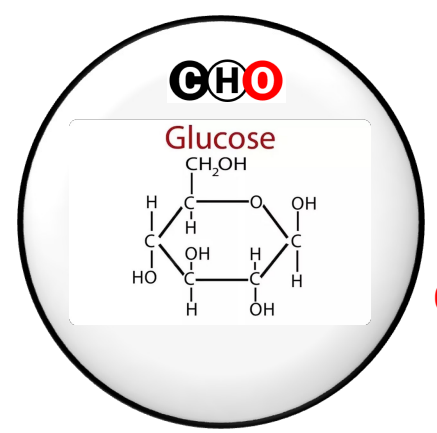
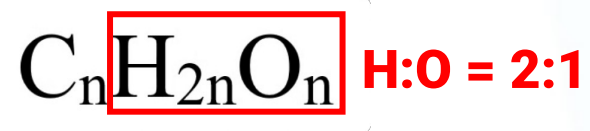




Nitrogen    Phosphorus    Sulfur

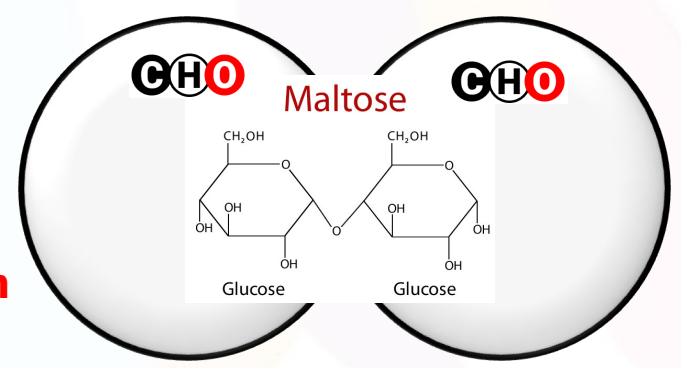
# 1. Carbohydrates

碳水化合物



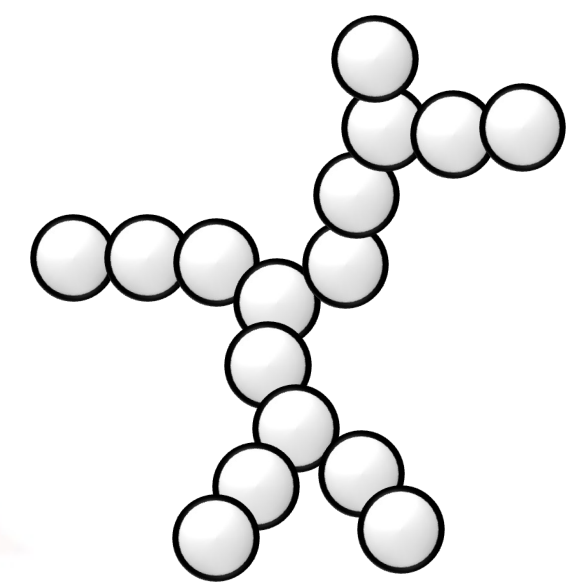
Monosaccharides  
(單糖)

Hydrolysis  
Condensation



Disaccharides  
(雙糖)

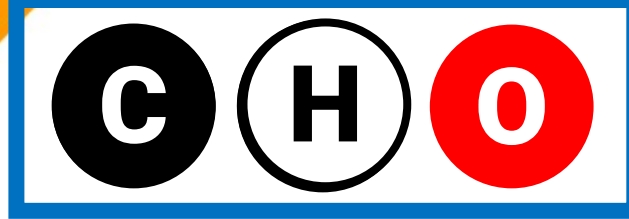
Hydrolysis  
Condensation



Polysaccharides (多糖)

# 1. Carbohydrates

碳水化合物



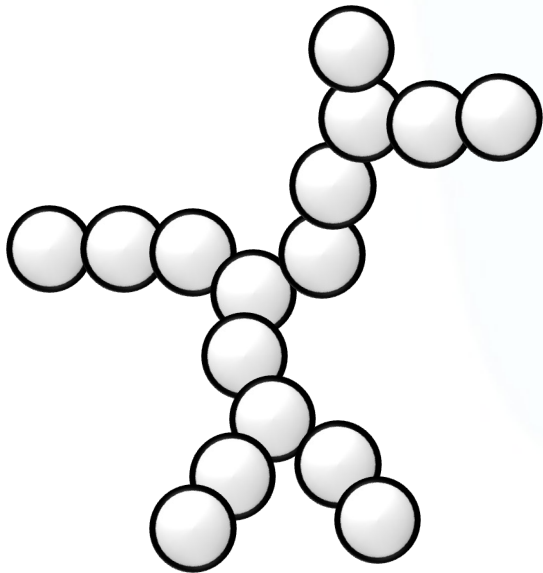
Nitrogen



Phosphorus



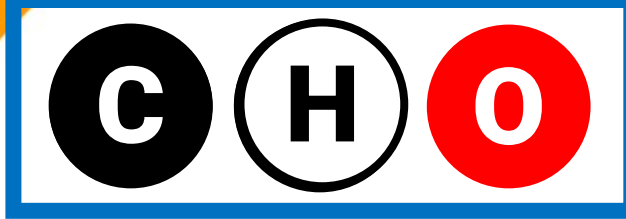
Sulfur



**Polysaccharides (多醣)**

# 1. Carbohydrates

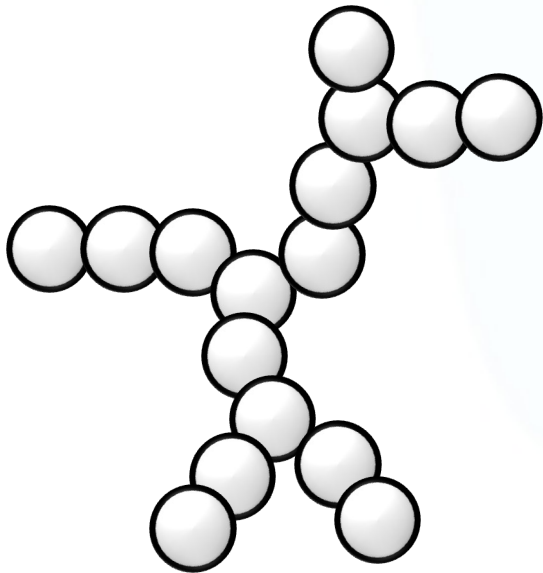
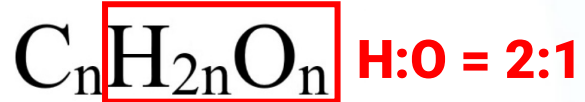
碳水化合物



Nitrogen

Phosphorus

Sulfur



**Polysaccharides** 多醣

- Insoluble in water
- Taste not sweet
- Non-reducing sugars  
(except sucrose)

Examples:

**Glycogen** 肝糖

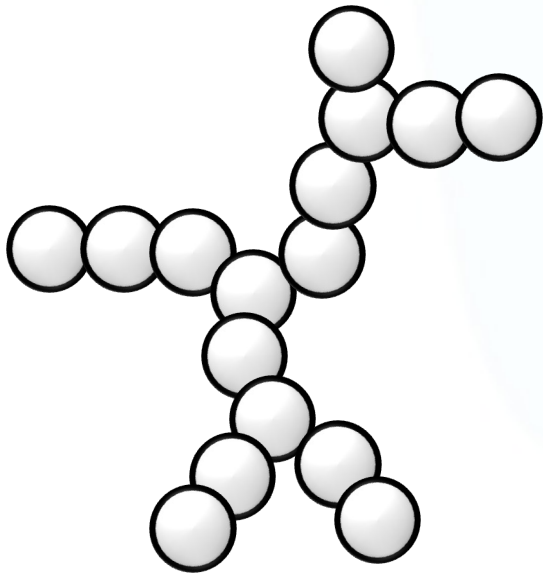
**Cellulose** 纖維素

**Starch** 澱粉

**Polysaccharides (多醣)**

# 1. Carbohydrates

碳水化合物



**Polysaccharides (多醣)**



Nitrogen

Phosphorus

Sulfur

**Polysaccharides 多醣**

- Insoluble in water
- Taste not sweet
- Non-reducing sugars  
(except sucrose)

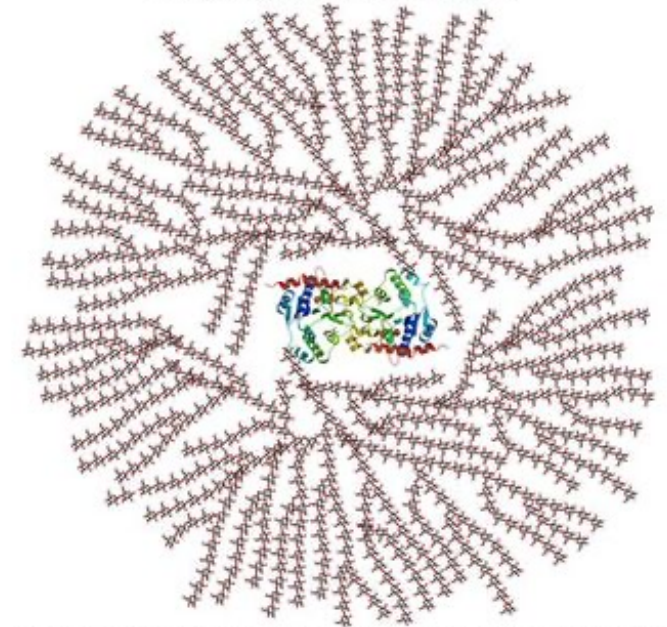
**Examples:**

**Glycogen 肝糖**

**Cellulose 纖維素**

**Starch 澱粉**

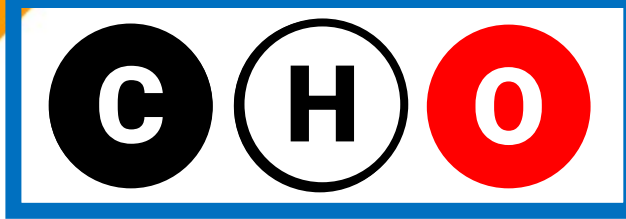
**Glycogen structure**



A core protein of glycogenin is surrounded by branches of glucose units.  
The entire globular complex may contain approximately 30,000 glucose units.

# 1. Carbohydrates

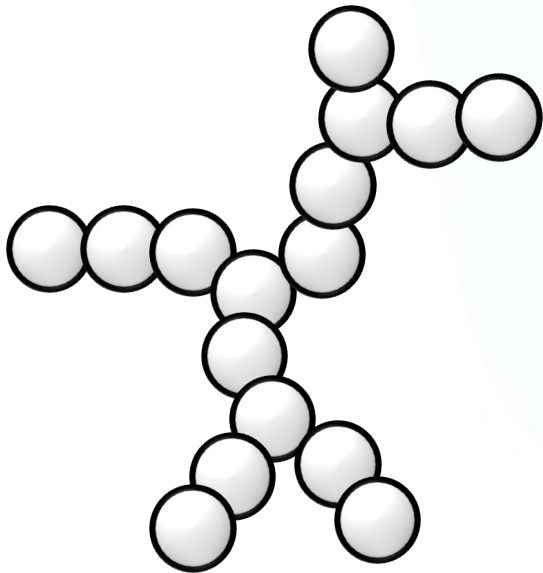
碳水化合物



Nitrogen

Phosphorus

Sulfur



**Polysaccharides** 多醣

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(except sucrose)

Examples:

**Glycogen** 肝糖

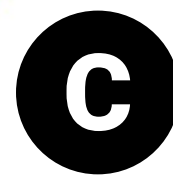
**Cellulose** 纖維素

**Starch** 澱粉

**Polysaccharides (多醣)**



# Organic biomolecules



Nitrogen

Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



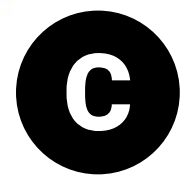
3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



Nitrogen

Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



3. Proteins 蛋白質

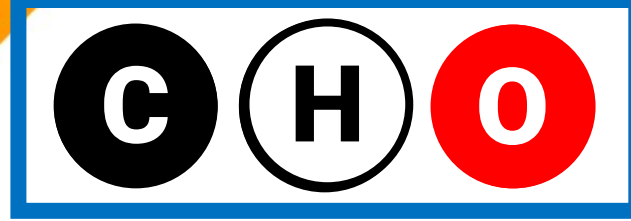


4. Nucleic acids 核酸





## 2. Lipids 脂質



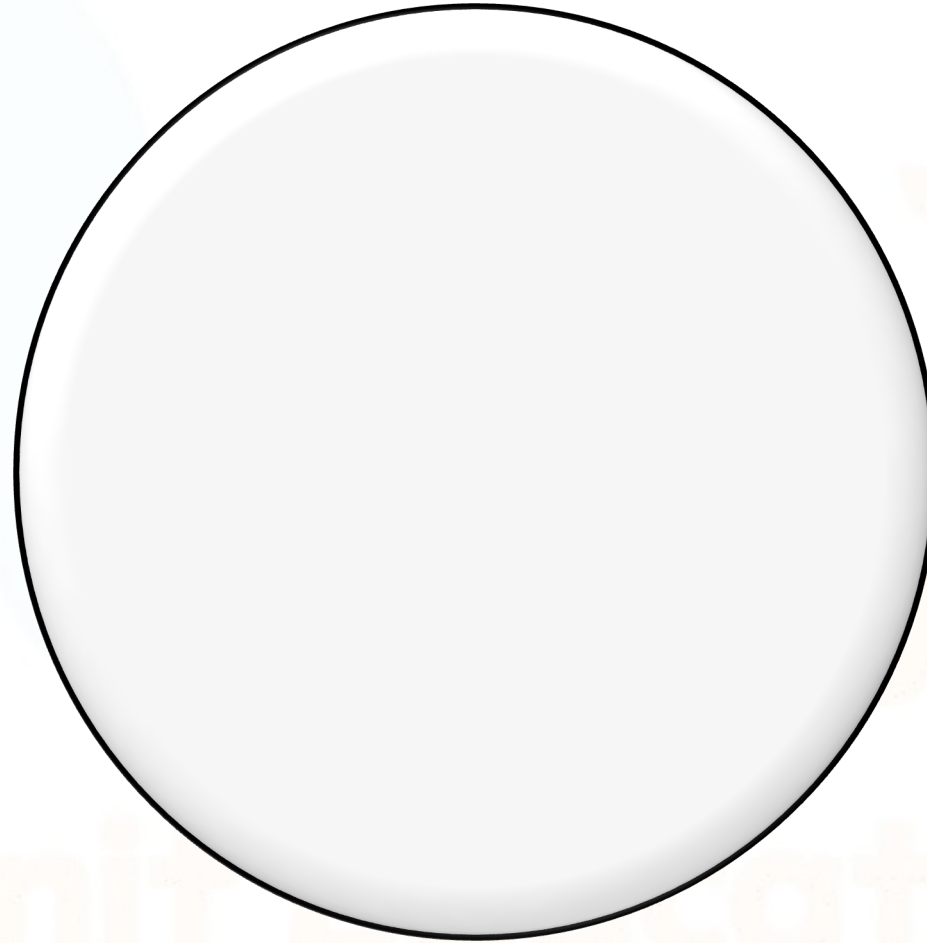
Nitrogen



Phosphorus



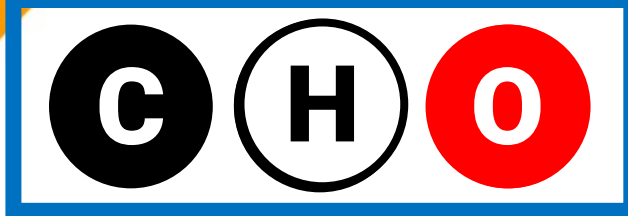
Sulfur



**Mono單-mer**



## 2. Lipids 脂質



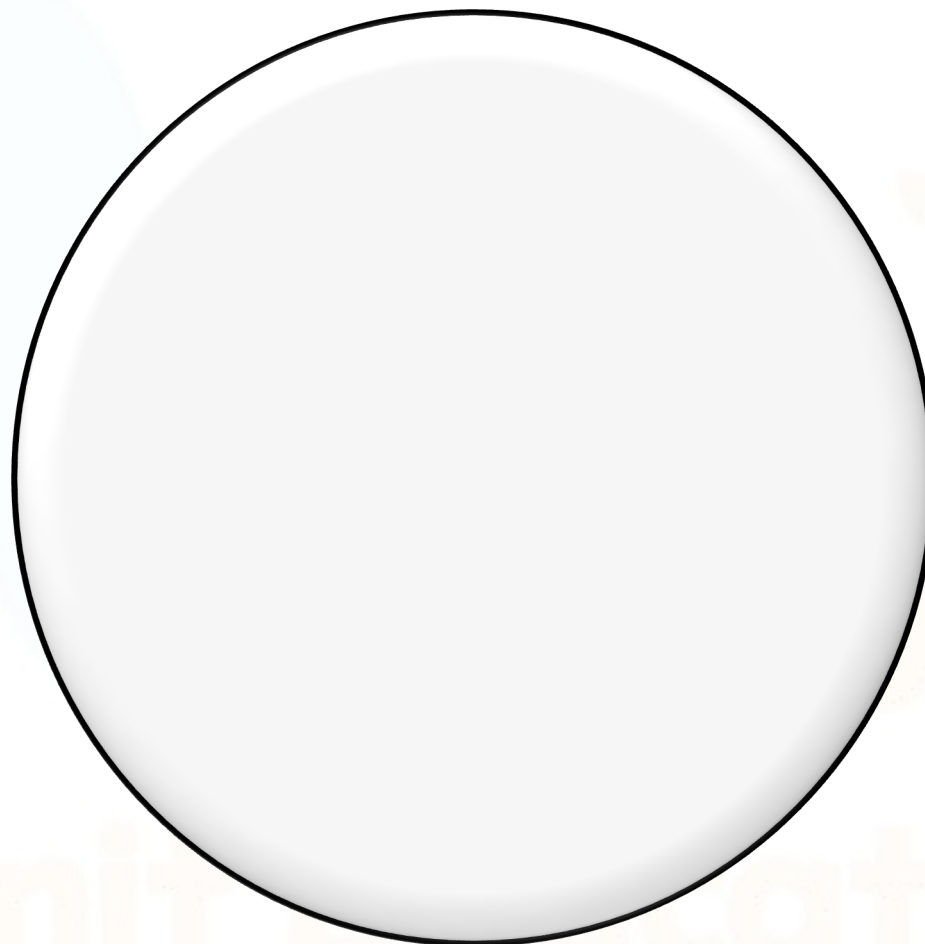
Nitrogen



Phosphorus

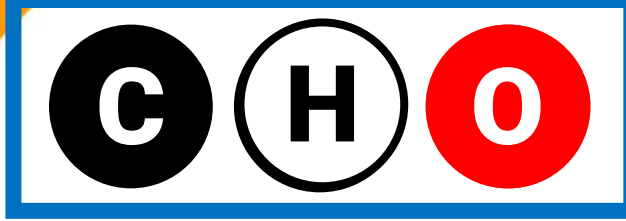


Sulfur



**Mono單-mer**

## 2. Lipids 脂質



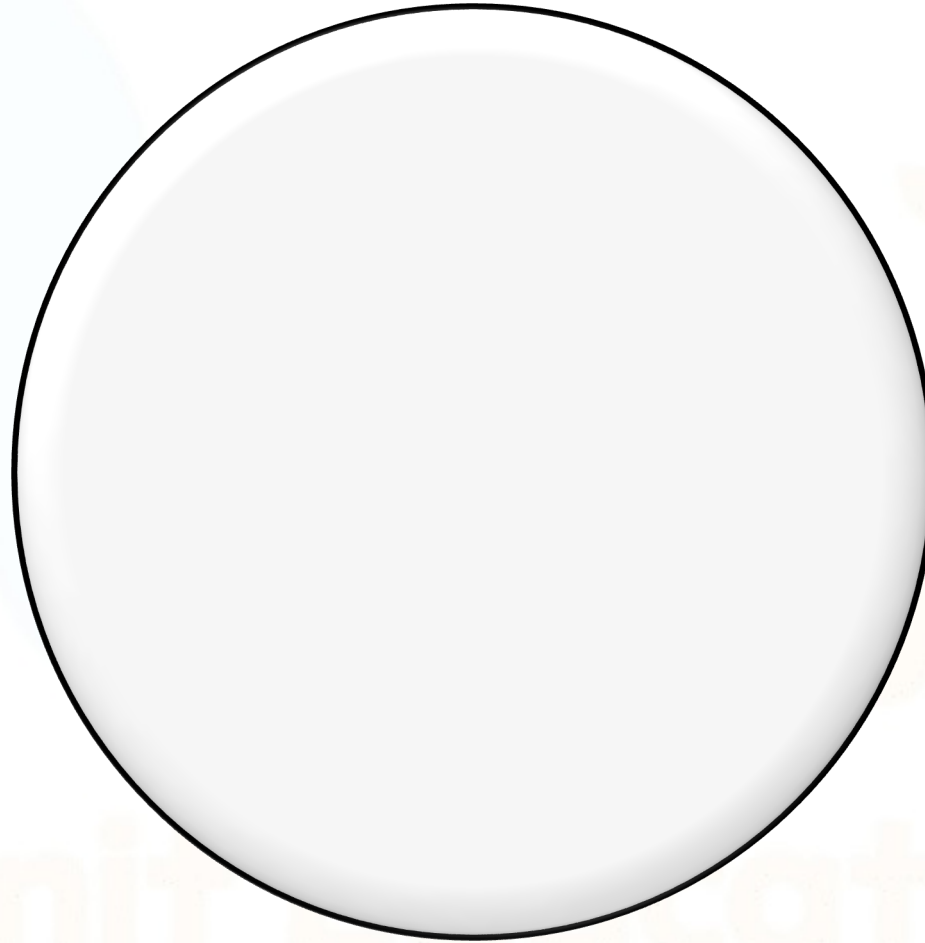
Nitrogen



Phosphorus



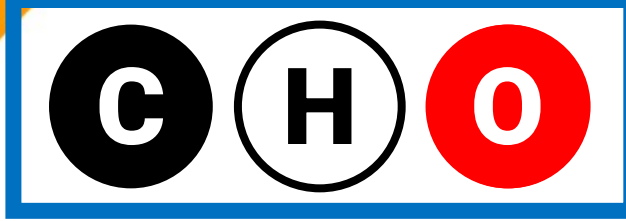
Sulfur



**Mono單-mer**

## 2. Lipids 脂質

$C_{51}H_{98}O_6$  **H:O > 2:1**



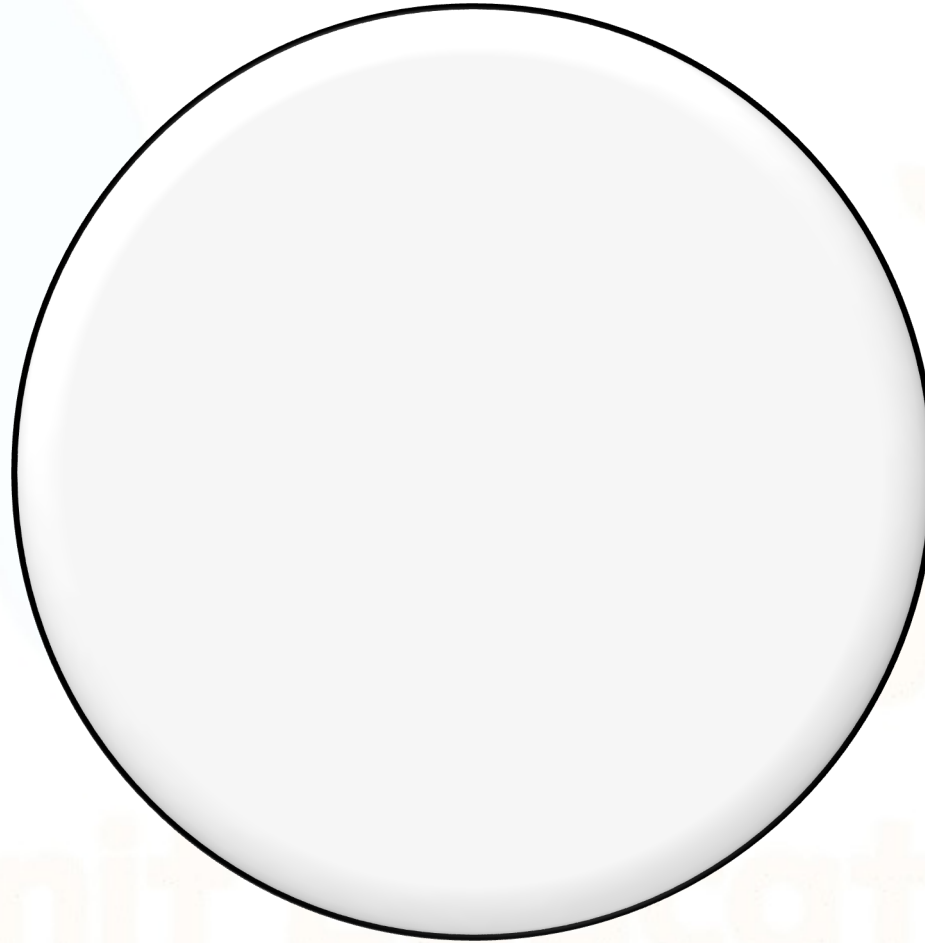
Nitrogen



Phosphorus



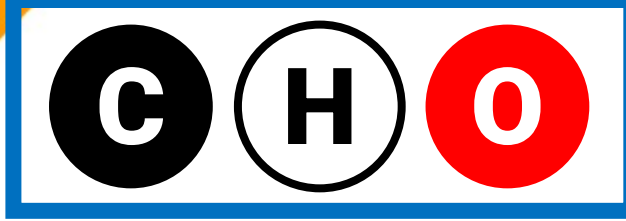
Sulfur



**Mono單-mer**

## 2. Lipids 脂質

$C_{51}H_{98}O_6$  **H:O > 2:1**



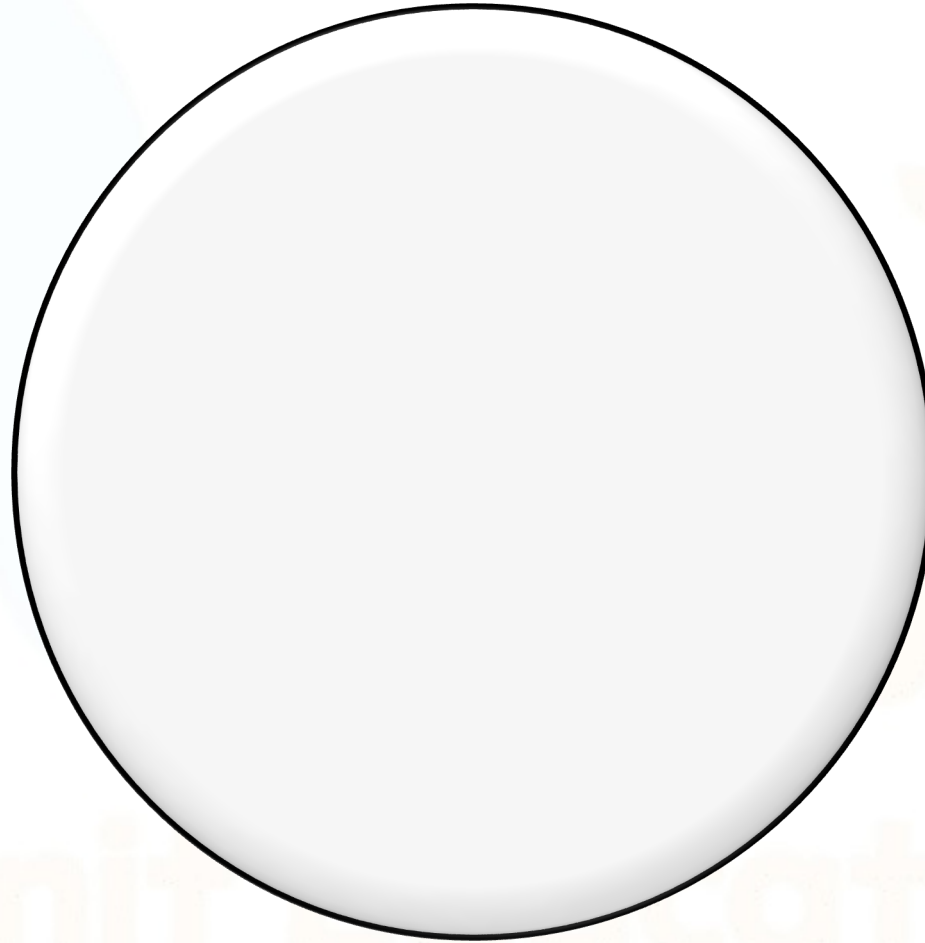
Nitrogen



Phosphorus

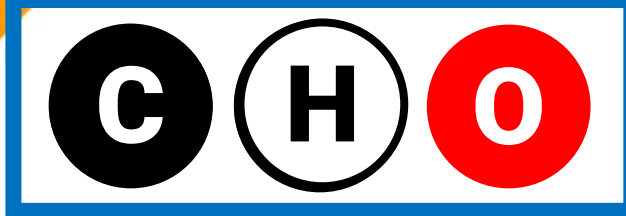


Sulfur



**Mono單-mer**

## 2. Lipids 脂質



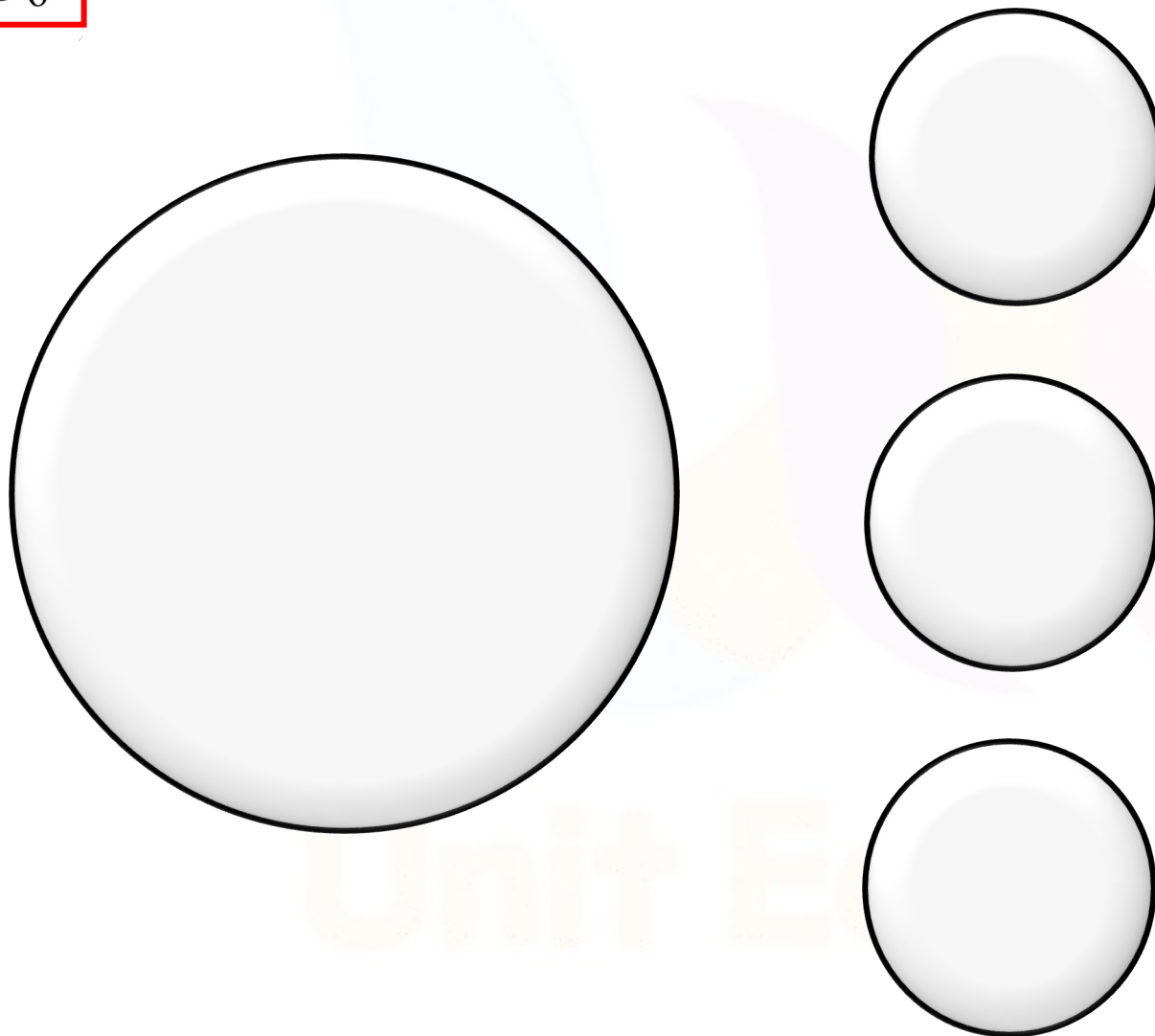
Nitrogen



Phosphorus



Sulfur



## 2. Lipids 脂質

$C_{51}H_{98}O_6$  **H:O > 2:1**



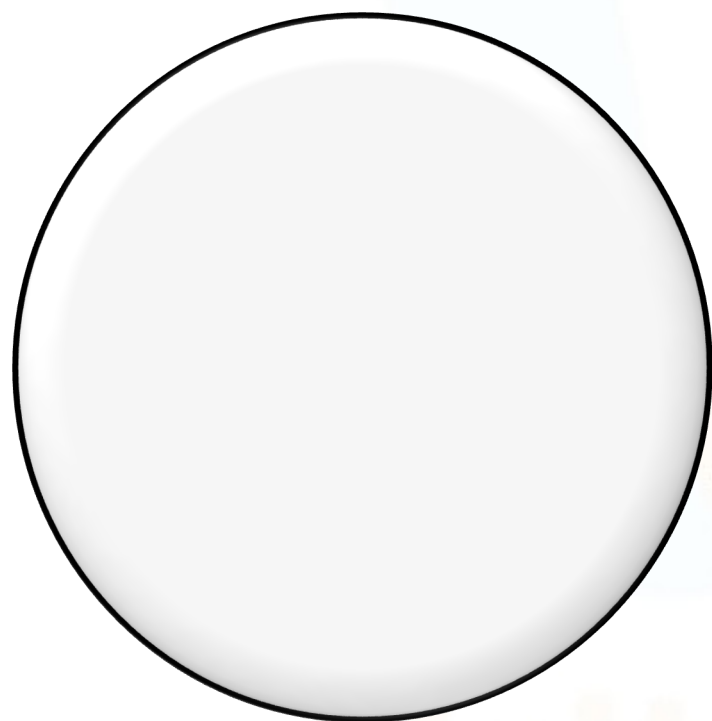
Nitrogen



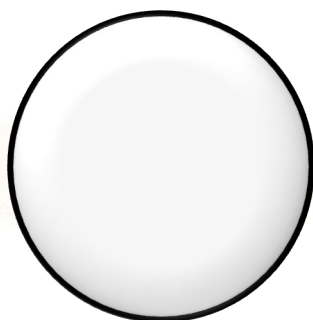
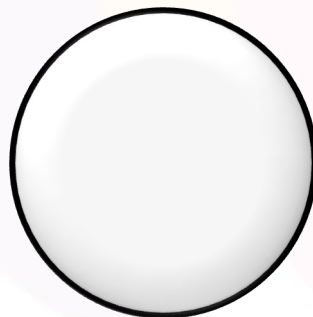
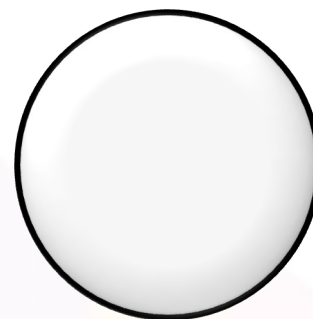
Phosphorus



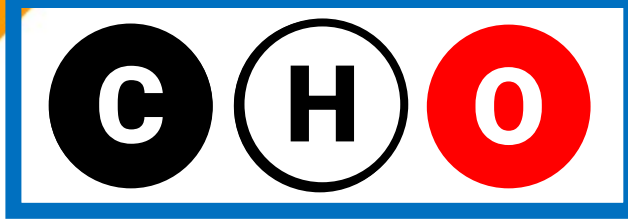
Sulfur



**Glycerol (甘油)**



## 2. Lipids 脂質



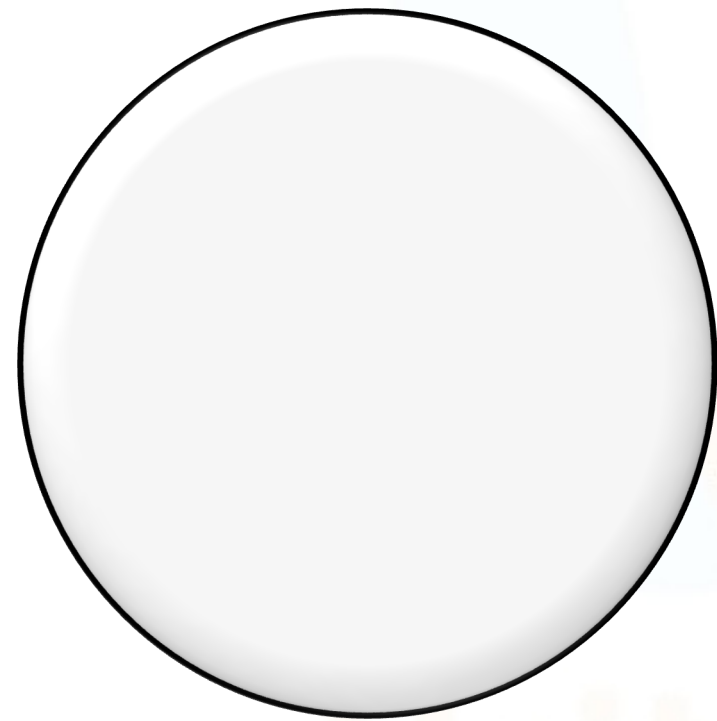
Nitrogen



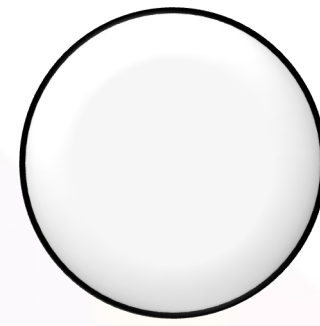
Phosphorus



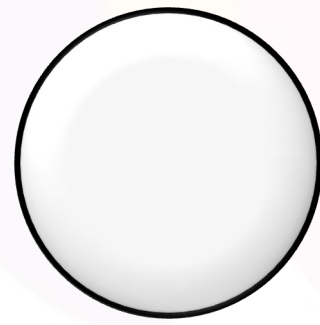
Sulfur



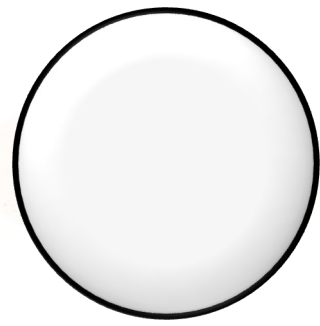
**Glycerol (甘油)**



**Fatty acid (脂肪酸)**



**Fatty acid (脂肪酸)**



**Fatty acid (脂肪酸)**

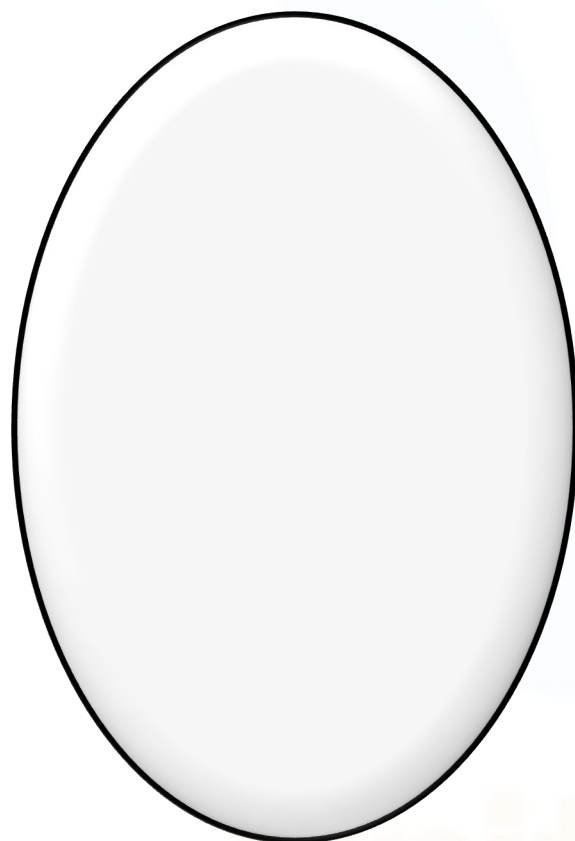
## 2. Lipids 脂質



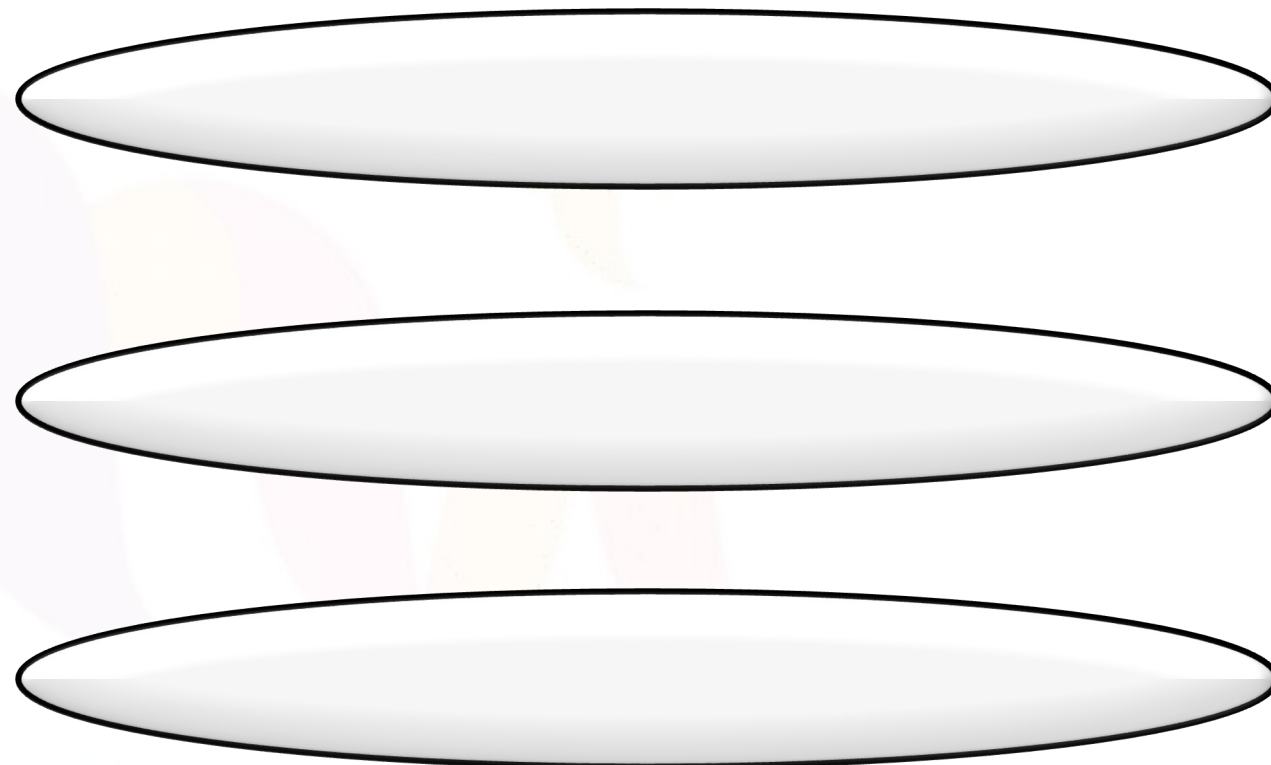
Nitrogen

Phosphorus

Sulfur



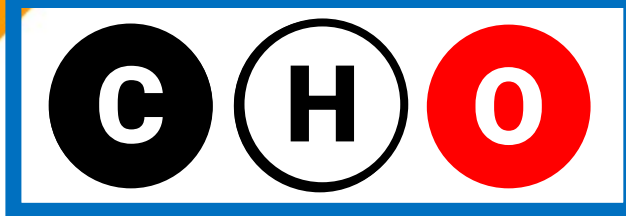
**1 Glycerol (甘油)**



**3 Fatty acids (脂肪酸)**



## 2. Lipids 脂質



Nitrogen

Phosphorus

Sulfur

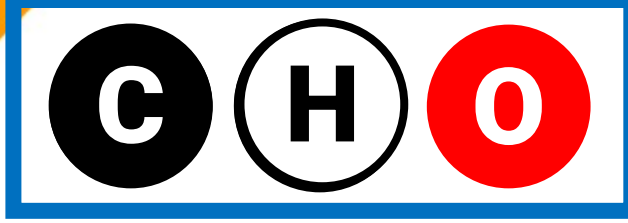


**1 Glycerol (甘油)**



**3 Fatty acids (脂肪酸)**

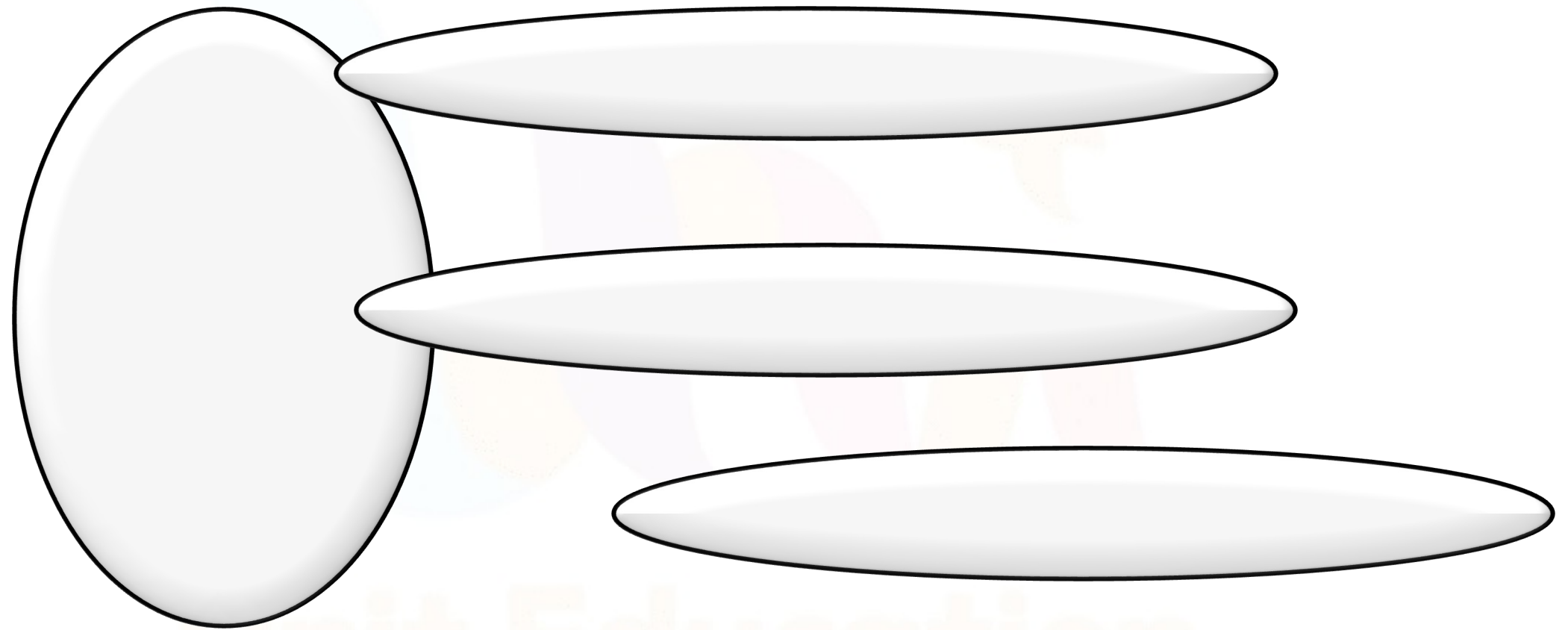
## 2. Lipids 脂質



Nitrogen

Phosphorus

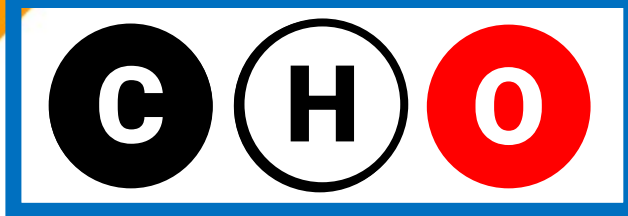
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

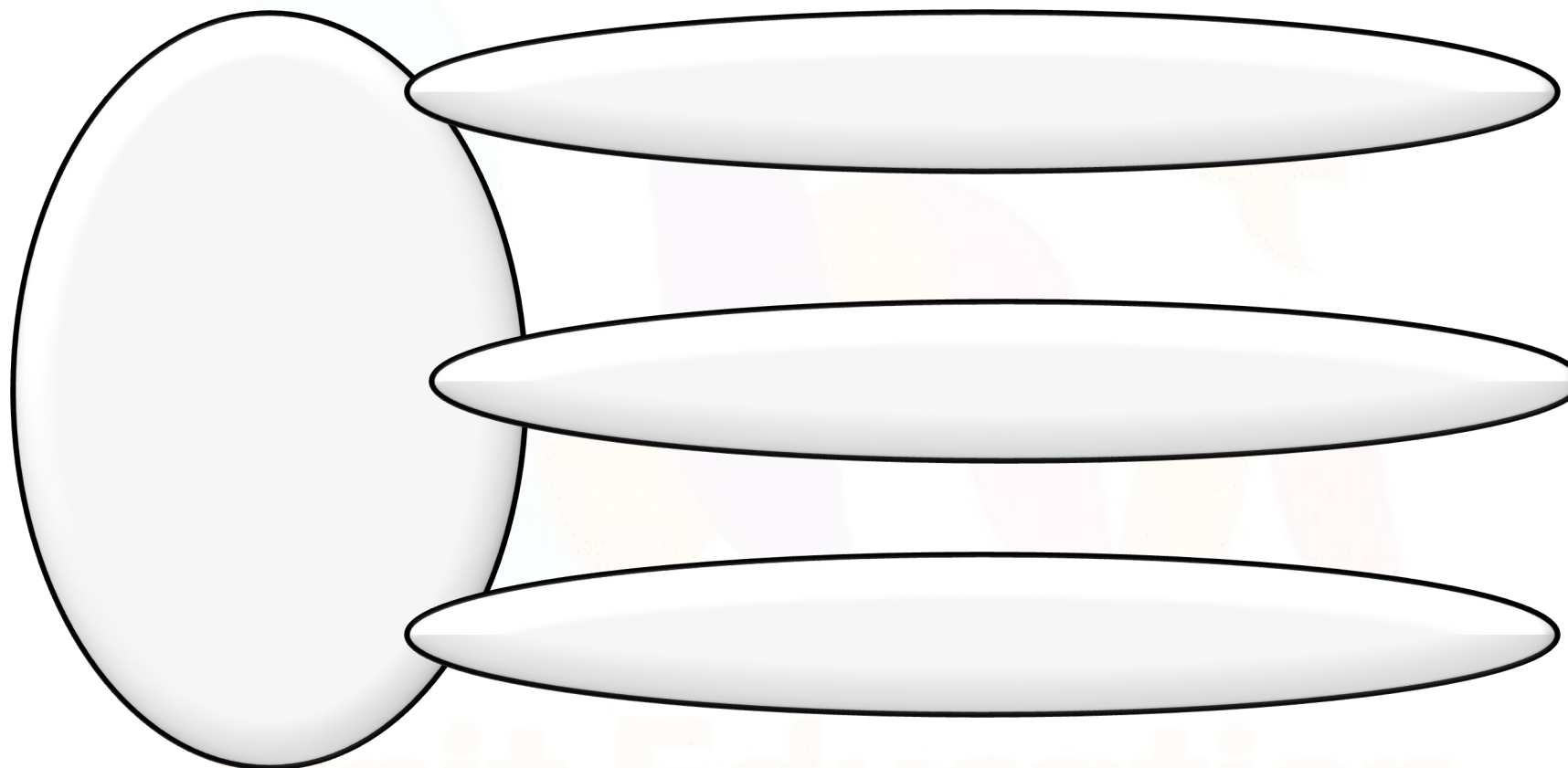
## 2. Lipids 脂質



Nitrogen

Phosphorus

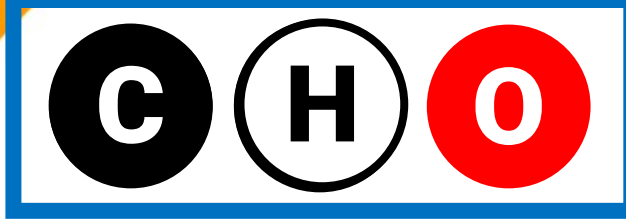
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

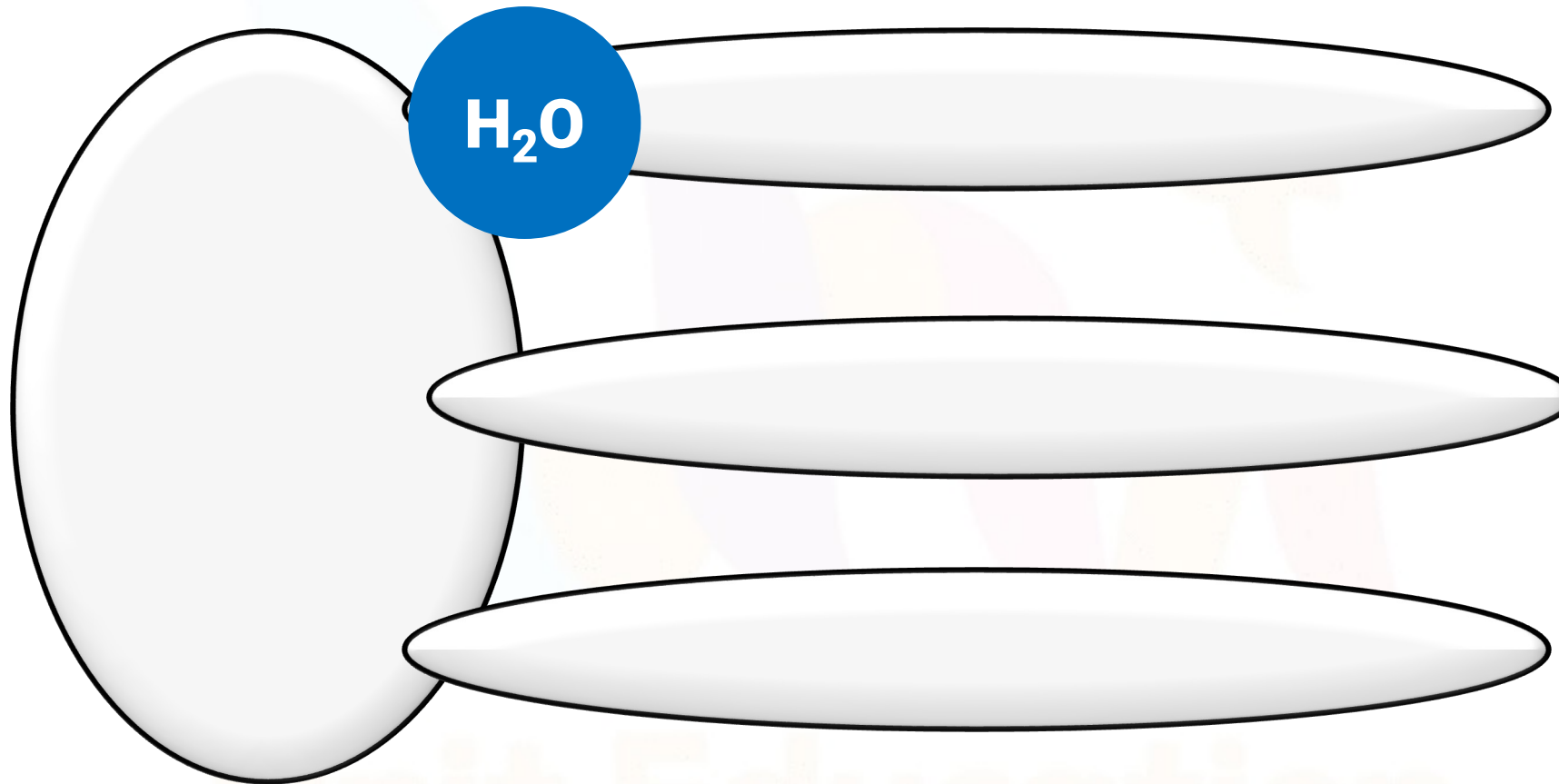
## 2. Lipids 脂質



Nitrogen

Phosphorus

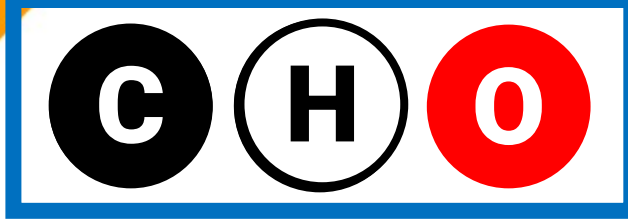
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

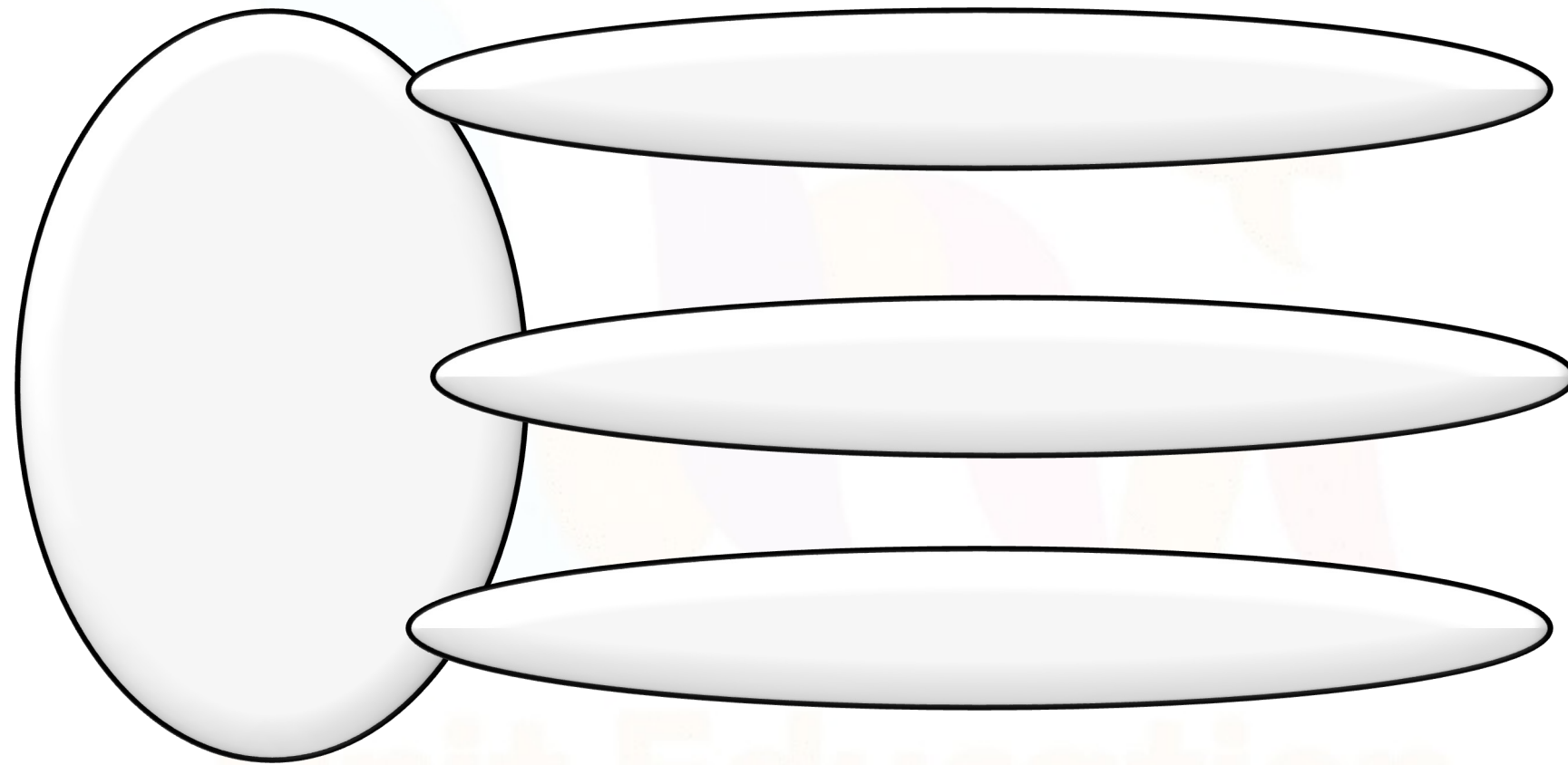
## 2. Lipids 脂質



Nitrogen

Phosphorus

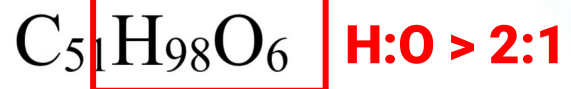
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

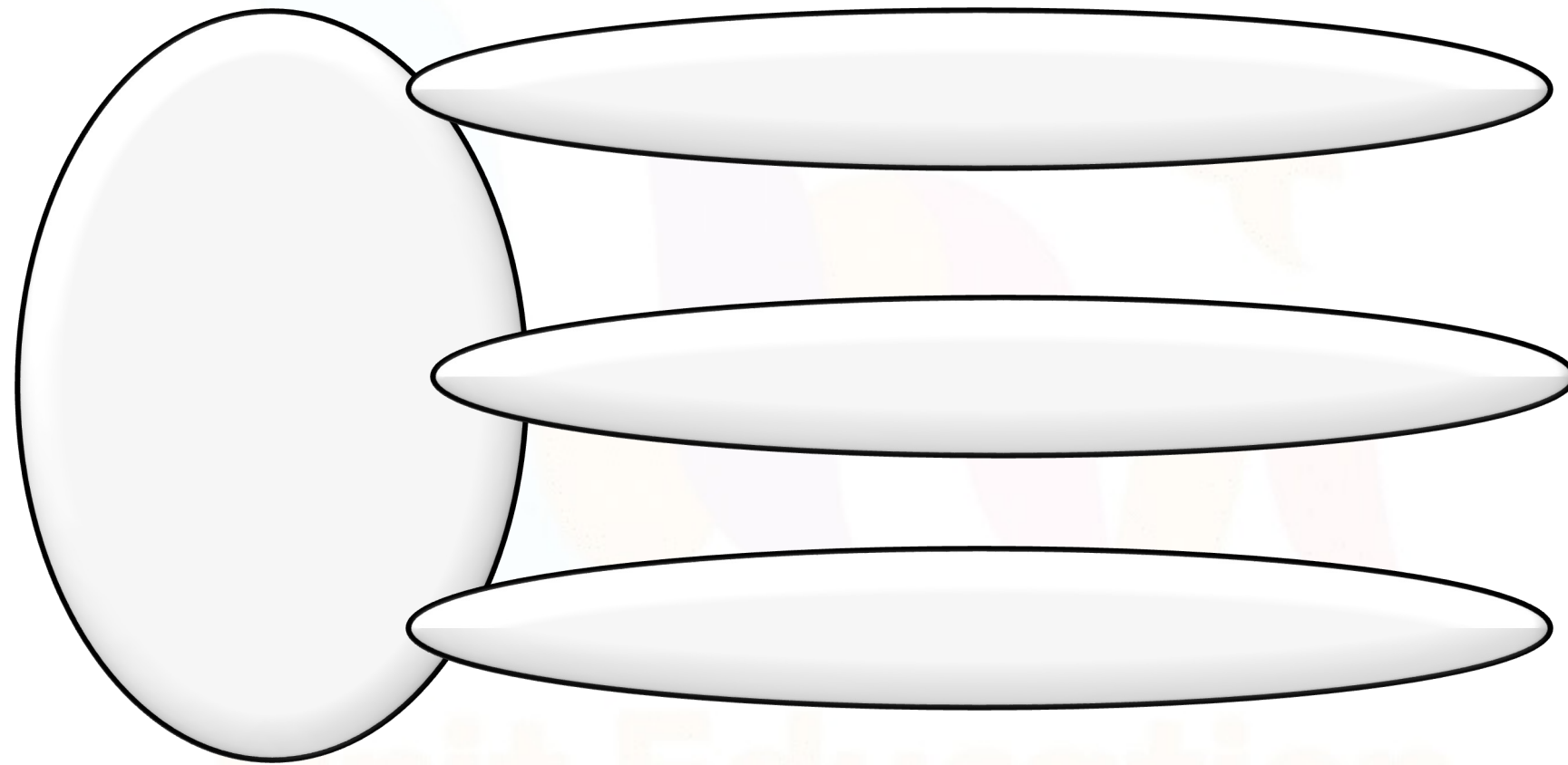
## 2. Lipids 脂質



Nitrogen

Phosphorus

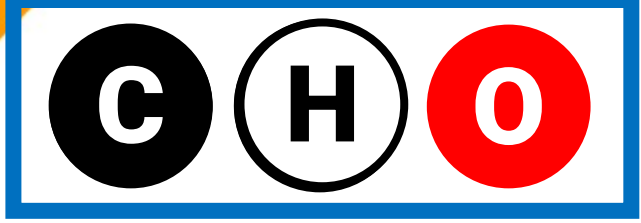
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

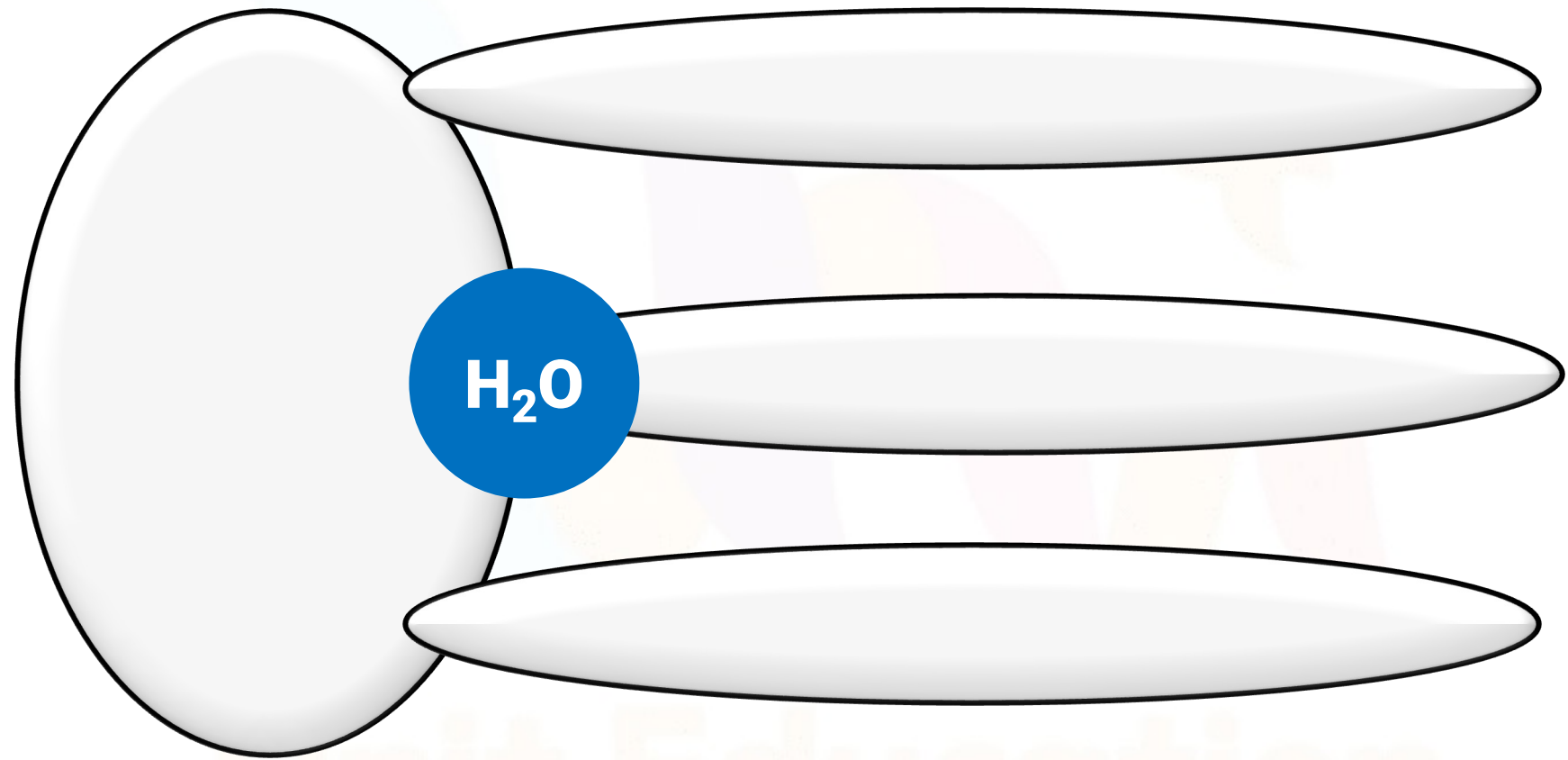
## 2. Lipids 脂質



Nitrogen

Phosphorus

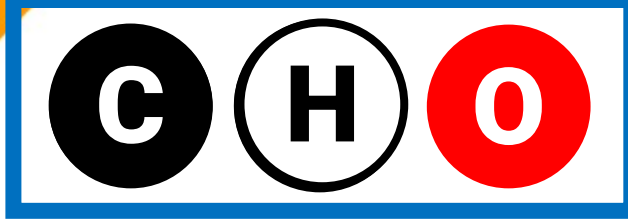
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

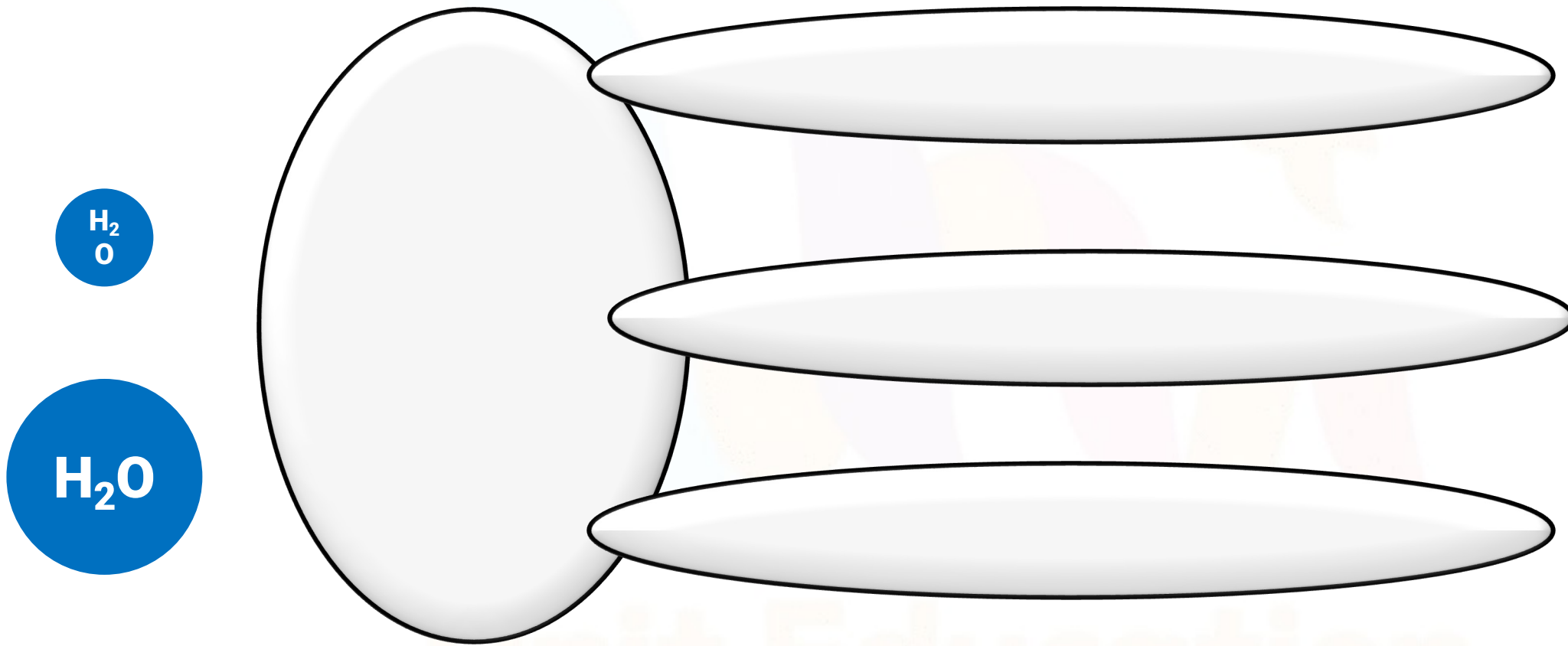
## 2. Lipids 脂質



Nitrogen

Phosphorus

Sulfur

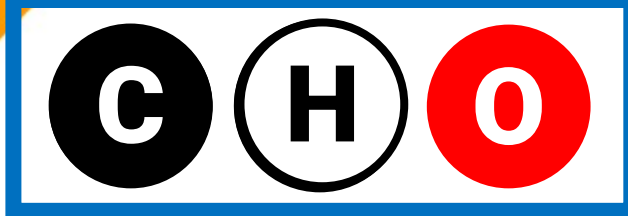


**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**



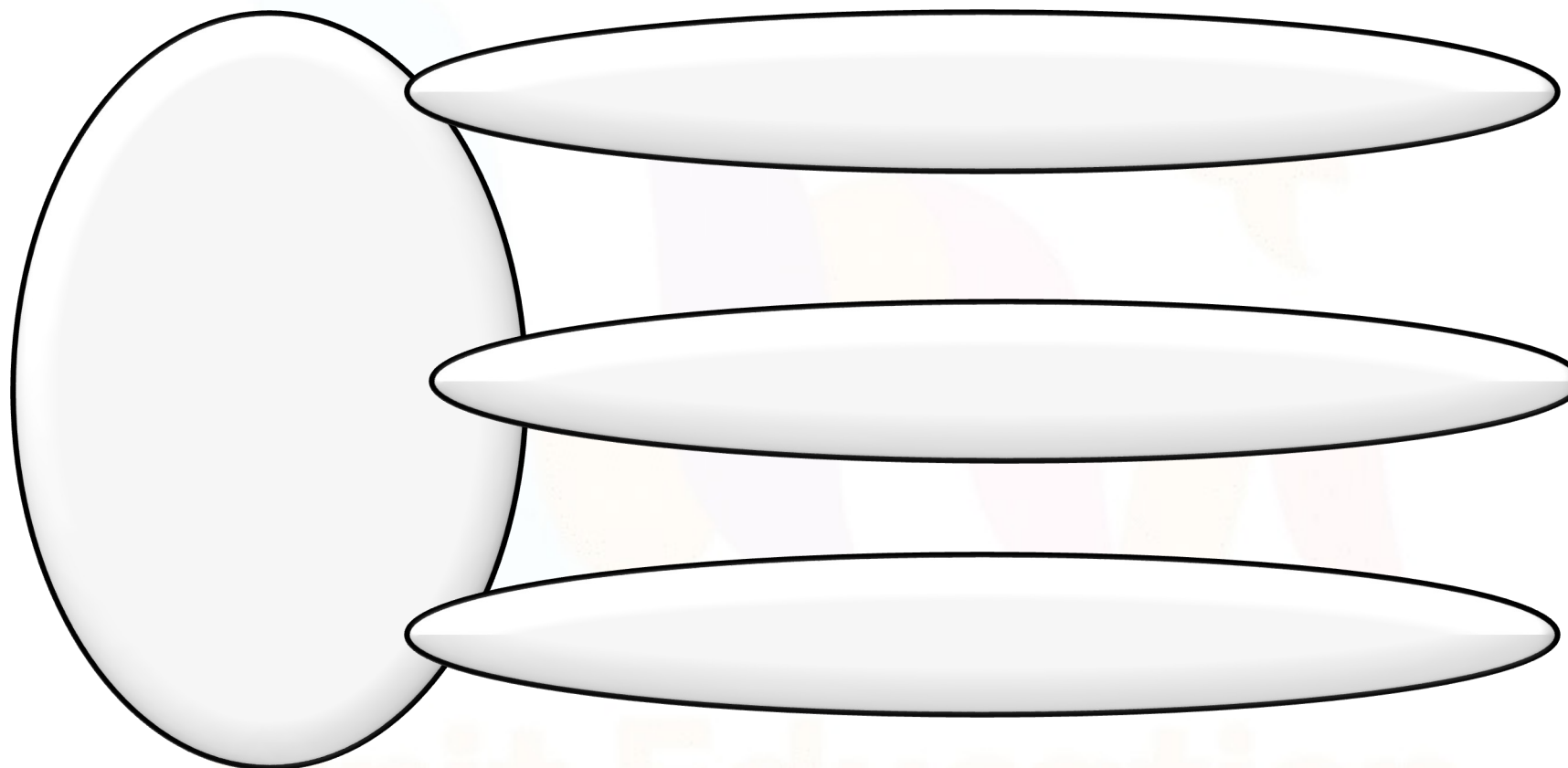
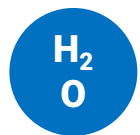
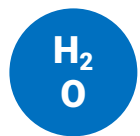
## 2. Lipids 脂質



Nitrogen

Phosphorus

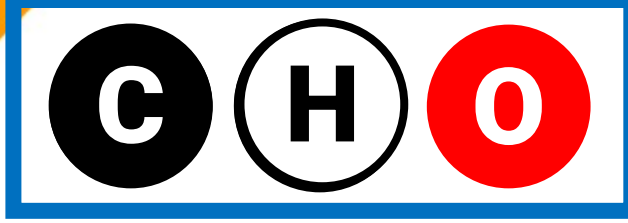
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

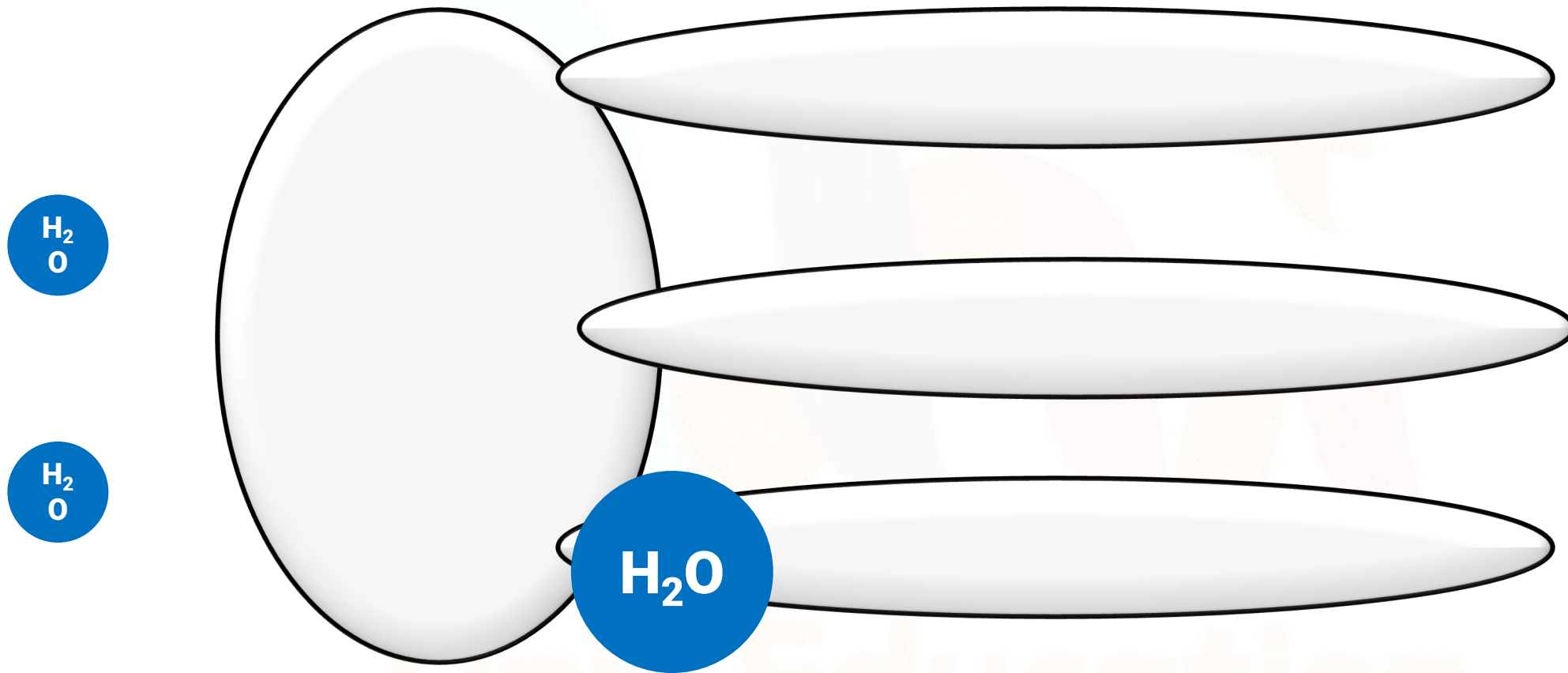
## 2. Lipids 脂質



Nitrogen

Phosphorus

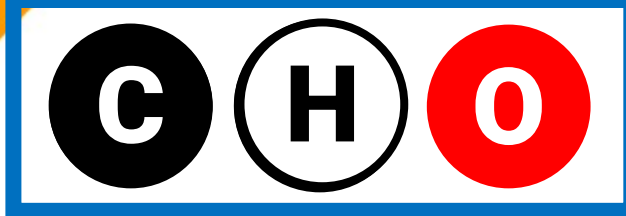
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

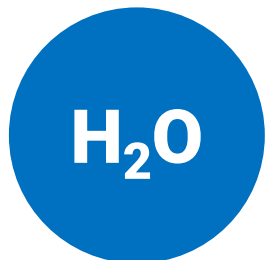
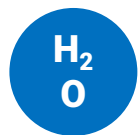
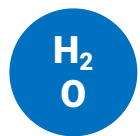
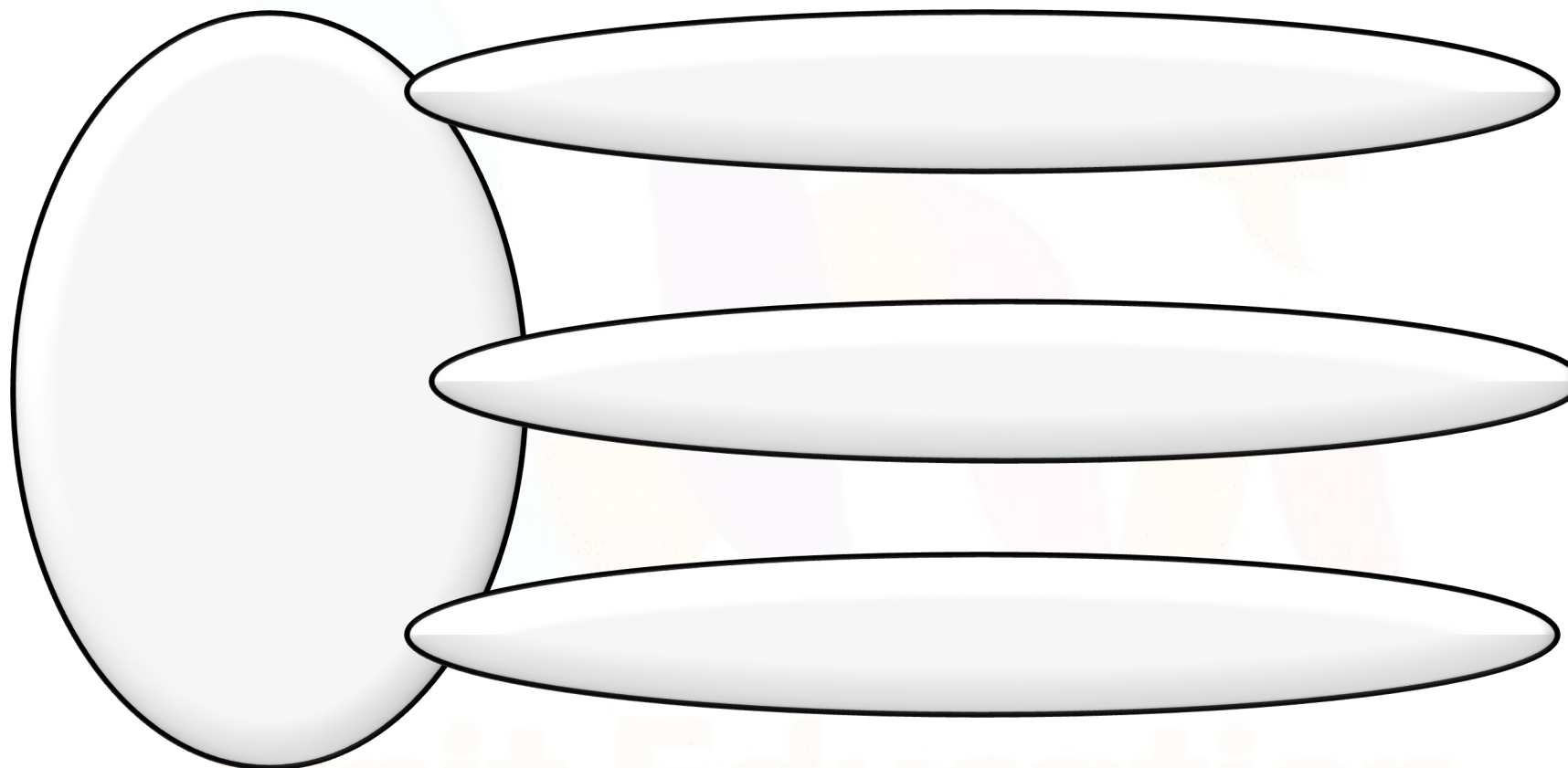
## 2. Lipids 脂質



Nitrogen

Phosphorus

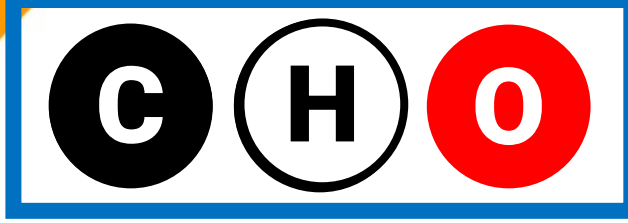
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

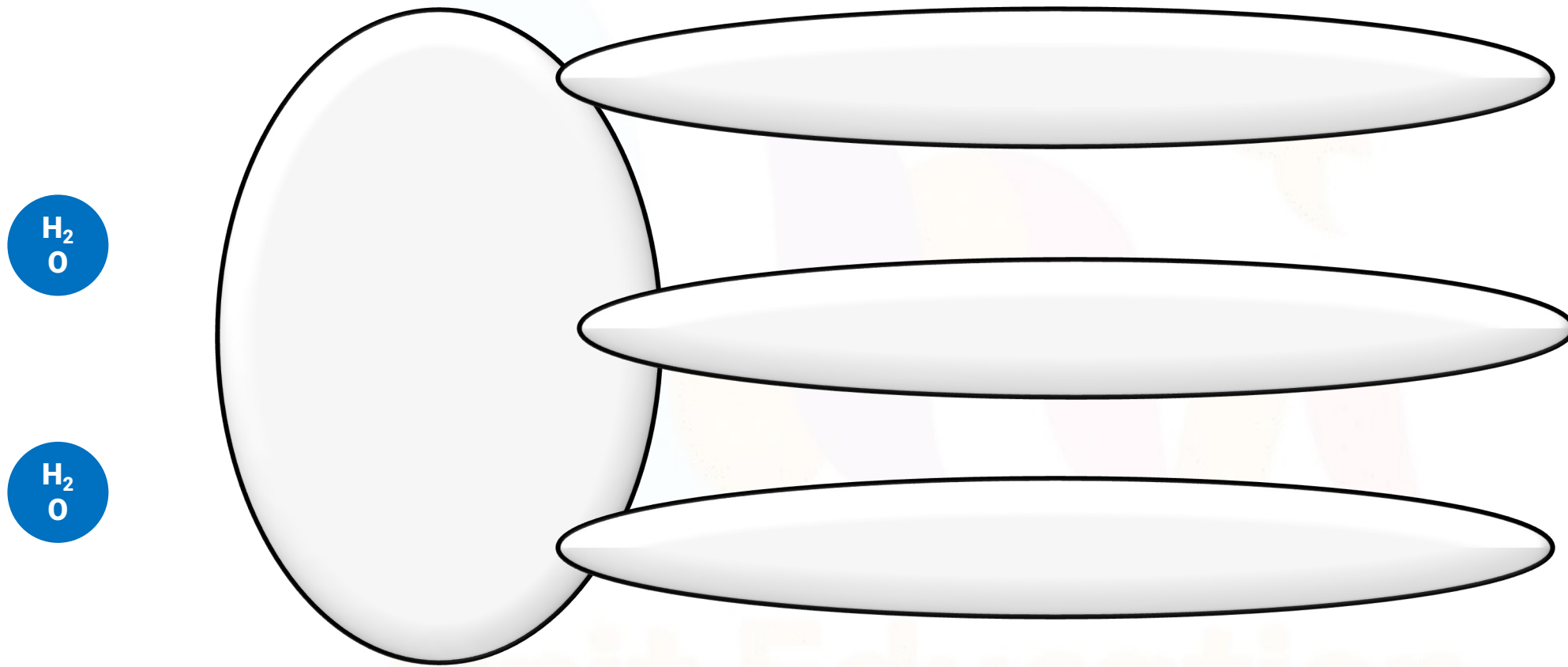
## 2. Lipids 脂質



Nitrogen

Phosphorus

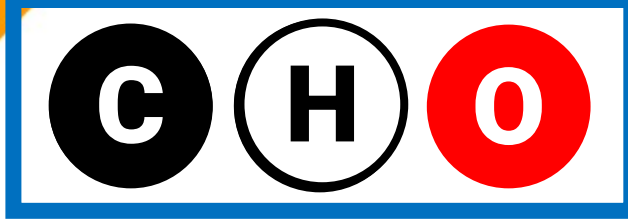
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

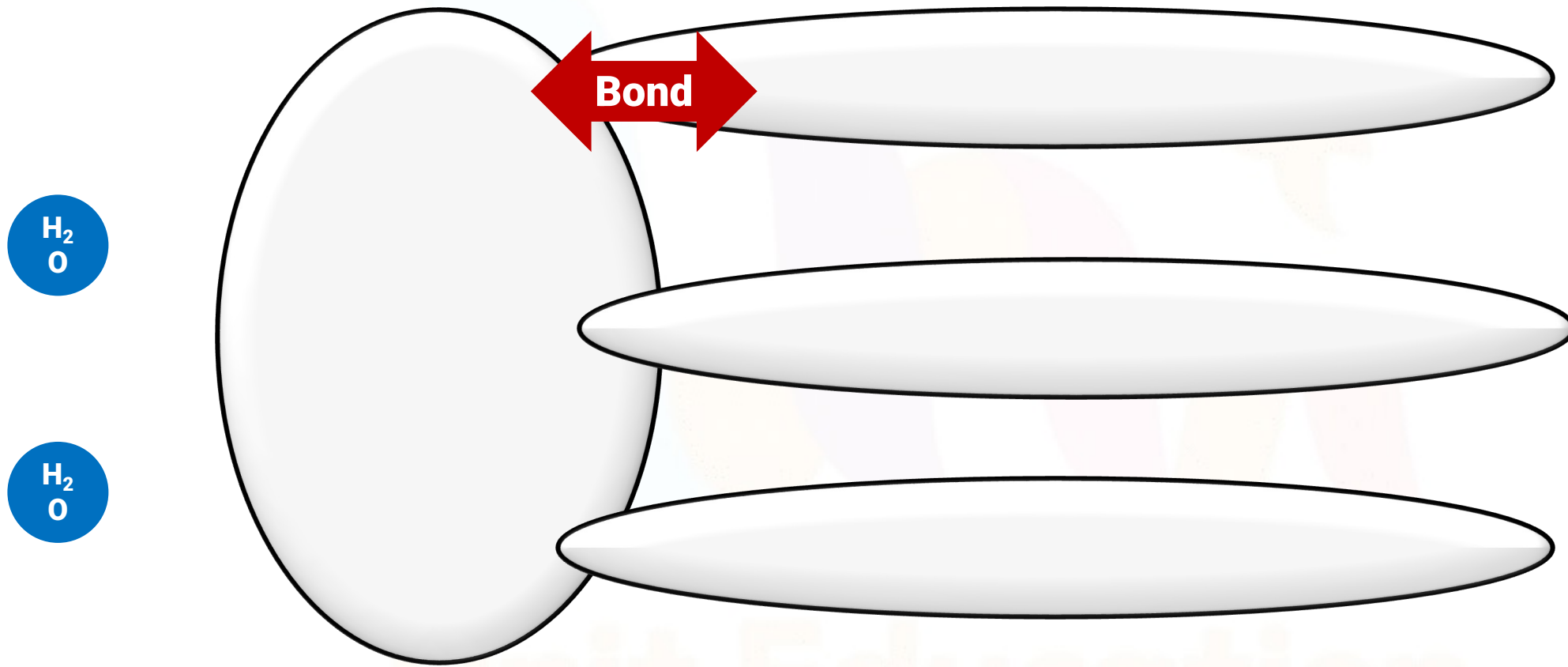
## 2. Lipids 脂質



Nitrogen

Phosphorus

Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

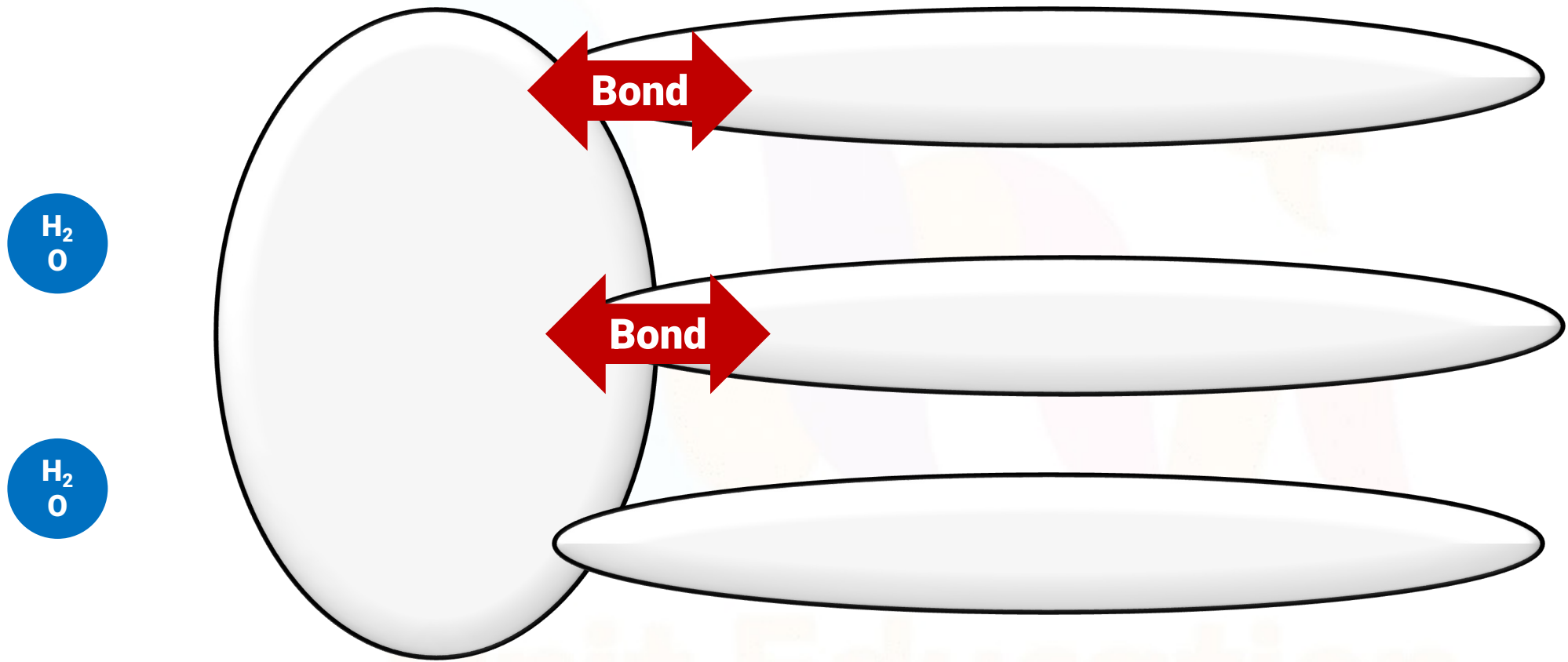


Nitrogen

Phosphorus

Sulfur

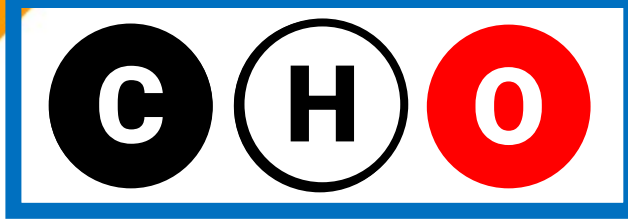
## 2. Lipids 脂質



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

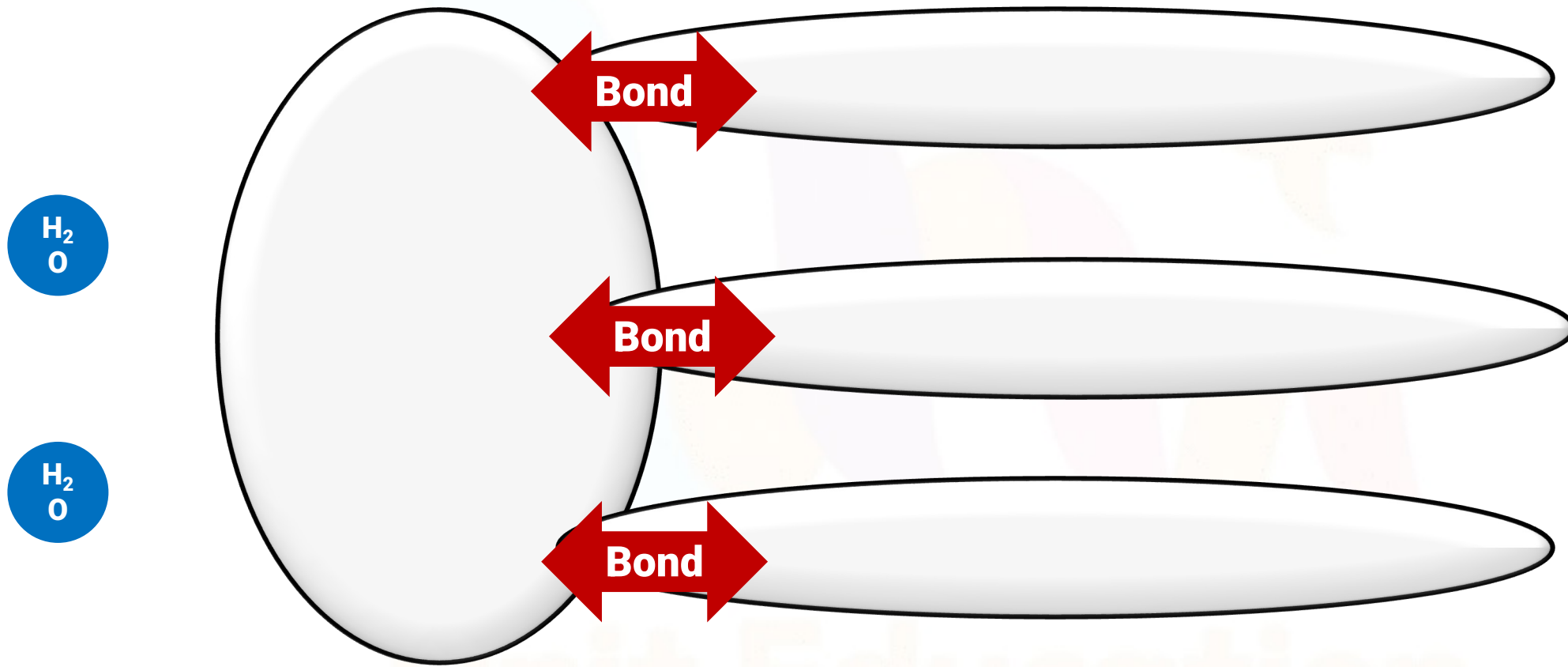
## 2. Lipids 脂質



Nitrogen

Phosphorus

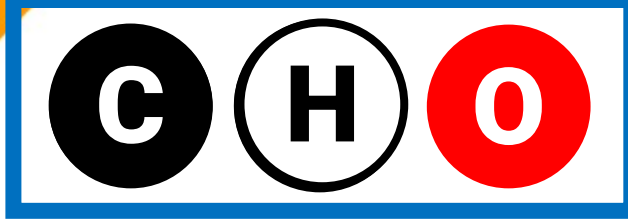
Sulfur



**1 Glycerol (甘油)**

**3 Fatty acids (脂肪酸)**

## 2. Lipids 脂質



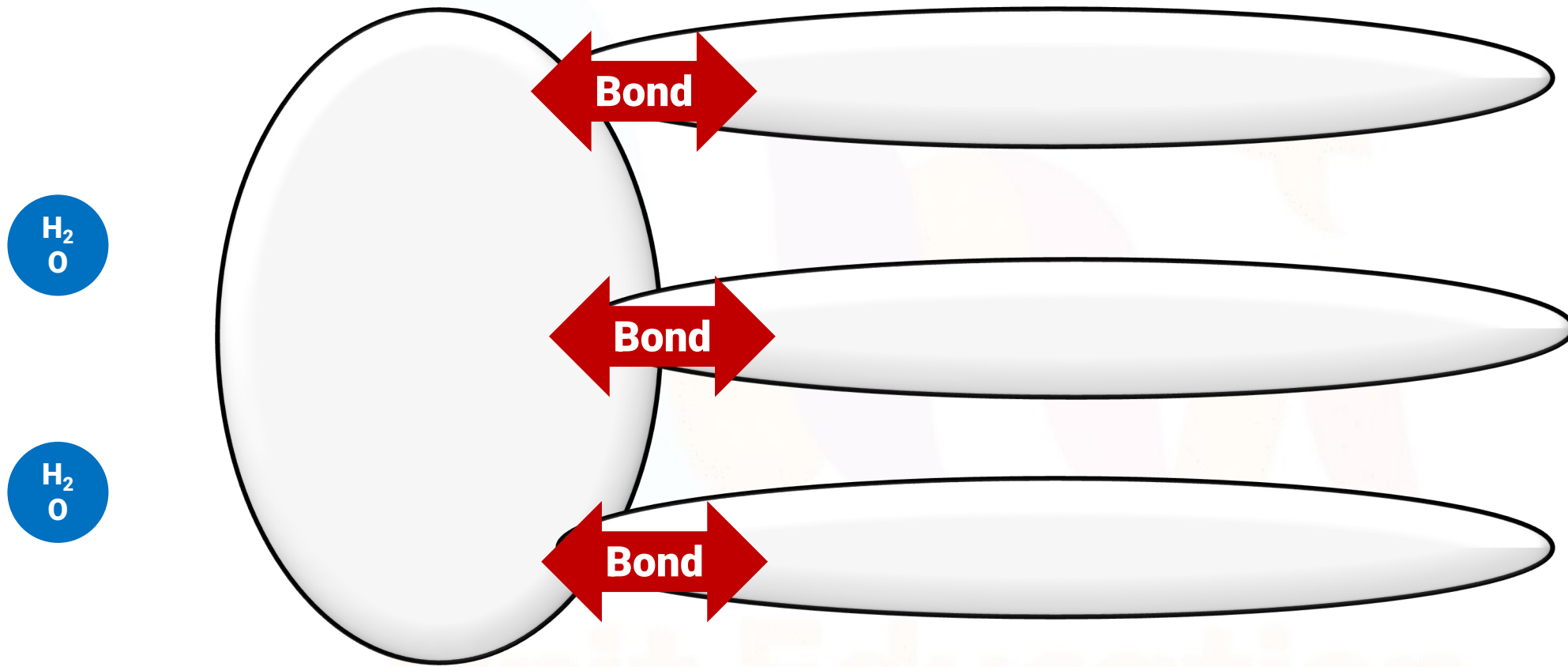
Nitrogen



Phosphorus



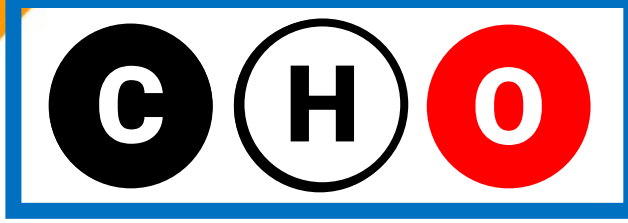
Sulfur



**3 Condensations 縮合!**



## 2. Lipids 脂質



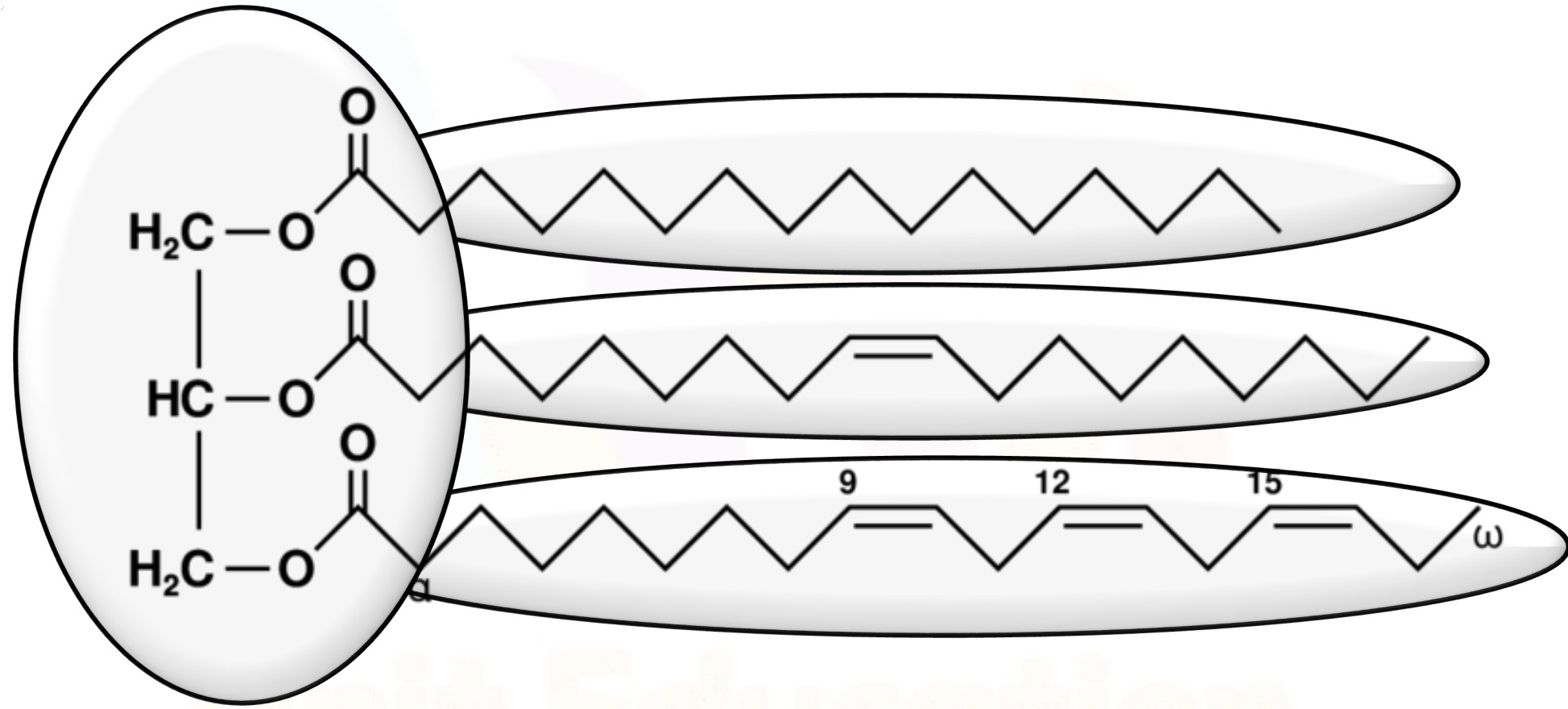
Nitrogen



Phosphorus

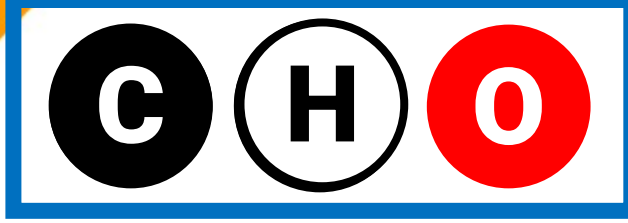


Sulfur



**Tri-glycer-ides (甘油三酯)**

## 2. Lipids 脂質



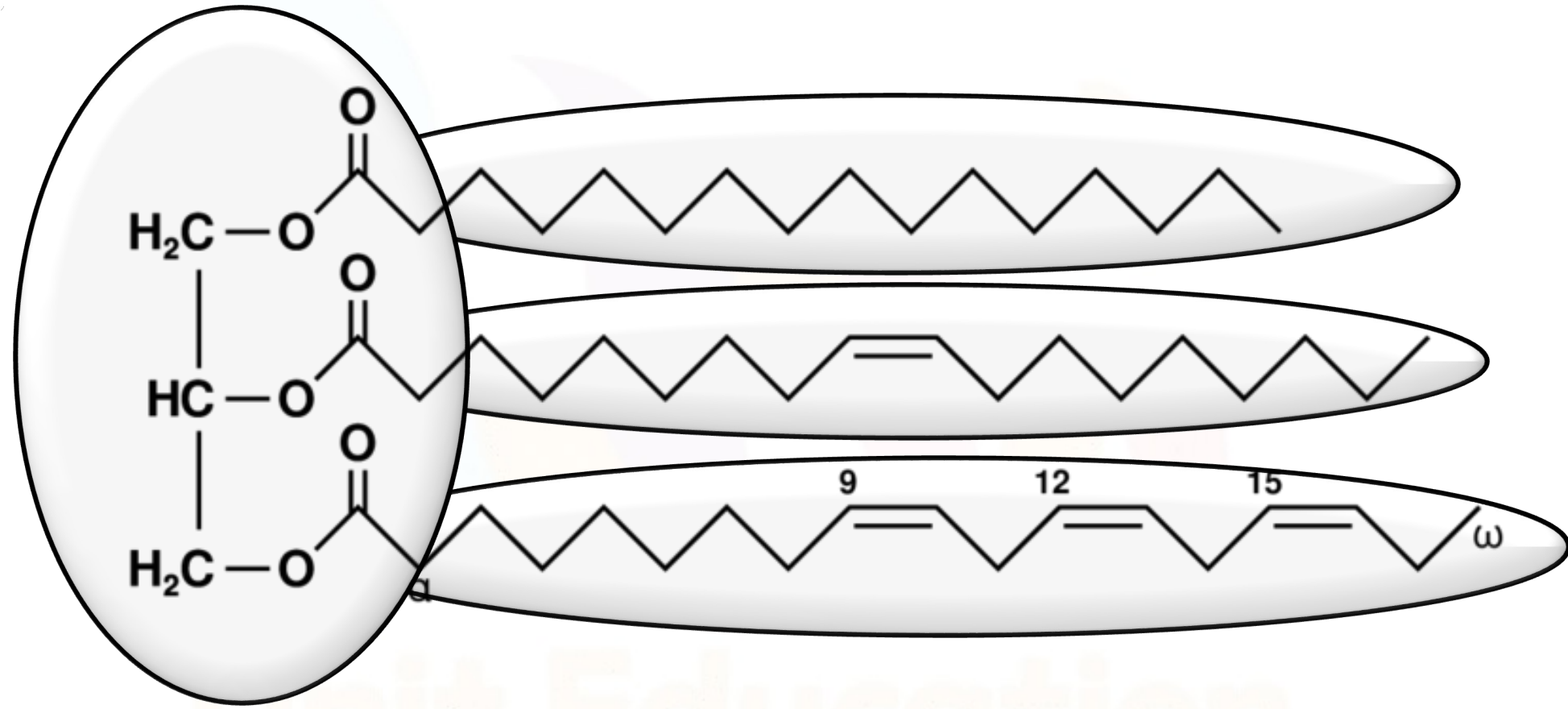
Nitrogen



Phosphorus

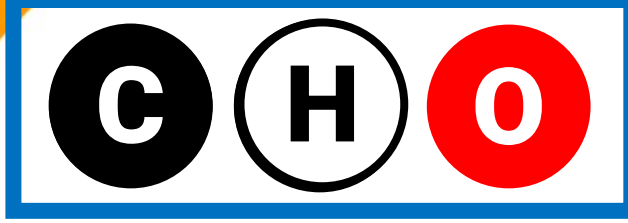


Sulfur



**Triglycerides (甘油三酯)**

## 2. Lipids 脂質



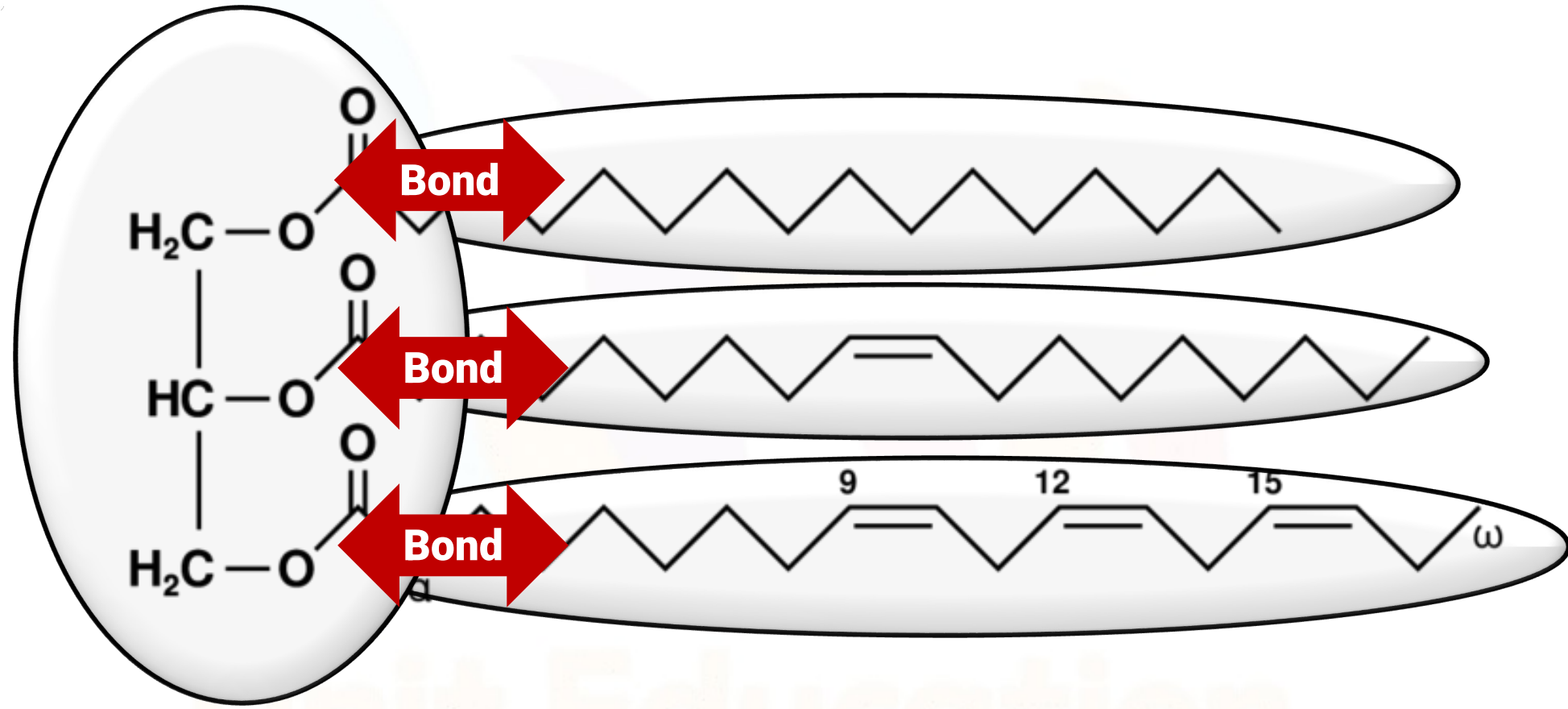
Nitrogen



Phosphorus

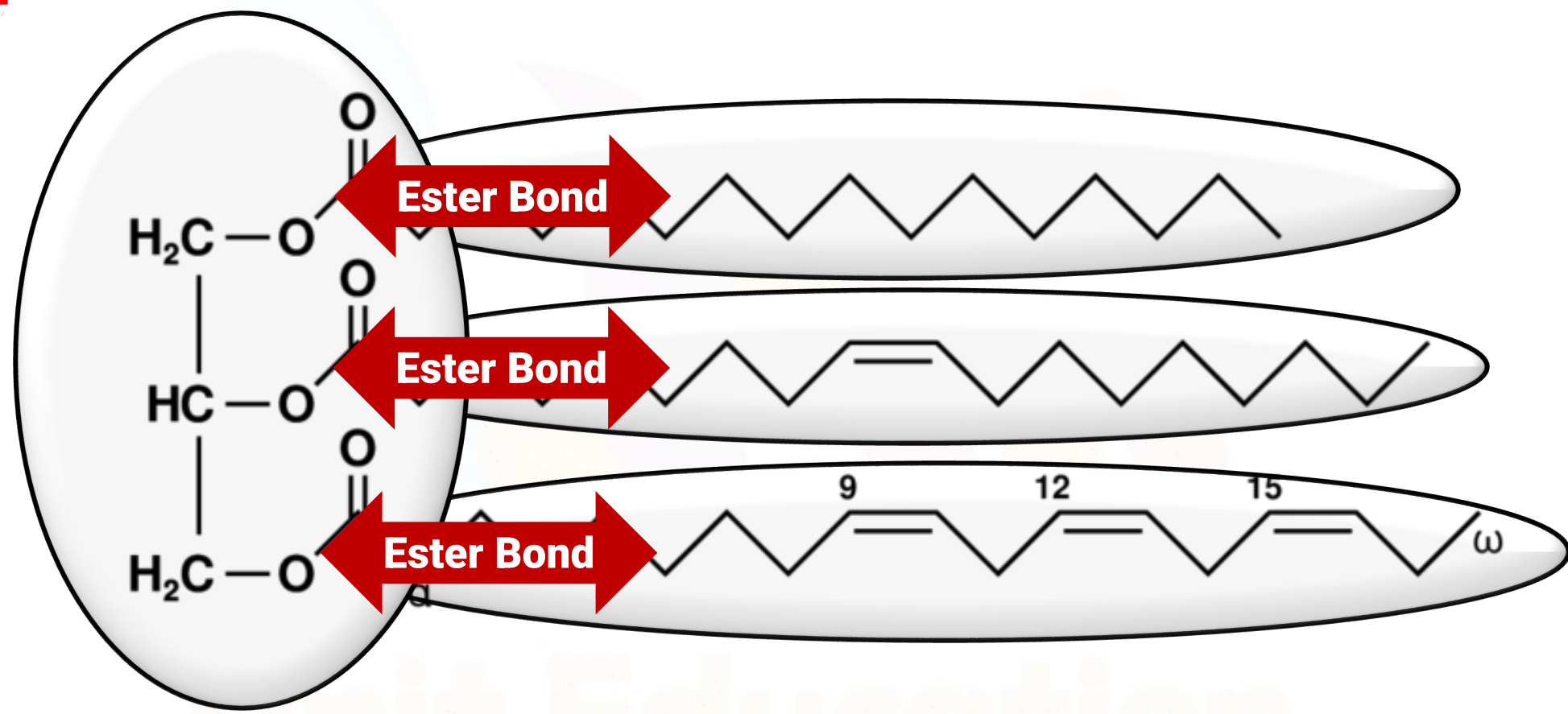
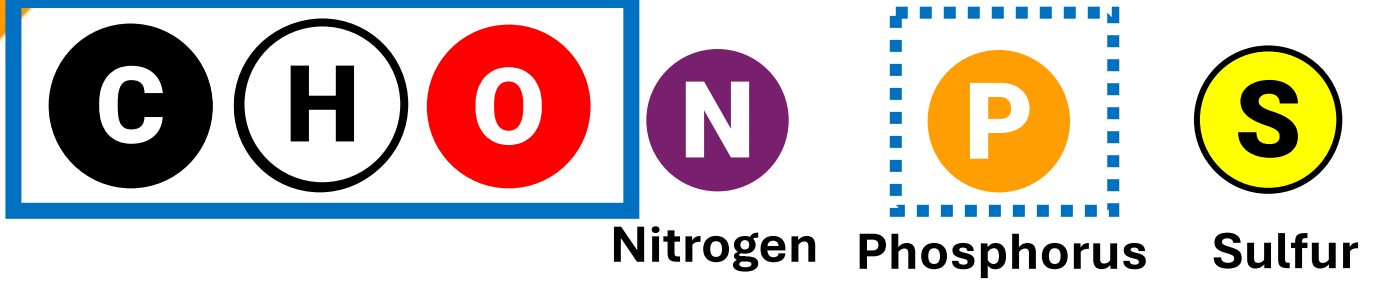


Sulfur



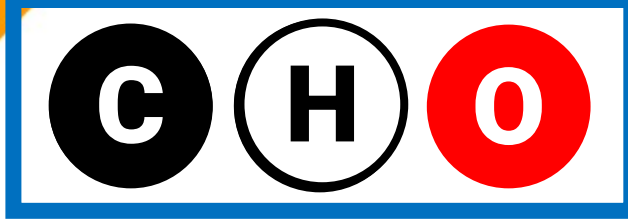
**Triglycerides (甘油三酯)**

## 2. Lipids 脂質



**Triglycerides (甘油三酯)**

## 2. Lipids 脂質



Nitrogen

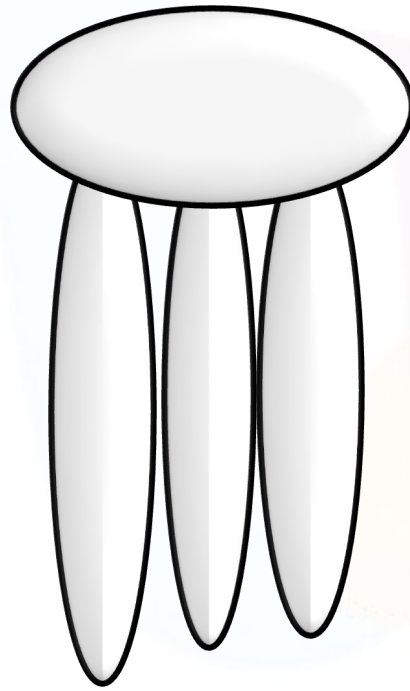


Phosphorus

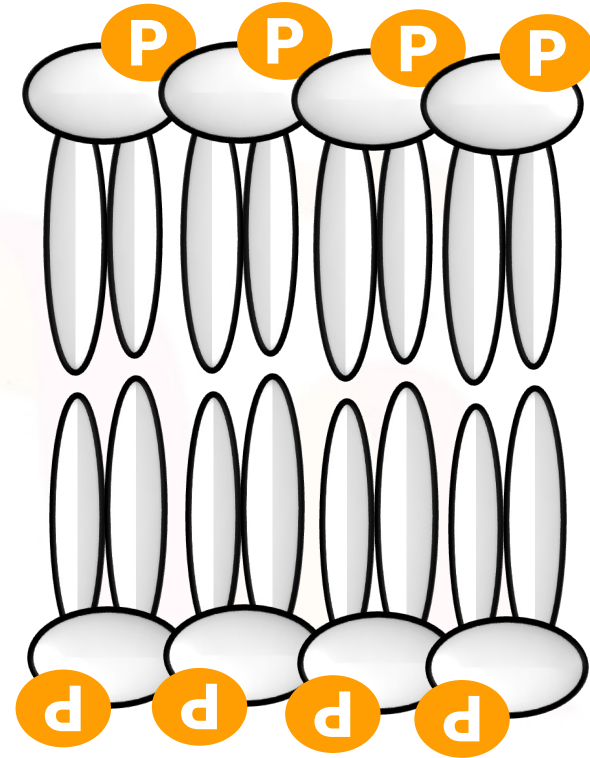


Sulfur

**Triglycerides**  
(甘油三酯)



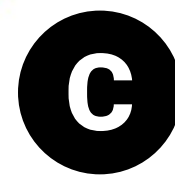
**1 Glycerol (甘油)**  
**+ 3 Fatty acids (脂肪酸)**



**Phospho-lipids**  
(磷脂)

**1 Phosphate group**  
**+ 1 Glycerol (甘油)**  
**+ 2 Fatty acids (脂肪酸)**

# Organic biomolecules



Nitrogen

Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



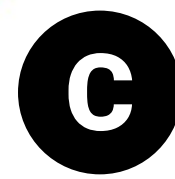
3. Proteins 蛋白質



4. Nucleic acids 核酸



# Organic biomolecules



Nitrogen

Phosphorus

Sulfur

1. Carbohydrates 碳水化合物



2. Lipids 脂質



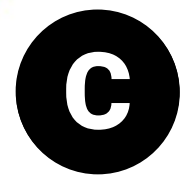
3. Proteins 蛋白質



4. Nucleic acids 核酸



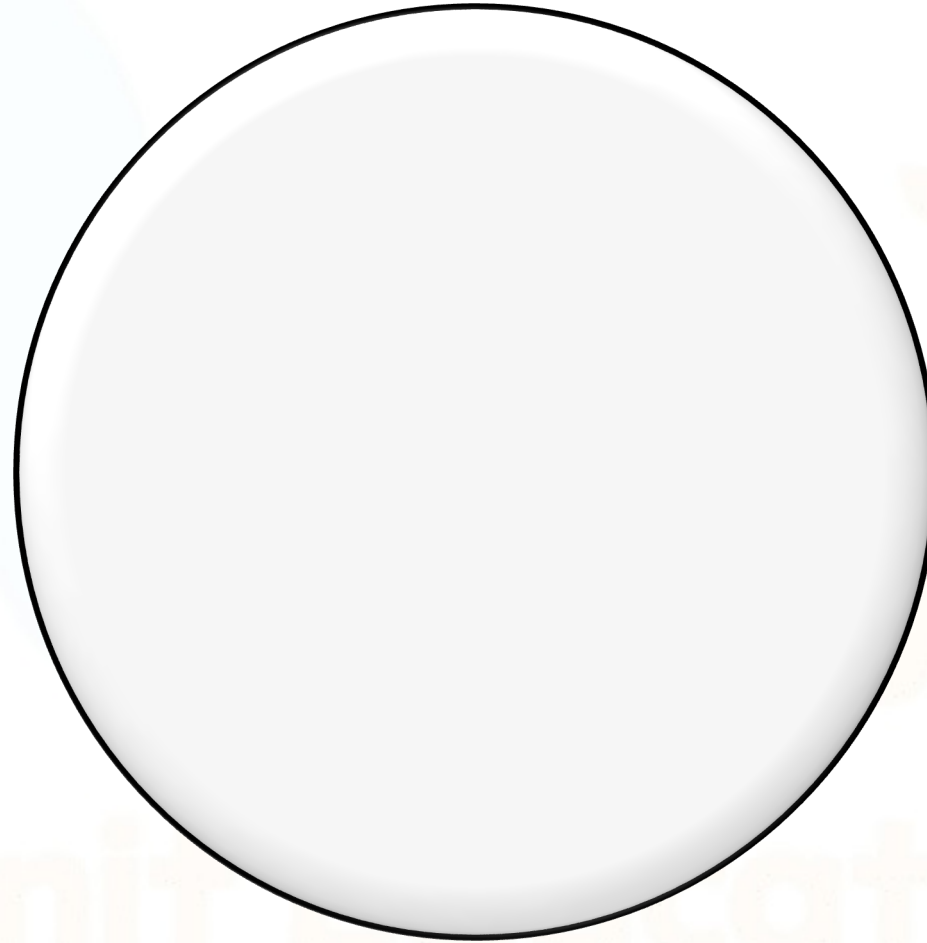
### 3. Proteins 蛋白質



Nitrogen

Phosphorus

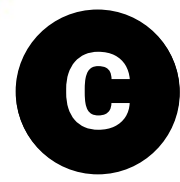
Sulfur



**Mono單-mer**



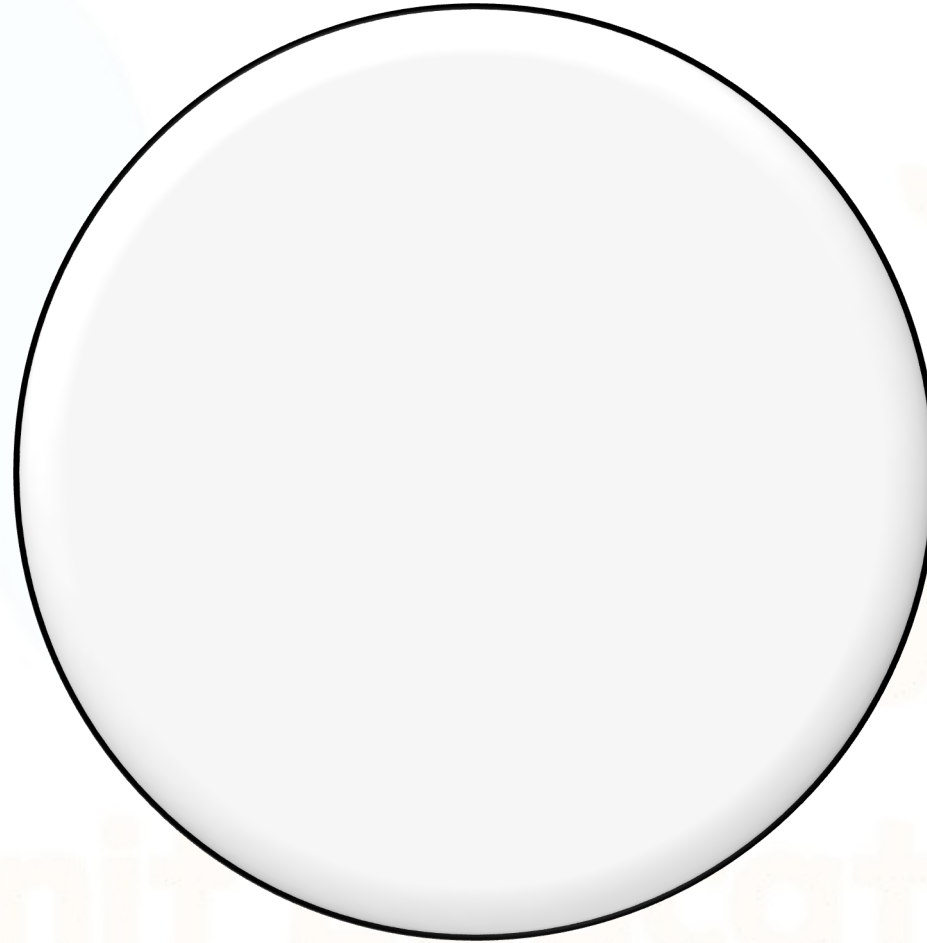
### 3. Proteins 蛋白質



Nitrogen

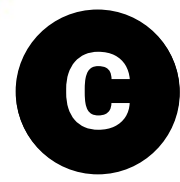
Phosphorus

Sulfur



**Mono單-mer**

### 3. Proteins 蛋白質



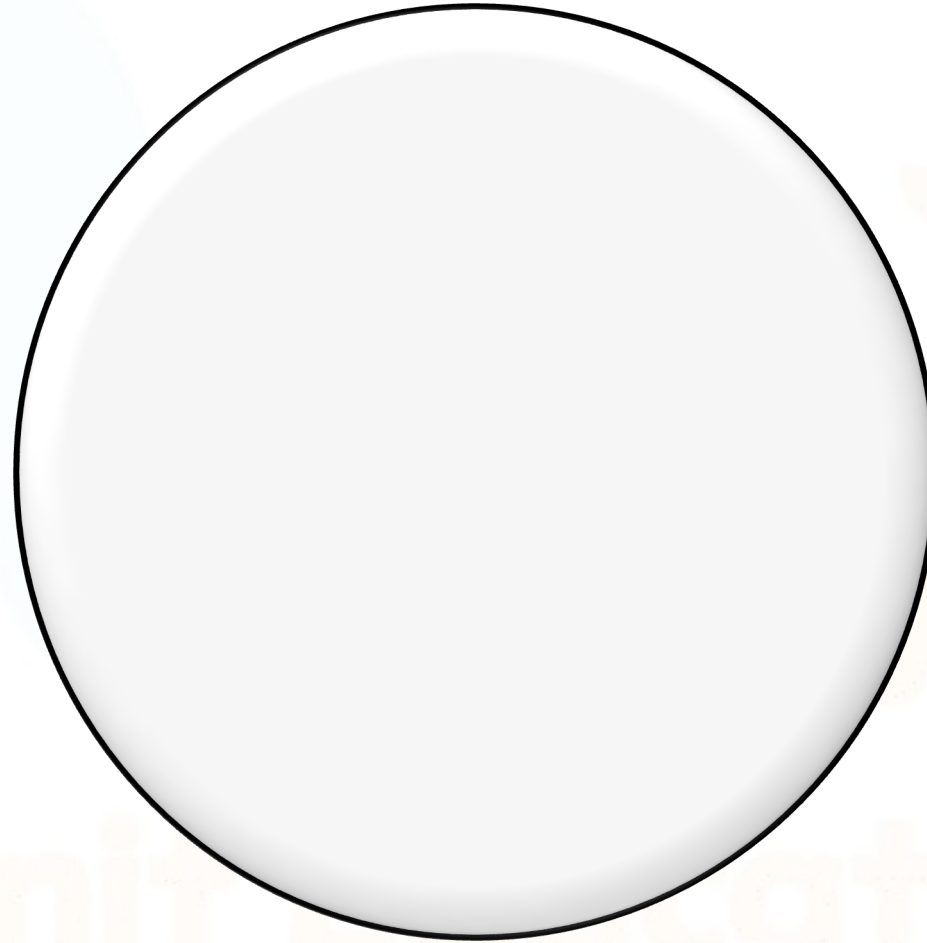
Nitrogen



Phosphorus

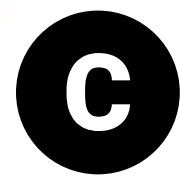


Sulfur



**Mono單-mer**

### 3. Proteins 蛋白質



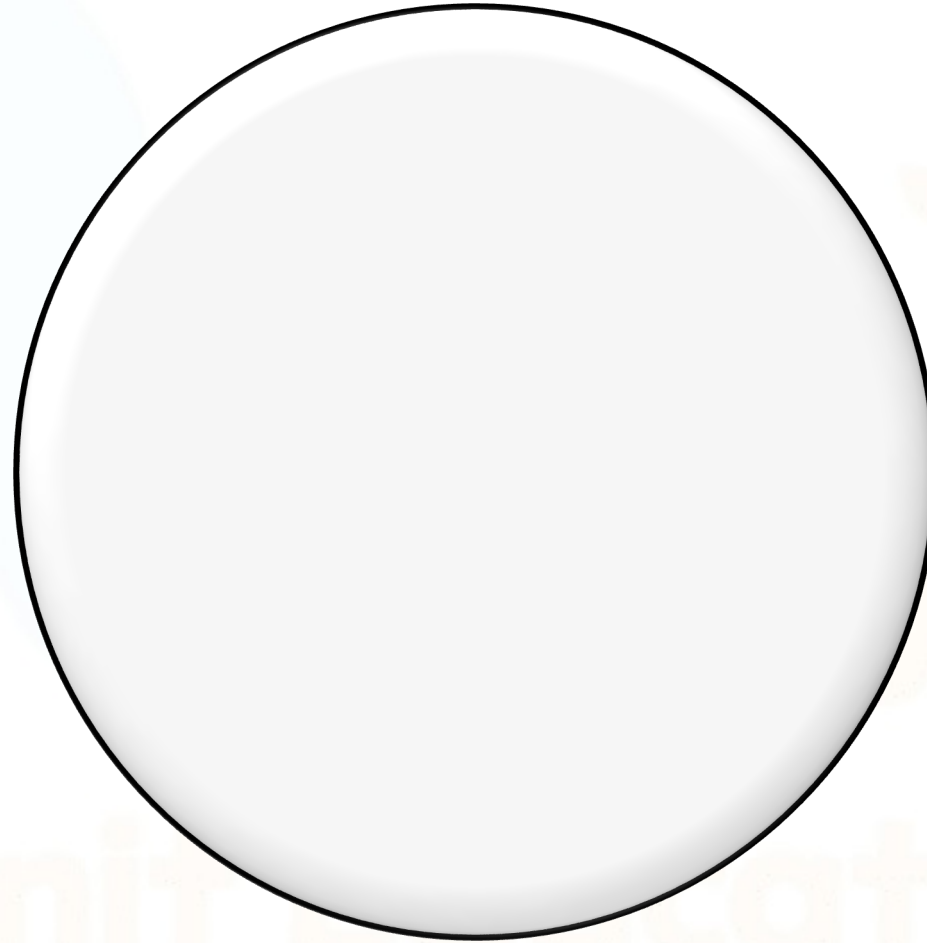
Nitrogen



Phosphorus

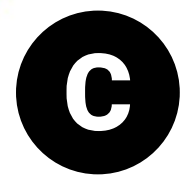


Sulfur



**Mono單-mer**

### 3. Proteins 蛋白質



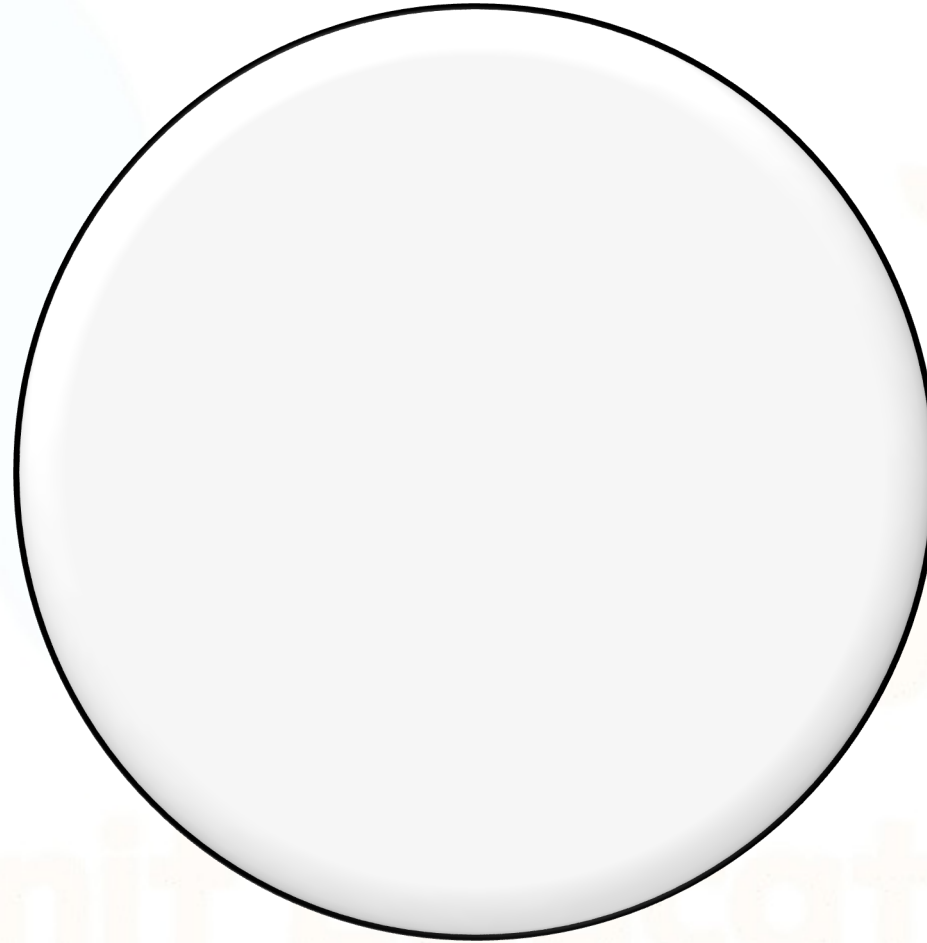
Nitrogen



Phosphorus

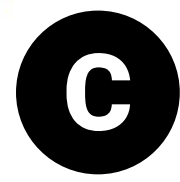


Sulfur



**Mono單-mer**

### 3. Proteins 蛋白質



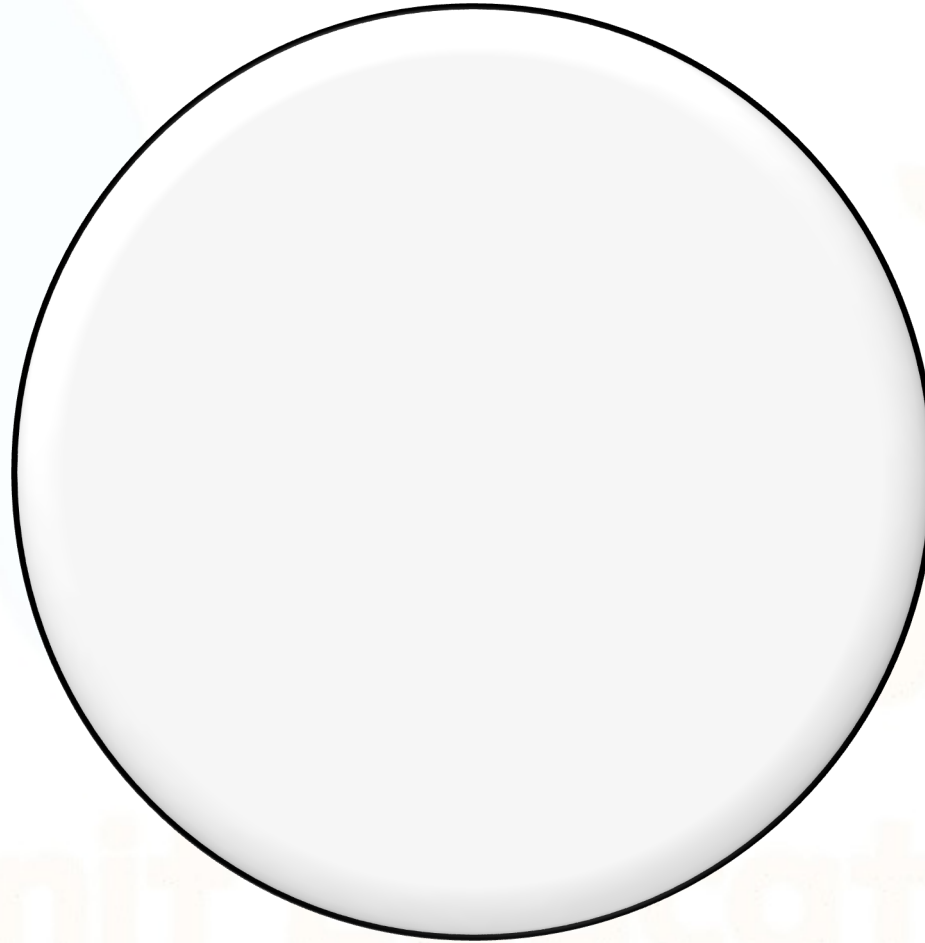
Nitrogen



Phosphorus

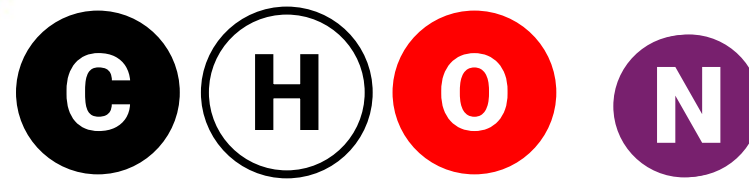


Sulfur



**Mono單-mer**

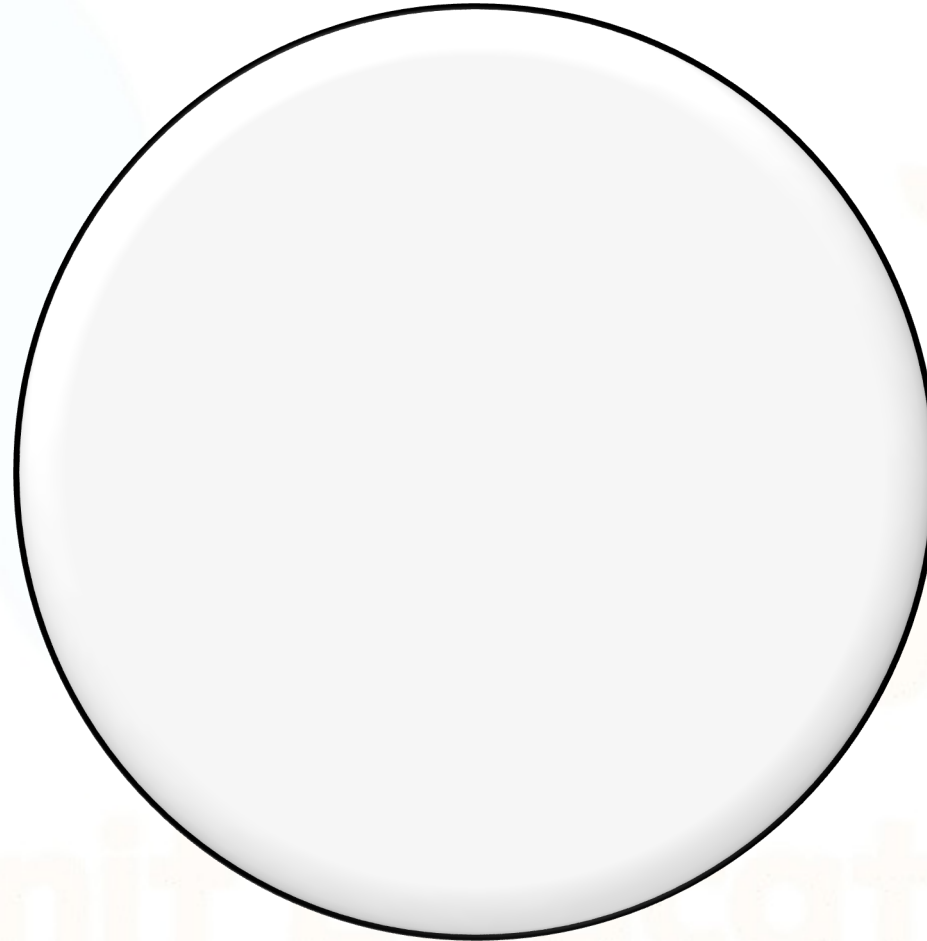
### 3. Proteins 蛋白質



Nitrogen

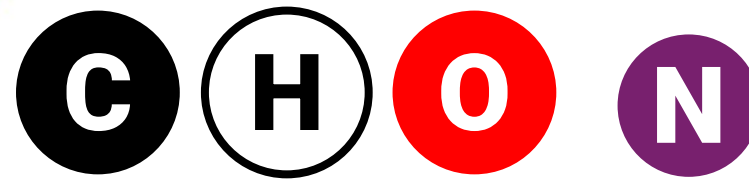
Phosphorus

Sulfur



**Mono單-mer**

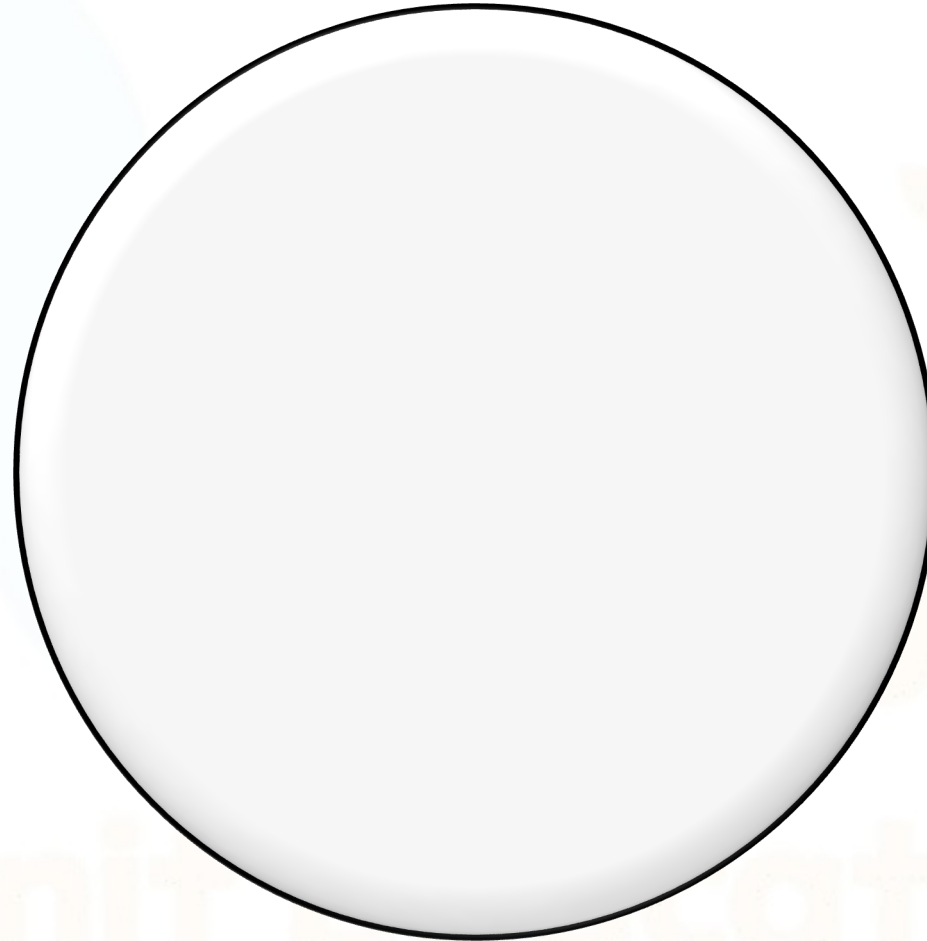
### 3. Proteins 蛋白質



Nitrogen

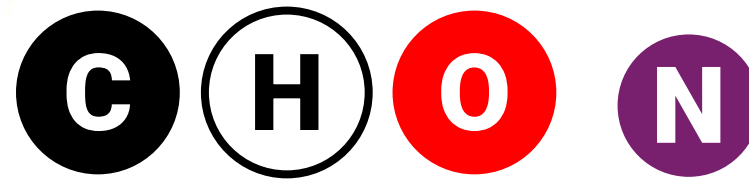
Phosphorus

Sulfur



**Mono單-mer**

### 3. Proteins 蛋白質



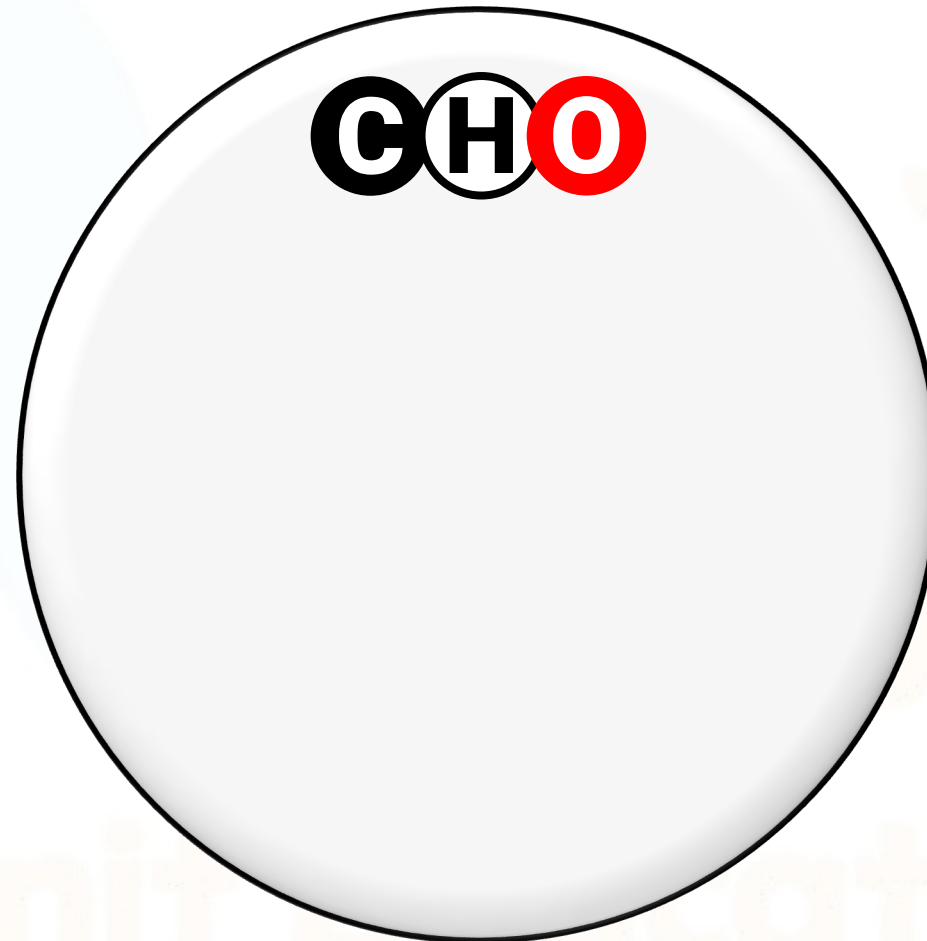
Nitrogen



Phosphorus



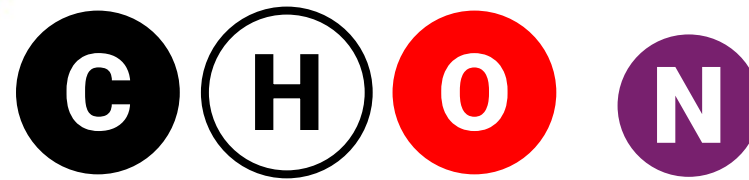
Sulfur



**Mono單-mer**



### 3. Proteins 蛋白質



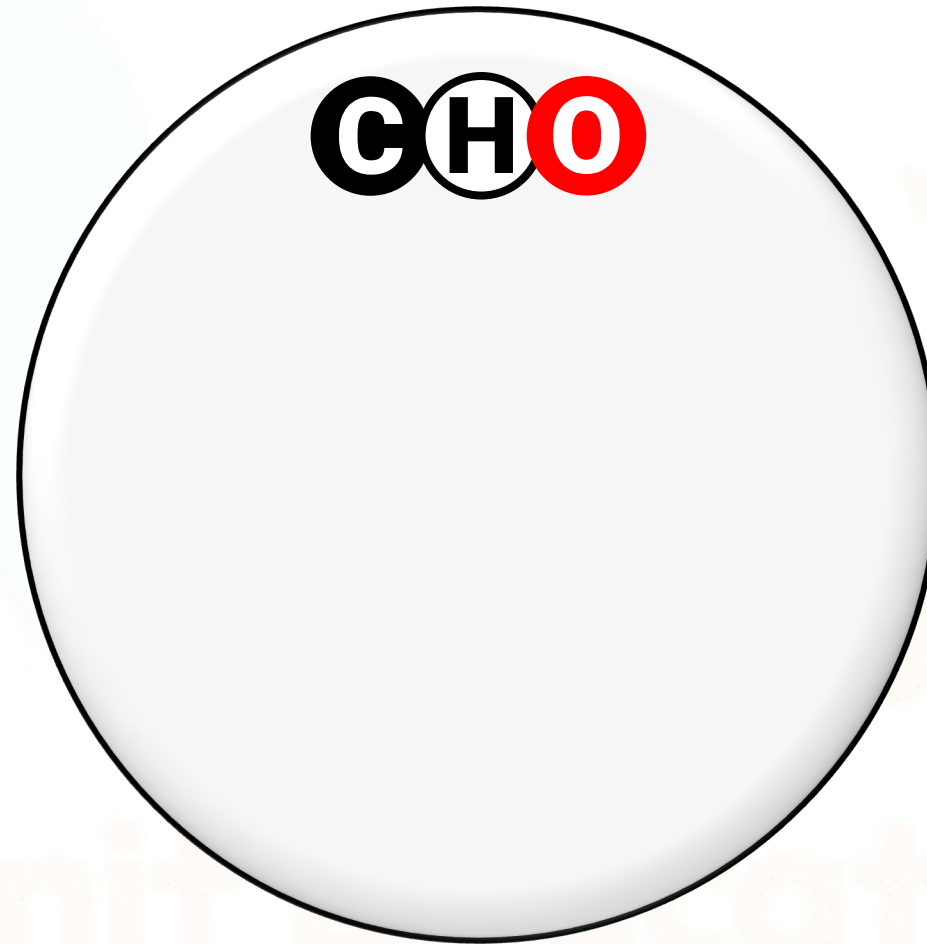
Nitrogen



Phosphorus

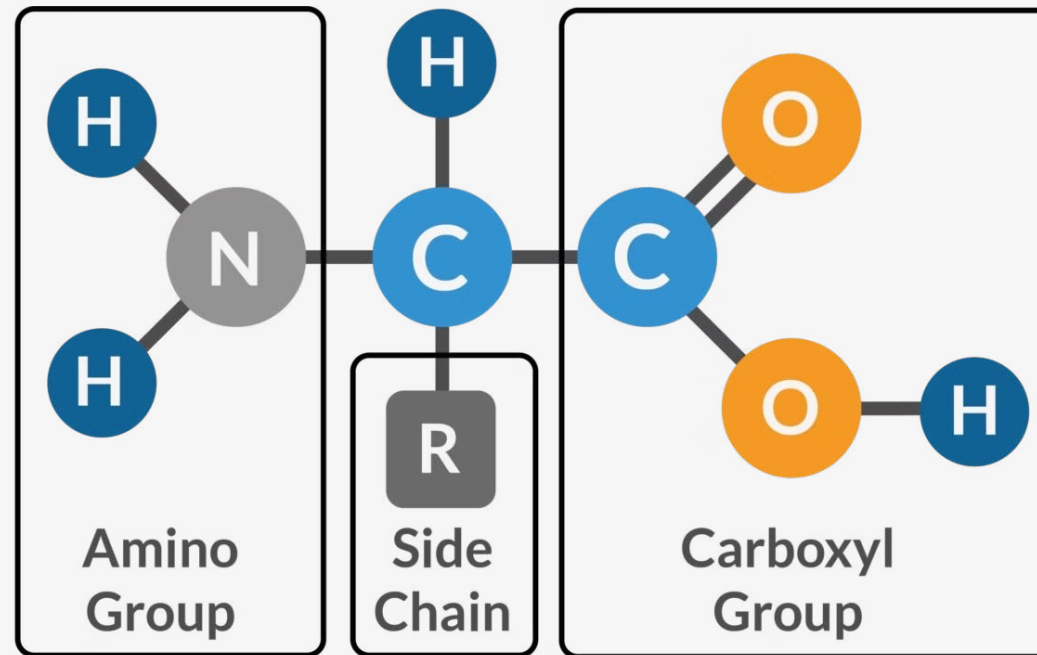
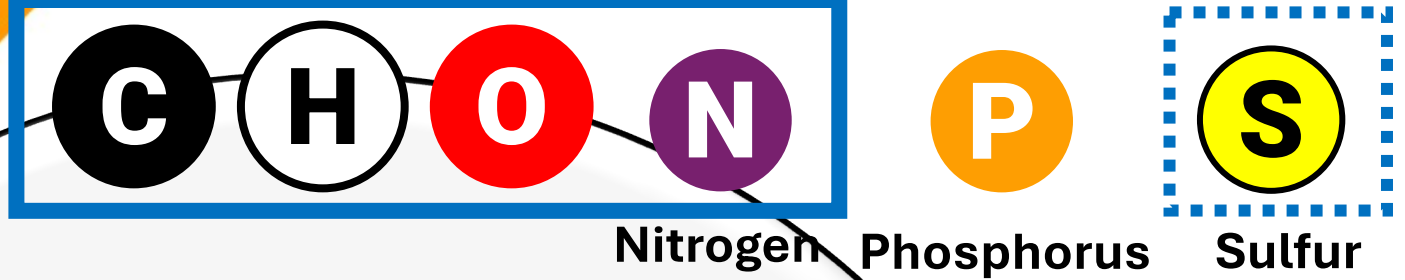


Sulfur



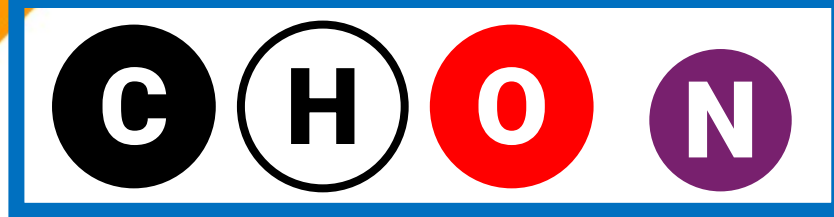
**Amino acids (胺基酸)**

### 3. Proteins 蛋白質



**Amino acids (胺基酸)**

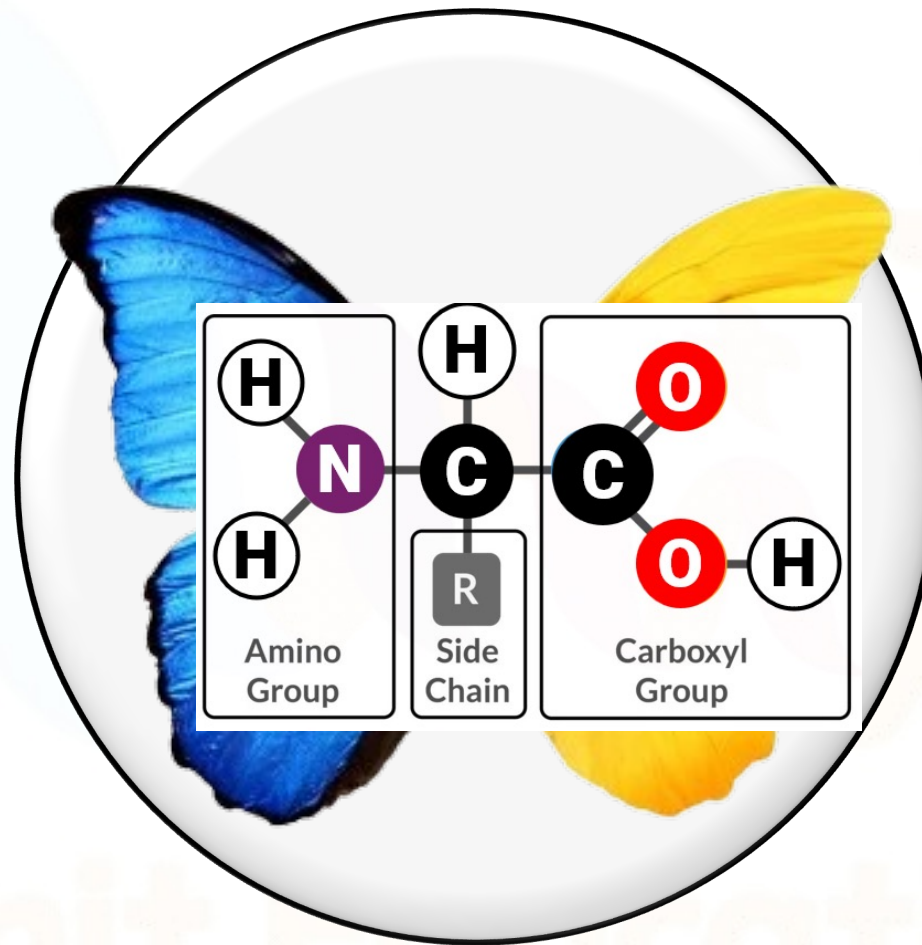
### 3. Proteins 蛋白質



Nitrogen

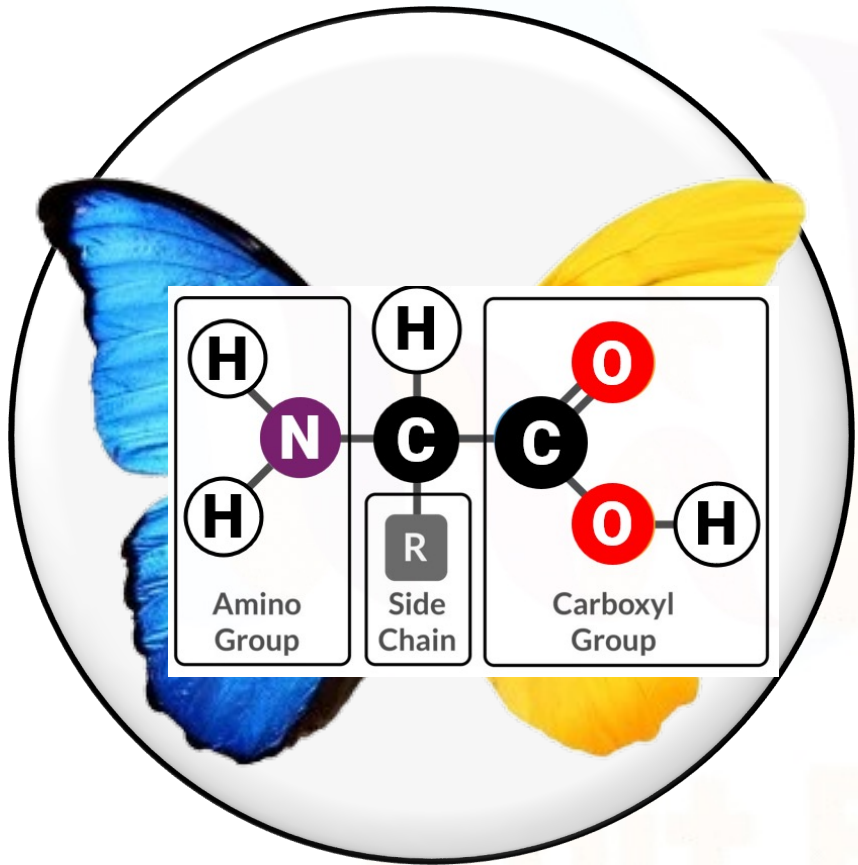
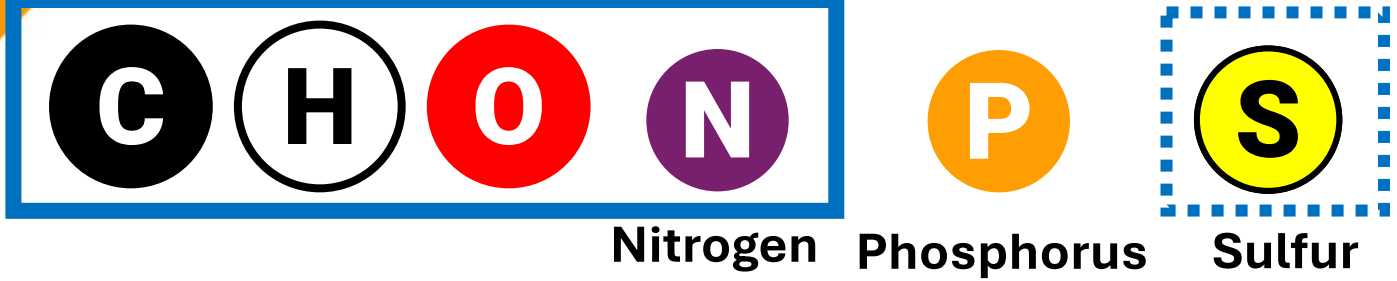
Phosphorus

Sulfur

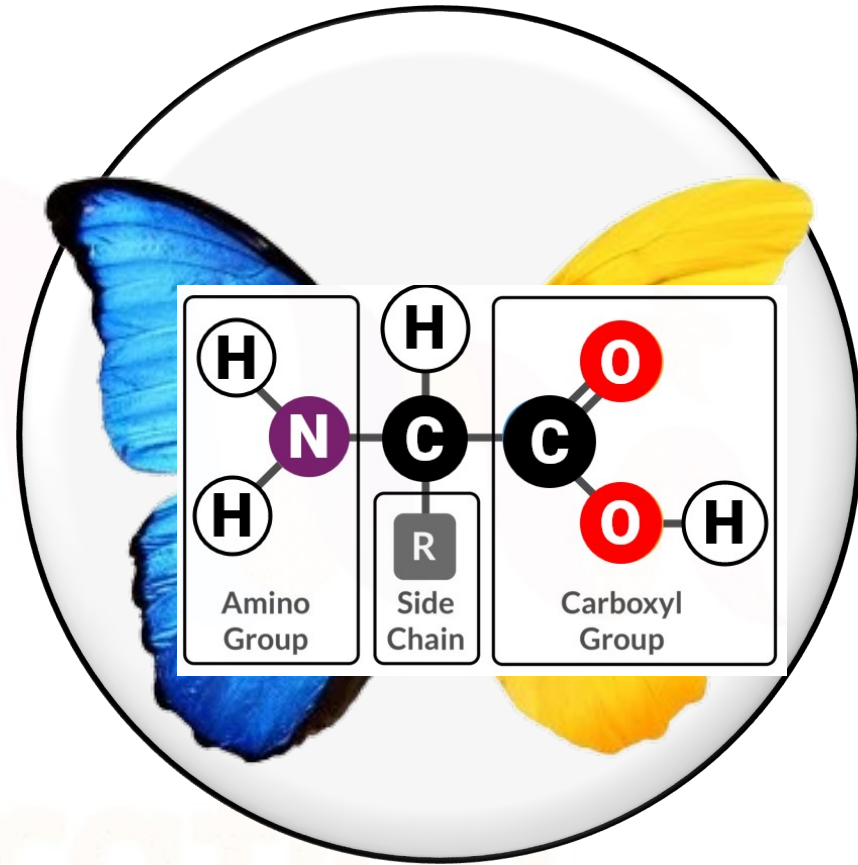


**Amino acids (胺基酸)**

# 3. Proteins 蛋白質

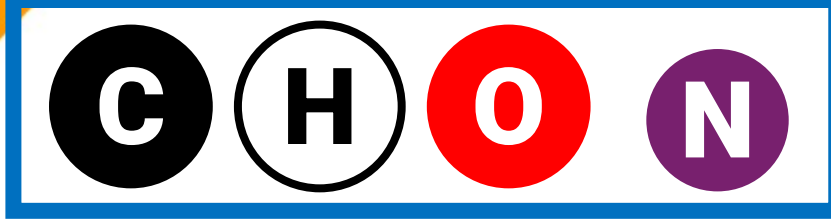


Amino acid



Amino acid

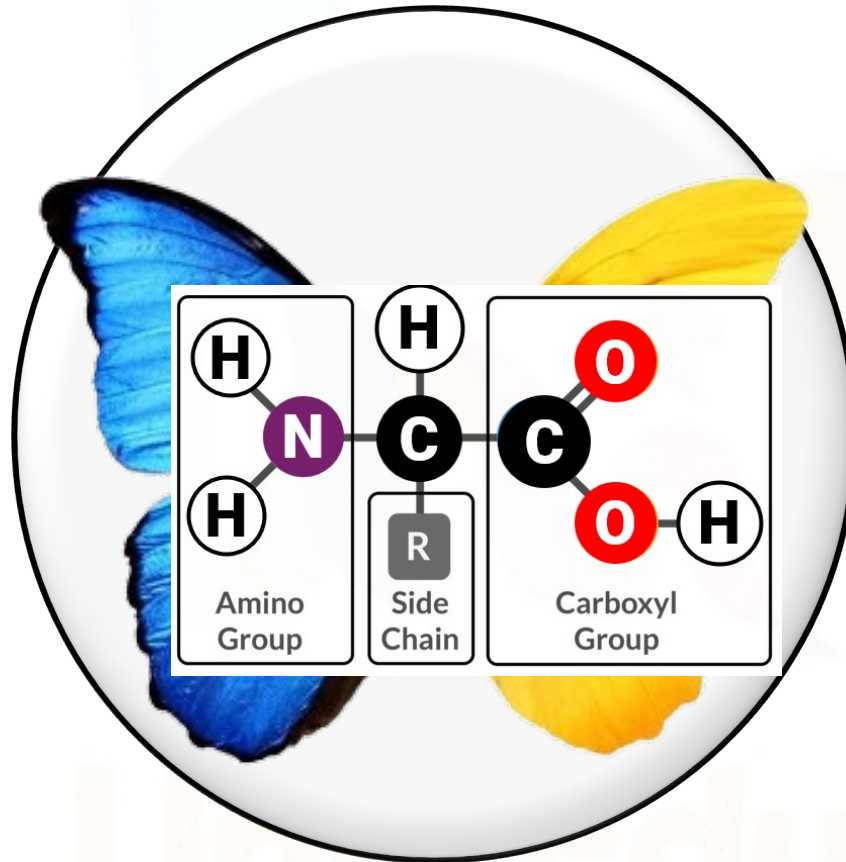
# 3. Proteins 蛋白質



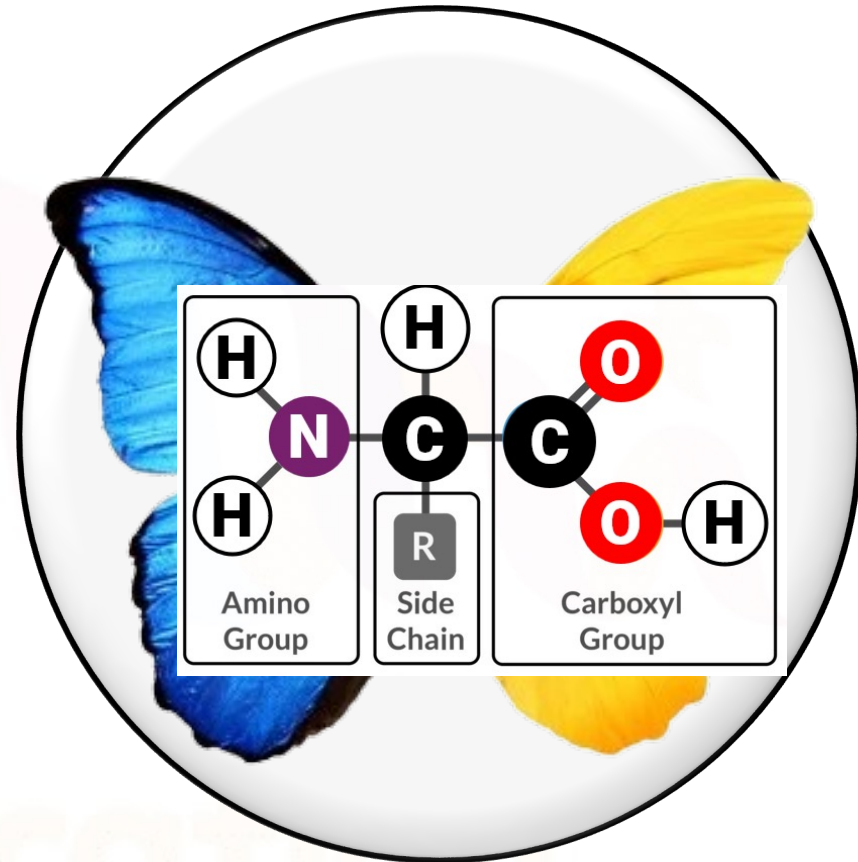
Nitrogen

Phosphorus

Sulfur

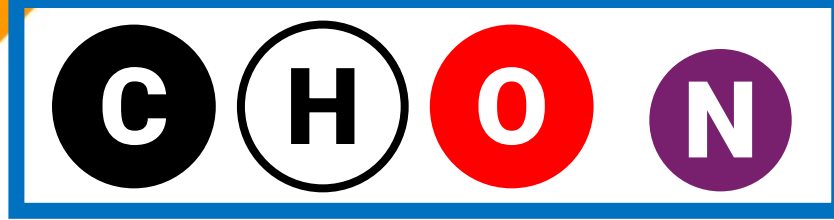


Amino acid



Amino acid

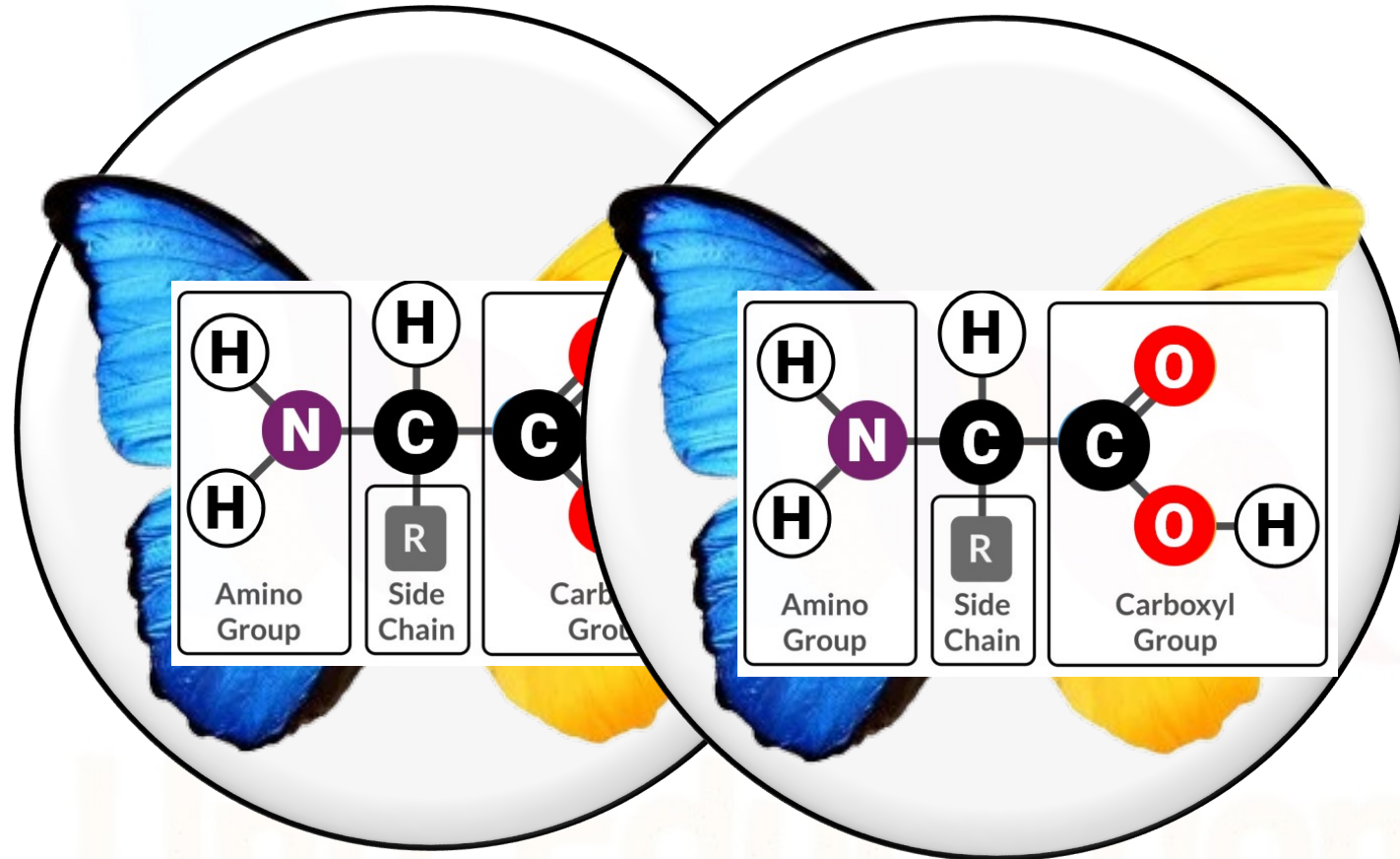
# 3. Proteins 蛋白質



Nitrogen

Phosphorus

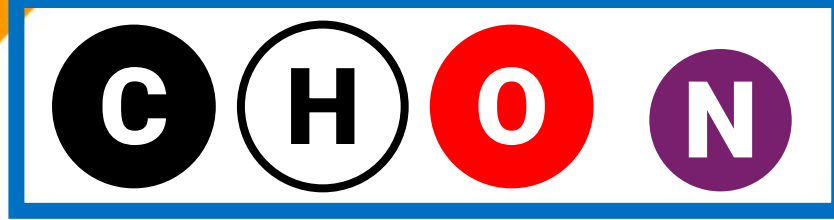
Sulfur



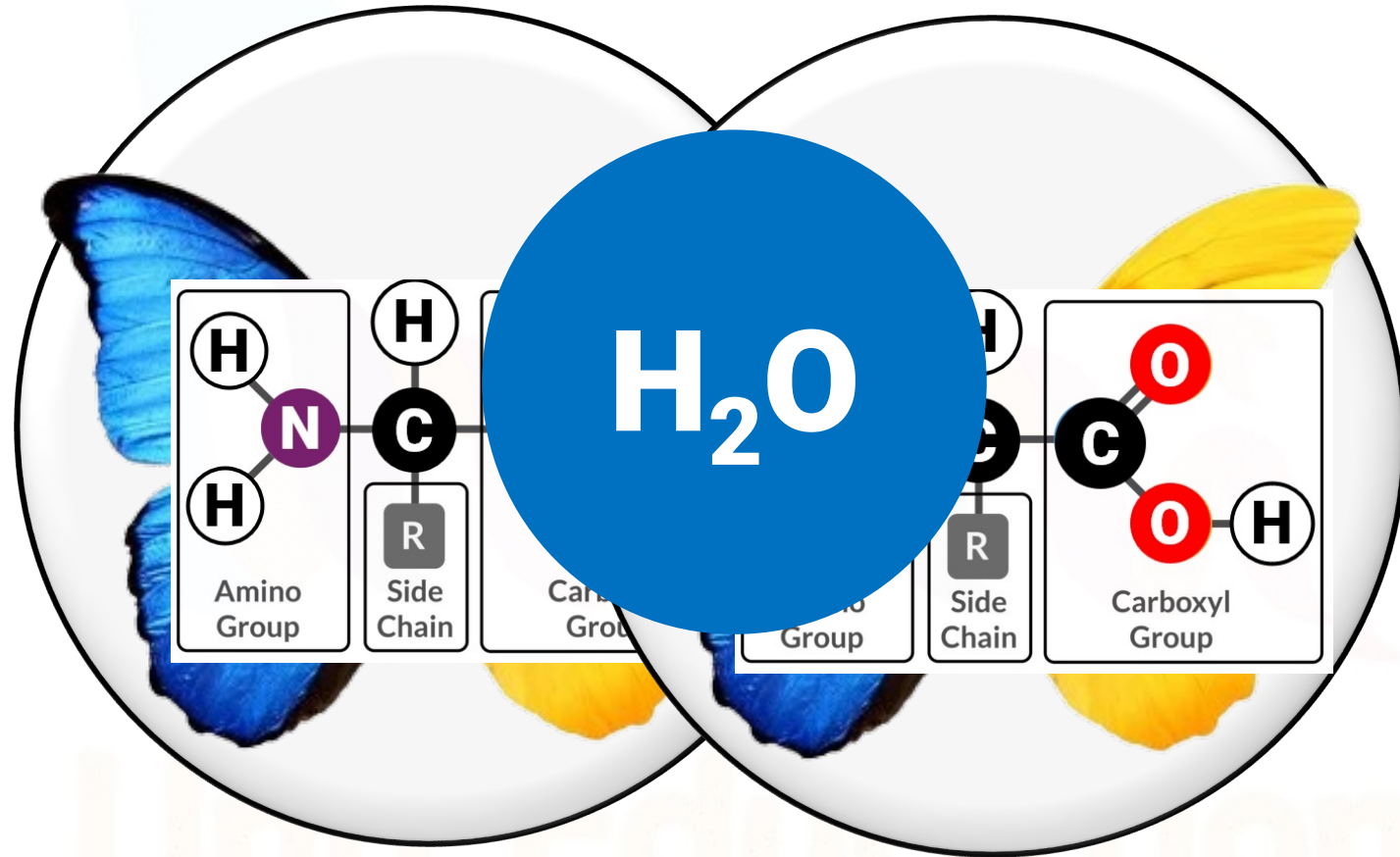
Amino acid

Amino acid

# 3. Proteins 蛋白質



Nitrogen    Phosphorus    Sulfur

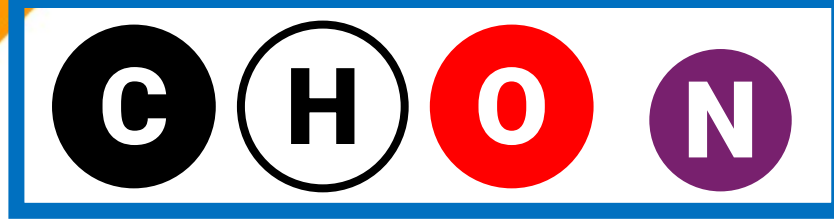


Amino acid

Amino acid



# 3. Proteins 蛋白質



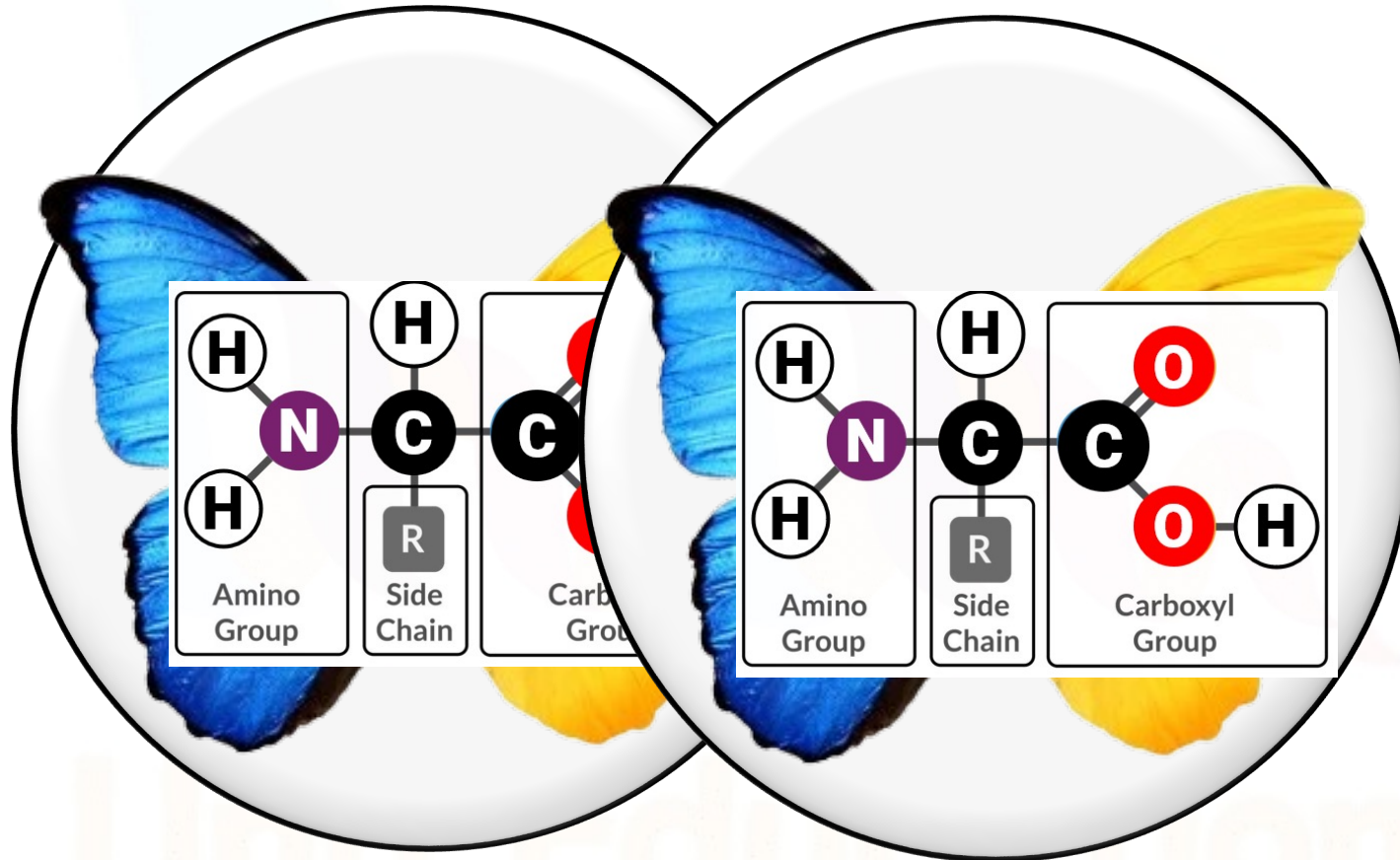
Nitrogen



Phosphorus



Sulfur



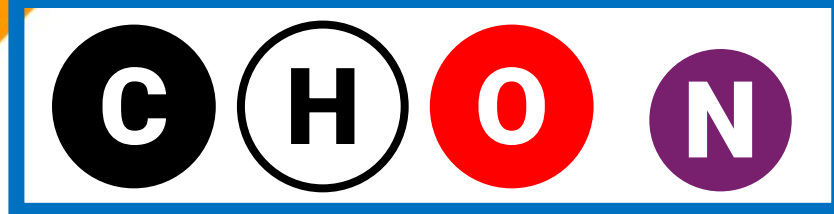
Amino acid

Amino acid

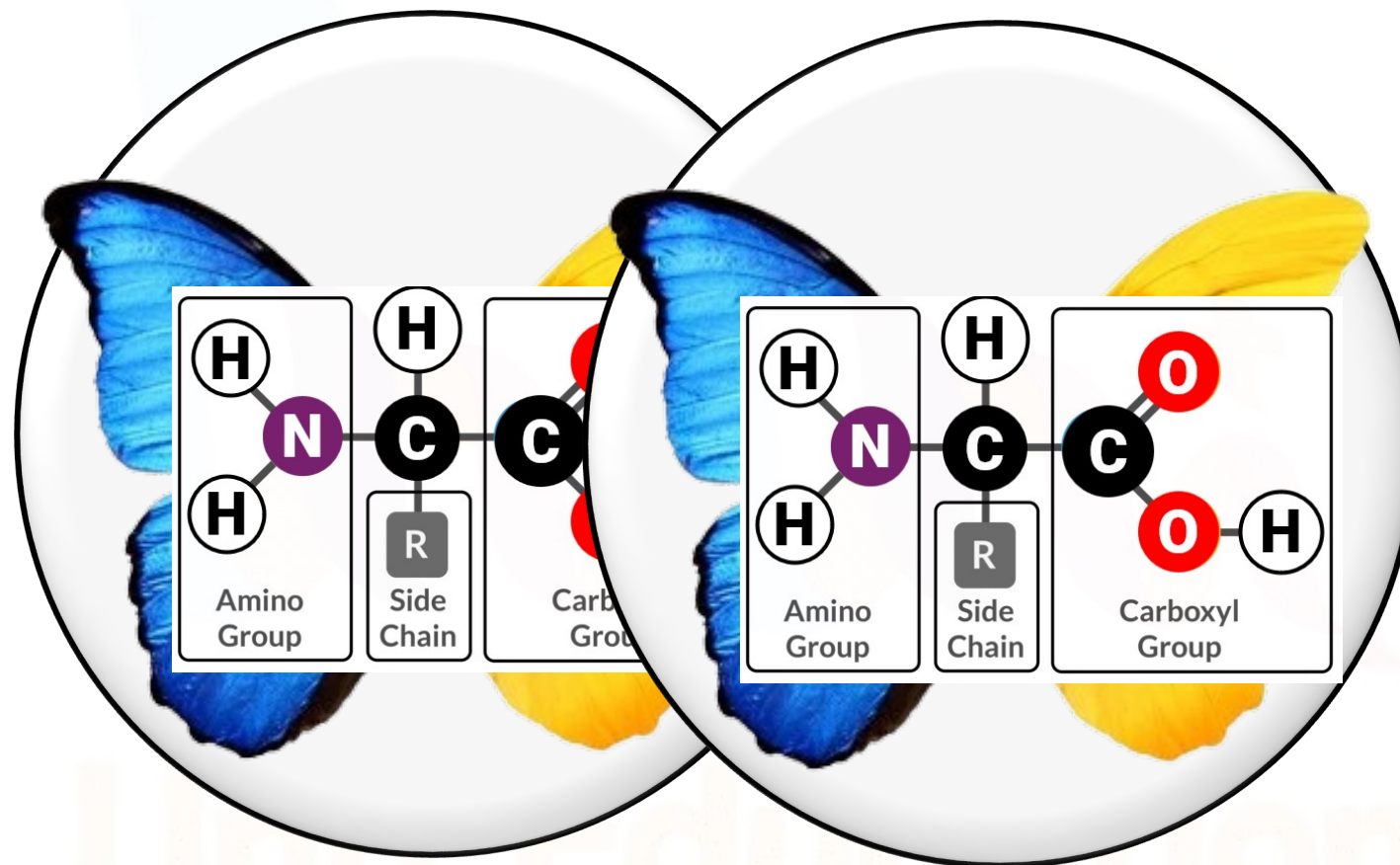




# 3. Proteins 蛋白質

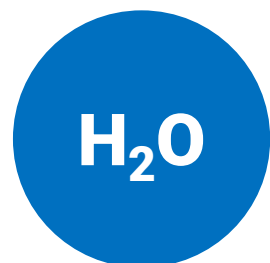


Nitrogen Phosphorus Sulfur

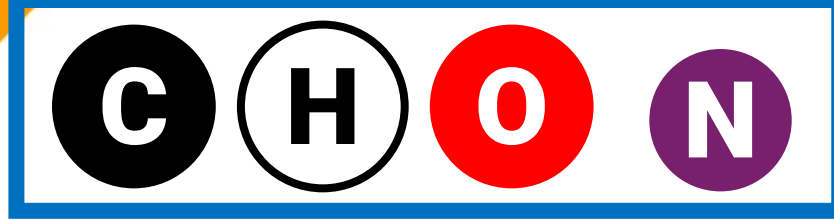


Amino acid

Amino acid



# 3. Proteins 蛋白質



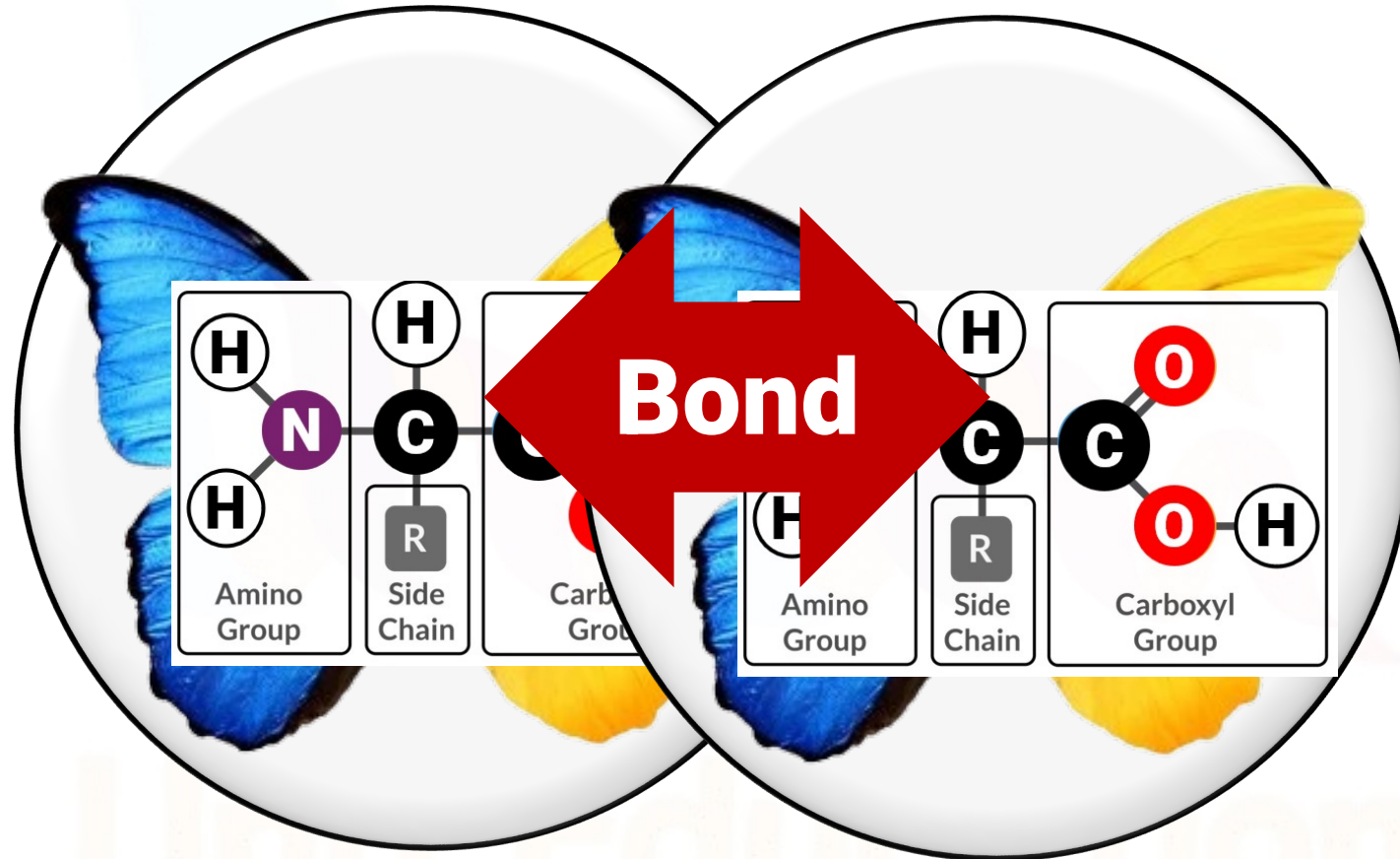
Nitrogen



Phosphorus



Sulfur

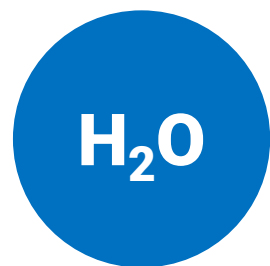
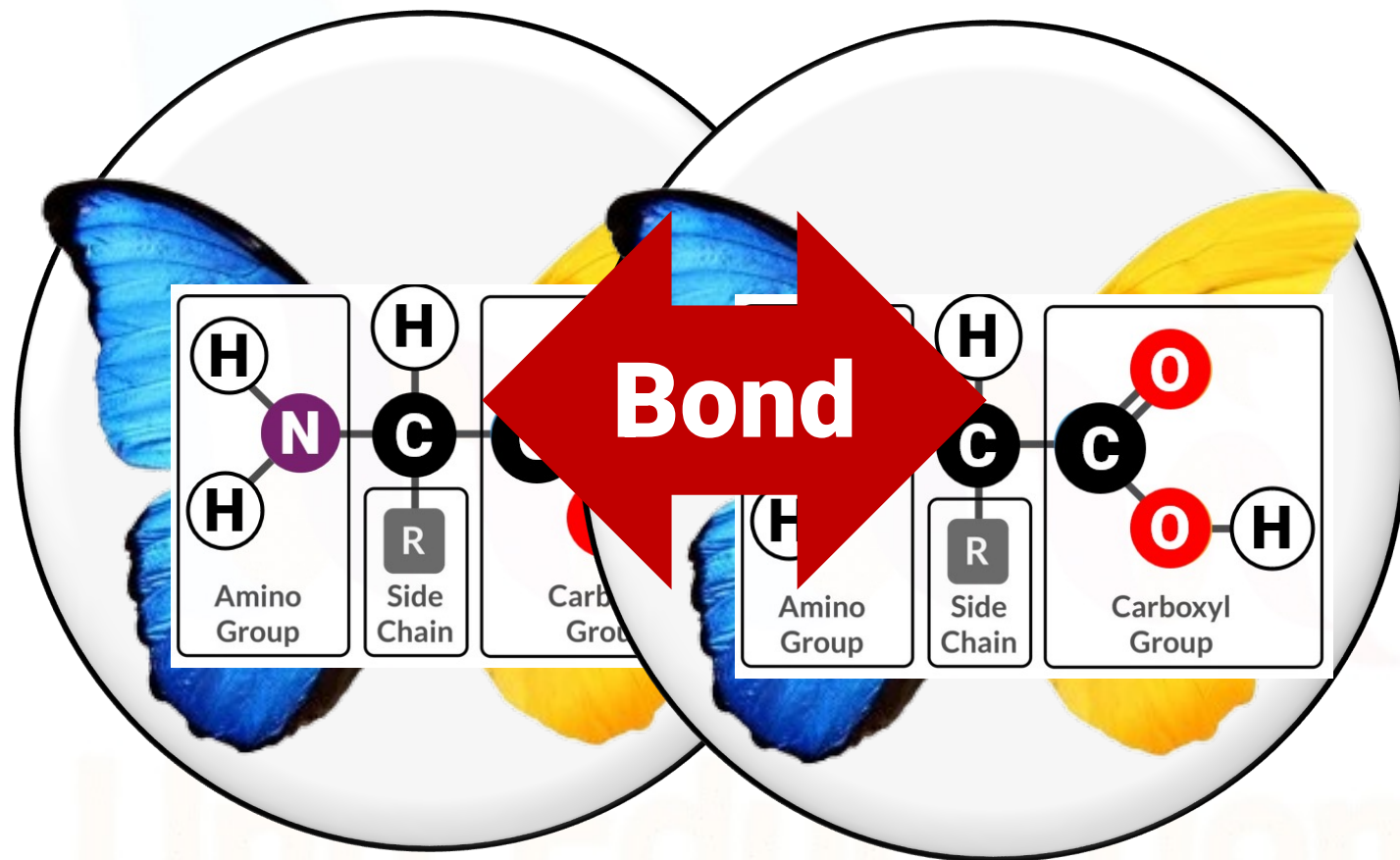
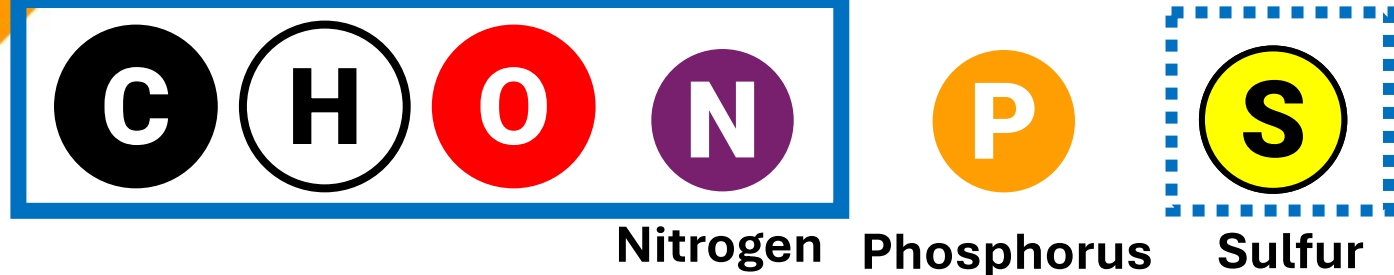


Amino acid

Amino acid



# 3. Proteins 蛋白質



**Dipeptide (二肽)**

### 3. Proteins 蛋白質

C

H

O

N

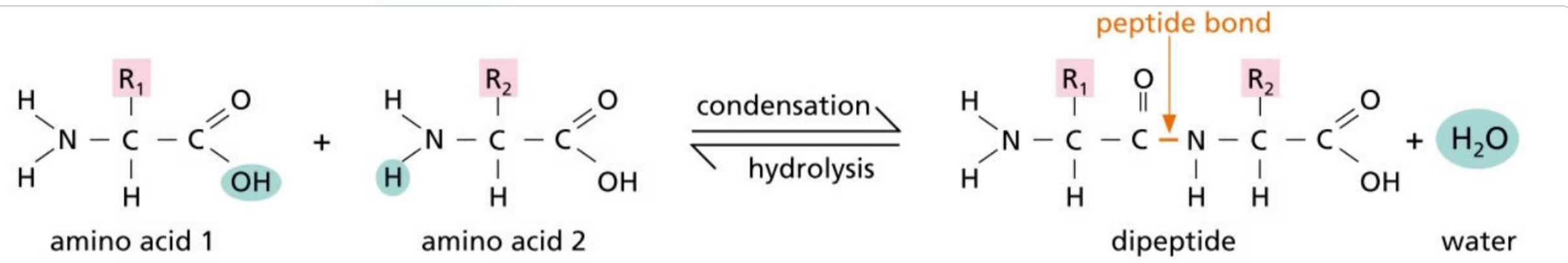
P

S

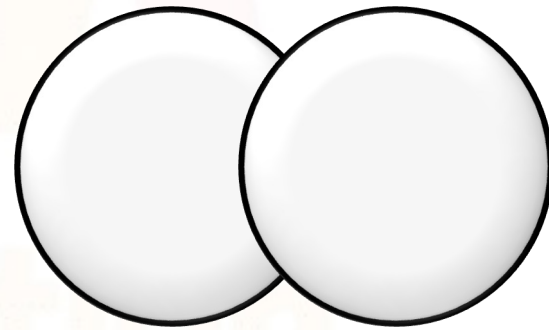
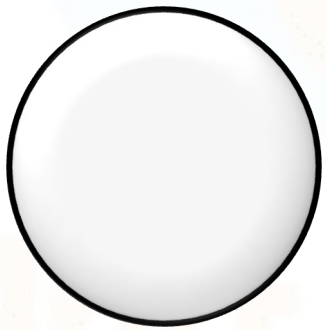
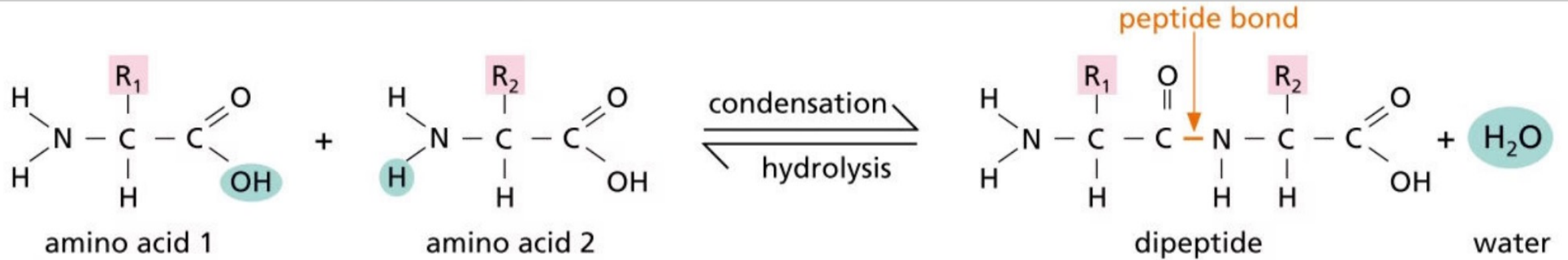
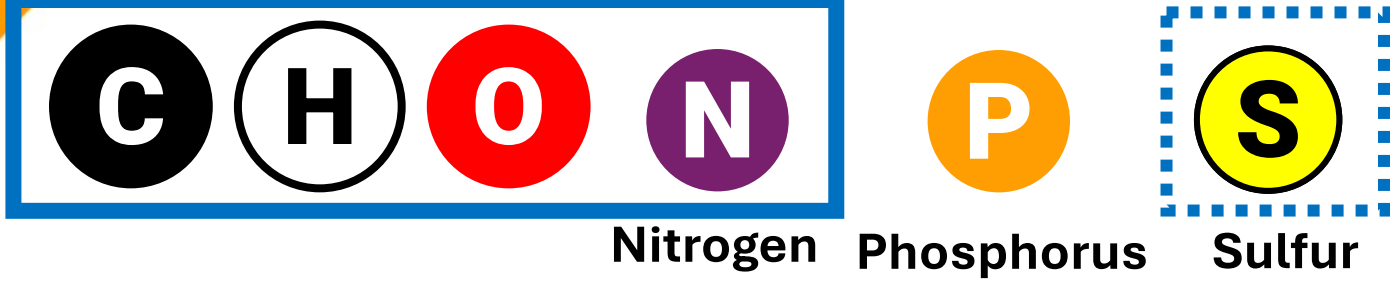
Nitrogen

Phosphorus

Sulfur

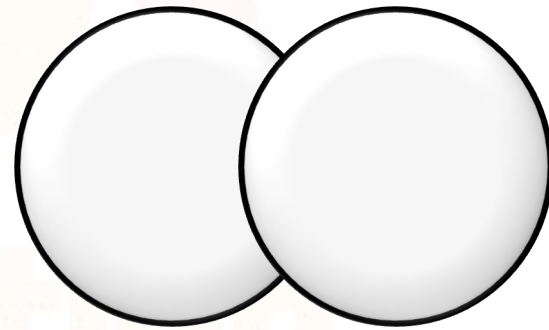
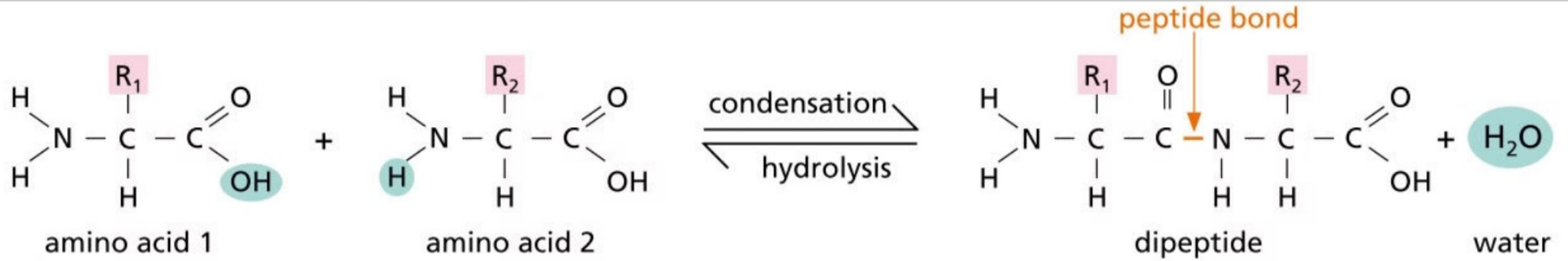
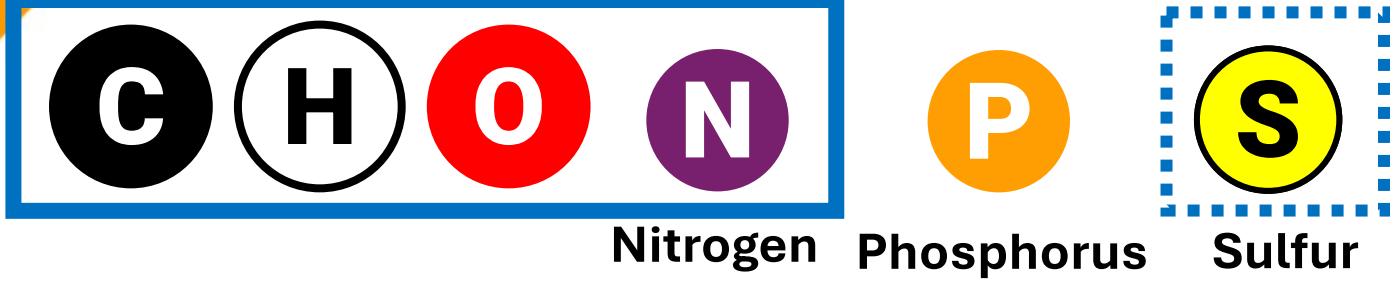


# 3. Proteins 蛋白質



Unit Education

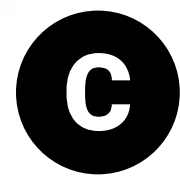
# 3. Proteins 蛋白質



Unit Education



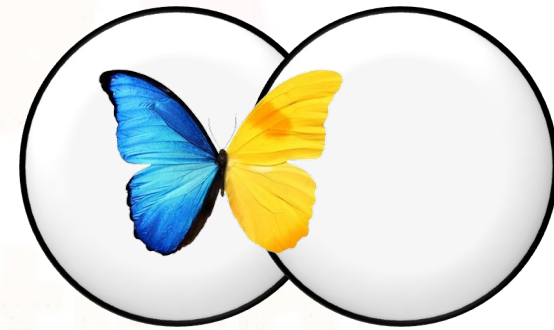
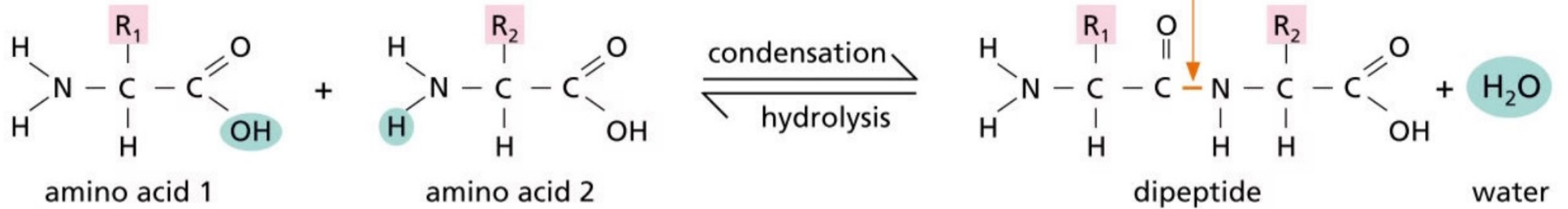
# 3. Proteins 蛋白質



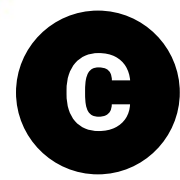
Nitrogen

Phosphorus

Sulfur



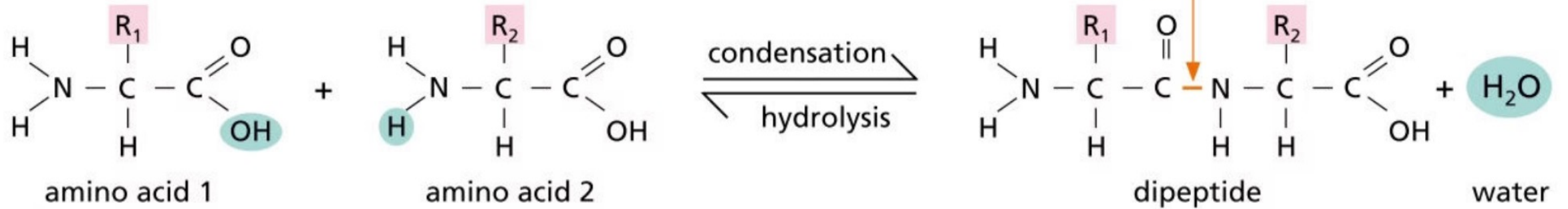
# 3. Proteins 蛋白質



Nitrogen

Phosphorus

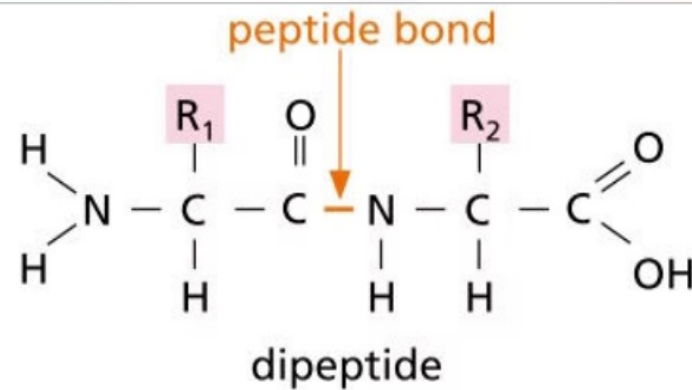
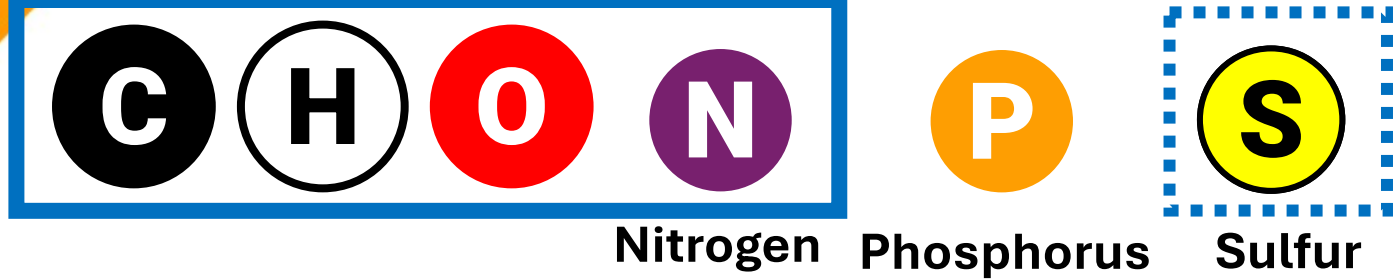
Sulfur



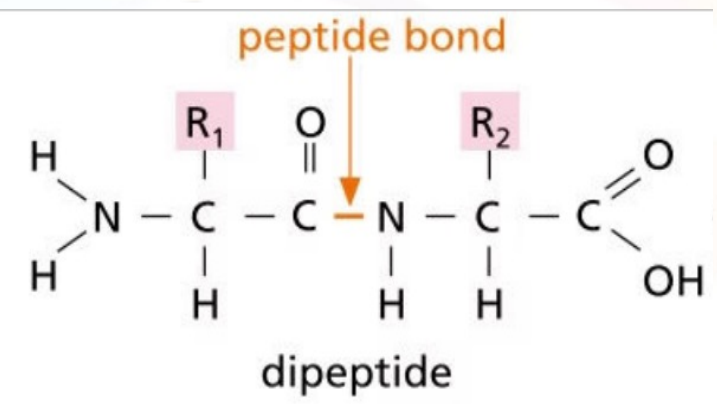
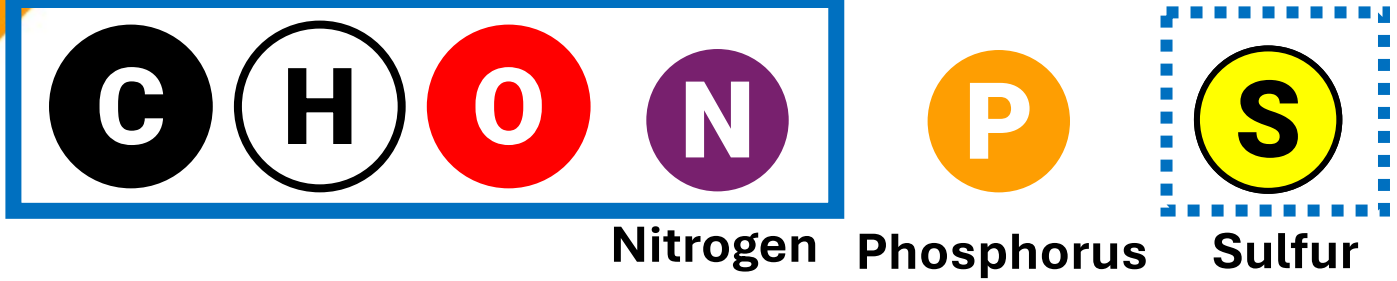
Unit Education



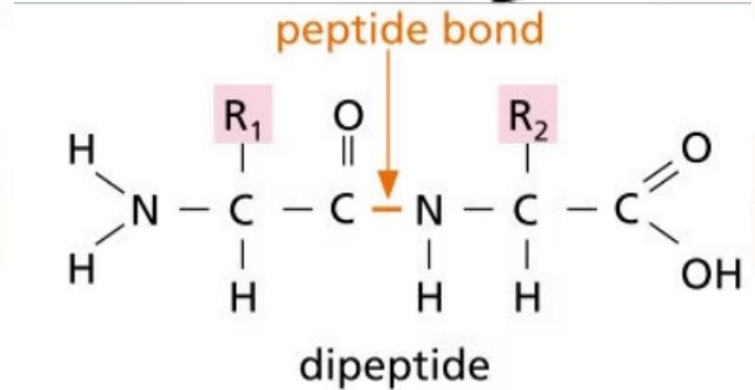
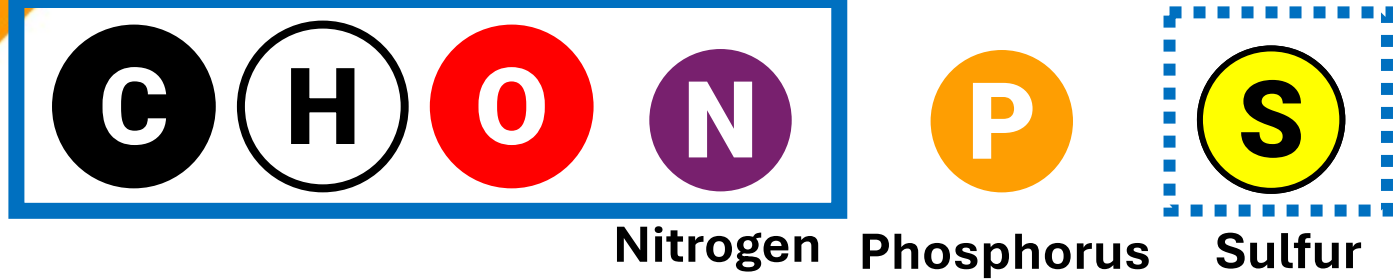
# 3. Proteins 蛋白質



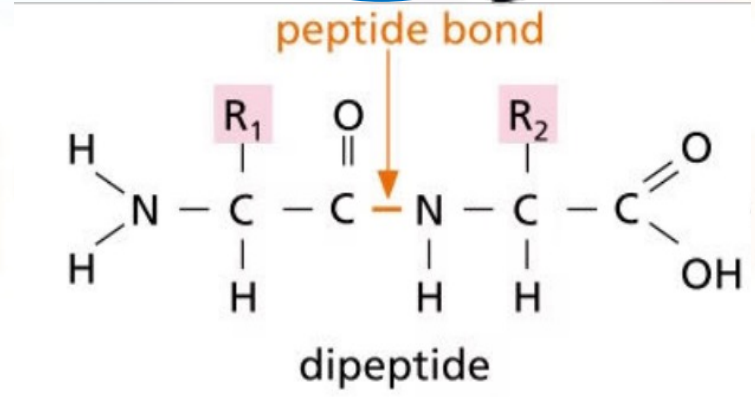
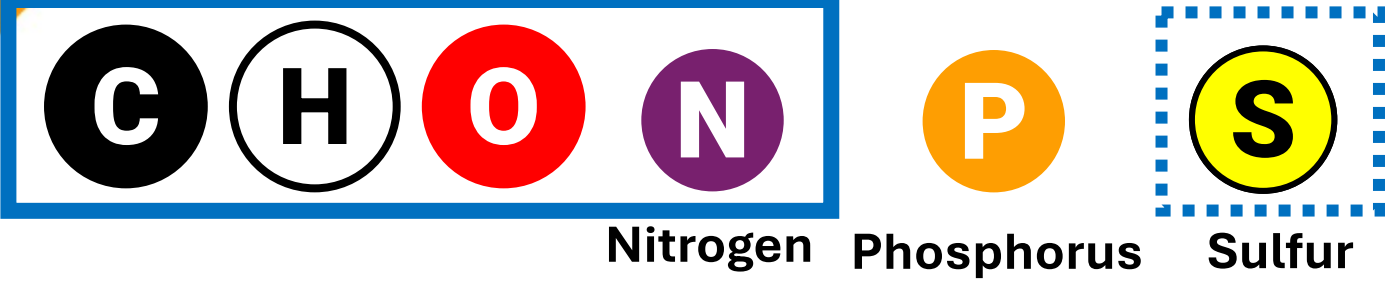
# 3. Proteins 蛋白質



# 3. Proteins 蛋白質

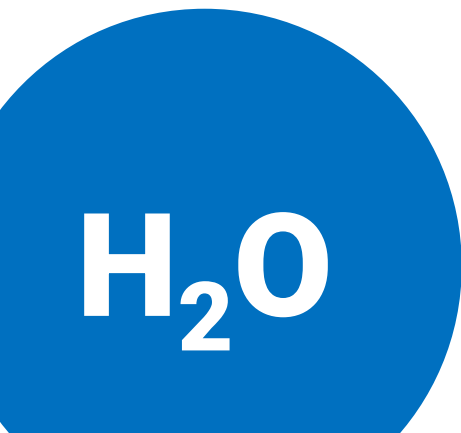
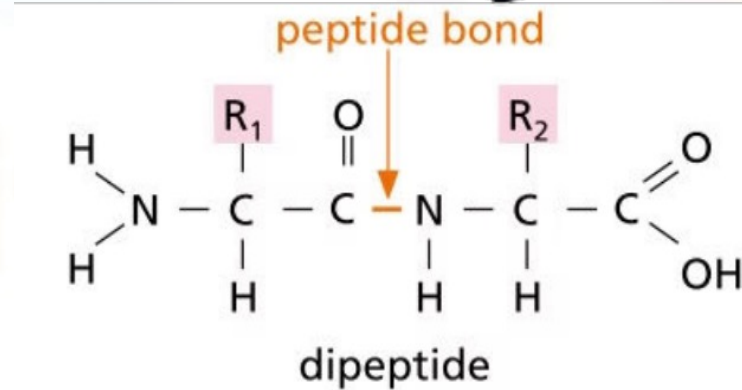
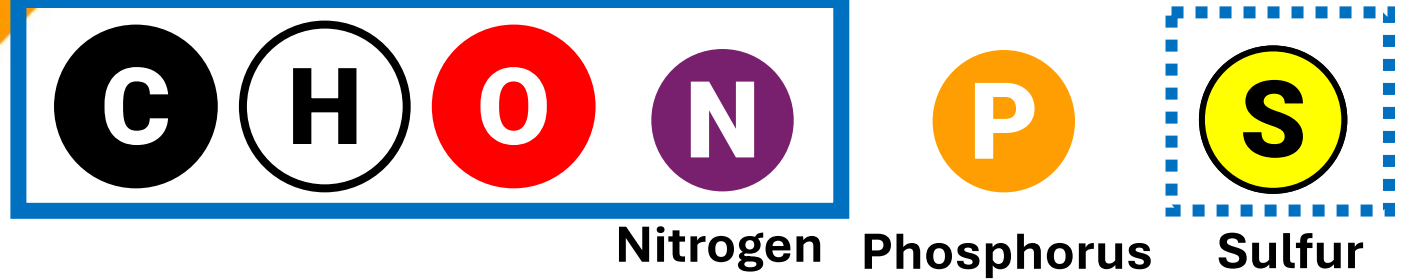


# 3. Proteins 蛋白質

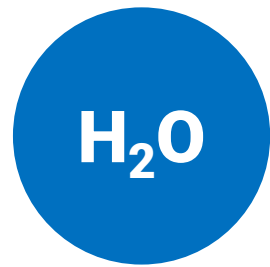
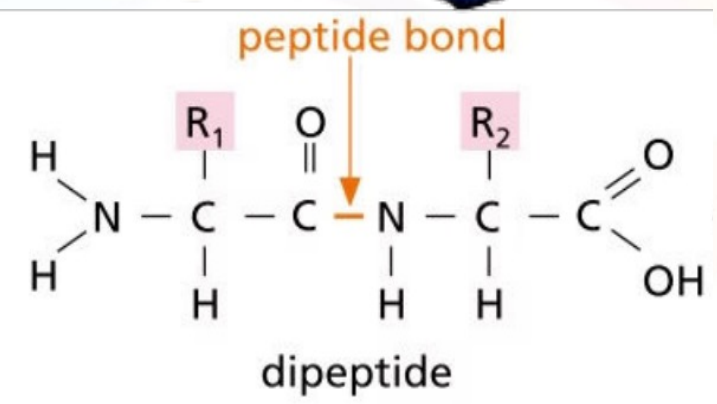
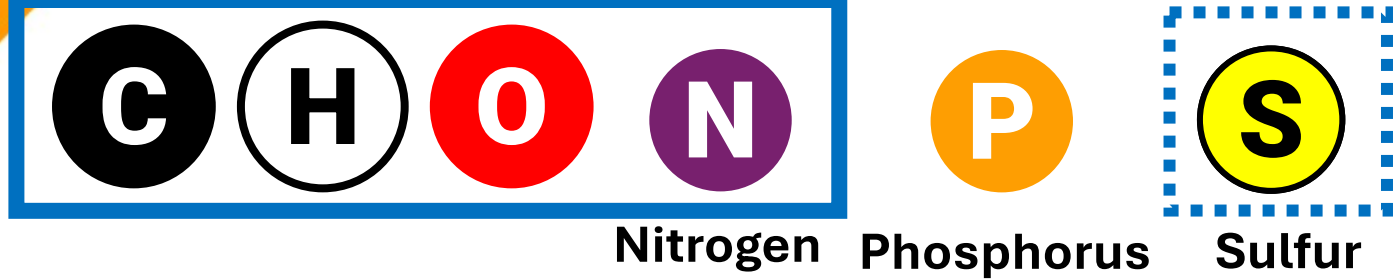




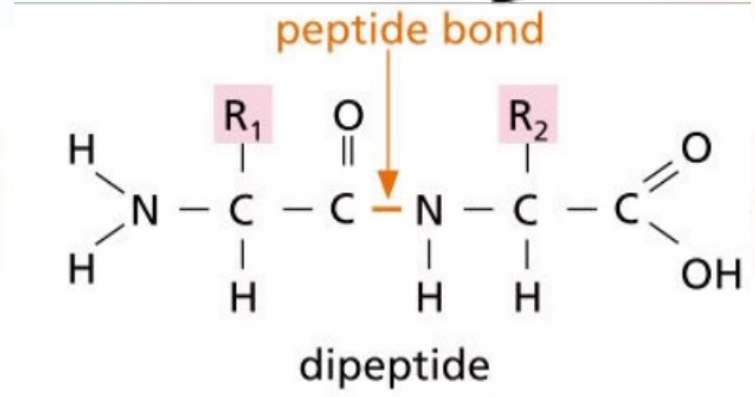
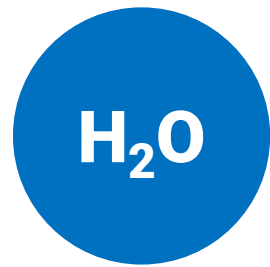
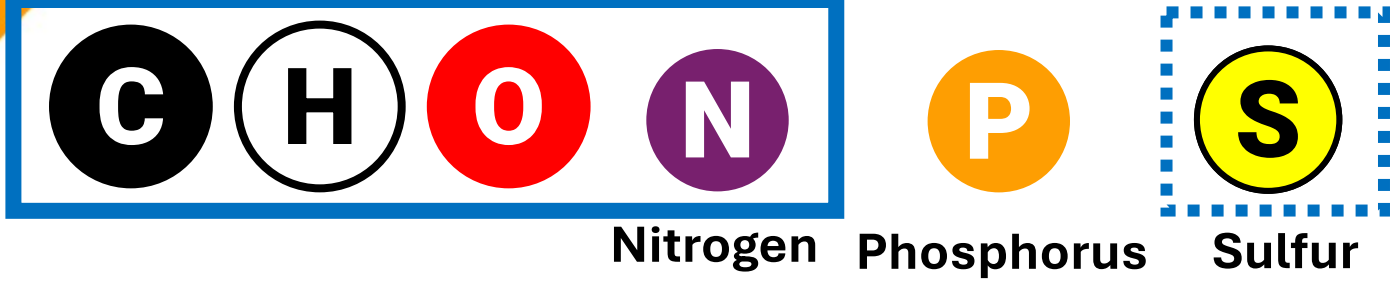
# 3. Proteins 蛋白質



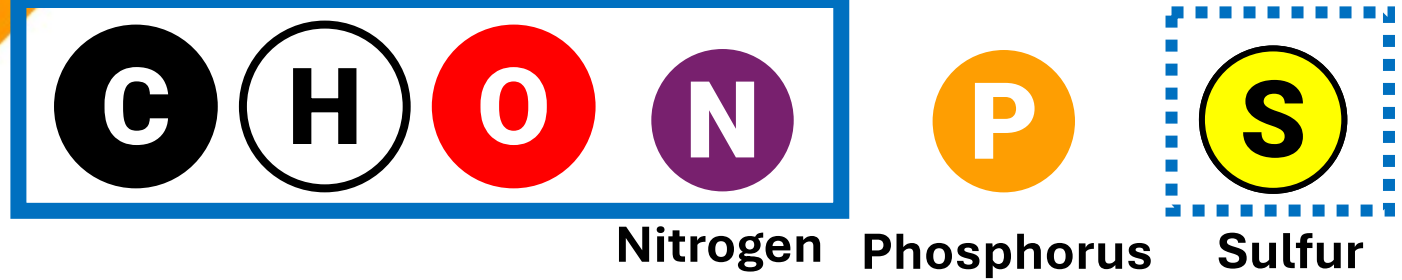
# 3. Proteins 蛋白質



# 3. Proteins 蛋白質



# 3. Proteins 蛋白質



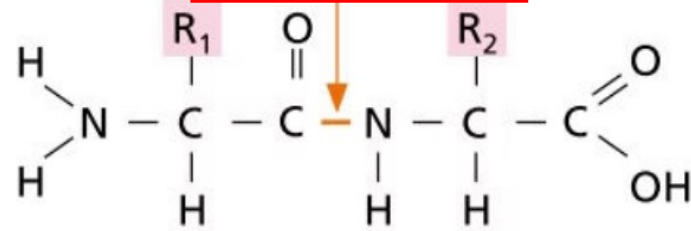
**Amino Group**

**Carboxyl Group**

**Amino Group**

**Carboxyl Group**

**peptide bond**



dipeptide



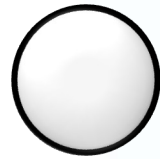
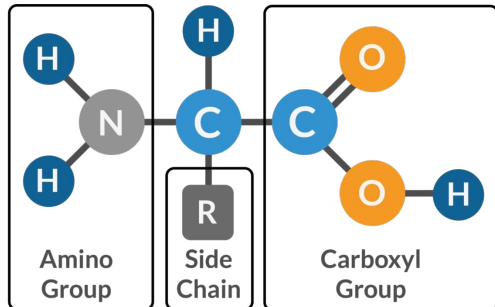


# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



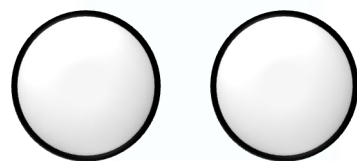
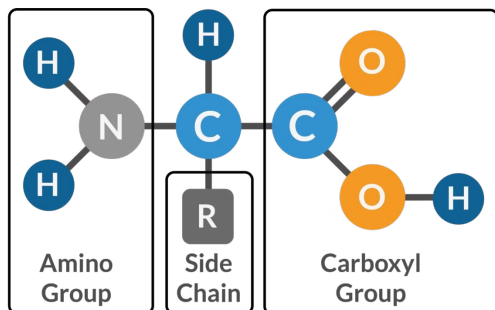
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group**   **Carboxyl Group**



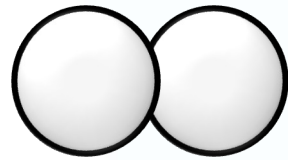
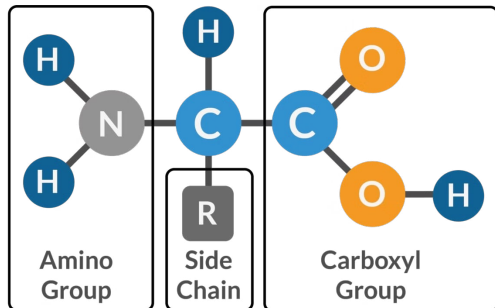
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group**   **Carboxyl Group**



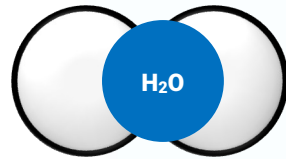
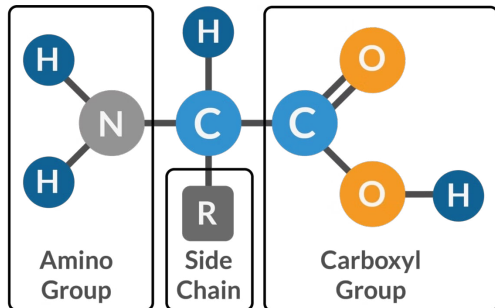
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group**   **Carboxyl Group**



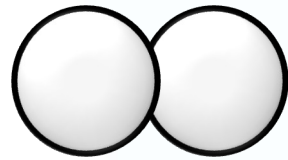
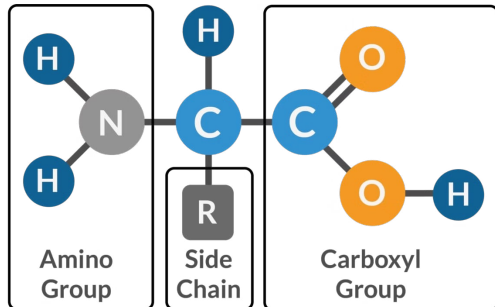
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



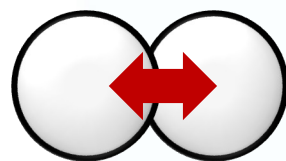
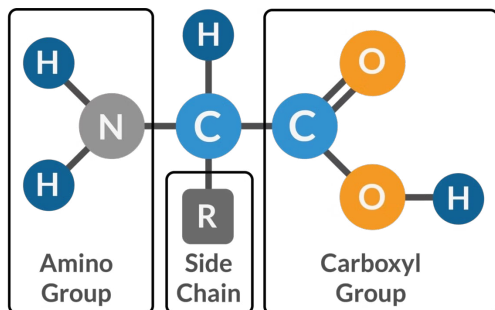
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group**   **Carboxyl Group**



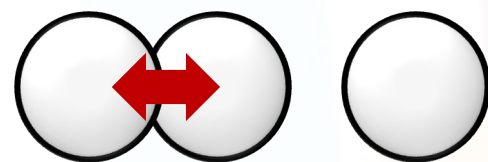
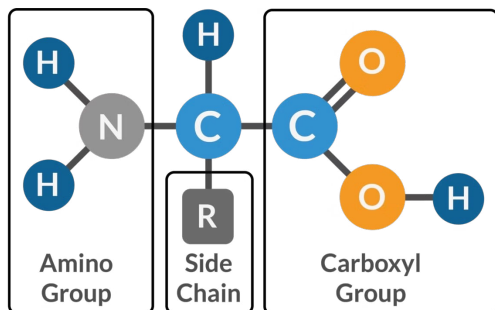
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



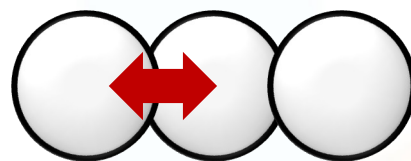
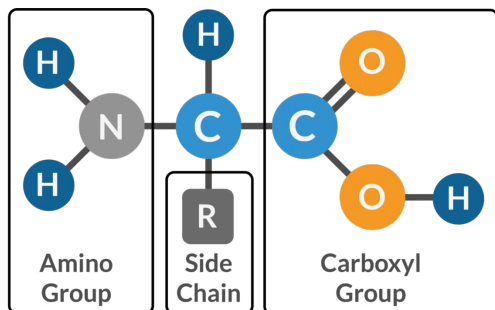
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



**Poly (多)-mer**

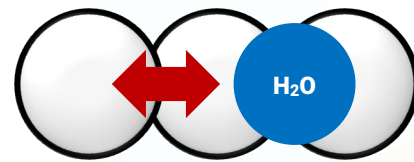
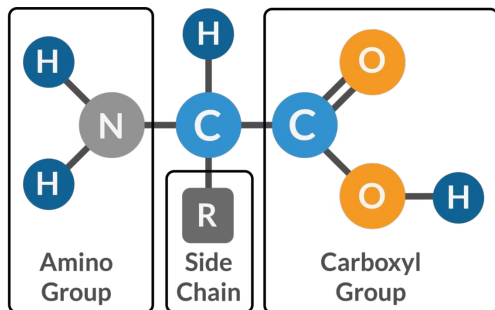


# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group**   **Carboxyl Group**



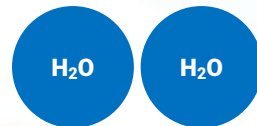
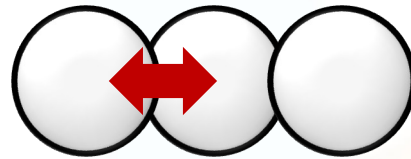
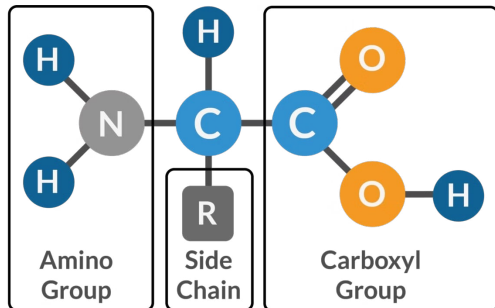
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



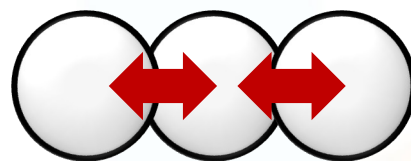
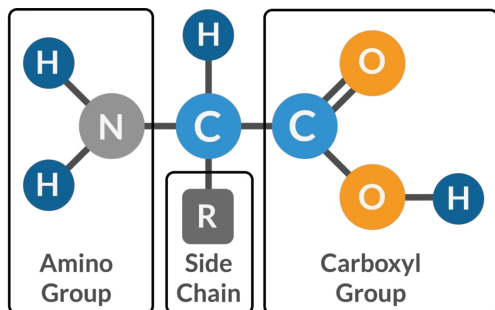
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



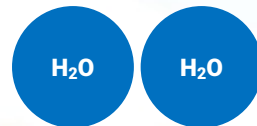
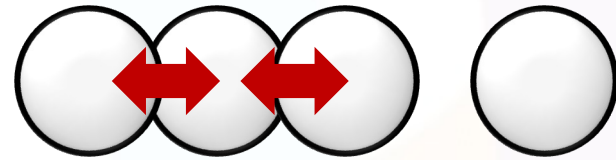
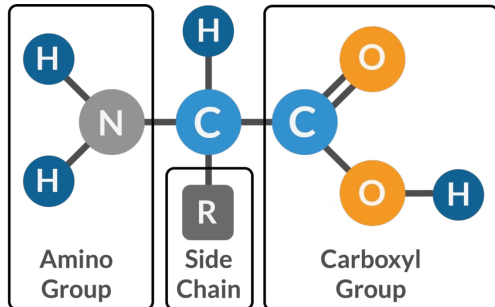
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



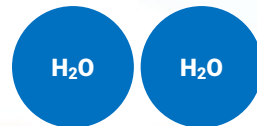
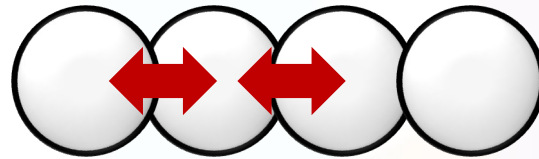
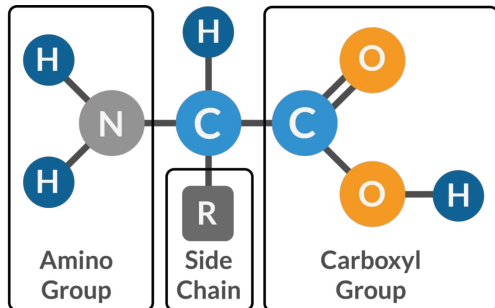
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



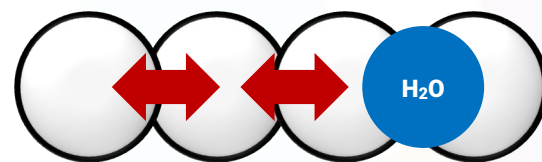
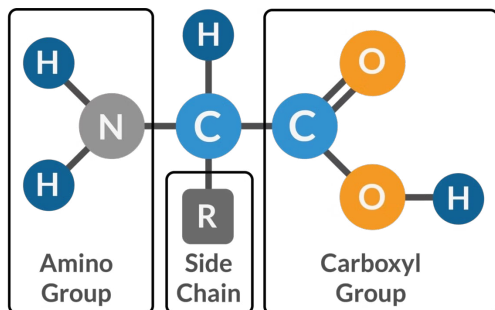
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group**   **Carboxyl Group**



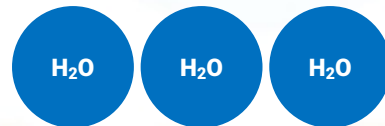
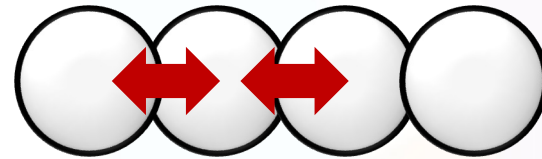
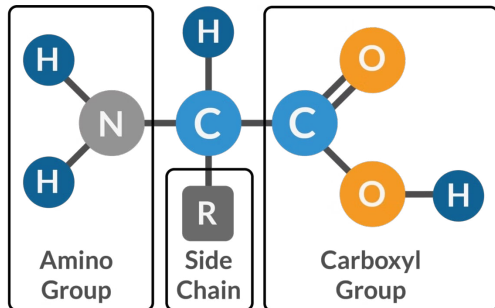
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group**   **Carboxyl Group**



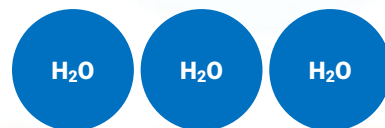
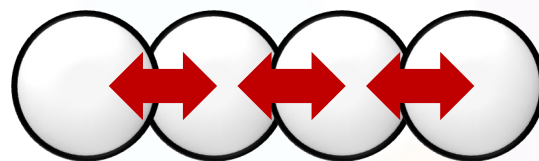
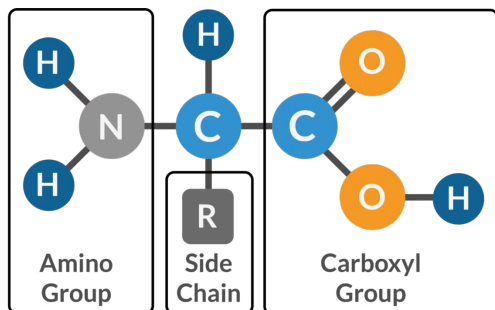
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



**Poly (多)-mer**

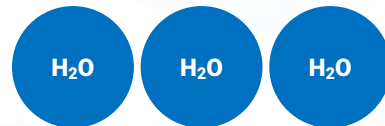
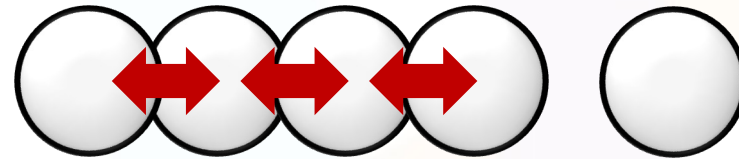
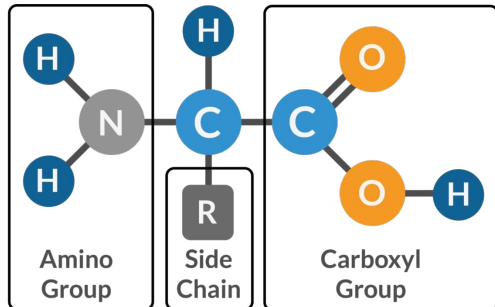


# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



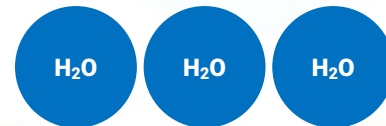
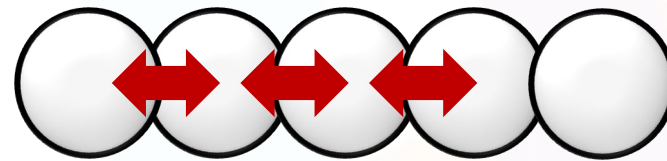
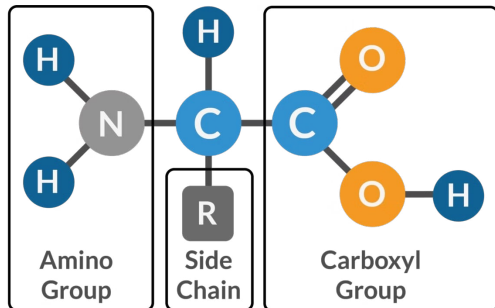
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



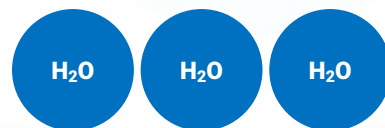
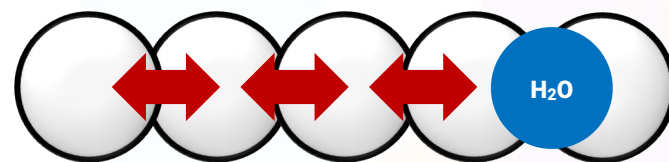
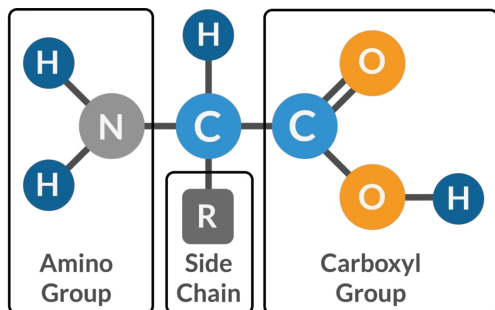
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group**   **Carboxyl Group**



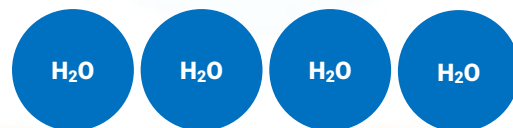
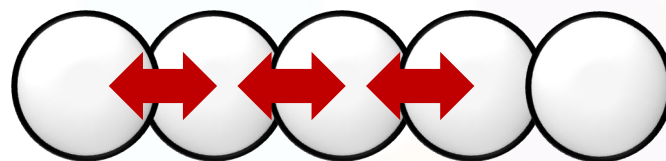
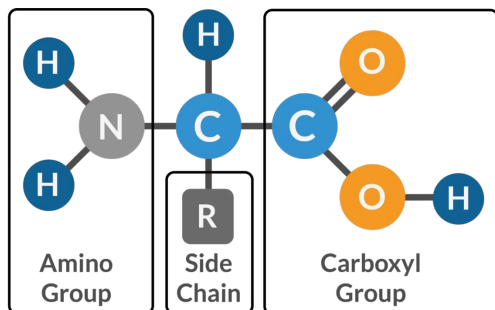
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



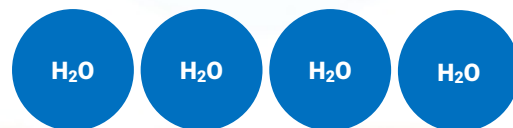
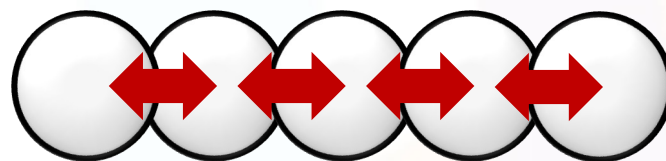
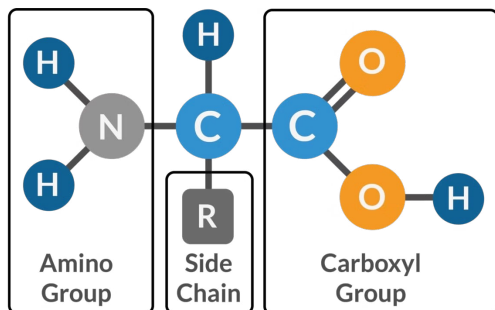
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



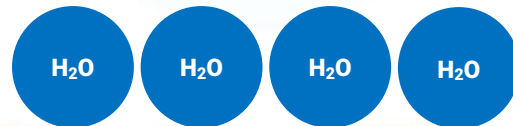
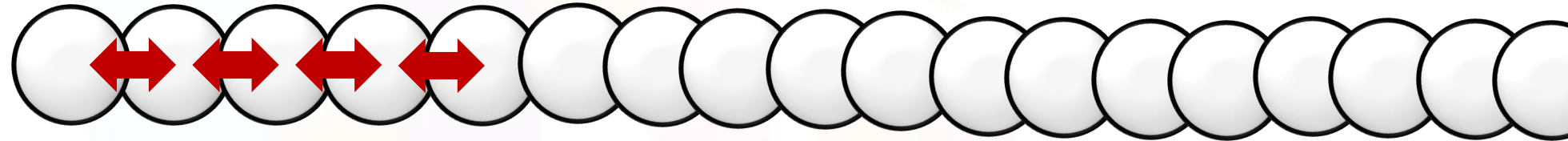
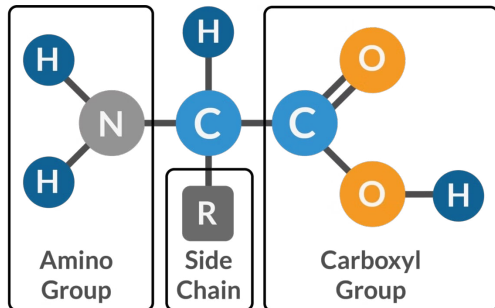
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



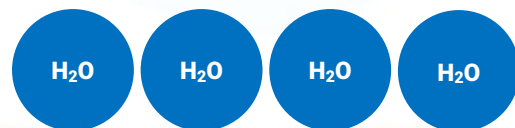
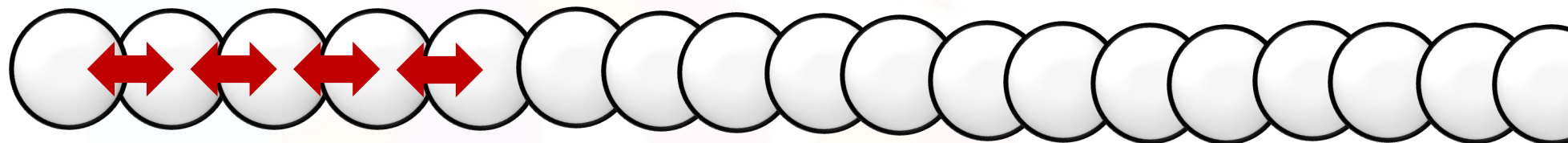
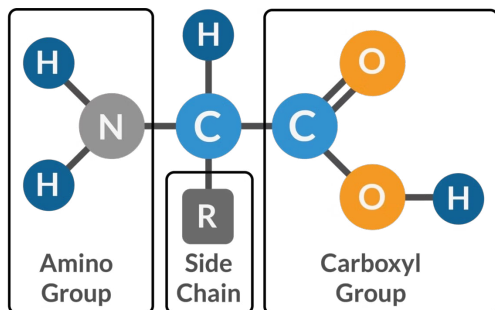
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



**Amino Group** **Carboxyl Group**



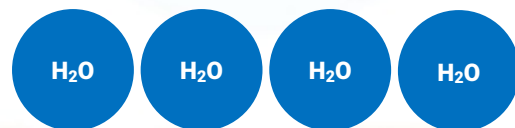
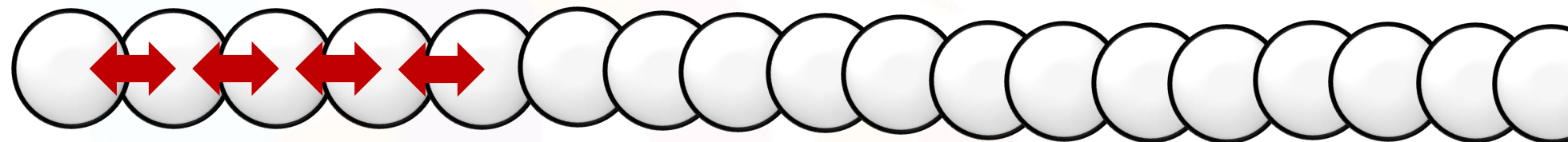
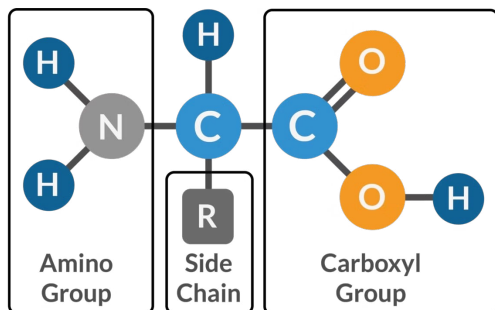
**Poly (多)-mer**

# 3. Proteins 蛋白質

## Condensation 縮合



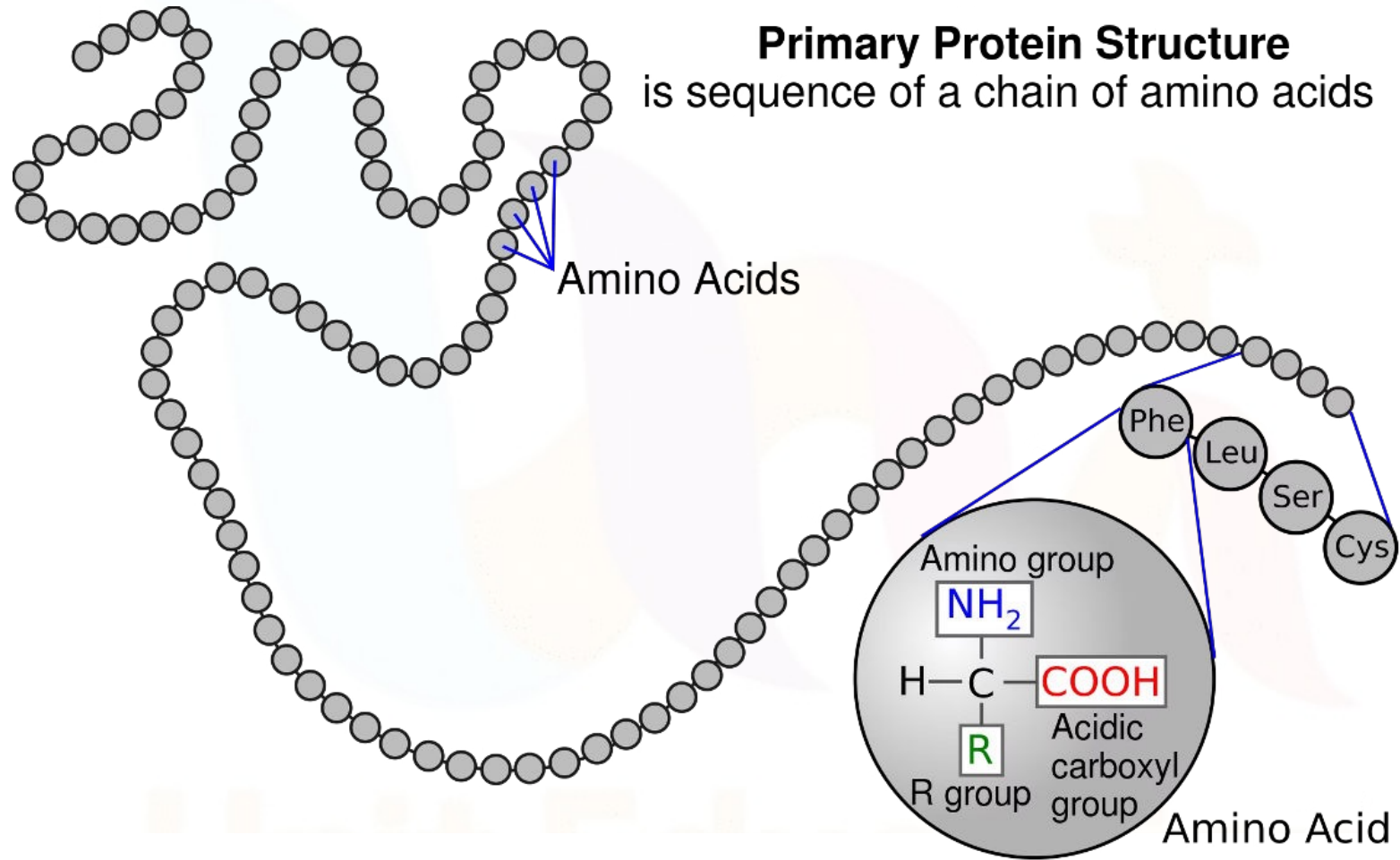
**Amino Group** **Carboxyl Group**



**Polypeptide (多肽) Chain**



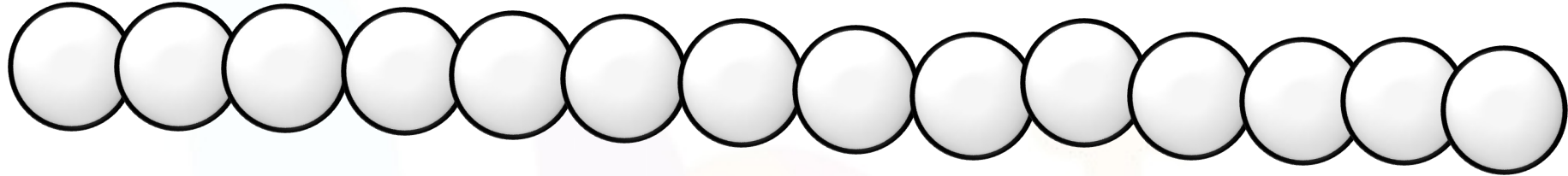
### 3. Proteins 蛋白質



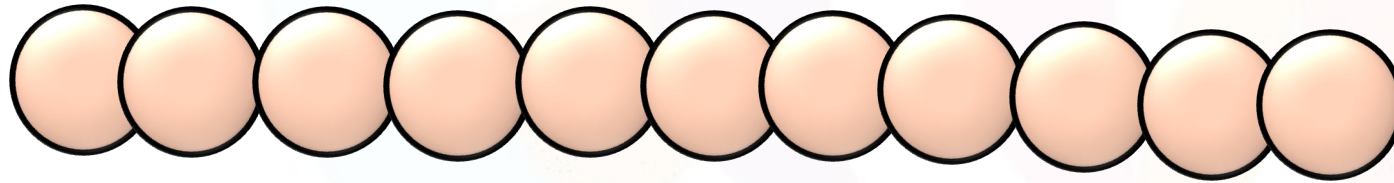
**Polypeptide (多肽) Chain**

### 3. Proteins 蛋白質

**Polypeptide chain 1**

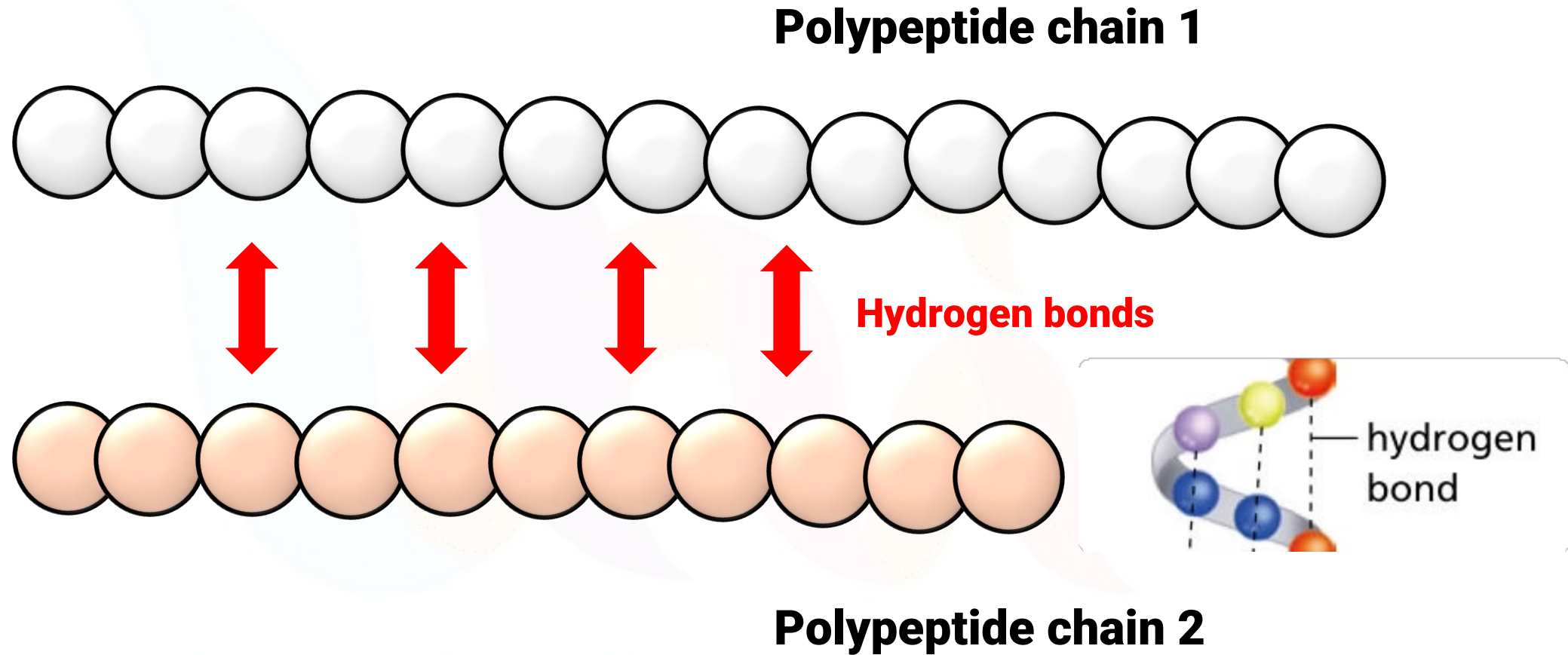


**Polypeptide chain 2**



**Polypeptide (多肽) Chains**

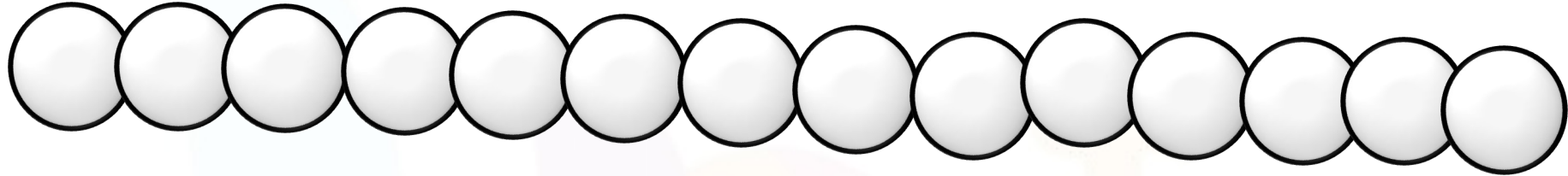
### 3. Proteins 蛋白質



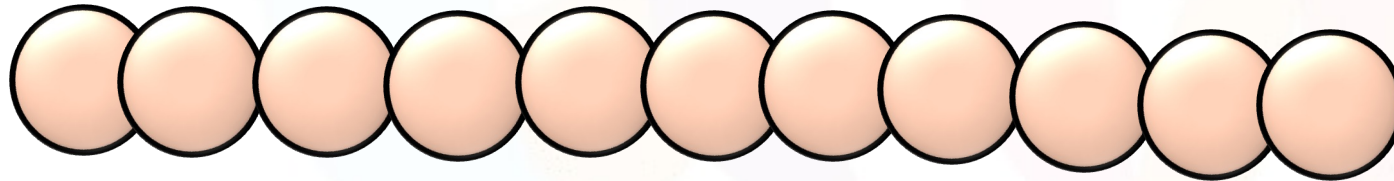
**Polypeptide (多肽) Chains**

### 3. Proteins 蛋白質

**Polypeptide chain 1**

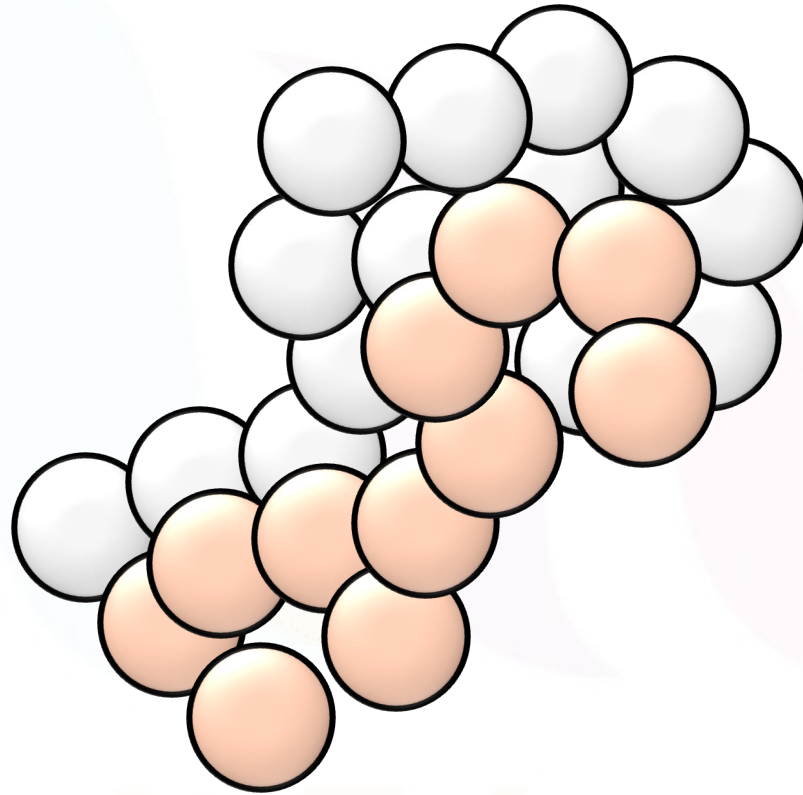


**Polypeptide chain 2**



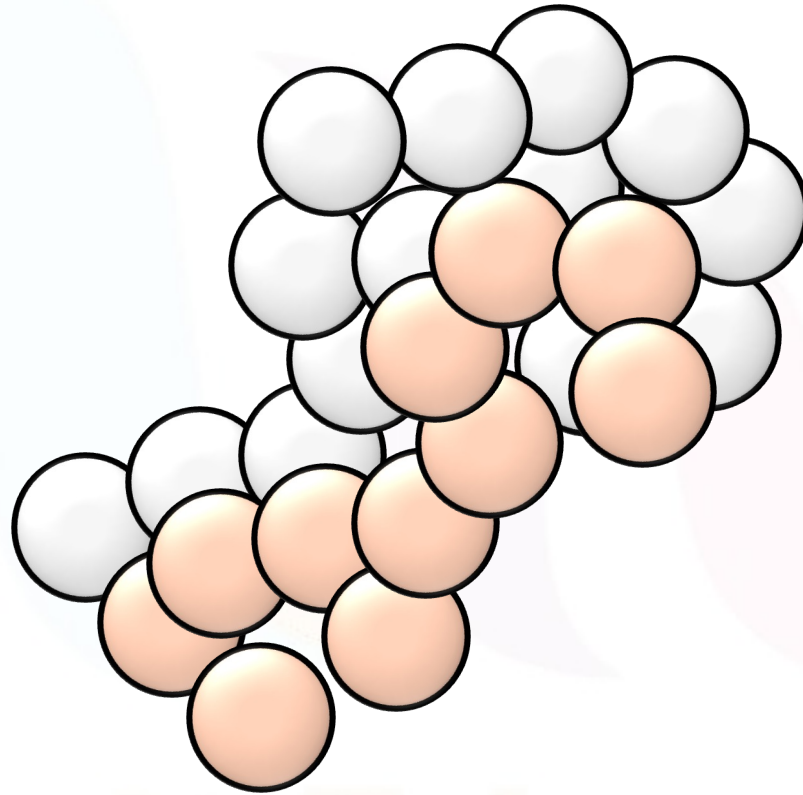
**Polypeptide (多肽) Chains**

### 3. Proteins 蛋白質



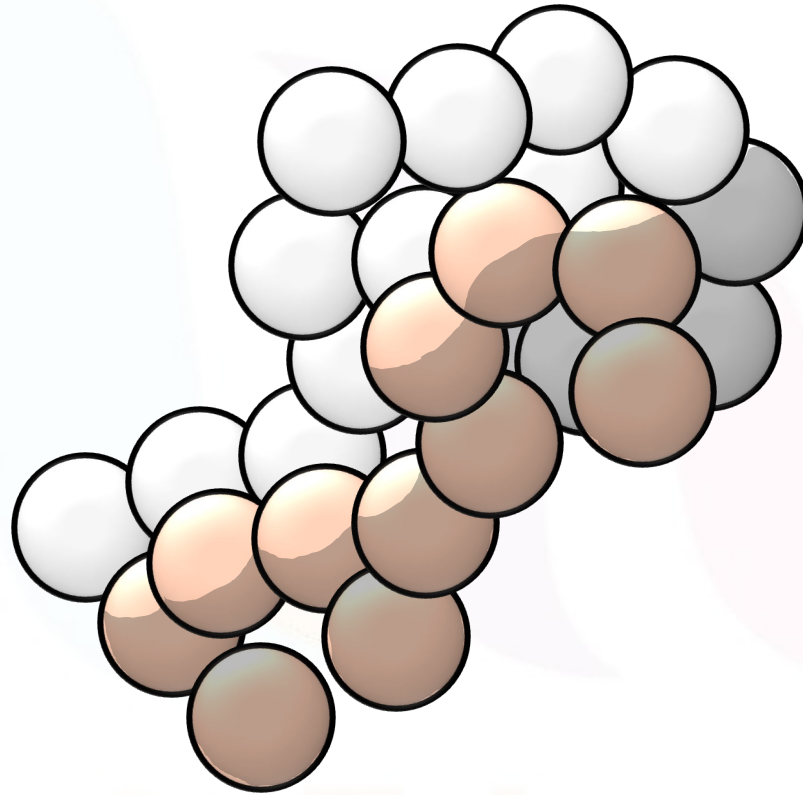
**Polypeptide (多肽) Chains**

### 3. Proteins 蛋白質



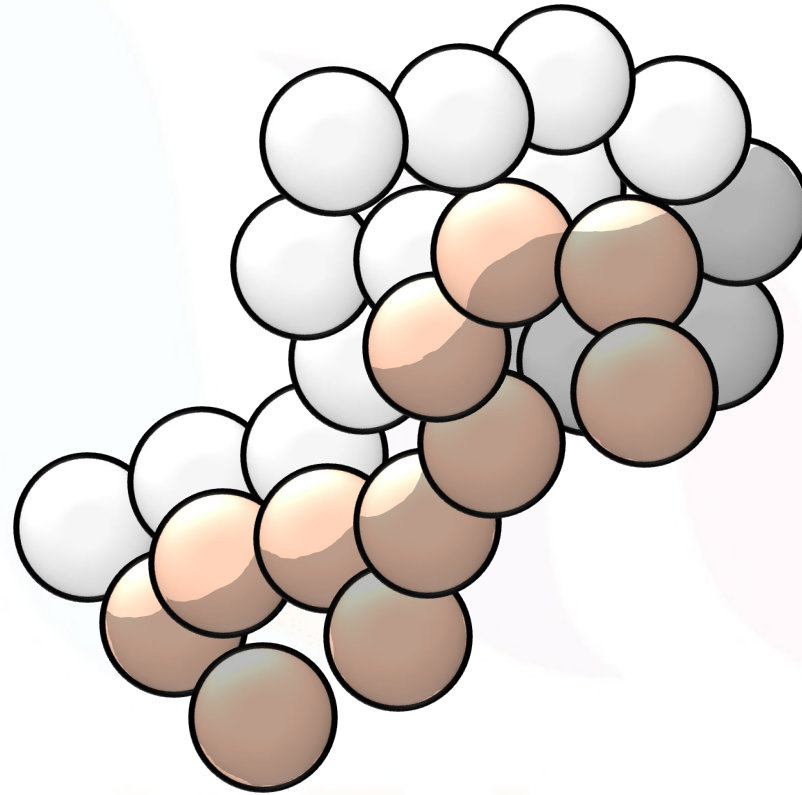
**Proteins 蛋白質**

### 3. Proteins 蛋白質



**Proteins 蛋白質**

### 3. Proteins 蛋白質



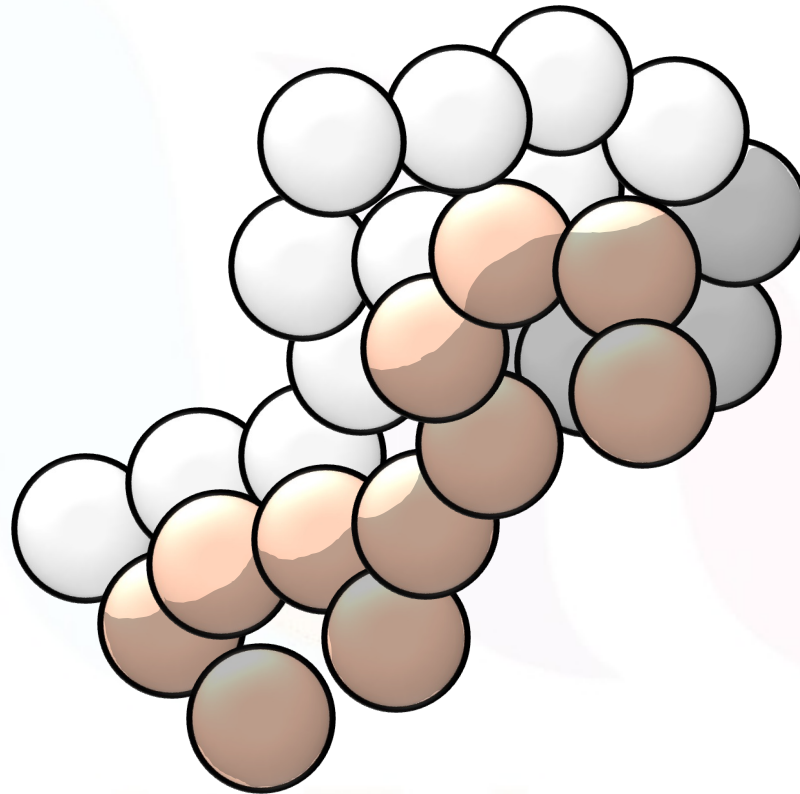
**Foldings give rise  
to 3D conformation**

**Proteins 蛋白質**



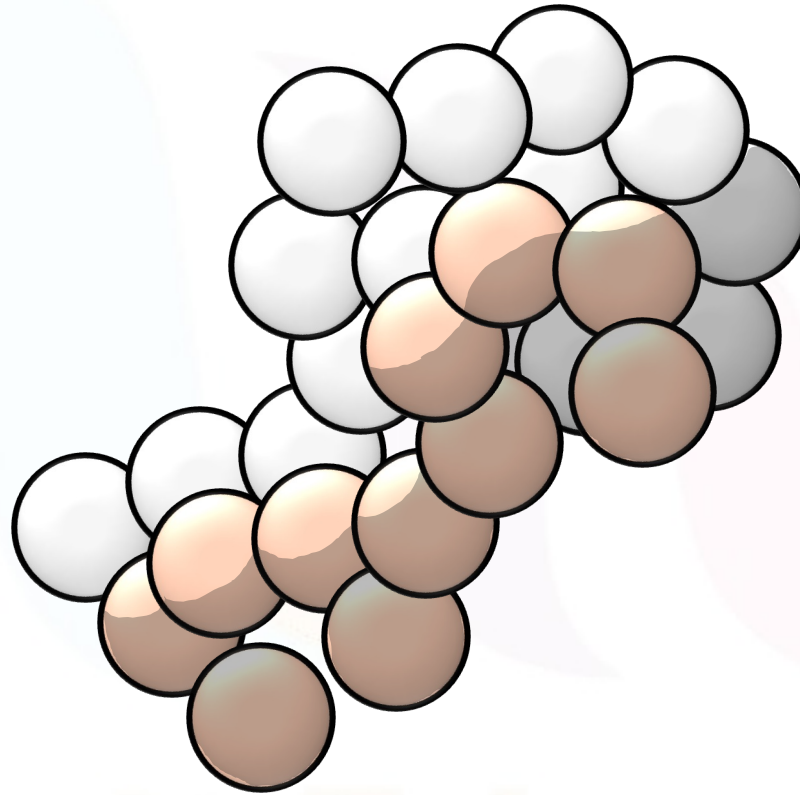
### 3. Proteins 蛋白質

**Foldings give rise  
to 3D conformation**



**Proteins 蛋白質**

### 3. Proteins 蛋白質



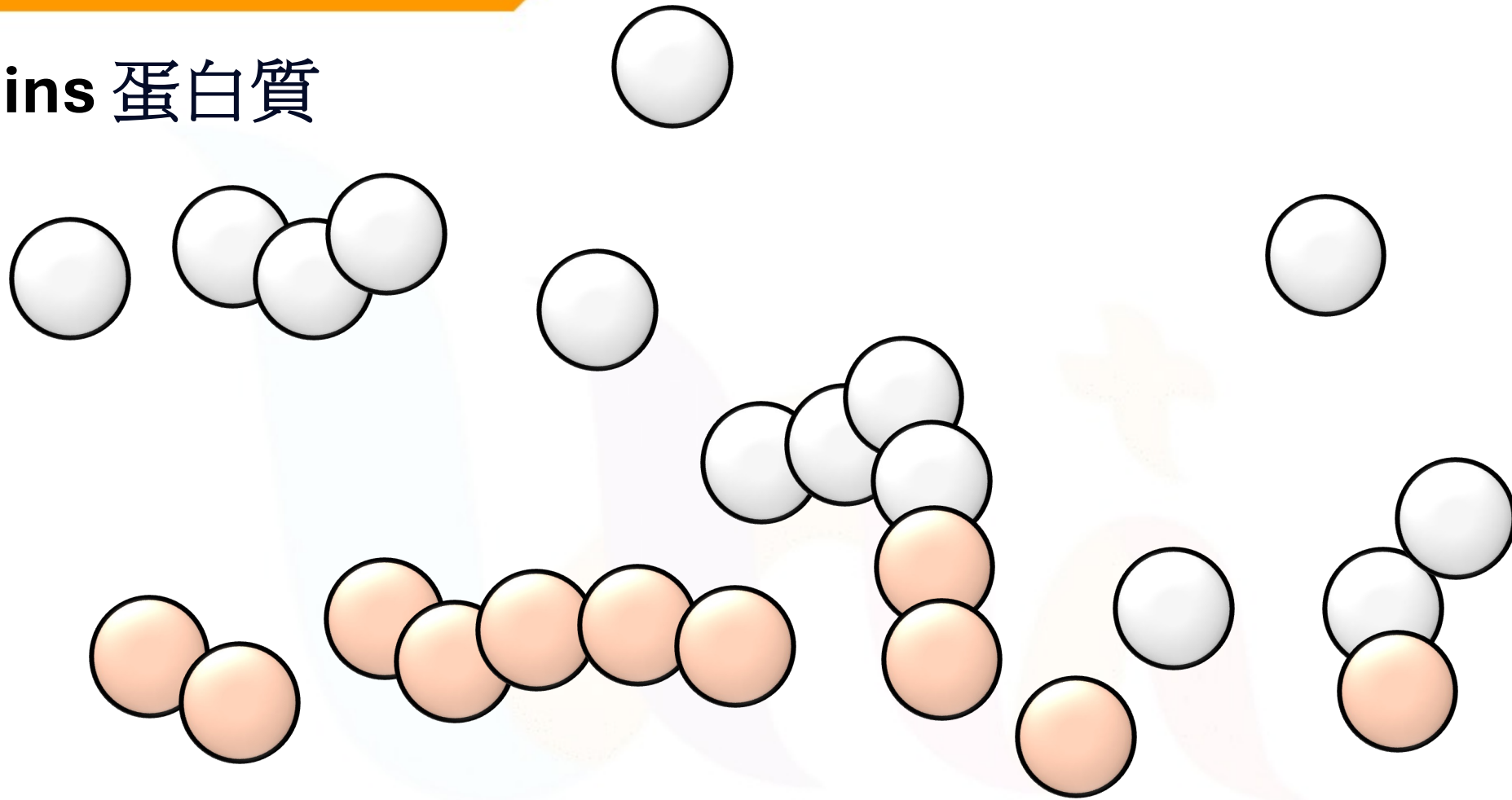
**Foldings give rise  
to 3D conformation**



**3D conformation  
determine protein  
functions**

**Proteins 蛋白質**

### 3. Proteins 蛋白質



**Suboptimal pH and temperature denature proteins**